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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Meaning, Rationale and Evolution

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Course Outline

❑ Module I

- Introduction: Meaning, Rationale and Evolution
- Basic concepts and tools from Micro and Welfare economics
- Comparing Environmental Economics, Ecology and Natural Resource economics

❑ Module II

- Types of Goods: Public; Private; and Environmental
- Problem of Collective Action: Tragedy of the commons, Prisoners' dilemma game and Olson's logic of collective actions

Contd…

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- Poverty-Economic Growth-Environment Nexus
- Environmental Kuznets Curve

❑ Module IV

- Concept of Environmental Sustainability
- Ecological Footprints and Environmental Performance Index

❑ Module V

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- Welfare Effects of Price Change

Contd…

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  - Non-Market Valuation: Revealed Preference Method
- Module VII
  - Non- Market Valuation: Stated Preference Methods
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  - Market- efficiency, Optimality, Consumer and Producer Surplus
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Contd…

❑ Module X

- Externality and market failure, Pigouvian fees

❑ Module XI

- Coase theorem and property right

❑ Module XII

- Environmental Regulation: Command and Control Approach and Market Based Approach

Contents

- Meaning
- Rationale
- Positive versus normative economics
- Evolution of environmental economics

Meaning

- A nascent sub-discipline of economics.
- It is the application of the tools of economics to analyse environmental and resource issues.
- Focuses on:
- How and why people make decisions that have consequences for the natural environment.

Contd…

- How economic institutions and policies can help in ensuring balance between human desires and the ecosystem.
- Impact of economy on the environment.
- The significance of the environment to the economy.
- The appropriate way of regulating economic activity.
- The objective is to achieve balance among environmental, economic, and other social goals.

Rationale

- Provides insights into:
- Why environmental problems exist and what are the best solutions for these problems?
- What are the institutional and economic causes of environmental problems?
- How can we assess the economic value of environmental degradation or improvements?
- What economic incentives or instruments can bring improvement in the quality of natural environment?

Contd…

- Determines the most reasonable trade-offs among various uses of the environment.
- Provides the solution for environmental problems through application of market based models.
- Provides solid foundation for most of the policy measures to address environmental problems.
- Example, should emissions be stopped by regulation or through taxes?

Contd…

- Compares the expected social costs and benefits of policy measures.
- Accommodates the supply constraint of environmental goods, the aspect ignored by traditional economics, to provide the solution for sustainable use.
- Evaluates the scarcity value of environmental goods for cost benefit analysis of environment management projects.

Positive versus normative economics

- Positive economics try to explain what we see around us or ‘what is’.
- Looks into how an economic system actually operates.
- For example, a study to show how the housing market reacts to changes in interest rates.
- Positive environmental economics; how economic factors influence the consumption and production of environmental goods and services.

Contd…

- Normative economics answers “what ought to be”.
- For example, to determine what kind of regulation we ought to adopt for a particular environmental problem.
- Involves value judgments.
- The normative environmental economics applies the principles of welfare economics to determine socially optimum production and allocation of environmental goods.

Contd…

- Addresses the problem of market failure in environmental goods.
- What kind of government intervention is required to correct the public provision of non-market?

Evolution of Environmental Economics

- Classical economists made no explicit mention of environmental issues but their theories provide the basis for the contemporary environmental debate.
- Classical theory highlighted the importance of market as a tool for stimulating growth and innovation.
- Neo-classical economists introduced marginal analysis, which deals with the study of the relationships between small or incremental change in input and output.
- Keynesian economists emphasized on the government intervention and deficit spending.

Contd…

- *Resource for Future* (1952), the first significant think tank started conducting research to improve the environmental and natural resource decisions.
- However, the issue got impetus during the 1960s following the recognition of harmful environmental impacts of expanding industrialization in the developed countries.
- Post-Energy Crisis of 1970s.
- Since 1970s, a number of ‘world views’ have crystallized within environmentalism, providing a rationale for the emergence of environmental economics as a sub-discipline of economics.

Three Major Views on Evolution of Environmental Economics

- A market and technology driven growth process (Julian Simons).
- Limiting the growth (Eco Preservationists).
- Middle path (Sustainable development).

Contd…

❑ Major Landmarks

- **1961**: WWF is created at Morges, Switzerland
- **1968**: UNESCO Biosphere Conference
- **1970**: Earth Day April 21
- **1972**: the Stockholm conference on Human Environment created formal international awareness of the need for maintaining the quality and integrity of the environment
- ***1972****: The Limits to Growth:* The Club of Rome, a group of leaders and thinkers from 40 countries, issues this report written by Donella and Dennis Meadows.

Contd…

- **1992**: The World Bank's the world development report highlighted the links between development and environment.
- **1992**: Agenda 21 of the United Nations conference on Environment and development (earth submit) in Rio de Janeiro.
- **1992:** Convention on Biological Diversity highlighted the need for biodiversity conservation.
- **1997:** Kyoto Protocol: Meeting in December in Kyoto, Japan, delegates to the third meeting of the Conference of Parties to the UN Climate Change Convention adopt a landmark agreement to reduce global emissions of greenhouse gases.

Contd…

- **2002:** The second Earth Summit, held in Johannesburg, was an attempt by the UN to review the progress of the expectations raised in Rio and to reaffirm the commitment of world leaders in continuing to pursue actions towards sustainable development.

- **2009:** Copenhagen Conference on Climate Change.

- **2010:** *Aichi Biodiversity Targets* (duration, 2011-2020), adopted at the 10th meeting of Conference of Parties held in Nagoya, Aichi Prefecture, Japan. This includes 20 targets which are to be achieved till 2020.

Contd…

- **2015:** SDGs, adopted by all United Nations Member States in 2015.

- **2015:** Paris Agreement, an agreement within the United Nations Framework Convention on Climate Change (UNFCCC) which deals with greenhouse gas emission, mitigation, adaptation and signed in 2016.

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- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. *OUP Catalogue*.
- Singh, K., & Shishodia, A. (2007). *Environmental economics: theory and applications*. SAGE Publications India.

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Environmental Economics

Basic Concepts and Tools from Micro and Welfare Economics

Prof. S.P. Singh

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



Contents

- Environmental economics and Economics
- The Economic concepts and environmental economics
- Marginalism
- Consumer’s surplus
- Producer’s surplus
- Externalities
- Subsidies
- Production possibility frontier
- Opportunity Cost
- Carrying Capacity
- Pareto optimality
- Tragedy of Commons

Environmental Economics and Economics

- The present scenario forces to acknowledge that environmental goods and services are also scarce and exhaustible.
- There is a trade-off between environmental goods and other goods.
- Economics is inevitable; It is a science that deals with scarcity.
- The microeconomic dimension of environmental economics deal with:
  - the behavior of individuals or small groups of consumers, polluting firms, and firms in the pollution-control industry.
  - how and why people make decisions that have consequences for the natural environment.

Contd…

- The environmental economics is one of the applied fields of microeconomics.
- Contributes significantly in nonmarket valuation.
- Involves adaption of economic tools to address the questions on environmental issues.
- The macroeconomics deals with the large aggregates of facts such as national income, GDP, Inflation, Money supply and others indicators of overall performance of an economy.
- The macro dimension of environmental economics deals with these questions:

Contd…

- What overall preferences do citizens have with respect to the balance between environmental protection and economic growth?
- Historically, how measures of environmental protection led to the lower economic growth rates?
- How to design environmental regulations to minimize their impacts on growth?
- When the perspective is broadened from growth to human welfare, how do environmental protection measures stack up?

Contd…

- Do the higher rates of growth, i.e. increase in our traditional measures such as GDP, imply greater environmental degradation, or might the opposite be true?

The Economic Concepts and Environmental Economics

❑ Marginalism

- For determining the optimum level of consumption and production of a commodity.
- Looks into the change in an economic activity, or in a system in terms of incremental or marginal costs and/or benefits associated with change.
- Marginal cost measures the change in cost over the change in quantity
- Marginal benefit is similar to marginal cost; it is a measurement of the change in benefits over the change in quantity.

Contd…

- In environmental economics, the marginal analysis can be applied for the solutions of problems like:
  - Whether clean air programme would be economically worthwhile?
  - Whether setting up of a waste treatment plant is economically or socially viable?
- The marginal benefit curve represents the trade off between environmental improvements and other things we could do with the resources needed to gain the improvement.

Contd…

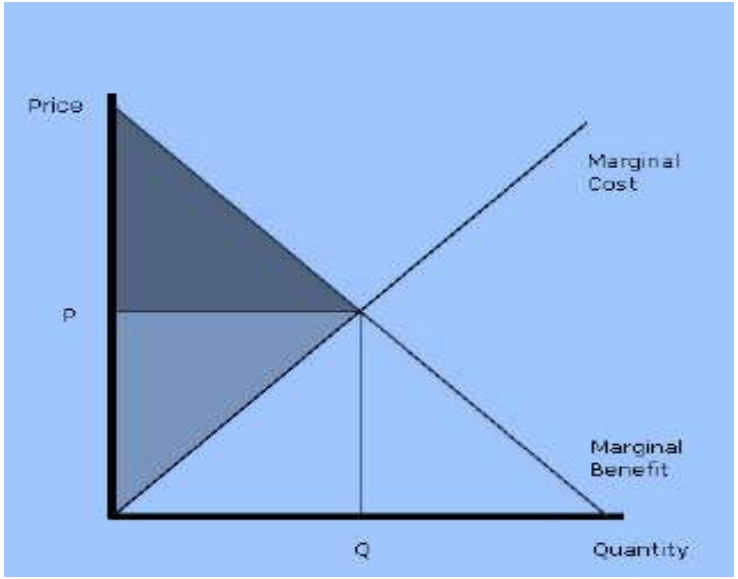


Figure 1: Marginal cost and Marginal Benefit

Contd…

❑ Consumer's Surplus:

- According to Marshal's definition, CS is "the excess of the price which the customer would be willing to pay rather than go without thing, over that which he actually does pay."
- It is imperative in environmental economics for evaluating the environmental goods and services and the impacts of a particular project or activity.
- For non-market goods, the willingness to pay (WTP) is considered as their price.
- WTP is referred to the amount that a consumer would pay for a hypothetical good, service or change in some state of environment.

Contd…

- CS is imperative in environmental economics:
  - For evaluating the environmental goods and services and the impacts of a particular project or activity.
  - For studying the effects of deviations from optimal level of production or consumption.

Contd…

❑ Producer's surplus

- It is the amount of revenue received by a producer of a good in excess of what is necessary to induce him to produce a specified quantity of good.
- Both PS and CS are useful in evaluating social welfare implications of environmental programmes and policies.

❑ Externalities:

- Unintentional and uncompensated side effects of an activity affecting people other than those directly involved in the activity.

Contd…

- Externality can be positive and negative.
- A negative externality is one that creates side effects that could be harmful to either the general public directly or through the environment.
- A positive externality is an unpaid benefit that extends beyond those directly initiating the activity.
- Leads to market failure.
- Almost all environment management projects have externalities, including afforestation, canals and dam construction, have externalities.
- Positive externalities of afforestation by NGO; reduction in soil erosion, increase in groundwater recharge and improvement in micro climate.

Contd…

- In such cases the private benefits of the NGO are lesser than the social benefits but the NGO cannot demand any compensation for it.
- Example of negative externality: canal irrigation.

❑ Subsidies:

- It is a direct or indirect payment by a government to households or firms in the form of grants or other aids.
- Government interventions through taxation, subsidies, private incentives and others leads to policy failure.

Contd…

- Policy failure means the government interventions that distorts the market functions or exacerbates the existing market failure.
- For example, the irrigation water provided to farmers at much lower price without reflecting its scarcity value or opportunity cost leads to over irrigation, soil salinity and water logging.

Contd…

❑ Production Possibility Frontier (PPF):

- PPF represents the point at which an economy is most efficiently producing and allocating its goods and services.
- It shows that there are limits to production.
- To achieve efficiency, an economy must decide what combination of goods and services can be produced.

Contd…

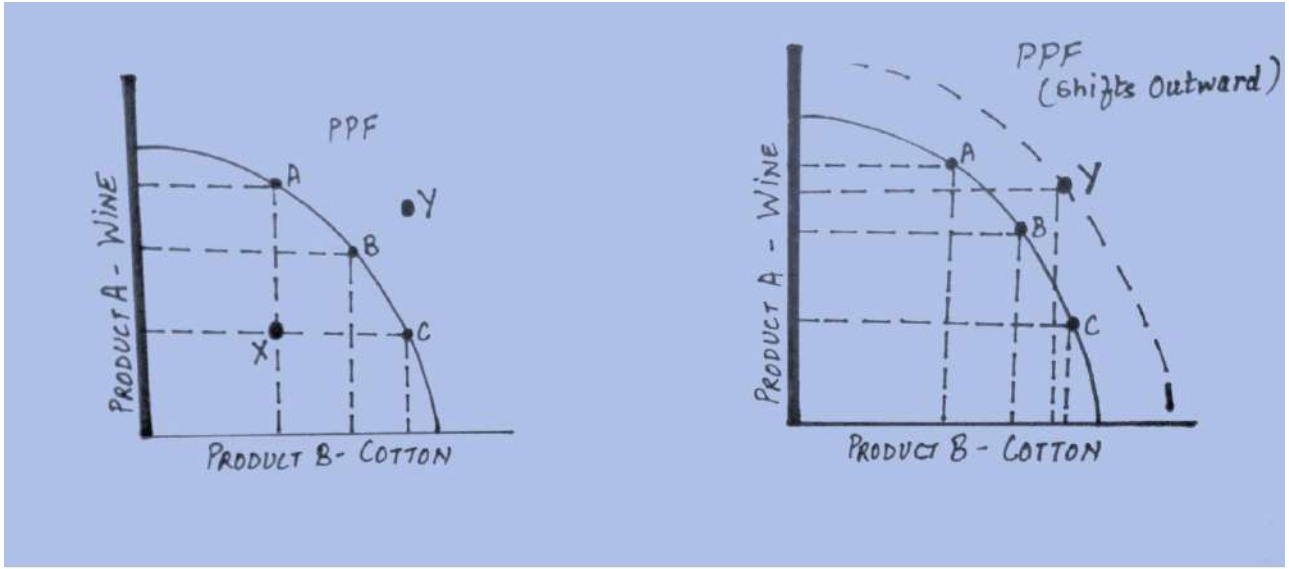


Figure 2: Production Possibility Frontier (PPF)

Contd…

❑ Opportunity Cost

- The Australian School of economics and its followers gave an alternative concept of real cost. According to them, the real cost of production of a given commodity is the next best alternative sacrificed in order to obtain that commodity.
- It is called opportunity cost or displacement cost.
- It can be defined as the value of the resource in its next best use, i. e, if it were not being used for the present purpose.

Contd…

❑ Carrying Capacity

- The number of individuals an ecosystem can support without having any negative effects.
- It can be affected by the size of the human population, consumption of resources and the level of pollution and environmental degradation.
- It is not fixed and can be expanded through good management and development resource-saving technologies.

Contd…

❑ Pareto Optimality:

- It refers to the "best that could be achieved without disadvantaging at least one group”.
- "A given economic arrangement is efficient if there can be no arrangement which will leave someone better off without worsening the position of others."
- Pareto improvement states that it is possible to move from one state to the second state without disturbing the well being of the others.

Contd…

❑ Tragedy of Commons:

- Hardin's tragedy of commons hypothesis explains why CPRs are over exploited, degraded and depleted.
- Fundamental to environmental economics.
- The tragedy arises from the non-cooperation among the co-users of a commons; the problem explained through PD.
- Here tragedy is the unregulated access to a CPR.
- Incremental private benefits far exceeds the incremental private cost.

Contd…

- This irrational consumption of CPRs results in degradation.
- The problem is aggravated by absence of defined property rights.

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- King, E. Singh, K., & Shishodia, A. (2007). *Environmental economics: theory and applications*. SAGE Publications India.
- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. *OUP Catalogue*.

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Environmental Economics

Environmental Economics and Other Sub-disciplines

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Contents

- Topics Covered so far
- Introduction
- Environmental and Traditional Economics
- Environmental and Ecological Economics
- Environmental and Natural Resource Economics
- Environmental and Energy Economics
- Environmental Economics and Environment Science

Topics Covered

- Meaning and rationale of EE
- EE as positive and normative economics
- Evolution of EE as a sub discipline of economics
- Difference in EE and economics
- Basic concepts and tools of economics applied in EE

Introduction

- Economics related sub disciplines:
  - ❑ Natural resource economics;
  - ❑ Environmental economics;
  - ❑ Ecological economics
- Common objectives:
  - ❑ Economic efficiency
  - ❑ Social equity
  - ❑ Ecological sustainability
- Similar intentions but different approaches and directions.

Contd…

- Natural resource economics matured as a discipline to deal with whether there are limits to economic growth and if so, what determines these limits.
- Pigou (1932) pointed out how externalities could add to economic scarcity and reduce economic welfare.
- Environmental economics evolved to deal with externalities, non-marketed goods & services, production & consumption efficiencies, market failures, intra and inter-generational equities and policy responses.
- Ecological economics evolved as a holistic approach stressing interdependence between economic activity, the natural world and social systems (interaction among ecosystem, economy and social systems).
- Ecological economists did not reject completely the type of microeconomic analysis used in environmental economics but emphasized its limits.

Environmental and Traditional Economics

Parameters	Environmental economics	Traditional economics
Target	Interrelationship and interaction between environment. and economic activities	All the economic activities to deal with scarcity
Goods dealt	Public or collective goods	Private goods
Externalities	Major concern	No association
Equity	More attention on Intergenerational equity	Less focus on time related decisions (intra-gen. equity
Environment concern	Explicit concern on limited environment capacity	Environment is secondary

Environmental and Ecological Economics

- Ecology: two Greek words—*Oikos*, meaning home or habitat and *Logos*, meaning study.
- It is a study of relationships and interactions of living organisms and its surrounding environment.
- Environment: a anthropocentric concept, which allows some primacy to mankind and put 'man' at the centre of complex set of relationship with other living and non-living objects.
- It has two components biotic (plants, animals and micro organisms) and abiotic (soil, sunlight, topography, water, atmosphere, nutrients etc.).
- Ecosystem is a functional unit. It includes environment (biotic+ abiotic) and ecology (how living organisms interact with each other and the environment).
- Ecosystem= Ecology+ Environment (such as river ecosystem, forest ecosystem)

Contd.

- The distinction between the two arises from the fact that:
  - Environmental economics tends to involve economists who have extended their discipline and paradigm to consider the environment.
  - Ecological economics tends to involve ecologists who have extended their discipline and paradigm to consider human and economy.
  - Environmental economics applies the theory, assumptions and paradigms of neoclassical economics to examine environmental issues.
  - Ecological economists measure the value in terms of energy content.
  - Ecological economics recognises natural environment materially finite and economy the sub set of this natural environment.
  - Environmental economics considers natural environment an important component of the economic system.

Environmental and Ecological Economics

Parameters	Environmental Economics	Ecological Economics
Approach	Intra-disciplinary	Interdisciplinary
Values of Goods	Energy, land, skilled labour, etc.	Energy Theory of Value
Evaluation Criterion	Cost effectiveness or efficiency	Multidimensional criterion: basic need, welfare, and poverty and environment linkages
Natural System	Natural environment as an important component of economic system	Economic system as a part of natural system
Interchangeability	Assumes environmental and man made goods are interchangeable	Assumes environmental and man made goods are not interchangeable

Environmental and Natural Resource Economics

Parameters	Environmental economics	Natural resource economics
Concern	Excessive pollution and over exploitation of resources due to market failure	Production and allocation of natural resources
Nature	Both static and dynamic	Dynamic
Major factor	Market	Time
Focus	Internalise the externalities; economic valuation of environment goods	Optimum production and use of natural resources
Objective	Sustainability plus valuation and qualitative management	Sustainability plus qualitative management

Environmental and Energy Economics

Parameters	Environmental Economics	Energy Economics
Commodities	All the environmental goods and services	Energy commodities and resources only
Scope	Broader concept, covers all the economic-environment related issues	Narrower concept, covers only energy related issues Sub-set of environmental economics
Objective	Internalize the externalities Optimize the impacts of economy on environment	Economically efficient use of energy commodities

Environmental Economics and Environmental Science

- Environmental Science:

- ❑ Interdisciplinary;
- ❑ Studies the interactions of physical, chemical, and biological components of the environment;
- ❑ Examines the relationships and effects of these components with the organisms in the environment;
- ❑ Studies environmental problems and human impacts on the environment;
- ❑ It is more of quantitative in nature. (For example, quantifying the impact of SO2 on acid rain)

References

- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. *OUP Catalogue*.
- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. *Sustainable Human Development Review*, *105*.

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Environmental Economics

Study Specific in Environmental Economics, Major Problems and Key Concerns

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Contents

- Study Specific in Environmental Economics
- Ten Major environmental Problems
- Key Concerns for environmental economists

Study Specific in EE

- Externalities
- Cost benefit analysis of environment projects
- Cap and trade system (it allows producers to trade emissions allowances under an overall cap, or limit, on those emissions)
- Economics of waste management
- Valuation of natural resources
- Carbon credit system
- Command & Control Versus Market-based Approach for environment protection and conservation

Ten major Environmental Problems

- Acid rain & regional scale air pollution
- Ozone depletion by CFCs and other industrial and agricultural chemicals
- Global warming & climate change due to GHGs in the atmosphere
- Deforestation
- Land degradation
- Fresh water pollution & scarcity
- Marine threats, including over fishing
- Decline in biodiversity and ecosystem services
- Excessive nitrogen production and over-chemicalisation of agriculture

Key Concerns for Environment Economists

- Inter-generational and intra-generational equity
- Development versus environment
- Institutions for natural resource management
- Community based versus market oriented regulation
- Ownership rights of the natural resources
- Population growth and environment
- Measurement of demand of non-market goods & Services
- Trade and Environment
- Climate Change as public bad (ecological footprint—US 12.22 ha; India: 0.9 ha, China 1.84 ha)

References

- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. *Sustainable Human Development Review*, *105*.
- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. *OUP Catalogue*.

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COMMONS AND COLLECTIVE ACTION PROBLEM: SEMINAL THEORIES-I

DIPTIMAYEE NAYAK

HUMANITIES AND SOCIAL SCIENCES



Contents

- Typology of Goods
- What is Collective Action Problem?

- Some Seminal Theories on Collective Action:
  - Hardins' thesis of the Tragedy of the Commons
  - Prisoners' Dilemma Game and
  - Olson's Logic of Collective Actions and Group Behavior
  - Ostrom's Governing Commons

Typology of Goods

- Public Goods: Non-excludable and non-rival

  Non-excludability: may be defined as the cost of keeping non-payers from enjoying the benefits of the good/service is too high that no one can be excluded from using the good.

  Non-rivalrous: it may mean consumption of a good by one individual does not reduce availability of that good for consumption by others.

- Example: Light house, national defence, knowledge, environmental goods-biodiversity, fresh air in the atmosphere

Contd…

- Pure Public goods: the quality and quantity will not be diluted or sacrificed. They are perfectly non-rivalrous in consumption and non excludable. For example- light house or national defence.

- Impure Public goods: that satisfy non excludability and no rivalrous conditions to some extent, not fully. Example- access to public road during any festive occasion and searching for an alternative route.

Contd…

- Private Goods: having ownership right/property right- Excludable and rival

  Example: Burger, Book, Shoes, house etc.

- Moreover, we can also identify the nature of the good whether public or private by the level of competition for the good and the possibility of excluding its consumption

- Pure Private Goods: which display the characteristics of excludability and rivalry

Contd…

- Common Goods/Collective Goods: They are non-Excludable but rival (to some extent)

  Example- Water, Air, Fishery, Forestry

- The first two goods can be polluted by so many sources like industries, vehicles etc.

- The last two common goods can be sustainably used through proper institutional/social arrangements; when common goods are used by institutional arrangement for their sustainable use, they are called as Common Pool Resources.

Contd…

- Based on institutional arrangement regarding common property rights, Common pool resources can be of three types, Viz., Common Property Resources, Private resources, Open Access Resources

- Common Property Resources: if common pool resources are owned and manged by any communal groups, they are known as Common Property Resources.

- Non-excludability within community and rival in consumption

- Examples: Village forests, pastures, ponds

Contd…

- If these resources are owned by no one, these are known as Open Access Resources.

- Example: fishery in the oceans or sea.

Contd…

- Club Goods: Excludable but non-rivalrous

  Example- toll road (generally non-congested), cable TV, social religious clubs, cinemas

Typologies of Goods in a Nutshell

Rivalrous consumption	Excludability: Yes	Excludability: No
Yes	Private Goods	Common Pool Resources
No	Club Goods	Public Goods

Collective Action

- By Collective Action we may mean

  any form of organised behaviour or actions of individuals in a group to work together for achieving their needs which is known as common group objectives .

What is a Collective Action Problem?

- A collective action problem may be defined as

  a situation in which all the participating individuals would be better off if they cooperate with each other for achieving a common/group objective; however they do not cooperate because of their own individual conflicting interests and hence they fail to achieve the common objective.

- They do not perform joint action because there exists a conflict between group interest and of their own vested interests.

Seminal Theories on Collective Action

- Olson's Logic of Collective Actions and Group Behavior (1965)
- Hardins' thesis of the Tragedy of the Commons (1968)
  - Analytical framework to explain the Tragedy of the Commons-
    - Prisoners' Dilemma Game by Merrill Flood and Melvin Dresser (1950)
    - Theory of Public Goods- 'The Pure Theory of Public Expenditure' by P A Samuelson (1954)

- Elinor Ostrom's Governing Commons and the Evolution of Institutions for collective Action (1990)

Selected Readings

- Hardin, G. (1968). The tragedy of the commons. *Science*, 162 (3859): 1243-1248.

- Hardin, Rachael. (1982). Collective Action. RFF., Chapter 2, 'Collective action and Prisoners' Dilemma'.

- Musgrave, R.A. and Musgrave, P. B. (1973). Public Finance in Theory and Practice. New York: McGraw Hill, First edition.

- Olson, Mancur. (1965). The logic of collective action- Public goods and the theory of groups. Cambridge: Harvard University Press.

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COMMONS AND COLLECTIVE ACTIONS PROBLEM: SEMINAL THEORIES-II

DIPTIMAYEE NAYAK

HUMANITIES AND SOCIAL SCIENCES



Contents

- Recapitulations of previous lessons
- Some Seminal Theories on Collective Action:
    - Hardins' thesis of the Tragedy of the Commons

Recaps

- Typology of goods
  - Modification to classification of goods by Vincent Ostrom and Elinor Ostrom. The modifications are-
    - Replacement of the term 'rivalry of consumption' with 'subtractability of use'
    - Instead of merely saying presence or absence of the attributes (excludability and subtractability of use) in a good, level from low to high can be assigned for the good
    - Addition of fourth types of good- i.e., common pool resources (subtractability and difficulty in exclusion)
    - Changing the name of 'club good' to 'toll good'

Issues of Typology of Goods

- Problems of Pure Public goods:
  - Market Failure, Externalities and lack of well defined Property Right
  - Market failure: A situation when market fails to provide efficient distribution of goods and services in the free market.
  - Externalities: Actions of one person benefits/costs another person, not at all related to the actions of first person, can be termed as Externalities.
  - Market fails because of the attributes (non-excludability and non-rivalry) and lack of property rights

Features of Property Right

- Exclusivity (no externalities)
- Enforceability (secured for involuntary seizure and encroachment)
- Transferability
- Comprehensibility (property rights be owned privately or collectively)
- Universality

Contd…

- Although there are many theories on each types of the goods, we will discuss seminal theories on Common/ collective actions.

Tragedy of Commons

- Garrett James Hardin (1968)
- An American Ecologist and Philosopher
- Revolves around the ‘Problems of Population Explosion’-which is said to be a ‘no technical solution problem’.
- Example- Dilemma of military power and national security (Wiesner and York, 1964)

Contd…

- Thomas Robert Malthus
- Jeremy Bentham, and
- Adam Smith

The Population Problem as a Metaphor of Commons

- Problem: Population naturally tends to grow geometrically (exponentially)
- Consequences: *Per capita* share of the world's goods must steadily decrease in the finite world.
- Solution: Technology?? Can technology solve this problem?
- Constraint: A finite world can support only a finite population.

- Conclusion deduced: And therefore, population growth must eventually equal zero.
- But However, in practical world this scenario is not found. Why??

Contd…

- If population growth is zero, what would happen to Jeremy Bentham's goal of welfare/utility maximisation 'the greatest good for the greatest *number*'?
- Two reasons for not achieving this Utilitarianism goal are:
  - It is not mathematically possible to maximise for two or more variables at the same time. (For example- D'Alembert's the theory of partial differential equations)
  - Biological facts

    Maximum good per person? What is good? Which one?

    Moreover, the goods are incommensurable (estuaries with factory lands??)

Contd…

- Adam Smith (1776, Wealth of Nations) contributed to a dominant tendency of thought that the tendency to assume that decisions reached individually will in fact, be the best decisions for an entire society.
- If this assumption is correct, laissez-faire in reproduction can be continued and we can assume that human being will control their fertility to produce optimum population.
- But, if the above assumption is not correct, we need to re-examine our individual freedom of reproduction.
- This may be termed as tragedy of Freedom in the commons to reproduce.

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- Musgrave, R.A. and Musgrave, P. B. (1973). Public Finance in Theory and Practice. New York: McGraw Hill, First edition.

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COMMONS AND COLLECTIVE ACTIONS PROBLEM: SEMINAL THEORIES-III

DIPTIMAYEE NAYAK

HUMANITIES AND SOCIAL SCIENCES



Contents

- Hardins' thesis of the Tragedy of the Commons (Cont)…

Population Problem as a Commons

- Everyone is allowed to breed, no one is excluded from the freedom to breed (non excludability).

- As a result, the per capital share of world's goods would decrease steadily in a finite world (rivalrous).

Tragedy of Freedom in a Commons

- Scenario of the Tragedy of the Commons narrated by English mathematician and economist William Forster Lloyd, in 1833.
- Picture a pasture open to all. It is to be expected that each herdsman will try to keep as many cattle as possible on the commons.
- After a point of time, the very practice generates tragedy.

- As a rational being, each herdsman seeks to maximise his gain.

- The question that he asks himself is 'What is the utility to me of adding one more animal to my herd?'

Contd…

- This utility has one negative and one positive component.

- The positive component is a function of the increment of one animal. Since the herdsman receives all the proceeds from the sale of the additional animal, the positive utility is nearly +1,

- Whereas, the negative component is a function of the additional overgrazing created by one more animal. Since, however, the effects of overgrazing are shared by all the herdsmen, the negative utility for any particular decision making herdsman is only a fraction of -1.

Evaluation of Benefits and Costs

- In this situation, now the rational herdsman concludes that the only sensible course for him to pursue is to add another animal to his herd., And another; and another….. This goes on.

- But this is the conclusion reached by each and every rational herdsman sharing a commons. And this is the tragedy.

What is Tragedy Here?

- Each herdsman is locked into a system that compels him/her to increase his/her herd without limit- in a world that is limited.
- Ruin is the destination towards which all herdsmen rush, each pursuing his own best interest in a society that believes in the freedom of the commons.
- In this process, Freedom in a commons brings ruin to all.
- Examples; Oceans, the national parks- open to all without limit.

Pollution as the Tragedy of the Commons

- The tragedy of the commons is also noticed in problems of pollution.
- However, the situation is bit different here from our earlier examples.
- Now, it is not a question of taking something out of the commons, but of putting something in- sewage, chemical, radioactive and heat wastes into water; noxious and dangerous fumes into the air, etc.

Contd….

- What is the share of benefits and costs?
- The rational man finds that his share of the wastes he discharges into the commons is less than the cost of purifying his wastes before releasing them.
- Since this is true for everyone, we are locked into a system of ‘fouling our own nest’, so long as we behave only as independent, rational, and free-enterprisers.

How to Fix the Problem?

- The most common pollution - the air and water surrounding us cannot be fenced

- Then how to fix these problems?
  - By coercive laws (Command and Control mechanisms)
  - Taxing devices that make it cheaper to treat his pollutants rather than to discharge them untreated (Economic Incentives)

- Pollution problem = f (population growth)
  = f(population density)
  = f (property right)

How to Legislate Temperance?

- Pollution problem should be a function of morality.
- What is morality of an act?

It is a function of the state of the system at the time it is performed.

- For example, using the commons as a cesspool does not harm the general public under frontier conditions, because there is no public; but the same behaviour in a metropolis is unbearable.
- Killing a bison hundred years ago

- So morality is system-sensitive escaped the attention of most codifiers of ethics in the past.

Contd…

- The laws of our society follow the pattern of ancient ethics, and therefore are poorly suited to governing a complex, crowded and changeable world.

- Prohibition is easy to legislate though not necessarily to enforce.

- Sometimes we face such situations that lead us to think- who will guard the guards themselves?

- Thus, the challenge is to invent the corrective feedbacks that are needed to keep custodians honest.

Freedom to Breed is Intolerable

- If over-breeding brings its own punishment like starvation or death, then there would be no public interest in controlling the breeding of families.

- But in a welfare state, it does not happen and the tragedy of the commons happens.
- The concept of freedom to breed with the belief that everyone born has an equal right to the commons is to lock the world into a tragic course of action. Eg.

  The Universal Declaration of Human Rights describes the family as the natural and fundamental unit of society. It follows that any choice and decision with regard to the size of the family must irrevocably rest with the family itself, and cannot be made by anyone else.

Conscience' is Self-eliminating And Has Pathogenic Effects

- "It is a mistake to think that we can control the breeding of mankind in the long run by an appeal to conscience."
- The argument assumes that conscience or the desire for children is hereditary.
- This argument applies equally well to any instance in which society appeals to an individual exploiting a commons to restrain himself for the general good- by means of his conscience.
- To make such an appeal is to set up a selective system that works toward the elimination of conscience from the race.

Contd…

- ‘A bad conscience is a kind of illness.’
- Appeal to conscience has long-term and short-term disadvantages.
  - Example: if we ask a man who is exploiting a commons to desist ‘in the name of conscience’, what are we saying to him? What does he hear?
  - He has received two communications, which are contradictory.
    - Intended communication: ‘if you don’t do as we ask, we will openly condemn you for not acting like responsible citizen’
    - The unintended communication: ‘if you do behave as we ask, we will secretly condemn you for a simpleton who can be shamed into standing aside, while the rest of us exploit the commons.’

Then what is the solution?- Mutual Coercion??

- Mutual coercion- mutually agreed upon by the majority of the people affected.

  Example: compulsory taxation

  - We institute and grumblingly support taxes and other coercive devices to escape the horror of the commons.
- An alternative to the commons need not be perfectly just to be preferable.

  Example: private property right with legal inheritance.

- We must admit that our legal system of private property plus inheritance is unjust- but we put up with it because we are not convinced, at the moment, that anyone has invented a better system.
- As alternative of the commons is horrifying to contemplate, therefore, injustice is preferable to total ruin.

Recognition of Necessity

- Educating all people about the tragedy of the commons, its consequences, and the alternatives to it, is necessary.
- Then various restraints and incentives for low reproduction can and must be instituted.
- "Freedom is the recognition of necessity- and it is the role of education to reveal to all the necessity of abandoning the freedom to breed. Only so, can we put an end to this aspect of the tragedy of the commons".

Selected Readings

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MANCUR OLSON'S THEORY OF COLLECTIVE ACTION

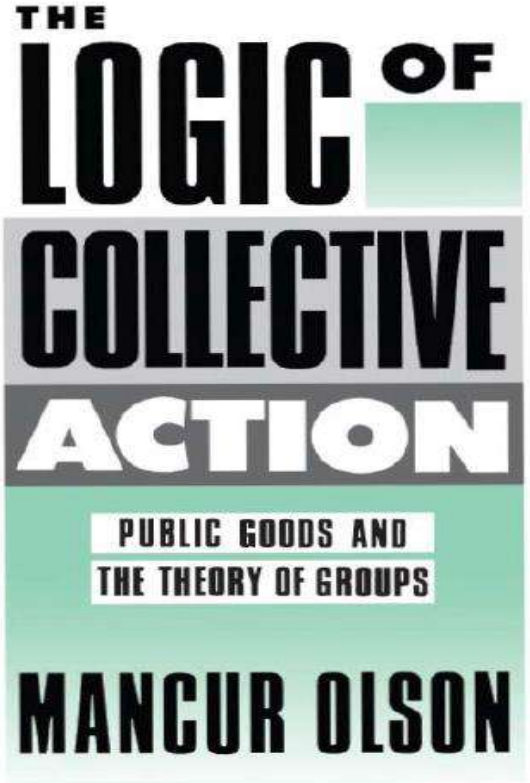
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Collective Action Theory: Mancur Olson

- Mancur Olson (1965)
- Trained in Economics and the founder of IRIS (Institutional Research and the Informal Sector) at the University of Maryland.
- Three Major Publications:
  - *The Logic of Collective Action* (1965)
  - *The Rise and Decline of nations* (1982)
  - *Power and Prosperity* (2000)



The Logic of Olson's Theory

- This piece of work is an application of economic reasoning to group behaviour .

- The theory emphasises that we need to perceive 'individual behaviour' differently from 'group behaviour' even though all the members of the group would get benefit out of a collective activity.

Popular Perceptions about Group Behaviour

- When economic objectives are involved, groups of individuals with common interests, usually attempt to further those common interests.
- Example: labour unions, in Marxian theories of class action, in concepts of countervailing power and in various economic institutions.



Contd…

- Olson states, however, that this is generally not the case.

- For the first time, he argued opposite to the conventional established characteristics of group behaviour that people having common interests automatically come together to form a group for achieving a common goal by the instinct of self-organisation.

Contd…

- In the very introduction of the book, he cited the example of labour union theories to state the common perception hold about group behaviour.

- “It is often taken for granted, at least when economic objectives are involved, that the groups of individuals with common interests usually attempt to further those common interests” (p.1).

- However, it is found to be very exceptional that individuals in a group disregard their personal welfare altruistically.

Contd…

- The assumption behind this view, is that individuals in groups act out of self-interest.

- If the individuals in a group altruistically disregarded their personal welfare, it would not be very likely that collectively they would seek some selfish common or group objective.

- However, such altruism is considered exceptional, and self-interested behaviour is usually thought to be the rule, at least when economic issues are at stake.

Contd…

- If, at all, groups tend to act in support of group interests is based on the logic that it follows from the premise of individual's rational and self-interested behaviour.

- Olson's Answer to the conventional Belief on Group Behavior:
- "But it is *not* in fact true that the groups will act in their self-interest follows logically from the premise of rational and self-interested behaviour." (p.2)
  - Rational and self-interested individuals will not act voluntarily to achieve their common or group interests.
  - Why??

Olson's View/Remark

- "It does *not* follow, because all of the individuals in a group would gain if they achieved their group objective, that they would act to achieve that objective, even if they were all rational and self-interested.

- Indeed, unless the number of individuals in a group is quite small, or unless there is coercion or some other special device to make individuals act in their common interest, *rational, self-interested individuals will not act to achieve their common or group interests*." (p.2)

Contd…

- Unless the number of individuals in a group is quite small,
- or unless there is coercion
- or some other special device to make individuals act in their common interest,
- rational, self-interested individuals will not act to achieve their common or group interest.

Contd…

- Why?
- “The larger the group, the less it will further its common interests” - Free Rider Problem.
- “The amounts of the collective good that a member of the group receives free from other members, will further reduce his incentive to provide more of that good at his own expense”.

- Olson’s argument of ‘Free Rider Problem’ become a central concern in public policy, political science or economic theory of politics and political economy.

The Main argument

- "The larger the group, the farther it will fall short of providing an optimal amount of a collective good" due to the free rider problem.

- "In small groups with common interests there is accordingly a surprising tendency for the "exploitation" of the great by the small".
  - It means that the small groups tend to devote too few resources to the satisfaction of their common interests and there is a surprising tendency for the 'lesser' members of the small group to exploit the 'greater members by making them bear a disproportionate share of the burden of any group action.

Group Size and Behaviour

- In Large Groups each member gets proportionally smaller benefit from a collective good.
  - the individual contributes less because of a smaller return.
  - One member of a large group is less likely to act for all, unless the individual return is more than its cost of action.
  - Need for coercion or some enforcers to provide a collective good.

Contd…

- As there is a tendency for large groups to fail to provide themselves with any collective good at all, so there is a tendency in small groups toward a suboptimal provision of collective goods.

- The larger the group, the farther it will fall short of providing an optimal amount of COLLECTIVE GOOD.

Contd…

- In Small Groups, each member gets proportionally more benefit from a collective goods than one of the larger groups.

  - The collective good provided by the voluntary, rational, and unilateral action of one or two members who find that their reward for providing the good is more than the costs they pay.
  - Since others pay the costs, some members of small groups see no incentive to provide the good and it leads to end of the exploitation by the small.

Group Size and Behaviour

- On the basis of size of the group, in small groups, there are two separate types of non-market groups .
  - Privileged groups
  - Intermediate groups

Contd…

- A privileged group: is a group where each of its members, or at least some of them, has an incentive to see that collective good is provided, even if he has to bear the full burden of providing it himself.

- In such a group, there is a presumption that the collective good will be obtained, and it may be obtained without any group organisation or coordination.

Contd…

- An intermediate group: is a group in which no single member gets a share of the benefit sufficient to give him an incentive to provide the good himself,

- but which does not have so many members that no member will notice whether any other member is or is not helping to provide the collective good.

- In such a group, a collective good may or may not, be obtained, but no collective good ever be obtained without some group coordination or organization.

Contd...

- Latent group or large groups: have no incentive to act to obtain a collective good because, however valuable the collective good might be to the group as a whole, it does not offer the individual any incentive to pay dues to any organisation working in the latent group's interest, or to bear in any other way any of the costs of the necessary collective action.
- Only a separate and selective incentive will stimulate a rational individual in a latent group to act in a group-oriented way.
- In such circumstances group action can be obtained only through an incentive that operates, not indiscriminately, like collective good, upon the group as a whole, but rather selectively toward the individuals in the group.

Contd…

- The incentive must be selective so that those who do not join the organisation working for the group's interest, or in other ways contribute to the attainment of the group's interest, can be treated differently from those who do.

- Mobilised Latent group: the latent group that has led to act in its group interest, either because of coercion of the individuals in the group or because of positive rewards to those individuals is called as a Mobilised latent group.

Economic and Social Incentives

- Small Group or Privileged and Intermediate Groups may lead to economic and social incentives which lead their members to obtain a collective good.

- Large and latent groups lead to no incentive and no social pressure lead their members to obtain a collective good.

- Demand for institutions

Contd…

- Size is one of the determining factors in deciding whether or not it is possible that the voluntary, rational pursuit of individual interest will bring forth group-oriented behaviour.

- Small groups are not only quantitatively, but also qualitatively different from large groups, and that the existence of large associations cannot be explained in terms of the same factors that explain the existence of small groups.

Effectiveness of Small Group

- ‘When the number of participants is large, the typical participant will know that his own efforts will probably not make much difference to the outcome, and that he will be affected by the meeting’s decision in much the same way no matter how much or how little effort he puts into studying the issues’.

- ‘Action taking’ groups and subgroups tended to be much smaller than ‘non-action taking’ groups and sub-groups. ‘Committees should be small when you expect action and relatively large when you are looking for points of view, reactions, etc’.

Social Incentives and Rational Behaviour

- Economic incentives are not the only incentive.

- People sometimes also motivated by a desire to win prestige, respect, friendship, and other social and psychological objectives.

Contd…

- What are the determinants of performing a successful group action?
  - The size of the group
  - The mechanisms of selective incentives

Selected Readings

- Olson. M. (1965). The logic of collective action: Public goods and the theory of groups.

- Czech. S. (2016). Mancur Olson's collective action theory 50 Years Later: A view from the Institutionalist perspective, *Journal of International Studies*, Vol. 9, No 3, pp. 114-123. DOI: 10.14254/2071-8330.2016/9-3/9

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COLLECTIVE ACTION AND PRISONER'S DILEMMA GAME

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Prisoner's Dilemma Game

- Two major analytical traditions that contributed to the understanding of Collective action are
  - The theory of Public goods (P A Samuelson)
  - Game theory (Jon von Neumann and Oskar Morgenstern, 1944)
    - Prisoners' Dilemma (Merrill Flood and Melvin Dresser, around 1950)

What is Prisoners' Dilemma?

- Prisoners' dilemma is a standard example of a game analyzed in game theory that shows why two completely "rational" individuals might not cooperate, even if it appears that it is in their best interests to do so.

- It was originally framed by Merrill Flood and Melvin Dresher in 1950.

- Albert W. Tucker (John F. Nash's Supervisor) formalized the game with prison sentence rewards and named it, "prisoner's dilemma".

Description of the Situation

- Two members of a criminal gang are arrested and imprisoned. Each prisoner is in solitary confinement with no means of communicating with the other.

- The prosecutors lack sufficient evidence to convict the pair on the principal charge. They hope to get both sentenced to a year in prison on a lesser charge.

- Simultaneously, the prosecutors offer each prisoner a bargain. Each prisoner is given the opportunity either to:

  – betray the other by testifying that the other committed the crime, or
  – cooperate with the other by remaining silent.

The Offer Alternatives

- The offers are:
- If A and B each betray the other, each of them serves 2 years in prison
- If A betrays B but B remains silent, A will be set free and B will serve 3 years in prison (and vice versa)
- If A and B both remain silent, both of them will only serve 1 year in prison (on the lesser charge)

Prisoners' Dilemma Payoff Matrix

- Betraying a partner offers a greater reward than cooperating with them (0,-3) or(-3,0)
- Purely rational self-interested prisoners would betray the other, and so the only possible outcome for two purely rational prisoners is to betray each other.
- Mutual cooperation is superior to mutual defection (-1,-1)>(-2,-2). But it is not a rational outcome from a self-interested perspective as choice to cooperate at individual level is irrational.

A \ B	B stays silent	B betrays
A stays silent	-1, -1	0, -3
A betrays	-3, 0	-2, -2

Contd…

- Defection is the dominant strategy (0,-3) and (-3,0) for both agents.
- The game involves strategic interaction because the players do not have complete control over their own payoffs.

Dominant Strategy

- One strategy is said to dominate another if the first strategy always yields a payoff at least as good as and sometimes better than the second, no matter what any other player does.

- So defection is the dominant strategy for both the players.
- Strategy of A : (-1,-3) < (0, -2), hence defection is a dominant strategy for A
- Strategy of B: (-1,-3) < (0,-2), hence defection is also a dominant strategy for B

Remarks

- The problem of collective action and Prisoner's Dilemma are essentially the same!!

- But how?



Collective Action as A Prisoner's Dilemma

- About Latent group: Individual effort to achieve individual interests will prevent their achievement, because if the collective good is not provided, the individual member fails to receive a benefit that would have exceeded the individual's cost in helping purchase that good for the whole group.

- A game theory analysis of collective action to demonstrate that the logic underlying it is the same as that of Prisoner's Dilemma.

Some Concepts

- Game theory is the study of interaction, and strategic decision-making among rational individuals.
- A game is any form of strategic intervention. The idea is that the players make decisions that affect each other.
- Payoffs are the reward or loss a player experiences.
- Dominant Strategy is a choice that always lead to a higher pay off, regardless of what the other player/s choose/s.
- Prisoners' Dilemma is a situation that shows how cooperation breaks down and non-cooperation is dominant strategy. This situation cannot get Pareto Optimal outcome.

Contd…

- Collective action Dilemmas are situations in which the production of some group benefit is limited or prevented by the temptation to free ride.
- Tragedy of the commons: Individuals acting independently and rationally according to each one's self-interest, behave contrary to the whole group's long-term best interests by depleting some common resource.

Individual Vs. Collective

Individual's and Collective's Payoff		
Individual	Collective	
	Pay	Not Pay
Pay	1,1	-0.8,0.2
Not Pay	1.8,0.8	0,0

Individual's Payoff		
Individual		
Pay	1	-0.8
Not Pay	1.8	0

Contd…

- Here, the row entries are the payoffs for individuals and the column entries are the per capita payoffs for collective and collective is defined as group less this individual.
- And payoffs are defined as benefits minus costs.

- Let's take that the group comprises of 10 members whose common interest is the provision of a collective good of value twice it costs.
- It means if market price of x = Rs. 6/unit
- We will pay Rs. 6 to enjoy 2 units of x.

Contd…

- Now there are two possible results, if one member of the group declines to pay a share, Viz.
  - Either the total benefits will be proportionately reduced or,
  - The costs to the members of the group will be proportionately increased.

Contd…

- Lets' take the former case and assume that there is no initial costs in providing the collective good and no differential costs as payments and resultant benefits rise, i.e., exactly 2 units of collective good (x) will be provided for each units paid by any member of the group.
- If one member is paying for 1 unit of x, then he will be provided 2 units of x.
- Now we have 10 members.
- If all members pay for 1 unit of x, then the total amount paid can cost for 10 units.
- However, the benefits each member will get is 2 units of x.
- Now, the amount of collective good x is 20 units
- The individual payoffs is benefits minus cost, or ., i.e. 1 unit

Individual Vs. Collective

Individual's and Collective's Payoff		
Individual	Collective (per capita)	
	Pay	Not Pay
Pay	1,1	-0.8,0.2
Not Pay	1.8,0.8	0,0

Individual's Payoff		
Individual		
Pay	1	-0.8
Not Pay	1.8	0

- The first row shows the payoffs to individuals if individual contributes
- The column represents per capita payoffs to the remaining members of the group, i.e.to the collective , if they pay.
- The second row shows the payoffs to individual if the individual does not pay and the second column indicates those for collective, if they do not pay.

Interpretations of Payoffs

- (1,1): individual pays for 1 units and gets 2 units.
- So, b-c= 2-1=1unit
- For group if everyone pays, then its payoff is also 1
- (1.8,0.8): individual is not paying but group pays. The amount paid will be Rs 9 and units procured is 18
- So the share of each= 18/10= 1.8
- For group per capita benefit=1.8 and cost =1
- So, group payoff = 1.8-1= 0.8

Contd…

- (-0.8,0.2): individual pays and group do not
- The total amount paid Rs. 1 and units procured 2
- Per capita share= 2/10=0.2
- So individual payoff= b-c= 0.2-1=-0.8
- Group pay off= b-c=0.2-0= 0.2

Contd…

- So the dominant strategy for both is not to pay for the purchase of collective good.

- Not pay, not pay is the only equilibrium for this game.(An outcome is an equilibrium if no player has an incentive unilaterally to switch strategies).

- However this equilibrium is not Pareto equilibrium.

Selected Readings

- Hardin, Rachael. (1982). Collective Action. RFF., Chapter 2, Collective action and Prisoners' Dilemma.

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Governing the Commons: The Evolution of Institutions for Collective Action-I

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Objectives of the Lesson

- To understand a third generation research agenda on the commons
  - What is Ostrom's theory on collective action?
  - Why is it important?
  - How is this theory different from other theories on collective action?
  - Pros and Cons of the theory

Elinor Ostrom

- **Elinor Ostrom**
- Governing the Commons: The Evolution of institutions for collective action (1990)
- Ostrom, Elinor; Schroeder, Larry; Wynne, Susan (1993). Institutional incentives and sustainable development: infrastructure policies in perspective.
- Ostrom, Elinor; Walker, James; Gardner, Roy (1994). Rules, games, and common-pool resources.

What is Ostrom's Theory on Collective Action?

- Ostrom's analysis was focussed on how to avoid exploitation of common pool resources.
- As we understand that when a resource is held in common and shared by many individuals, well-defined individual property rights over the same is absent.
- Consequences?
- Overexploitation…
- Solution/strategy?
- Privatisation or enforcement imposed by outside force (State/Leviathan)

Ostrom's Argument…

- Is there any alternative solution?
  - The problem can be solved by stable institutions of self-governance which can be possible if problems related to supply, credibility and monitoring are solved.
  - She describes some key elements for successful common-pool management schemes.
- She was the first woman to receive Nobel Prize in 2009 for her 'analysis of economic governance focussing on common property'

Contd…

- Her arguments has challenged the established theories like ‘tragedies of the commons,’ ‘collective-action problems,’ ‘prisoner's dilemmas’.
- The prior theories underestimated the capacity of the community or groups to manage the collective goods.
- Hence, she argued that common resources can be well managed under the condition that those who benefit most from these resources must be in close proximity to the resources;
- And, tragedy in those commons may occur in the case when external groups exerted power socially, economically or politically.

Contd…

Ostrom explored the following questions:

- Why have some efforts to solve commons problems failed while others have succeeded?
- What can we learn from experience that will help stimulate the development and use of a better theory of collective action?
- What are the key variables that can enhance or detract from the capabilities of individuals to solve problems?

Contd…

- In her analysis of the group dilemma or collective action problem, she stated that the participants themselves design their own contracts.

- The self-interest of those who negotiated the contract will lead them to monitor each other and to report observed infractions so that the contract is enforced.

- Whereas, a regulatory agency/authority, always needs to hire its own monitors. The regulatory agency then faces the principal-agent problem of how to ensure that its monitors do their own job.

Contd…

- The optimum results are obtained in such self-governing groups because of following parameters:
  - Communication among the participants.
  - Better understanding of the structure of the game or collective goods.
  - Reliability on the others' actions.

Contd…

- In her investigation into ‘an institutional approach to CPR self-governance’ to identify the underlying design principles of the institutions used by those who have successfully managed their own CPRs over extended periods of time, she has found that

  – (a) appropriators have devised, applied, and monitored their own rules to control the use of their CPRs and
  – (b) the resource systems, as well as the institutions, have survived for long periods of time.

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- Araral, E. (2014). Ostrom, Hardin and the commons: A critical appreciation and revisionist view. *Environmental Science and Policy*, 36: 11-23.

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Governing the Commons: The Evolution of Institutions for Collective Action-II

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8 Design Principles for Self- Governing CPR Institutions

- These principles are essential elements that account for success of CPR institutions in sustain the CPRs:

- 1. The CPR institutions has clearly-defined boundaries: individuals or households who have rights to withdraw resource units from the CPR must be clearly defined, as must the boundaries of the CPR itself.
- 2. Congruence between appropriation and provision rules and local conditions: Appropriation rules restricting time, place, technology, and/or quantity of resource units are related to local conditions and to provision rules requiring labour, material, and/or money.

Contd…

- 3. Collective-choice arrangements: Most individuals affected by the operational rules can participate in modifying the operational rules.
- 4. Monitoring: Monitors, who actively audit CPR conditions and appropriator behaviour, are accountable to the appropriators or are the appropriators.
- 5. Graduated sanctions: Appropriators who violate operational rules are likely to be assessed graduated sanctions (depending on the seriousness and context of the offence) by other appropriators, by officials accountable to these appropriators, or by both.

Contd…

- 6. Conflict-resolution mechanisms: Appropriators and their officials have rapid access to low-cost local arenas to resolve conflicts among appropriators or between appropriators and officials.
- 7. Minimal recognition of rights to organize: The rights of appropriators to devise their own institutions are not challenged by external governmental authorities.
- For CPRs that are parts of larger systems:
- 8. Nested enterprises: Appropriation, provision, monitoring, enforcement, conflict resolution, and governance activities are organized in multiple layers of nested enterprises (multi faceted interests/boundaries/conditions monitoring etc.).

Collective action and common pool resources problems in present scenario

- Curbing the emissions of greenhouse gases by developed and developing countries.
- Terrorism-related collective action; gathering intelligence on common terrorist threats .
- Health-based collective action issues possess an intergenerational component.
- Protecting the biodiversity of the planet.

Contd…

- International agreements on environment protection
- Blood donation
- Public sanitation
- Elections
- Pollution

Some Pros and Cons of Ostrom's Argument

- Her challenge to the first generation theory on commons.

- Her contribution to institutional analysis and second generation theories on common suggesting the possibility of self-governance as a viable solution to collective action problem.

- The second generation theories point to the centrality of trust and reciprocity as the main determinants of collective action.

Contd…

- Her critique of Hardin's Tragedy of the Commons may be valid for special cases like small-scale, locally governed commons; whereas Hardin's thesis seems justified for macro scales like regional, national and global commons (Araral, 2014).
- Validity of her institutional design principles for long-lived commons: Managing global commons requires a different set of governance principles compared with the local commons (Ibid; Stern, 2011)
- Question on generalising eight institutional design principle(Araral, 2014; Cox eta la., 2010)
- Failure to distinguish between a commons and partnership arrangements: in case of the commons outsider can be excluded from entry but not for the partnership arrangements.

References

- Araral, E. (2014). Ostrom, Hardin and the commons: A critical appreciation and revisionist view. *Environmental Science and Policy*, 36: 11-23.

- Ostrom, E. (1990). Governing the Commons: The Evolution of institutions for collective action. Cambridge University Press: Cambridge.

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Environmental Goods and Ecosystem Services-I

Prof. S. P. Singh

Humanities and Social Sciences



Contents

- Characteristics of environmental goods
- What is ecosystem?
- Ecosystem services
- Categories of ecosystem services

Environmental Goods

- Characteristics:

- ❑ Goods: Excludable/non-excludable and Rival/non-rival
- ❑ A good is excludable/non-excludable if it is feasible/not feasible and practical/not practical to selectively allow consumers to consume the good (Technology and Regulations). Market operates due to excludability of goods.
- ❑ A good is rival/non rival in consumption if the act of consumption reduces/does not reduce the amount of the good that might be available for other consumers.
- ❑ Pure public/collective goods: solar radiation, weather, clean air, etc.
- ❑ Mixed collective goods: marine fish, wildlife, watershed services, etc.
- ❑ Public bads: air and water pollution, ozone depletion, etc.
- ❑ Lack of well-defined property rights leads to environmental degradation.

What is Ecosystem?

- The term was first used by Tansley in 1935 in his article 'The Use and Abuse of Vegetational Concepts and Terms'.
- An ecosystem is the set of organisms living in an area, their physical environment, and the interactions between them.
- It is a complex and dynamic combination of plants, animals, micro-organisms and the natural environment, existing together as a unit, and depending on one another.
- Components: Biotic (all living organisms) and abiotic (non-living elements such as soil, water, sunlight, minerals).
- The health and wellbeing of human beings depends upon the services provided by ecosystems and their components — organisms, soil, water, and nutrients.

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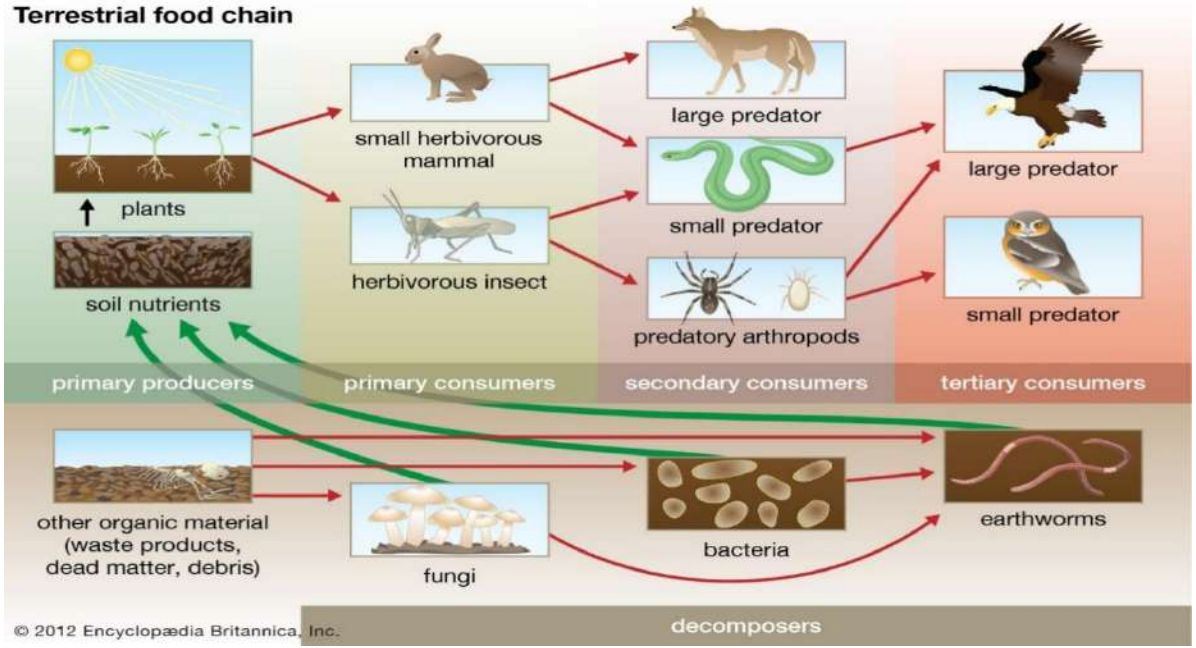
- Three types of ecosystems:

❑ Freshwater ecosystems : They are a subset of Earth's aquatic ecosystems which includes lakes, rivers, ponds springs, wetland etc.

❑ Terrestrial ecosystems: It includes organisms and their environment that occurs on the land masses of continents and islands.

❑ Ocean ecosystems: Ocean ecosystem covers about 75% of the planet. It is also known as Marine ecosystems .

Demonstration of a Terrestrial Ecosystem



The biotic and the abiotic components are linked by the flow of energy through the ecosystem, and the cycling of nutrients within the ecosystem.

Ecosystem Services

- Ecosystem services are the conditions and processes through which natural ecosystems, and the species that make them up, sustain and fulfil human life.
- ES are the processes by which the environment produces resources that we often take for granted such as clean water, timber, habitat for fisheries, pollination of native and agricultural plants etc.
- Classifications of ES, given by The Economics of Ecosystem and Biodiversity (TEEB), Millennium Ecosystem Assessment (MEA), and Common International Classification of Ecosystem Services (CICES).

Classifications of Ecosystem Services

S.N	MEA (2005)	TEEB (2009)	CICES (2017)
1	Provisioning Services	Provisioning Services	Provisioning Services
	Food, medicinal resources, crops raw materials, animal based resources, energy etc.	food, water, raw materials, and genetic, medicinal and ornamental resources etc.	animal and their product, cultivated food, drinking and non-drinking water, animal & plant based energy.
2	Regulatory Services	Regulatory Services	Regulation and maintenance
	Carbon sequestration, waste decomposition, purification of water and air etc.	air purification, climate regulation, C-sequestration and storage, disturbance prevention etc.	mediation of waste, toxics and other nuisances, mediation of flows, and maintenance of physical, chemical and biological condition etc.
3	Cultural Services	Cultural and Amenities	Cultural Services
	Spiritual, recreation and tourism, and aesthetic etc.	aesthetic, recreation and tourism, spiritual experience etc.	experiential use of plants and animal, spiritual, aesthetic recreational
4	Supporting Services	Habitat Services	
	Nutrient cycle, soil formation pollination and gene pool protection etc.	gene pool protection and lifecycle maintenance etc.	

List of Some Key ES

- Moderation of weather extremes
- Seeds dispersal
- Drought and floods mitigation
- Nutrients cycling and movement
- Protection of streams, river channels and coastal shores from erosion
- Waste absorption and decomposition
- Controlling agricultural pests
- Maintaining biodiversity
- Generating and preserving soils and renewing their fertility
- Contribution to climate stability
- Purification of air and water
- Regulating disease carrying organisms
- Pollination of crops and natural vegetation

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- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. OUP Catalogue.
- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. Sustainable Human Development Review, 105.

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Environmental Economics

Environmental Goods and Ecosystem Services-II

Prof. S. P. Singh

Humanities and Social Sciences



Contents

- Complementary and competitive nature of ES
- Intermediate and final ES
- Value and measurement of ES
- Issues in sustaining ES
- Measures to sustain ES

Complimentary and Competitive Nature of ES

- The trade off or competitive nature:

- It is generally not possible to manage ecosystems to simultaneously maximize all services, and as a result trade-offs could occur.
- Trade-offs appear to be particularly common between provisioning services and regulating services,
- The integrity of ecological processes are compromised as provisioning services are gained.
- For example, the cutting of forests for wood leads to reduction in temperature managing service of the forest.
- Excessive discharges of waste materials into the oceans will reduce their capacity to provide a habitat for fish stocks.

Contd…

- Complementary nature:

❑ Environmental functions can be complementary.

❑ An ecosystem service can perform different functions simultaneously.

❑ An appropriate forestry policies can provide a sustainable source of timber (a natural resource function) and reduce soil erosion (an improvement in the life-support function).

❑ Cultural services are often connected to provisioning and regulating services.

❑ Example, small scale fishing is not only about food and income, but also about fishers' way of life.

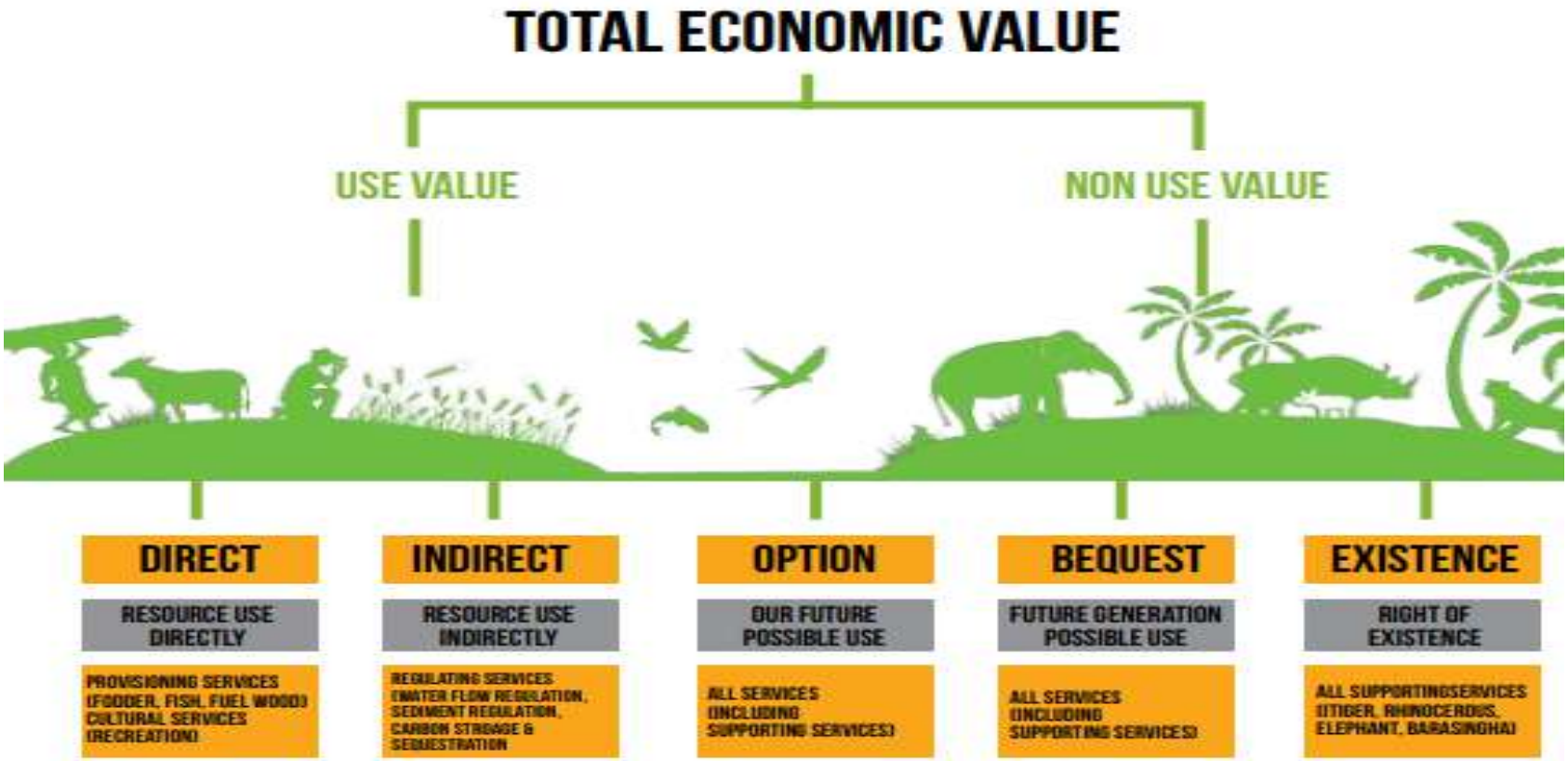
Intermediate and Final Ecosystem Services

- The terms have evolved from economics discipline for accounting of ecosystem services but is not very clearly understood concept still.
- As defined by Millennium Ecosystem Assessment(MEA):
- **Final ES**, also known as Final Ecosystem Goods and Services.
- Benefit people explicitly and can be accounted for in biophysical or monetary terms.
- Such as fisheries output per season (provisioning ES) or number of lake visitors per year (cultural ES).

Contd.....

- **Intermediate ES** are often described as those ES with little or no direct benefits to people.
- Consist of the biophysical structures and processes that maintain ecosystems in a favorable state for the provision of final ES.
- Correspond roughly to the regulating and supporting ES categories.
- Example, pollination service by the pollinators.
- However, a service can be both intermediate and final depending upon circumstances.
- For example, clean water from a water body as a drinking water is a final service and it is intermediate in case of managing the fish population for fishing.

Value and Measurement of Ecosystem Services



Contd…

❑ Direct, consumptive use value

- Market Price

❑ Direct, Non-Consumptive

- Travel Cost Method – Entry fee,
  - ➢ Travel cost, and Time or opportunity cost
- Contingent Valuation Method –
  - ➢ Willingness to pay (WTP)
  - ➢ Impact of socio-demo-economic variable on WTP.

Contd…

❑ Non-Use Values (Option, Bequest and existence)

- Contingent Valuation Method

❑ Cost Based Approach

- Previous approaches are based on the benefits that one derives from ecosystem services.
- Valuation can also be done on the basis of cost involved to achieve it.
- Cost approach states that the total value of a service is equal to the cost incurred to achieve.
- For example, if total cost to create a park is 2 million dollar then its total value would the 2 million value.

Issues in Sustaining Ecosystem Services

- ES are severely threatened due to two pressures:
  - Growth in the scale of human enterprise (population size, per-capita consumption, energy intensive life-style, and effects of technologies to produce goods for consumption).
  - A mismatch between short-term needs and long-term societal well-being.
- Some of the human activities that affect ecosystem services are:
  - runoff of pesticides, fertilizers, and animal wastes
  - pollution of land, water, and air resources
  - introduction of non-native species
  - destruction of wetlands & over-fishing
  - deforestation and soil erosion
  - urbanization & Industrialization

Measures to Sustain ES

- Payment for ecosystem services (PES) to incentivise the involvement of local stakeholders in the conservation and management of ecosystem.
- Market-based approach such as tradable pollution permits
- Developing the institutional framework and social capital for an effective self-governing common-property rights regime.
- Community Participation
- Internalization of externalities

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ECOLOGICAL FOOTPRINT-I

DIPTIMAYEE NAYAK

HUMANITIES AND SOCIAL SCIENCES



Contents

- What is Ecological Footprint?
- Origin and Development of the Concept
- Indicators of Ecological Foot Print

Meaning and Definition

- The Ecological Footprint is the only metric that measures 'how much nature we have and how much nature we use' (Global Foot Print Network).

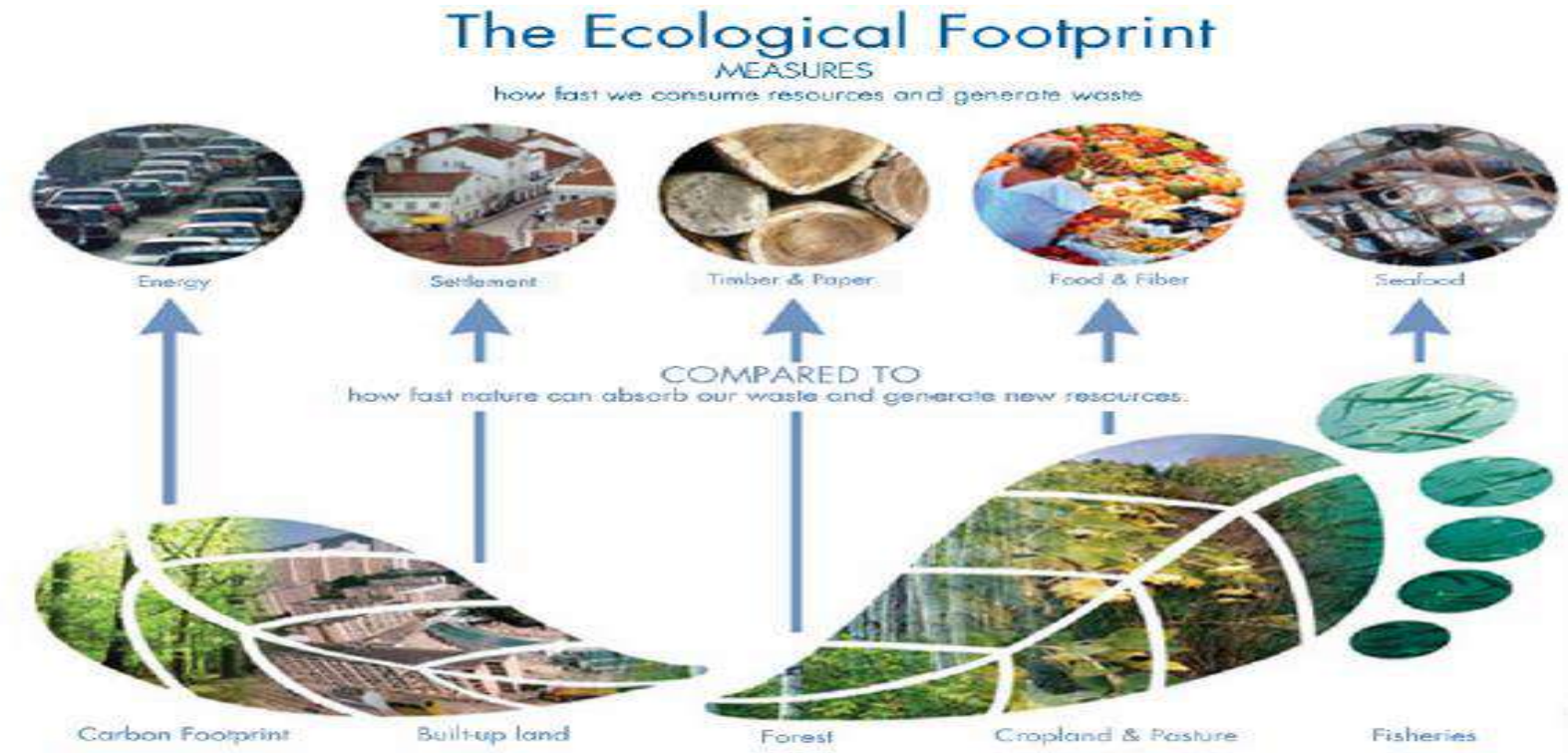
- It is an accounting that measures the dependence of whole mankind on natural resources by assessing demand on nature and supply of nature.

- Measuring Demand on Nature: EF quantifies 'the ecological assets (like six categories of productive surface areas- cropland, grazing land, fishing grounds, built-up land, forest area, and carbon demand on land) that a given population requires to produce the natural resources for consumption and to absorb its waste, especially carbon emissions'.

Contd…

- On the supply side: EF measures the bio-capacity of geographical boundary (state/nation) which represents the productivity of its ecological assets like crop land, grazing land etc.

Contd…



https://www.footprintnetwork.org/our-work/ecological-footprint/

Origin and Development of the Concept

- Canadian ecologist William Rees created the EF concept in form of 'regional capsule' and 'appropriated carrying capacity' in 1992, which Swiss urban planner Mathis Wackernagel further developed in his dissertation under Rees's supervision.

- Together, Wackernagel and Rees wrote *Our Ecological Footprint* (1996), which describes the concept and criticizing the economic models of resource use, arguing that focus should be given to the land area required by urban areas to sustain themselves.

Two Other Contemporary Concepts

- Environmental space: ‘ the share of the planet and its resources that the human race can sustainably take’.
- This concept is operationalised though the measurement of **material inputs.**
- Hence this concept is said to be a more complex approach.
- Ecological Rucksack: This is a refinement of ‘material inputs approach’ of Environmental space.
- It calculates ‘what would normally be hidden behind our material inputs as it includes all of the resources used in our consumption of a particular material.’

Ecological Deficit and Ecological Reserve

- Ecological Deficit: when EF of population in an area is greater than the bio-capacity of that geographical area.

- The demand of population for goods and services exceeds what ecosystems of that area/region can renew.

- Ecological Reserve: When the bio-capacity of a region exceeds ecological foot print of population of that area.

Indicators of Ecological Footprint

- Assuming earth to be a single ecosystem, there would be six sub-ecosystems.

- Total EFs is estimated by summing up EFs of all the available indicators of a particular ecosystem.

Contd…

- Cropland foot print: It consists of the area of land required to grow all crop products- including livestock feed, fish meal, oil crops and rubber.
- The footprint of each crop type is calculated as *the area of cropland that would be required to produce the harvested quantity at world-average yields.*

Contd…

- The grazing land footprint: It measures the area of grassland used to support livestock.
- Grazing land comprises all grasslands = cultivated pastures + wild grasslands + temperate grass lands like prairies/savanna.

- Forest Land footprint: It measures the annual harvest of fuelwood and timber to supply forest products.
- It takes into account the footprint of forest land according to the production quantities of primary timber products and wood fuel products.

Contd…

- Fishing ground footprint: It is calculated on the annual primary production required (PPR) to sustain all harvested aquatic species.
- PPR is defined as the mass ratio of harvested fish to annual primary production (e.g. Plants, algae) needed to sustain that species, based on its average trophic level.

- Built upland footprint: It is calculated based on the area of land covered by human infrastructure including roads, carparks, buildings etc.

Contd…

- Carbon land footprint: It is the uptake land to mitigate carbon dioxide and other greenhouse gas emissions. Also, it is the only land use type for which bio capacity is not explicitly defined.
- Many different ecosystem types have the capacity for long-term storage of CO2, including cropland and grassland.
- Carbon uptake land is assumed to be forest land by the ecological footprint methodology in order to avoid overestimation.

- Hence, carbon land is considered to be a subcategory of forest land.

Example

- It has been estimated that in 2014 the per capita global footprint was 2.8 gha and global bio-capacity that year was 1.7 gha per person.
- Hence, the EF of humanity overshot Earth's biocapacity by 1.1 gha.
- In other words, 1.7 "Earths" would be needed to sustain current resource demands.

Selected Readings

- https://www.footprintnetwork.org/our-work/ecological-footprint/
- Calculation Methodology for NFAs, 2010

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ECOLOGICAL FOOTPRINT-II

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HUMANITIES AND SOCIAL SCIENCES



Contents

- How to calculate EF?

How to Calculate Ecological Footprint?

- Global Footprint Network (GFN)
- National Foot Prints Accounts, 2003
- **Fundamental Assumptions of 'Ecological Footprint Accounting':**
- The majority of the resources people consume and the wastes they generate can be quantified and tracked.
- An important subset of these resource and waste flows can be measured in terms of the biologically productive area necessary to maintain flows.
- Resource and waste flows that cannot be measured are excluded from the assessment, leading to a systematic underestimate of humanity's true Ecological Footprint.

Contd…

- Weighting each area in proportion to its bio-productivity, different types of areas can be converted into the common unit of global hectares, hectares with world average bioproductivity.

- Because a single global hectare represents a single use, and each global hectare in any given year represents the same amount of bio-productivity, they can be added up to obtain an aggregate indicator of Ecological Footprint or bio-capacity.

Contd…

- Human demand, expressed as the Ecological Footprint, can be directly compared to nature's supply, bio-capacity, when both are expressed in global hectares.

- Area demanded can exceed area supplied if demand on an ecosystem exceeds that ecosystems regenerative capacity.

Contd…

- Three ways to calculate EF:
- Top Down
- Bottoms Up
- Input Output Model

Contd…

- In the first two methods, a land consumption matrix is constructed for a broad set of economic activities and each entry is filled with the amount of land used for a particular activity.

- The final sum of the entries does give the total ecological footprint.

- The top down approach involves use of data collection from national and regional statistics like per capita consumption data of energy, food etc. This data is then filled in the matrix.

Contd…

- In the bottoms up approach, the data is collected individually. This is predominantly conducted at a lower scale like cities, institutions etc.
- Major disadvantage with this kind of calculation is the assumption of the system boundary that the researcher believes to contribute to the ecological footprint.
- This system boundary may not be able to capture the entire ecological footprint, for example the embodied emissions associated with manufacture (processes and sub-processes) of car might not be reflected in the ecological footprint calculated (Lenzen, 2001).

Contd…

- One can estimate the EF, measured in “global hectares” (gha), at various scales —for individuals, regions, countries, and humanity as a whole (One hectare equals 2.47 acres.)

- ‘gha’ is defined as ‘the annual productivity of one hectare of biologically productive land or sea with world average productivity’ (Wiedmann and Lenzen, 2007).

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Example

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Environmental Economics

Poverty and Environment Linkages-I

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- Poverty: concept and measurement
- Effects of environmental change on poverty
- Poverty and environment linkages: Some hypotheses
- Some myths related to poverty- environment nexus
- Strategies to improve environment and poor's livelihood
- Let's sum up

Introduction

- The environment is not only source our survival- air, water and food but also a source of the materials we require to take our lives from pure survival to subsistence and beyond.
- The absence or denial of these basic necessities constitutes absolute poverty.
- Their unequal access is the foundation of relative poverty.
- Poor households rely more on environment for their livelihoods.
- They are more vulnerable to natural disasters and environmental degradation than rich people.

Contd…

- The environment-poverty nexus is a two-way relationship:
- Environment affects poverty situations in three distinct dimensions by :
  - providing sources of livelihoods to poor people;
  - affecting their health; and
  - influencing their vulnerability.
- While poverty affects environment in various ways:
  - by forcing poor people to degrade environment;
  - by encouraging countries to promote economic growth at the expense of environment; and
  - by inducing societies to downgrade environmental concerns.

Poverty: Concept and Measurement

- There are two concepts of poverty— absolute and relative.
- Absolute poverty refers to the minimum basic needs of human beings, such as, food and nutrition, clothing, shelter and preventive and protective healthcare, necessary for their survival and physical health.
- It involves defining a minimum standard of living below which an individual or household is considered to be poor. Based on the income required for maintaining this minimum level of living, a poverty line is estimated.
- A distinction is then made between the poor and non-poor depending on persons who lie either below or above this defined line of poverty.

Contd…

- Absolute poverty is not related to the income/consumption distribution while relative poverty depends on it.
- A relative poverty line is an explicit function of the income distribution – namely, a constant fraction of some income standard.
- It is defined in terms of a certain percentage (say 40%) of a country's median or mean level of income. For instance, a household may be called relatively poor if its income is below 40% of the national average.
- Selection of cut-off percentage may vary across time .
- It is more relevant in developed economies, where the number of absolute poor may be negligible.

Measurement of Poverty

- Three Methods:
  - The Headcount Ratio
  - The Poverty Gap Ratio
  - The Squared Poverty Gap Ratio

Head Count Ratio

The headcount ratio of measuring poverty is defined as the percentage of the total population that is poor. Thus, it is measured as the proportion of the population that is counted as poor. It is estimated as:

$$P_0 = \frac{N_P}{N}$$

Where P_0 is headcount ratio; N_p is the number of poor; and N is the total population. If the monthly per-capita consumption expenditure (MPCE) of 60 million households out of 179 million households of rural India is below the cut-off point, the headcount poverty ratio is = 60/179 = 0.3352. This means 33.52 percent of rural households in India are below the poverty line. If the value of MPCE (say Y_i) in a sample of household is below the poverty line (say Z), i.e. $Y_i < Z$ then the ith household is counted as poor. If the value of $Y_i > Z$, then the ith household is counted as non-poor. Thus, the aggregate value of 'proportion of poor household' by the headcount method (P_0) in a region can be represented as:

$$P_0 = \frac{1}{N}\sum_{i=1}^{N} \text{Yi} \quad (if\ Y_i < Z)$$

Poverty Gap Ratio (PGR)

- For studying the intensity of poverty, estimation of PGR is useful. It tells us the extent to which individuals, on average, fall below a poverty line.
- It is a key indicator that measures how far the extremely poor fall below the poverty line reflecting both the depth and incidence of poverty.

It is estimated as: $$P_1 = \frac{1}{N}\sum_{i=1}^{N}\frac{G_i}{z}$$

- Where *P1 is PGR; N= number of hhs; Gi = poverty gap i.e. the* difference between the MPCE Yi and the level of *Z taken as the minimum income* required for basic sustenance.
- If Yi > Z, then Gi is taken as equal to 0 and if Yi < Z, Gi is taken as equal to Z- Yi.

Squared Poverty Gap Ratio

- The SPGR, which considers the sum of squares of Gi/Z values, measures the severity of poverty. As a weighted sum of poverty gaps, where the weights are the proportionate poverty gaps themselves, it is sensitive to the prevalence of the poor by the extent to which their incomes fall below the poverty.
- This is in contrast with the PGR method, where the gaps are all weighted equally. Hence, by squaring the poverty gap index, the measure implicitly puts more weight on observations that fall below the poverty line. The SPGR (P2) is:

$$P_2 = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{G_i}{z}\right)^2.$$

Poverty Estimate: An Example

Region	MPCE in Rs.								Poverty Ratios
	1	2	3	4	5	6	7	8	Z=Rs. 800
Region I	950	1100	1000	975	750	775	790	1400	P0: 3/8=0.375
Poverty Ga	0	0	0	0	50	25	10	0	
Gi/Z	0	0	0	0	0.0625	0.03125	0.0125	0	P1: 0.1063/8=.0133
$(Gi/Z)^2$	0	0	0	0	0.0039	0.0010	0.0002	0	P2:0.0051/8=0.00064
Region II	1250	1150	1400	1100	550	600	490	1200	P0: 3/8=0.375
Poverty Ga	0	0	0	0	250	200	310	0	
Gi/Z	0	0	0	0	0.3125	0.25	0.3875	0	P1: 0.95/8=0.119
$(Gi/Z)^2$	0	0	0	0	0.0977	0.0625	0.1502		P2:0.31036/8=0.039

Multidimensional Concept of Poverty

- In measuring multidimensional poverty, various dimensions and indicators like education, health, employment status and household living conditions are used.
- Some of these dimensions may have more than one indicator. For instance, education dimension can consist of school enrolment and years of schooling and health consists of child mortality and malnutrition.
- Likewise, household living conditions may comprise access to electricity, safe drinking water, sanitation, flooring and clean cooking fuel, assets, etc.
- The Oxford Poverty and Human Development Initiative (OPHI) estimates MPI for different countries, using equally weighted three dimensions :
  - Education ; Health ; Living standard

Effects of Environmental Change on Poverty

- Environmental change may relate to the various dimensions of poverty – namely

❑ Opportunity

❑ Capability

❑ Security

Opportunity

- A majority of poor live in rural areas and depend on natural systems for their livelihood (such as agriculture, fishing, forestry, etc.)
- Poor people are often impoverished by a declining resource base and in turn often forced by their circumstances to degrade the environment further.
- They are usually viewed either agents or victims of environmental degradation.
- Poverty and environment relation is more complex and mediated through various macro and micro-level factors, such as policy measures, markets and prices, local institutional arrangements, gender relations, property rights, entitlements to natural resources, etc.

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Environmental Economics

Poverty and Environment Linkages-II

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Contents

- Dimensions of Poverty
  - Opportunity
  - Capabilities
  - Security

Opportunity

- A majority of poor live in rural areas and depend on natural systems for their livelihood (such as agriculture, fishing, forestry, etc.)
- Poor people are often impoverished by a declining resource base and in turn often forced by their circumstances to degrade the environment further.
- They are usually viewed either **agents or victims** of environmental degradation.
- Poverty and environment relation is more complex and mediated through various macro and micro-level factors, such as policy measures, markets and prices, local institutional arrangements, gender relations, property rights, entitlements to natural resources, etc.

Contd…

- **Time horizons** further complicate the relationship between poverty and environment. For instance, reducing income poverty can have a positive or a negative effect on natural resource management in the long term.
- For example, if income of livestock holder increases, he may purchase more cattle having negative impact on CPR (grassland) and long run income.
- Conversely, subsistence or semi-subsistence farmers whose land is prone to erosion can, when their income rises, use the increases to invest in soil conservation measures.
- There is also a **spatial dimension** to the nature of tradeoffs involved. For example, construction of Dam displaces the people in one region and provides the benefits of power and irrigation to other region.

Contd…

- Since CPRs tend to be more important to the poor than to the rich, the effects of natural resource degradation are also more pronounced on their livelihoods.
- CPRs also play an important role indirectly, by helping poor farmers adjust factor proportions in farming. For example, by providing alternative space for grazing, they allow farmers to devote all their land to the production of cash or food crops.
- There is also **gender dimension**.
- Rural poor women are disproportionately affected by the degradation of natural resources. They can be severely affected by environmental degradation in with respect to time, income, nutrition, health, etc.

Contd…

- **Urban Dimension**: A majority of urban poor engage in informal activities and live in poor environmental conditions in slums with inadequate sanitation and water facilities and greater risk of exposure to air and water borne diseases.
- Most of them migrate from the rural areas.
- While the poor consume less water and energy and contribute less to environmental pollution compared to the rich, they often bear a disproportionate burden of environmental hazards.

Capabilities

- Environmental pollution, degradation of natural resources, natural disasters and inadequate environmental services, impose costs to society in the form of ill health, lost income, and increased poverty and vulnerability.
- Capability is impaired when poor people's health is damaged by dirty water, dirty air or by diseases related to environment (such as malaria).
- Improvement in environmental services, including sinking function (pollution absorption) plays an important part in people's well-being.
- Burden of diseases : respiratory infections and diarrhoeal diseases are the two biggest causes of death among

Condt…

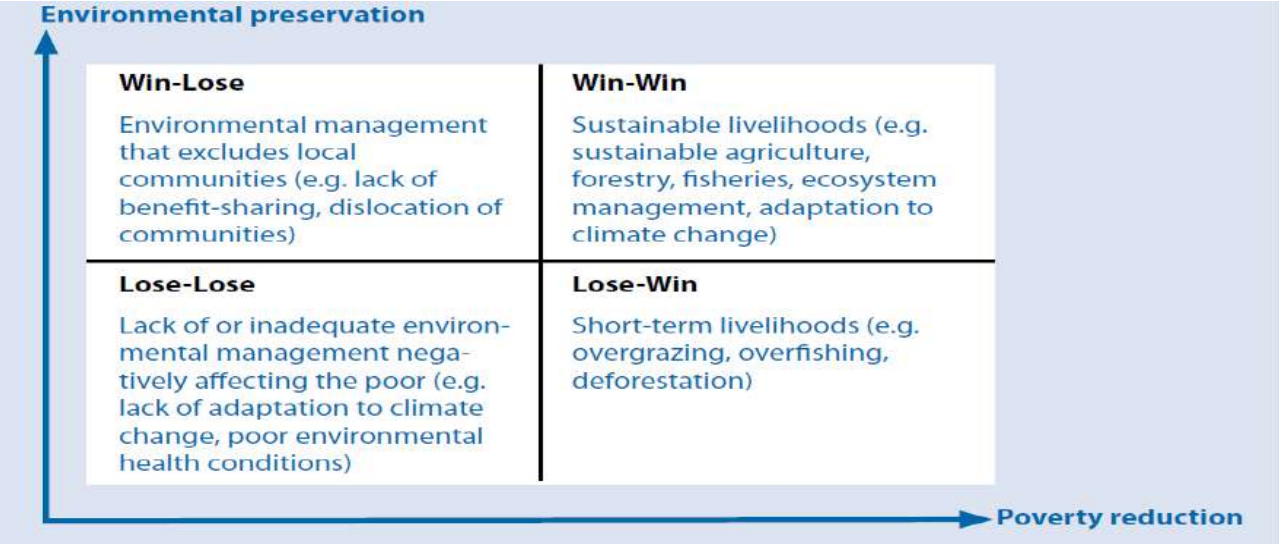
- The losses related to water, sanitation and hygiene are likely to be concentrated among the poor who most often do not have access to clean environment (both rural and urban poor).

- Projects related to reduction in pollution, clean water, sanitation (toilets), underground sewerage and drainage, would improvement the health condition of poor and enhance their capabilities.

Security

- Poor people have less resilience to environmental risks (livelihood and food insecurity).
- Their security is threatened by natural disasters and climatic variation. The poor tend to be more physically vulnerable to natural disasters and have fewer resources to enable them to ride out the shocks.
- Access to credit is lower than that of better off households and they have fewer assets to sell or consume in times of hardship.
- Improving the ways in which environmental resources, such as forests, are managed increases the resilience of poor people and their livelihoods to environmental risks.

Figure 1: Environment Improvement and Poverty Reduction



Source: UNDP-UNEP (2009): *Mainstreaming Poverty-Environment Linkages into Development Planning: A Handbook for Practitioners*

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Poverty and Environment Linkages-III

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Contents

- Introduction
- Poverty and environmental linkages: Some hypotheses
- The downward hypothesis
- The beckerman hypothesis
- The porter hypothesis
- Strategies to improve environment and reduce poverty
- Some myths
- Sum up

Introduction

- Preceding Lecture Explained:
  - Effects of Environmental changes on poverty
    - Three dimensions of poverty—opportunity, capability and security
- This Lecture Explains:
  - Poverty –environmental linkages related some hypotheses
    - Downward Spiral Hypothesis
    - The Beckerman Hypothesis
    - Porter Hypothesis
  - Some Myths
  - Strategies to improve environment and reduce poverty

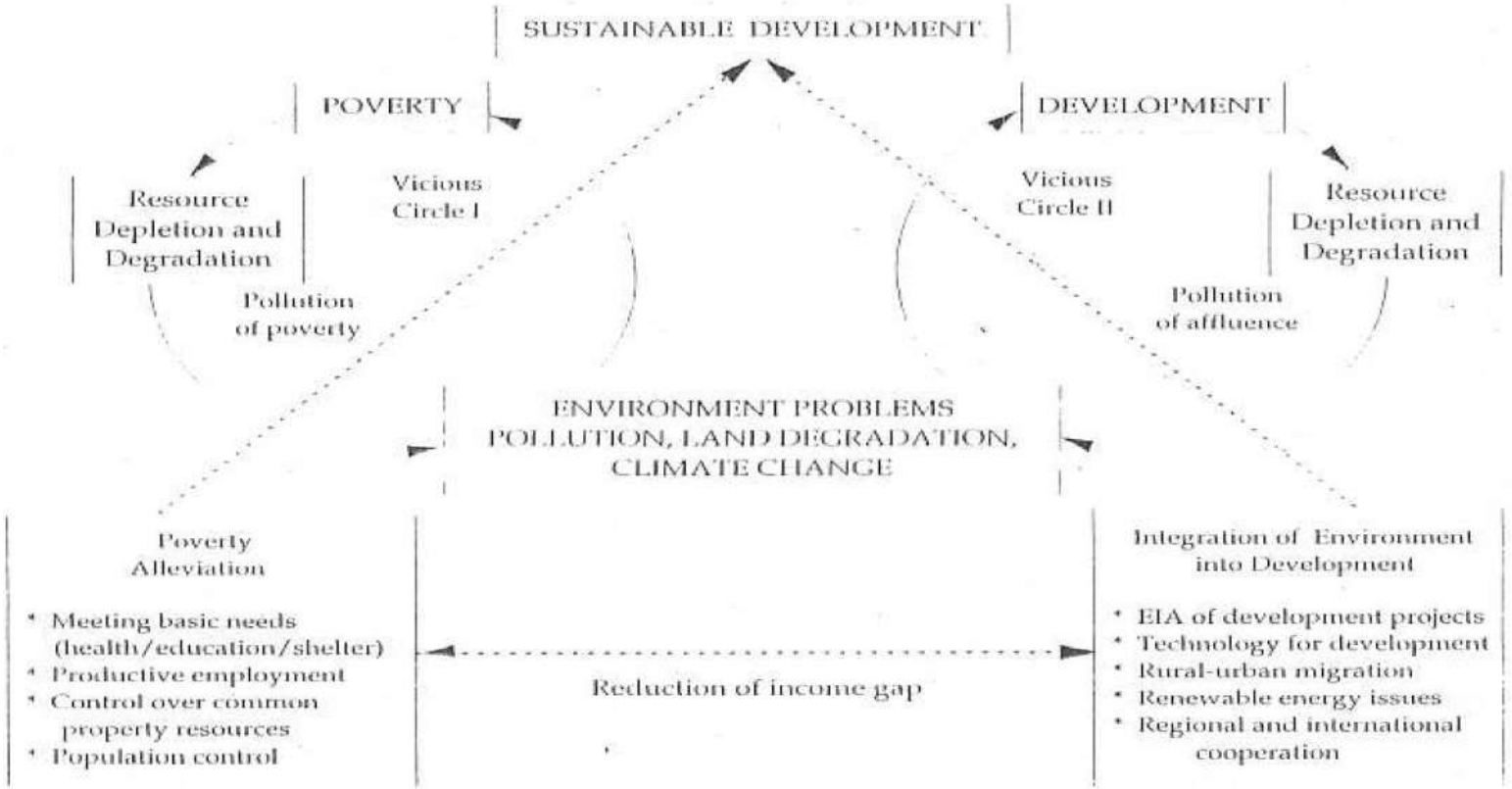
Poverty and Environment Linkages: Some Hypotheses

- The environment-poverty nexus is a two-way relationship.
- Environment affects poverty situations in three distinct dimensions : by providing sources of livelihoods to poor people, by affecting their health and by influencing their vulnerability.
- On the other hand, poverty affects environment in various ways: by forcing poor people to degrade environment, by encouraging countries to promote economic growth at the expense of environment, and by inducing societies to downgrade environmental concerns.
- Poverty  environment degradation
- Environment degradation  poverty

Contd…

- It is a popular claim that poverty leads to environmental degradation (deforestation, shifting cultivation, depletion of CPRs, indoor pollution due to fire wood, poor sanitation, etc.)
  - Poor people have shorter time horizons
  - Poverty increase risk-aversion and discount rates
- Environmental degradation leads to poverty
  - By declining their livelihood base
  - By impacting their health and productive capability
  - By enhancing their vulnerability to environmental and climatic risks

Linkages Between Sustainable Development, Environment and Poverty



The Downward Spiral Hypothesis

- The hypothesis maintains that poor people and environmental damage are often caught in a downward spiral (Jehan and Umana, 2003).
- People in poverty are forced to deplete resources to survive, and this degradation of environment further impoverishes people.
- Environmental degradation can sometimes be associated with poverty, but there is not necessarily a direct causal relationship.

The Beckerman Hypothesis

- The hypothesis maintains that as growth provides accumulated assets that can be used to ameliorate environmental degradation, it makes sense to degrade now and pay later to put things right.
- There is no guarantee that the accumulated assets to improve environment.
- ‘Grow now and clear up later’ is not a good strategy
- This hypothesis seems to undermine the need for conscious policy interventions.
- It takes a simplistic approach towards the intergenerational equity issue. It basically says that there will be physical degradation at present, but that monetary compensation will be made in future.

The Porter Hypothesis

- It states that high levels of environmental protection are compatible with high levels of economic growth and may encourage innovation that supports growth (Jehan & Umana, 2003).
- The hypothesis makes two fundamental points. First, environmental protection is justified not only for pure environmental reasons, but because such protection makes economic sense as well.
- Second, if people see the economic value of environmental protection, initiatives may be undertaken for innovations in technology, input-mix, and management to increase resource-use efficiency and also to minimise resource waste and the adverse environmental impact of production.
- But the hypothesis can lead to an extreme situation whereby environmental standards are imposed on trade.

Some Myths

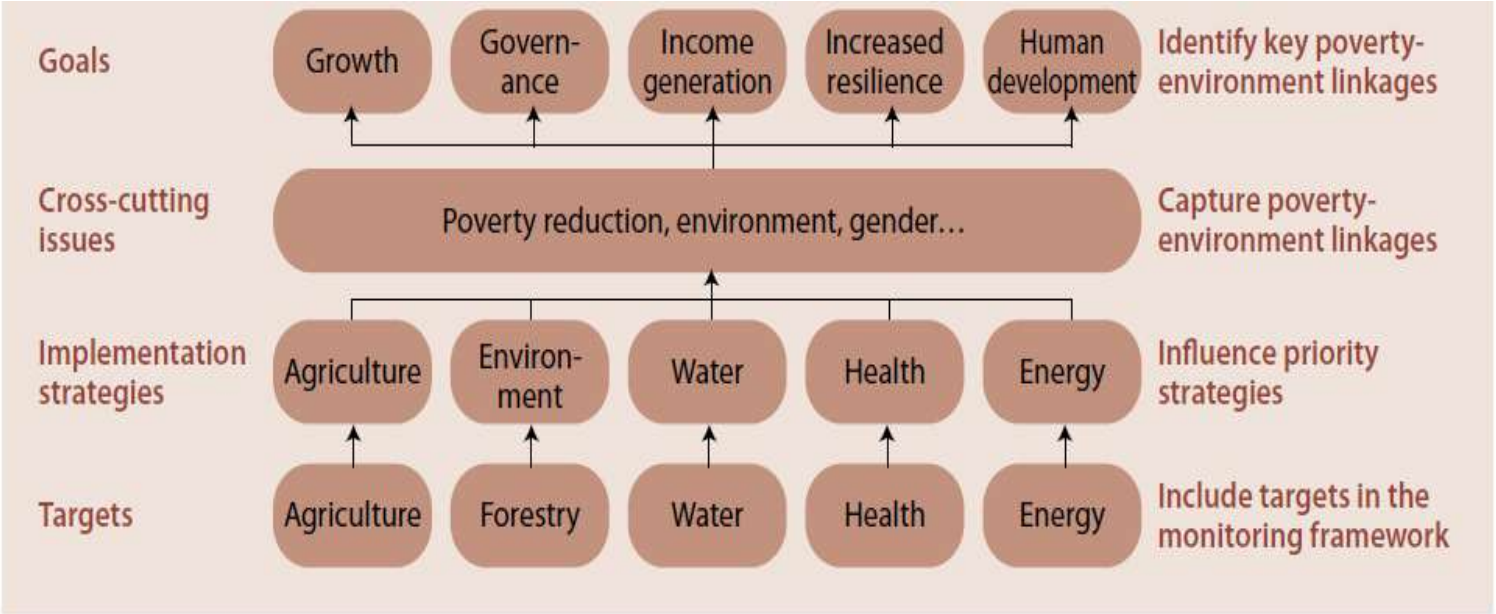
- Poor people are the principal creators of environmental damage
- Population growth leads to environmental degradation

- The poverty-environment nexus basically stems from low incomes

Strategies to improve environment and reduce poverty

- Need to understand:
  - who are the poor and where they are?
  - the extent to which they depend on natural resources for their livelihoods and where and how those natural resources are being degraded;
  - their coping strategies;
  - what sorts of institutions mediate entitlements to various assets and CPRs.
  - how pollution and other environmental factors (dirty drinking water, dirty air, lack of sanitation) affect the health of the poor; and
  - the ways in which the poor are vulnerable to natural shocks or disasters (droughts, floods, earthquakes, etc.).

Figure-2: Aligning the Analytical Approach with the Overall Policy Framework (Source: UNDP-UNEP, 2009)



Let's Sum Up

- Poverty-environment nexus is a two-way relationship:
  - Environment affects poverty situations in three distinct dimensions: Livelihood, health and vulnerability.
  - Poverty affect environment by forcing poor to degrade environment , encouraging countries to increase growth leading to pollution, and inducing societies to downgrade environmental concerns.
- Poverty and environment relation is more complex and mediated through various macro and micro-level factors, such as policy measures, markets and prices, local institutional arrangements, gender relations, property rights, entitlements to natural resources, etc
- Lack of well-defined property rights, poor regulatory mechanism and weak governing institutions are rather more responsible for environmental degradation than poverty.

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Environmental Economics

Environment and Economic Growth Linkages-I

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CONTENTS

- Introduction
- Environment and economic activity
- Environment and economic growth
- Environment cost of economic growth

Introduction

- Development is must, but environment is first.
- The environment is vital to supporting life, absorbing waste and providing inputs for production.
- Over the last 30 years, here has been increasing concern about the effects of economic activities on the environment.
- Economic growth has caused serious environmental damage.
- The current state of the environment will constrain future development.
- The poor in developing countries are often dependent on the natural environment for their livelihood, and existence.
- Damage to the environment and the relationship between the environment and the economy are more relevant to developing countries.

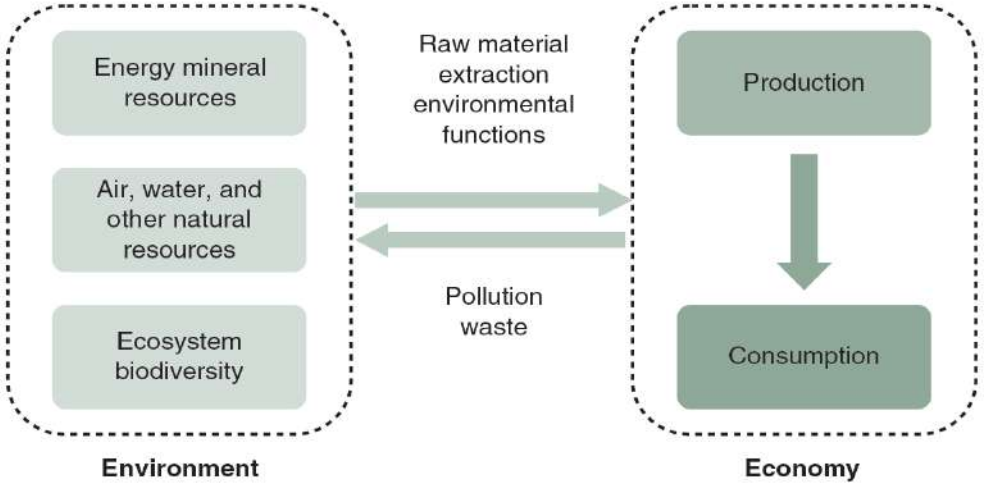
Environment and Economic Activity

Four functions of Environment:

- Life-support
- Supply of natural resources
- Absorption of waste products
- Supply of amenity services.

- The environment provides a biological, chemical and physical system that enables human life to exist.
- The system includes the atmosphere, river systems, the fertility of soil and a diversity of plant and animal life.
- These environmental goods & services are used by households and are essential to life.

Contd…



Links between the environment and the economy

Contd…

- Economy consists of production and consumption of goods and services.
- The environment provides raw materials and energy for production and household activity.
- These resources are either renewable or non-renewable.
- The use of a non-renewable resources reduces the finite stock of the resource forever.
- The waste products of production and consumption are absorbed by the environment.
- This sinking function allows much of such waste to be disposed of safely.

Contd…

- Some wastes are difficult for the environment to dispose of safely.
- Waste absorption capacity of environment is not infinite.
- Also provides amenity and cultural services (education recreation etc.).
- These functions may be competitive as well as complementary.
- Examples:
  - Excessive discharges of waste materials into the oceans will reduce their capacity to provide habitat for fish stocks (competitive).
  - Appropriate forestry policies can provide a sustainable source of timber and reduce soil erosion (complimentary).

Economic Growth and the Environment

- High economic growth may lead to depletion of non-renewable resources.
- If market operates efficiently, increased scarcity of these resources would increase their prices, which will give incentives for changes in the economic behaviour:
  - The direct consumption of these resources may fall.
  - The use of higher priced non-renewable resources in production will decline through substitution towards techniques of production that are less intensive in these inputs.
  - Incentives for search for new supplies of these resources (e.g. exploration of new oil fields).
  - Higher prices will encourage the development of new technologies that provides substitutes for the scarce resources (Increase in R&D).

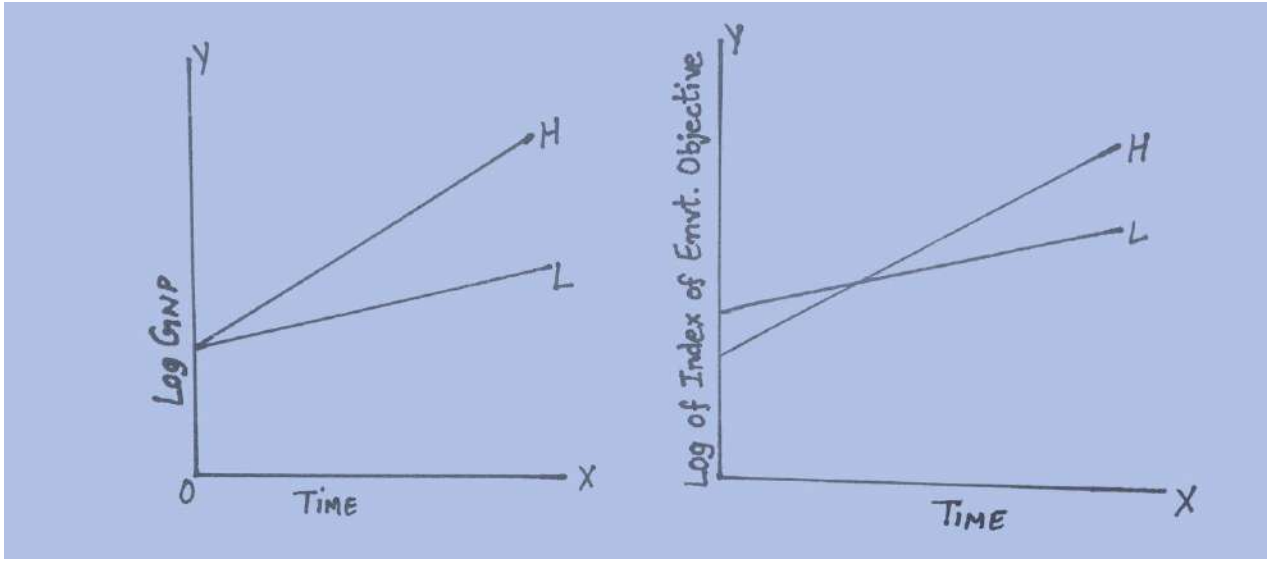
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- Supporters of economic growth argue that its contribution to environmental degradation is overestimated.
- They argue that economic growth release more resources, which can be used to reduce environmental degradation.
- This view is also supported by the World Bank.
- Grossman and Krueger (1995) suggest that in the early stages of economic development the level of environment degradation increased, but after this phase the environment improved with economic development.

Environmental Cost of Economic Growth

- Many economists argue that economic growth is necessary for achieving greater human welfare.
- However, measurement of economic growth in terms of output of goods and services is considered deficient as it does not take externality into account.
- Air and water pollution, volume of solid waste due to production of goods and services are the real cost of economic growth, which are not accounted for.
- Environmentalists challenge unlimited growth. To them, continued growth of output and population will eventually lead to losses of many species of flora and fauna.
- “Effluent reduces affluence”.

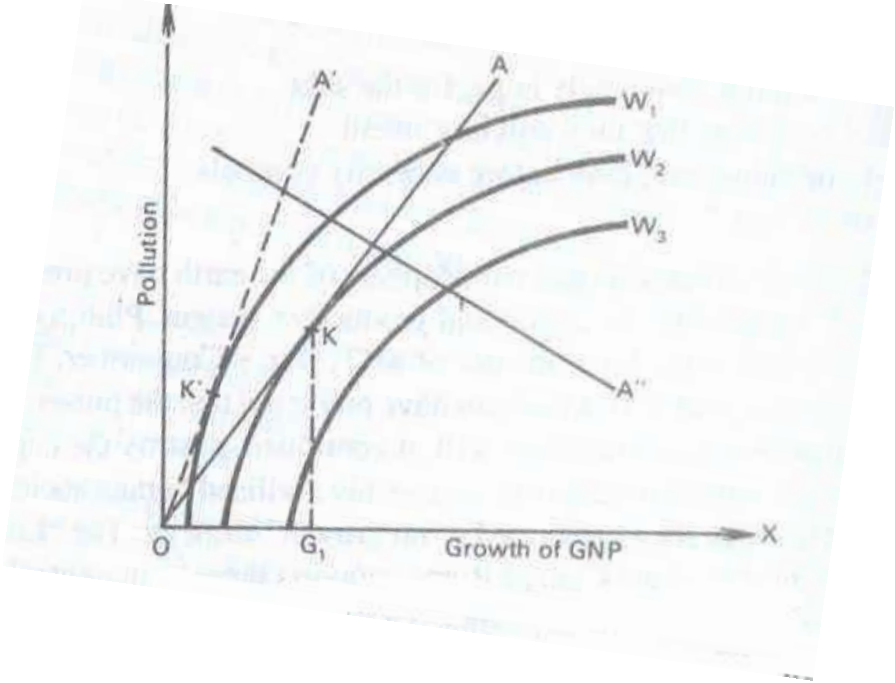
Trade off between Environmental objectives and Economic Growth



Contd…

- Which path is chosen depends on the rate of time discount for environmental purity, compared with that of GNP.
- Since the marginal utility of consumption declines with rising income, while relative value attached to reducing pollution increases with rising income, the rate of discount for the environment is likely to be lower than that of GNP.
- If this is so, those who give priority to environment ought to advocate a higher, not lower growth strategy and thereby promote rapid development of anti-pollution techniques, process and products.

Contd…



•OA: GNP growth path and pollution associated with it

•W1…w3 are Social Indifference curves, showing alternative combination of pollution and GNP

•If GNP increases faster than pollution, w will increase

•If grow path is steeper as shown OA', w will decline

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Environmental Economics

Environment and Economic Growth Linkages-II

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CONTENTS

- Environmental Kuznets Curve (EKC)
- Explanations for EKC

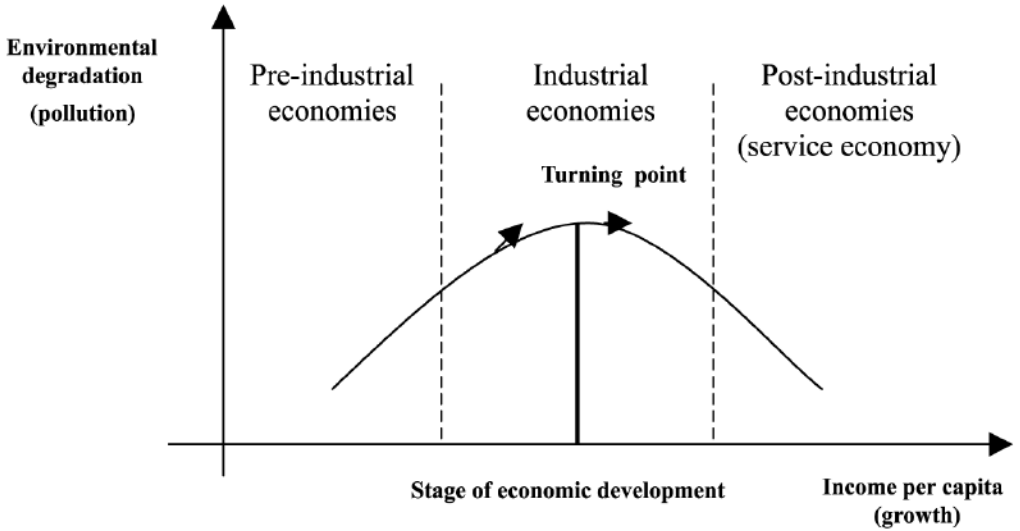
Environmental Kuznets Curve

- Following the analogy of income-inequality relationship postulated by Simon Kuznets (1955), an inverted-U relationship has been hypothesized between environmental degradation and economic growth.
- Hypothesis: Environmental degradation first rises as PC GNP rises due to inadequate attention to environmental concerns and then falls as more resources and greater attention are given to environmental concerns.
- If this hypothesis is true, then EKC suggests that environmental degradation in a country like India during the economic take-off stage would be a temporary phenomenon.
- Many cities in India are more polluted today than 3 decades ago, while reverse is true in case of many cities in developed nations (Tokyo, Paris, Stockholm).

Contd…

- A critical threshold level of PCI after which environmental condition improves.
- Absence of long time series data and a single measure of environmental quality index makes it difficult to determine how the EKC behave under the influence of various stages of environment and development.
- For example, it would be interesting to explore how the shape of EKC changes under the influence of population control, withdrawal of some subsidies, restructure of industries, etc.

Environmental Kuznets Curve



Source: Panayotou (1993)

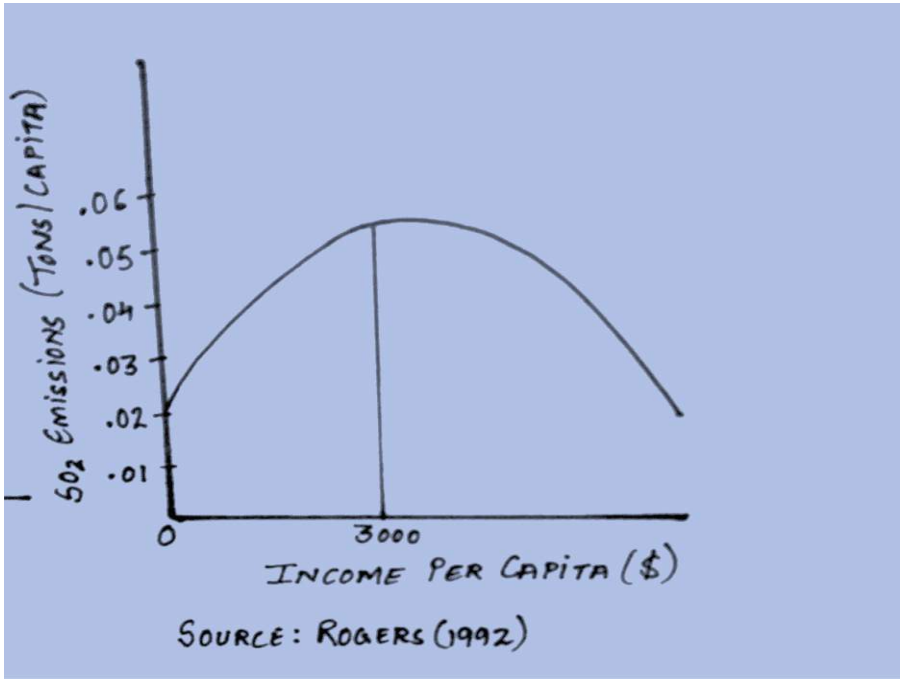
Explanations for EKC

- The transition of economy from agri. to industry and then to services.
- Three effects—scale, composition and the technology effects.
- The scale effect has a negative effect (more output more pollution).
- The composition effect may have a positive influence on the environment offsetting the adverse effects.
- High growth → New technology → Low Pollution
- EKC suggests that while scale effect prevails in the initial stages of economic growth, the composition and technology effects dominate in the later stages, resulting in improved quality of environment.

Contd…

- High income elastic environmental goods (market-driven forces).
- Increase in government resources to invest in environment protection.
- Influence of environmentalist lobby in public polices.
- Role of judiciary.
- International influences (trade & environment).

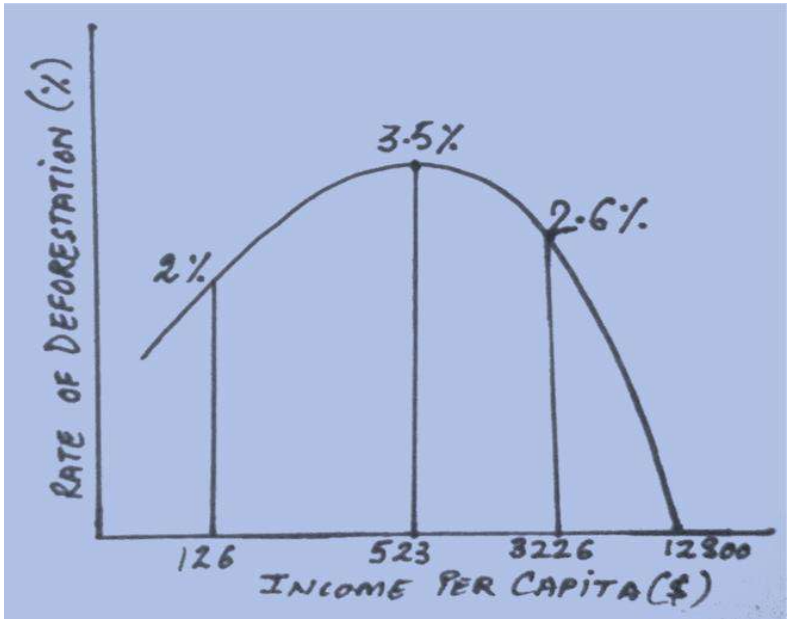
Contd…



- Rogers uses world bank data and identifies $3000 as critical PCI at which SO2emission in major cities decreases.

- Depending on the nature of environmental problem, there is variation in the critical PCI at which the environmental condition begins to improve.

Contd…



Source: Panayatou (1992)

- Panayotou (1992) establishes an empirical relationship betwee economic growth and deforestation as a measure of environmental degradation taking sample data from developing countries.

- Applied OLS to estimate EKC.

- Max rate of deforestation (3.5%) at $823, after that it declines slowly.

Let's Sum Up

- Growing economic activity (production and consumption) require larger inputs of energy and material, and generate larger quantities of waste by-products.
- Increased extraction of natural resources, accumulation of waste and concentration of pollutants adversely affect the carrying capacity of environment and human welfare, despite rising income. (beyond GDP)
- Trade-offs between the goals of achieving high rate of economic growth and attaining high standards of environmental quality.
- Relationship between economic growth and environmental degradation change sign from positive to negative as a country reaches a high level of income (less evidence of NKC in case of CO_2 emission).

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ENVIRONMENTAL SUSTAINABILITY-I

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DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



What is Sustainability?

- Sustainability has its roots in all cultures of the world and is understood as a 'survival strategy'.
- Etymologically, the concept sustainability is derived from the Latin word '*sustinere*' which mean 'to hold/maintain/support/endure.'
- The dictionary meaning of sustainability is **'the quality of being able to continue over a period of time'** (Cambridge dictionary).
- It may be defined as the ability to continue a definite behaviour for an infinite time.

Contd..

- '…Every generations should leave water, air and soil resources as pure and unpolluted as when it came on earth' .

  (Solow, 1991, from a UNESCO Documents).

- 'Each generation should leave undiminished all the species of animals it found existing on earth'.

Contd…

- Definition by Daly (1990)

  1. Renewable resources shall not be used faster than they can regenerate. (sustainable yield)

  2. Pollution and wastes shall not be put into the environment faster than the environment can recycle them or render them harmless.

  3. Nonrenewable resources shall not be used faster than renewable substitutes (used sustainably) can be developed.

- However, this definition is said to be incomplete because it does not talk about ‘level of quality of life’.

Contd…

- The earliest instance of sustainability can be found in ‘Limits to Growth’ by Meadows, Meadows, Randers and Behrens III (1972).
  - This book talked about the limits of ‘earth’s carrying capacity’ in relation to long term global trends in population, economics and the environment.

- ‘It is possible to alter these growth trends and to establish a condition of ecological and economic stability that is sustainable far into the future. The state of global equilibrium could be designed so that the basic material needs of each person on earth are satisfied and each person has an equal opportunity to realize his individual human potential.’ (Meadows et al., 1972, p.24).

Contd…

- However, the concept of ‘sustainability’/sustainable development was first promulgated in the framework of World Strategy of the International Union for Conservation of Nature and Natural Resources (IUCN) in 1980.

- Further, it was elaborated and defined at World Commission on Environment and Development (WCED) by the Norwegian PM Gro Harlem Brundtland in ‘Our Common Future’ Report in 1987.
- Sustainable Development is defined as which ‘meets the needs of the present generation without compromising the ability of the future generations to meet their own needs’ (WCED, 1987).

Contd…

- This definition was further extended by the Earth Summit, 1992 for Agenda 21.

- The world Summit on sustainable development in 2002 emphasised three pillars of sustainability- Social, Environmental and Economic as per the summit motto- ‘People , Planet and Prosperity’.

- However, these two words sustainability and sustainable development are not same, although, the fundamental sense of both these words are the same.

Sustainability Vs. Sustainable Development

- "While sustainability denotes a system property referred to as quality, we believe that the key to the sustainable development concept is provided by … Brundtland definition and Article 1 of the Rio Declaration (UNCED, 1992): human beings are at the centre of concerns for sustainable development. They are entitled to a healthy and productive life in harmony with nature.'' (Moldan, Janouskova and Hak, 2012, P.4).
- The term sustainability implies a characteristic of a system, a programme or a resource that will last forever (Singh and Shishodia, 2007, p.55).

What Needs To Be Sustained?

- Schools of thought:
- Economists view: maintaining and improving human welfare through long-term stability in economic output, income or consumption.
- In this perspective, Robert Solow (1991) defined sustainability as making sure the next generation is as well off as the current generation and ensuring that this continues for all time (Kolstad, 2011, P. 46).
- Ecologists view: focus is on sustaining the ecosystems. That is, long-term preservation of biosphere.
- Anthropologists view: they focus on protecting and preserving the cultures and languages.

Environmental Sustainability

- Originally, ‘environmentally responsible development’ was used by World Bank in 1992.

- Then ‘environmentally sustainable development’ was employed (Serageldin and Streeter, 1993) ;
- and the concept of ‘environmental sustainable development’ was developed (Goodland, 1995).



Contd…

- Environment sustainability: It may refer to – the rates of renewable resource harvest, pollution creation, and nonrenewable resource depletion that can be continued indefinitely.
- It is synonym with sustainable development.
- It implies responsible utilization of the natural resources such that the present generation's needs are fulfilled without jeopardizing the needs of the future generation.
- It can be defined in terms of the carrying capacity of the natural systems.
- Carrying capacity: The maximum population size of species that a natural system can sustain indefinitely or for a defined period for a given standard of living.

Contd…

- According to Goodland (1995), ‘environmental sustainability seeks to improve human welfare by protecting the resources of raw materials used for human needs and ensuring that the sinks for the human wastes are not exceeded, in order to prevent harm to humans.’

Readings

- Moldan, B., Janouskova, S. and Hak, T. (2012). How to understand and measure environmental sustainability: Indicators and targets. *Ecological Indicators*, 17:4-13.

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ENVIRONMENTAL SUSTAINABILITY-II

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Economic Sustainability

- Sustainable capitals- man-made, natural, human and social
- Sustainable use of renewable natural resources
- Use of non-renewable resources by future generations
- Slow-rate of depletion of non-renewable energy sources
- Optimal resource management (Markandya and Pearce, 1988)

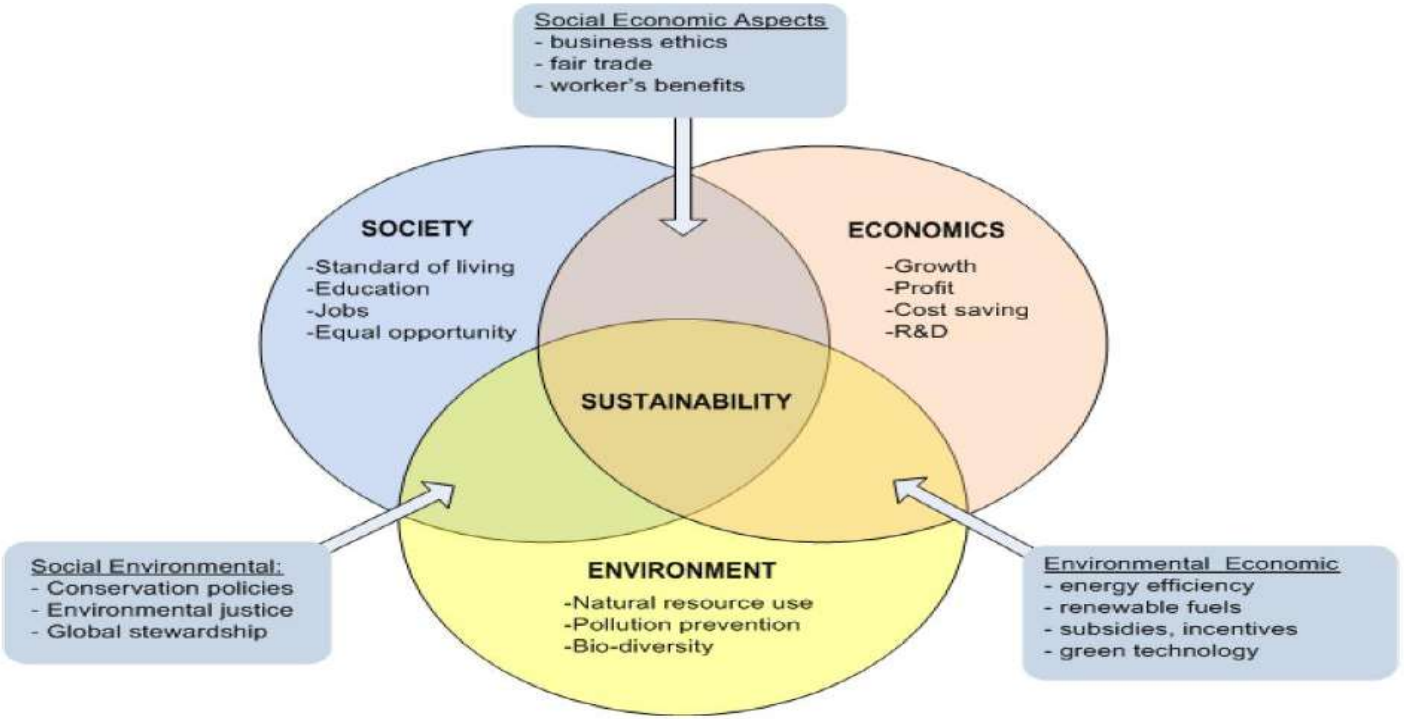
Social sustainability

- It may be defined as ‘the extent to which social values, social identities, social relationships and social institutions can continue into the future’ (Black, 2004).

- ‘Social sustainability requires that the cohesion of society and its ability to work towards common goals be maintained. Individual needs, such as those of health and well-being, nutrition, shelter, education and cultural expression should be met’ (Gilbert, 1996).

The Three Pillars of Sustainability

Social, Environment, Economic



https://www.e-education.psu.edu/eme807/node/575

Weak Vs. Strong Sustainability

- On the basis of different schools of thought, sustainability can be categorized as weak sustainability and strong sustainability.
- Weak Sustainability: It is based on the idea of pioneering works of two neoclassical economists- Robert Solow (1974) and John Hartwick (1977).
- Solow supported the view that man-made capital and natural capital are substitutes.
- The depletion in the natural capital can be accommodated by investment in man-made capital.

Contd…

- It is based on the belief that ‘ what matters for future generations is only the total aggregate stock of man-made, human and natural capital, but not natural capital as such.’ (Neumayer, 2013, p. 1).

- It is based on ‘substitutability paradigm’ and is an extension to neoclassical welfare economics. Natural capital and man-made capital are substitutable.

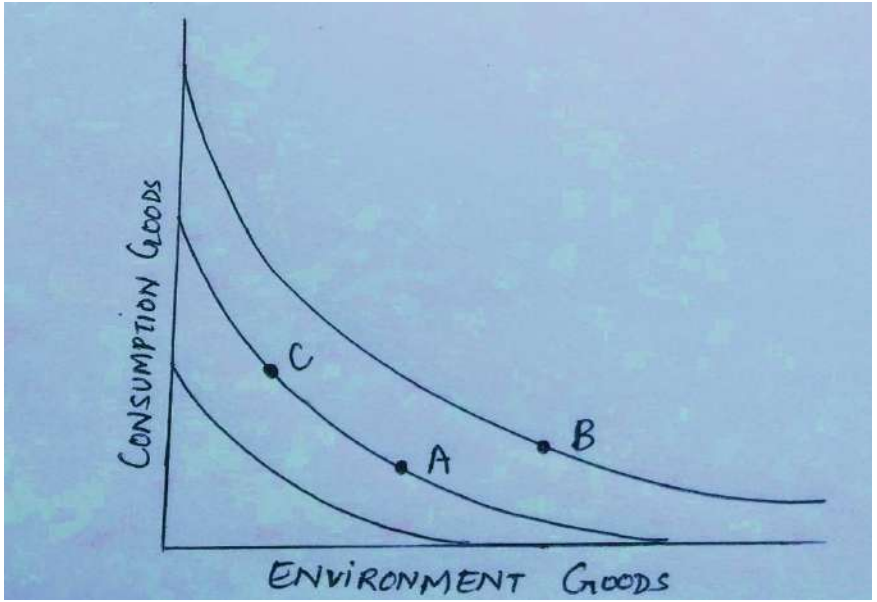
- It implies that economic growth can manage environment quality by substituting natural capital with human capital.

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- **Strong sustainability** states that the natural capital and man-made capital are complementary and cannot be substituted for any economic activity.
- Strong sustainability is based on ‘non-substitutability paradigm’ where natural capital is taken as non-substitutable to other capitals in production consumption economy.
- This implies that under this criterion both the utility level and environment quality must not decline (Berck and Helfand 2011, P. 476).

Weak Vs Strong Sustainability

- This can be explained with the help of following figure.
- The movement from A to B or staying at A shows strong sustainability and
- Moving from A to C is an example of weak sustainability.



Source: Berck and Helfand (2011), p. 475

Criteria for Environmental Sustainability

- The OECD Environmental Strategy for the First Decade of the 21st century (2001) stated four criteria for environment sustainability (P.6):

- Regeneration: rate of exploitation of the renewable resources must not exceed long-term rate of natural regeneration.

- Substitutability: non-renewable resources to be utilized to the level that can be substituted with the renewable resources or other form of capital.

Contd…

- Assimilation: the waste discharged or pollution in the environment should not exceed its assimilative capacity.

- Avoiding irreversibility: Irreversible adverse effects of anthropogenic activities on different aspects of nature's systems and processes like -- ecosystems, biochemical and hydrological cycles need to be avoided.

Measuring Environmental Sustainability

- Many environmental sustainable composite indicators have been proposed. Viz.
- Environmental Sustainability Index (ESI)/
- Environmental Performance Index
- Ecological Foot Print
- State of the Environment
- Environmentally adjusted GDP
- Environmental Vulnerability Index
- Sustainable livelihood security index
- Living planet Index
- Wellbeing index



Readings

- Moldan, B., Janouskova, S. and Hak, T. (2012). How to understand and measure environmental sustainability: Indicators and targets. *Ecological Indicators*, 17:4-13.

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ENVIRONMENTAL PERFORMANCE INDEX

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HUMANITIES AND SOCIAL SCIENCES



What is Environmental Performance Index?

- EPI is the basis for comparison of countries based on the extent to which they preserve their environment.
- It is a metrics that helps to analyze targets set by UN sustainable development goals and Paris climate change regarding- how the countries are meeting their established environmental policy making goals.
- Therefore, EPI metrics identify problems, track trends, highlight policy successes and failures, identify best practices, and optimize the gains from investments in environmental management.

History, Conception and Evolution

- The Pilot EPI 2002 is built on the pervious work collaborated by Yale and Columbia University, i.e., Environmental Sustainability Index (ESI) which was published between 1999-2005.

- Although both are related with distinct purposes.

ESI Vs EPI

- Similarities: Both the indices try to bridge the gap between governmental commitments concerning environmental goals and the weak ability to measure conditions with respect to those environmental goals.
- Differences:
- So far target audience is concerned:
  - The ESI primarily targets at decision makers, public and analysts for comparing long term environmental trajectories of nations.
  - It makes use of the best available current data to determine which nation/s is/are comparatively well situated to achieve lasting environmental sustainability.

Contd…

- On the other hand, EPI helps in national comparisons on recent efforts to manage common policy objectives concerning a narrow set of parameters- Air quality, Water quality, climate change and Ecosystem protection (EPI, 2002).
- Both EPIs 2006 and 2008 are outcome –oriented performance index (EPI, 2006 & 2008).
- Both of these indices 2006 and 2008 EPIs attempt to assess current environmental conditions to provide information to policy makers for assessing policy responses to environmental challenges.

Chronology of EPI Reports

- Pilot EPI 2002
- Pilot EPI 2006 (Period taken 1994-2006)
- EPI 2008 (1994-2007)
- EPI 2010 (1994-2009)
- EPI 2012 (2000-2010)

Published in 2012

- EPI 2014 (2002-2014)
- EPI 2016 (1950-2016)
- EPI 2018 (1950-2018)

Contd…

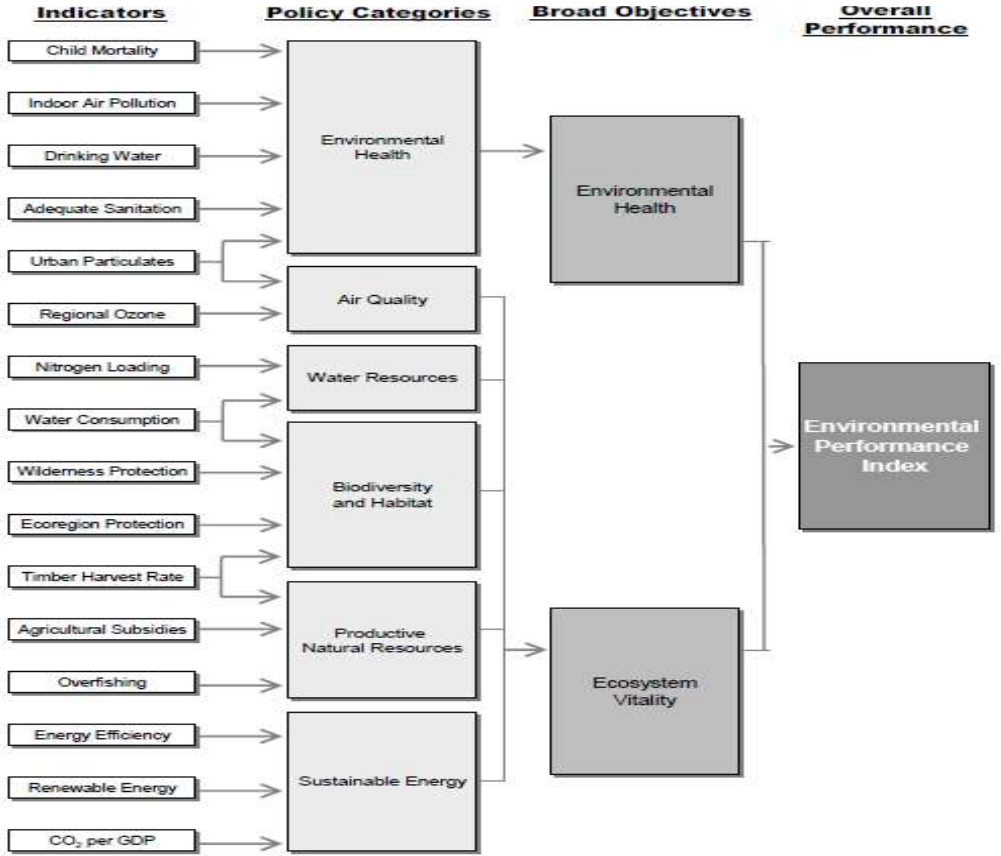
- The EPI focusses on two broad environmental protection objectives:

(1) reducing environmental stresses on human health, and

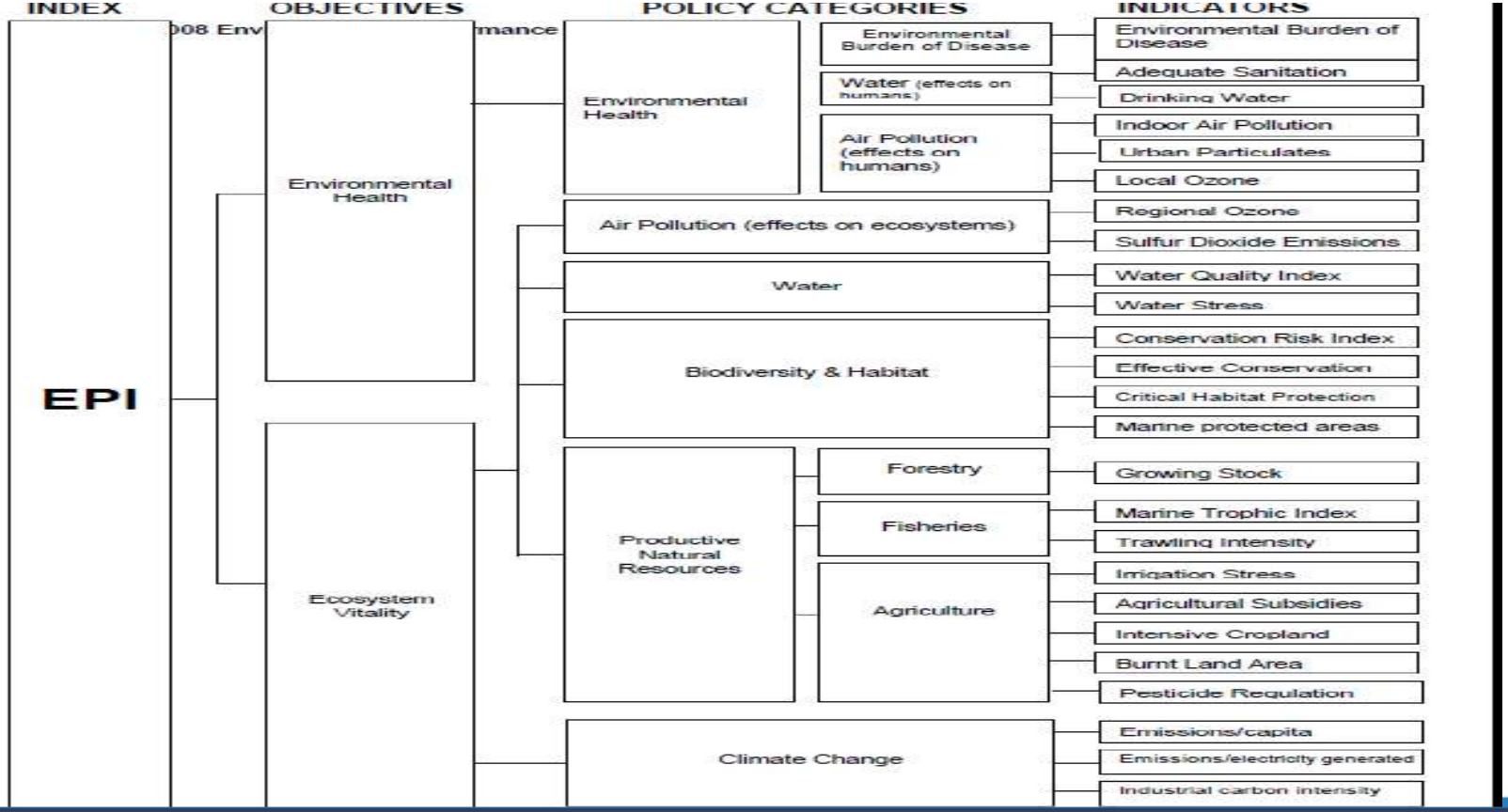
(2) promoting ecosystem vitality and sound natural resource management.

Methodology: Components of EPI 2002 and 2006

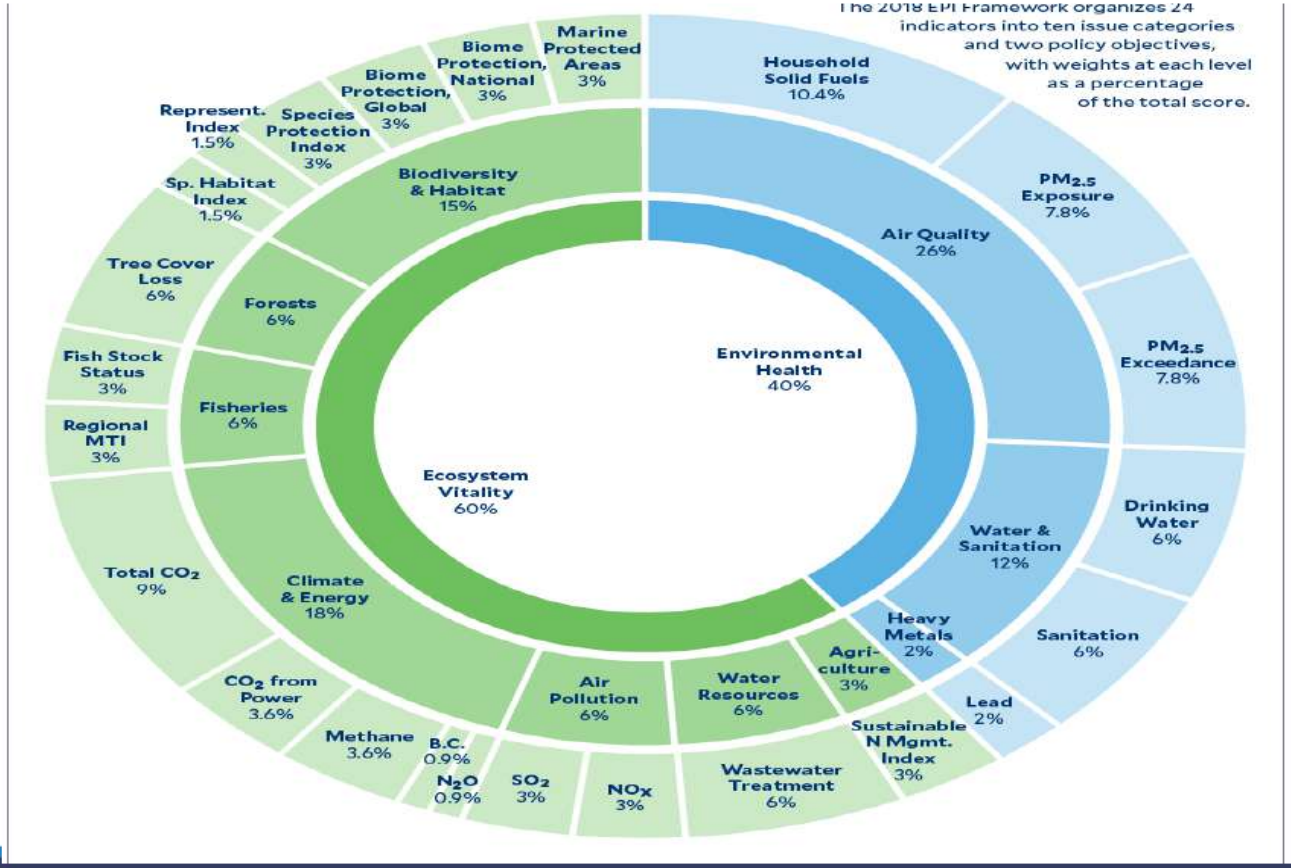
Indicator	Variables
Air Quality	Concentration of Sulfur Dioxide
	Concentration of Nitrogen Oxide
	Concentrations of Lead
	Total Suspended Particulates
Water Quality	Dissolved Oxygen
	Phosphorous concentrations
	Water pollution (Biological Oxygen Demand)
Climate Change	Carbon Economic Efficiency (CO_2 emissions per GDP)
	Carbon Lifestyle Efficiency (CO_2 per capita)
Land Protection	Protected Areas (percent of territory)
	Waste Disposal at Landfills per populated land area
	Paper Recycling Rate
	Glass Recycling Rate



EPI 2008


INDEX
OBJECTIVES
POLICY CATEGORIES
INDICATORS
EPI
Environmental Health
Ecosystem Vitality
Environmental Health
Air Pollution (effects on ecosystems)
Water
Biodiversity & Habitat
Productive Natural Resources
Climate Change
Environmental Burden of Disease
Water (effects on humans)
Air Pollution (effects on humans)
Forestry
Fisheries
Agriculture
Environmental Burden of Disease
Adequate Sanitation
Drinking Water
Indoor Air Pollution
Urban Particulates
Local Ozone
Regional Ozone
Sulfur Dioxide Emissions
Water Quality Index
Water Stress
Conservation Risk Index
Effective Conservation
Critical Habitat Protection
Marine protected areas
Growing Stock
Marine Trophic Index
Trawling Intensity
Irrigation Stress
Agricultural Subsidies
Intensive Cropland
Burnt Land Area
Pesticide Regulation
Emissions/capita
Emissions/electricity generated
Industrial carbon intensity

2018 EPI


The 2018 EPI Framework organizes 24 indicators into ten issue categories and two policy objectives, with weights at each level as a percentage of the total score.
Environmental Health 40%
Ecosystem Vitality 60%
Air Quality 26%
Water & Sanitation 12%
Heavy Metals 2%
Biodiversity & Habitat 15%
Climate & Energy 18%
Air Pollution 6%
Water Resources 6%
Agri-culture 3%
Forests 6%
Fisheries 6%
Household Solid Fuels 10.4%
PM2.5 Exposure 7.8%
PM2.5 Exceedance 7.8%
Drinking Water 6%
Sanitation 6%
Lead 2%
Sustainable N Mgmt. Index 3%
Wastewater Treatment 6%
NOx 3%
SO2 3%
N2O 0.9%
B.C. 0.9%
Methane 3.6%
CO2 from Power 3.6%
Total CO2 9%
Regional MTI 3%
Fish Stock Status 3%
Tree Cover Loss 6%
Sp. Habitat Index 1.5%
Represent. Index 1.5%
Species Protection Index 3%
Biome Protection, Global 3%
Biome Protection, National 3%
Marine Protected Areas 3%

ESI and EPI Objectives and Design

- Table 1

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BENEFIT-COST ANALYSIS-I

And THE ENVIRONMENT

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

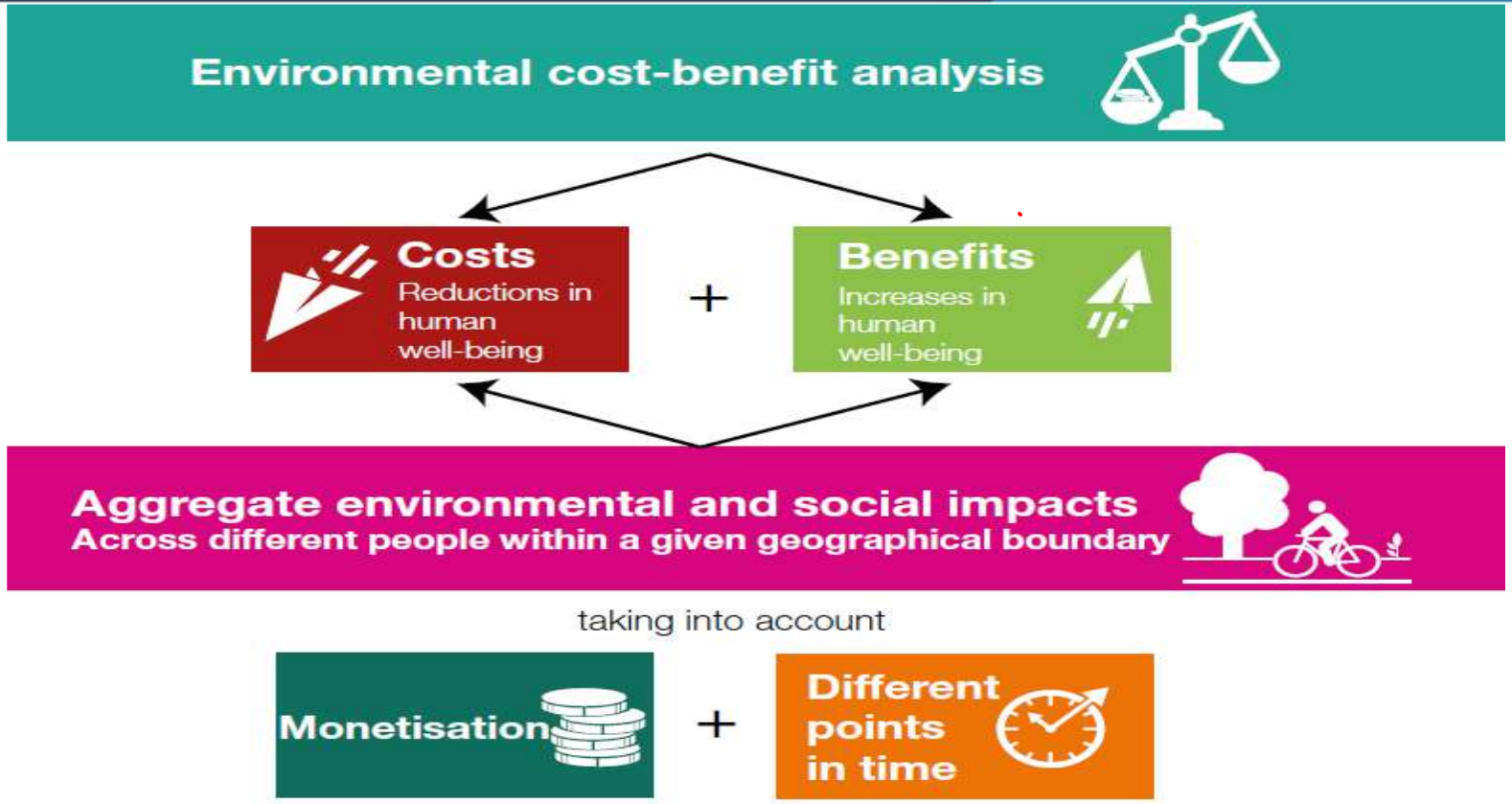


Contents

- What is CBA?
- Rationality of using CBA
- History and Theoretical Foundations
- Basic Concepts

What is CBA?

- Cost –Benefit analysis (CBA) is a practical tool which is used in actual policy formulation and decision-making.
- Where all the benefits and costs are measured in terms of monetary units.
- Since 1960s, CBA has been the major appraisal technique for public investment projects and public policy.
- In CBA, benefits are defined as increases in human wellbeing (utility) and costs are defined as reductions in human wellbeing.


Environmental cost-benefit analysis
Costs
Reductions in human well-being
+
Benefits
Increases in human well-being
Aggregate environmental and social impacts
Across different people within a given geographical boundary
taking into account
Monetisation
+
Different points in time

OECD (2018)

Rationality- Why Use CBA?

- CBA provides a model of rationality: It helps the decision-maker to look at who are the beneficiaries and losers of a potential decision/action across different social groups by focusing on both the spatial and temporal dimensions (by treating time though discounting factors).
- By taking into account all gains and losses, it forces wider view on decision-makers.
- CBA is clear in its requirement: Setting the alternatives for achieving the chosen goal is a fundamental prerequisite of this tool. Therefore, it has the capacity to determine the optimal scale of the policy by appraising the alternatives in getting maximised net benefits.
- It seeks explicit preferences rather than implicit ones: it directly looks what people want, what is their preference.

Theoretical Foundations of CBA?

- Its theoretical origins date back to issues in infrastructure appraisal in France much earlier in the 19th century (Jules Dupuit, 1844, 1853).
- In its application to environmental issues, the focus of CBA is to address 'externality' or third part effect. For e.g. Pollution- by weighing the benefits of pollution against the damage done in monetary units.
- Thinkers on externality- Henry Sidgwick (1883); Alfred Marshall (1890); A. C. Pigou (1920)
- Modern Welfare Economics- Hicks (1939,1943), Kaldor (1939);
- Pareto optimality conditions; Kaldor- Hicks compensation principle
- Pigou's notion of the divergence between private and social cost = the value of the externality.

Contd…

- Valuation procedure for externality involves monetary valuation, where the money value of benefits and/costs reflects individual preferences which is captured through Willingness to pay.

- These preferences are revealed through choices in the market place.

Basic Concepts

- The criterion for selecting a project is social benefits must exceed its social costs.

- Social benefits- are the aggregates of benefits across different social groups and

- Social costs- are the aggregates of costs of social groups.

Contd..

- The two basic aggregation rules:
  - First, aggregating benefits across different social groups or nations involves summing willingness to pay for benefits, or willingness to accept compensation and for losses (WTP, WTA respectively), regardless of the circumstances of the beneficiaries or losers.
  - Second, aggregation rule requires that higher weights be given to benefits and costs accruing to disadvantaged or low income groups.
- (Rationale for this second rule is that marginal utilities of income be higher for the low income group)

- WTP: An amount of money an individual is willing to pay for obtaining a benefit or avoiding a loss. It is an expression of individual values.

- WTA: It is an amount given towards compensation for foregoing a benefit or tolerating/incurring a loss.

- In case of valuing environmental gain,
  - Individuals reveal their preferences for an environmental gain by their WTP for it or WTA to forego the benefit.

- In case of environmental loss,
  - Individuals are willing to pay to prevent loss or are willing to accept compensation in order to tolerate the loss.

Contd…

- Aggregating over time involves discounting. Discounted future benefits and costs or Present values of benefits or costs are calculated at constant prices.
- Concepts of WTP and WTA are used for measuring compensating and equivalent variations.
- For deciding the appropriateness of a government intervention in the economy, B-C analysis is used as a tool.
- We know that Total Surplus= CS+PS
- TS is equivalent to B–C
- Max TS= Pareto Optimum

Selected Reading

- OECD.(2018). *Cost-benefit analysis and the environment: Further Developments and Policy use*. OED Publishing: Paris.
- OECD. (2006).Cost- benefit analysis and the environment: Recent Developments.
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- https://www.oecd-ilibrary.org/sites/9789264085169-5-en/index.html?It_emId=/content/component/9789264085169-5-en
- http://www.lse.ac.uk/lse-player?id=4568

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BENEFIT-COST ANALYSIS-II

And THE ENVIRONMENT

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



Content

- Major Steps
- Issues in Estimation of Costs and Benefits
- Recent developments in environmental CBA
- Taxonomy of Environmental Values and Non- Market Valuation

Major Steps

- Asking the relevant questions: What policy or project is being evaluated? What alternatives are there?
- For an initial screening of the contribution that the project or policy makes to social wellbeing to be acceptable, the present value of benefits must exceed the present value of costs.
  - PV (B) > PV (C)
- Whose costs and benefits are to count?
  - Time preferences and discounting
  - Capturing ‘relative prices’ (through time), risk and uncertainty
  - The distributional incidence of costs and benefits

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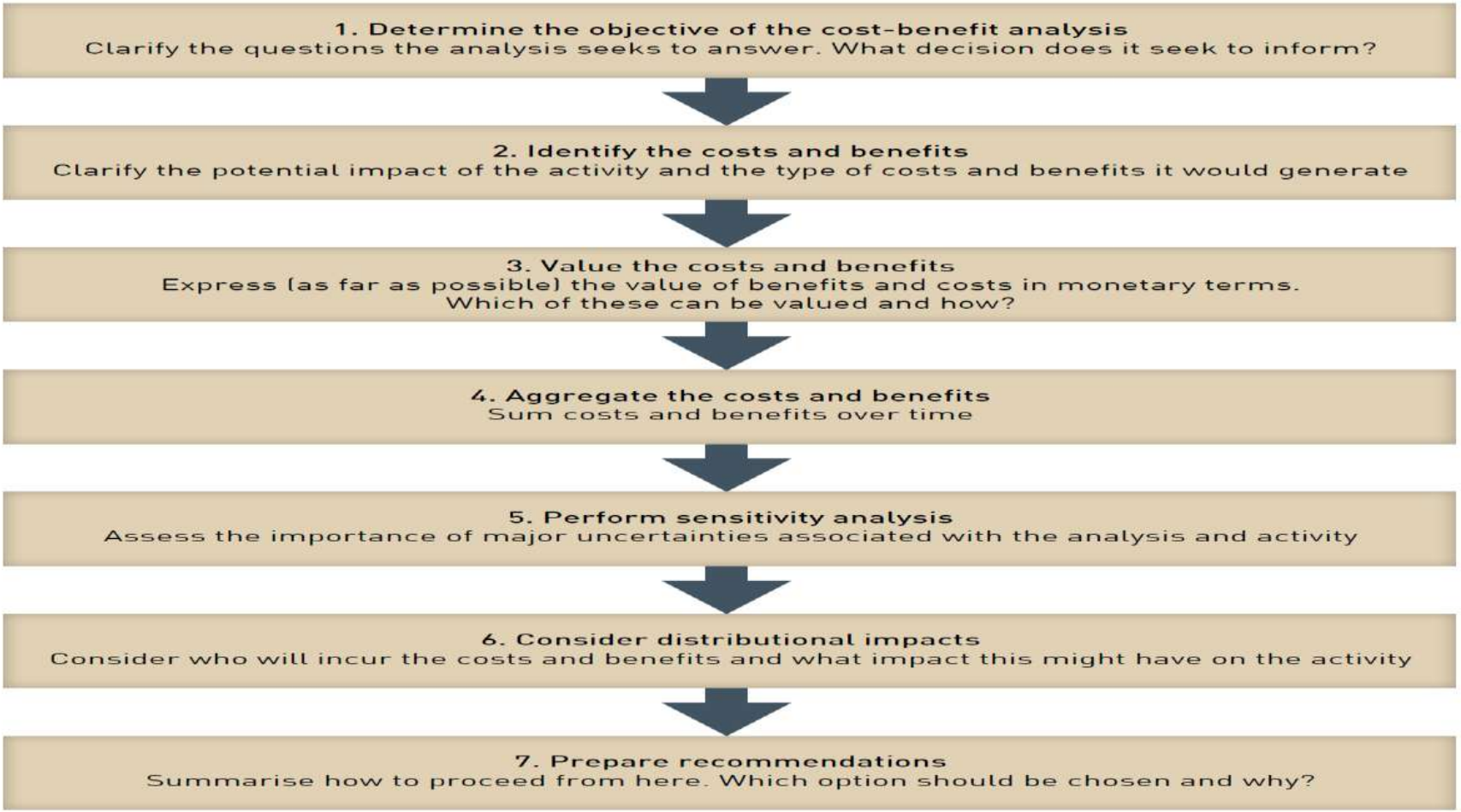
- Decision rules:
  - NPV (Net present value)>0
  - B/C ratios (in case of budget constraints)
  - IRR

$\frac{B}{C} > 1$ ✓, ✗

NPV = 0

IRR > MARR ✓

< = ✗


1. Determine the objective of the cost-benefit analysis
Clarify the questions the analysis seeks to answer. What decision does it seek to inform?
2. Identify the costs and benefits
Clarify the potential impact of the activity and the type of costs and benefits it would generate
3. Value the costs and benefits
Express (as far as possible) the value of benefits and costs in monetary terms.
Which of these can be valued and how?
4. Aggregate the costs and benefits
Sum costs and benefits over time
5. Perform sensitivity analysis
Assess the importance of major uncertainties associated with the analysis and activity
6. Consider distributional impacts
Consider who will incur the costs and benefits and what impact this might have on the activity
7. Prepare recommendations
Summarise how to proceed from here. Which option should be chosen and why?

Issues in Estimation of Costs and Benefits

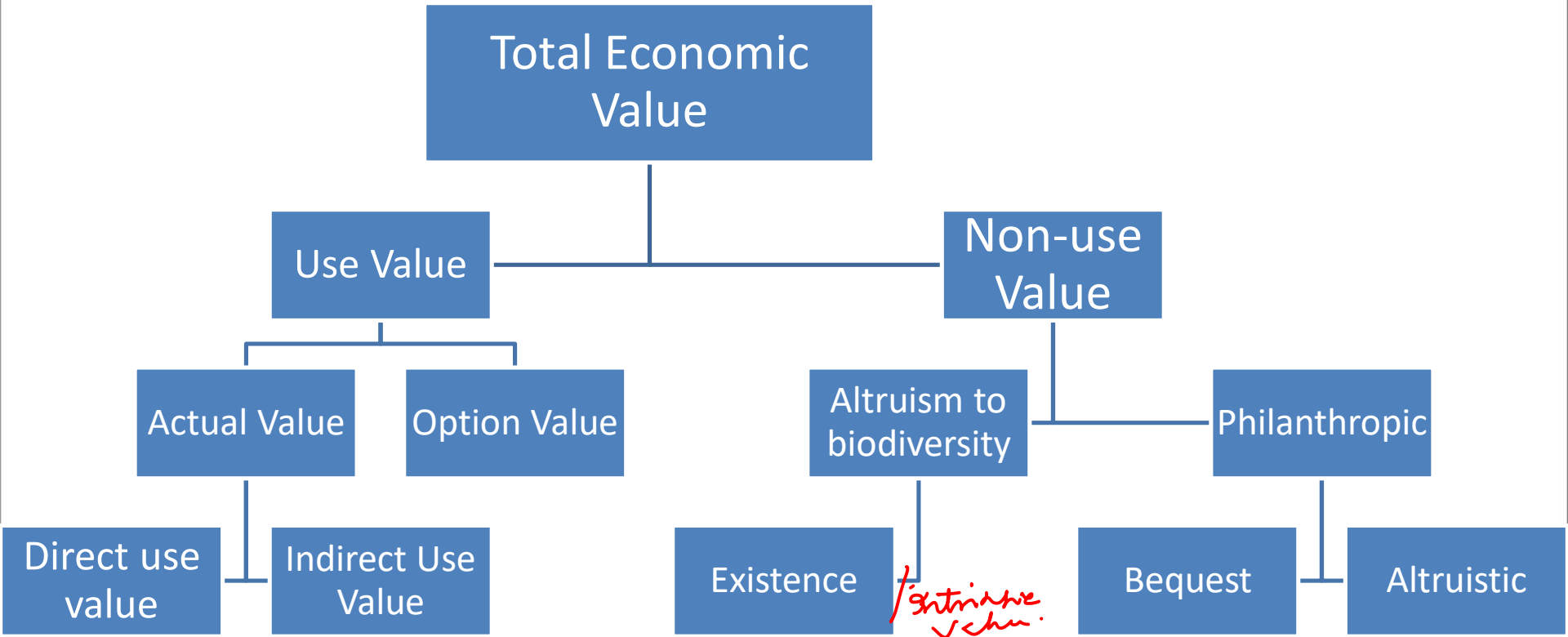
- Identification of costs and benefits
- Expected costs
- Expected benefits
- Putting weightages (on the basis of distribution incidence of a change in policy)
- Capturing opportunity costs
- Choosing a rate of discounting
- Valuating costs and benefits
- Reliability and validity of the numbers

Recent Developments in Environmental CBA

Finding money values:

- By principle, we know that the net sum of all the relevant *WTP*s and *WTA*s for a project outcome or policy change defines the *total economic value* (TEV) of any change in well-being due to a project or policy.
- TEV can be different according to the type of economic value arising.
- It is usual to divide TEV into *use* and *non-use* (or *passive use*) values.

Taxonomy of Environmental Values



Contd..

- Use value: relates to
  - direct actual use of the good
    - which may be extractive (timber) or non-extractive (recreation), planned use (a visit planned in the future) or possible use,
  - Indirect Use Value: indirectly used by individuals like carbon storage in forest
- Among Actual, planned and possible use- for the last category, people may be willing to pay to maintain a good in existence in order to preserve the *option* of using it in the future.

Contd…

- Option value: the value of environment as a **potential benefit** as opposed to actual present use value, anticipating their use at future date.

- It is a form of use value. E.g. value of maintain the option for direct or indirect use of forest in future.

- **Total use value**= actual direct/indirect use value+ potential use value

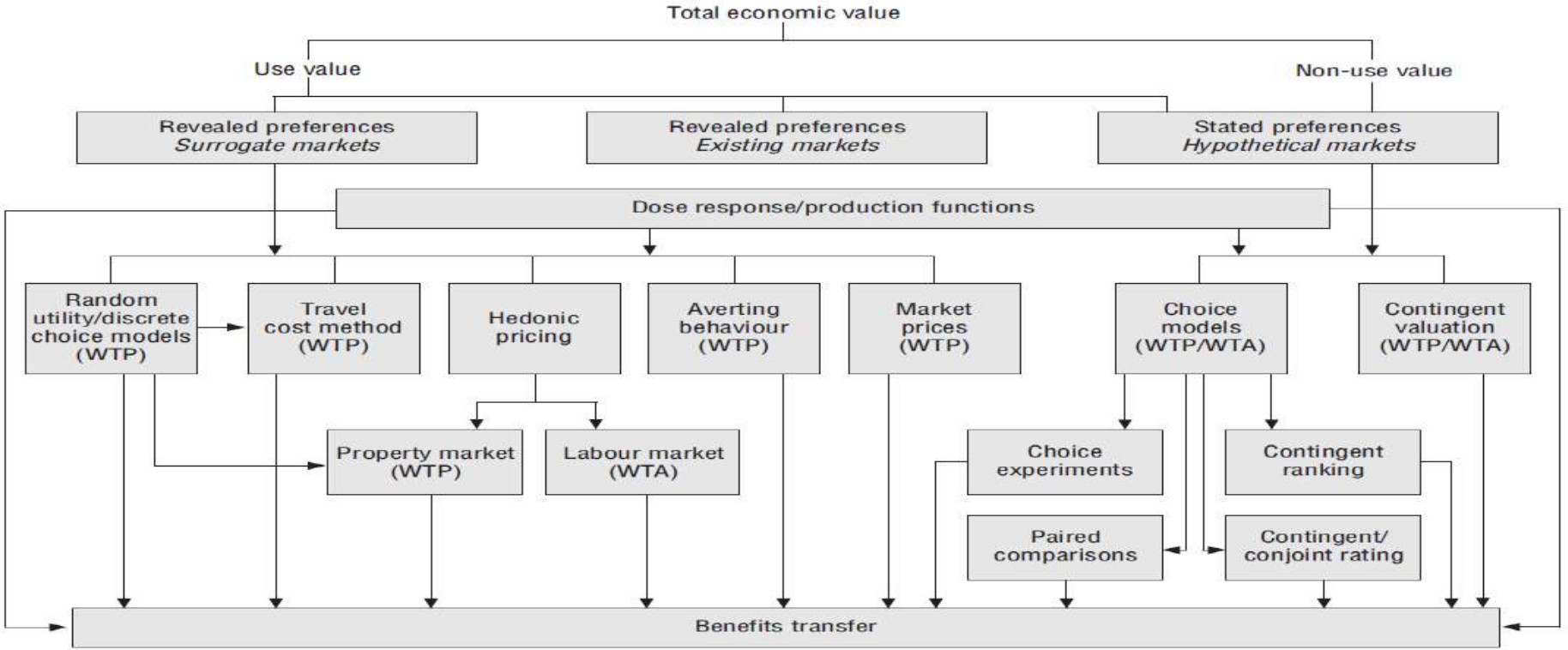
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- Non-use value/Passive value: It is taken to be entities that reflect people's preferences, but include concern for, sympathy with, welfare of non-human beings and the values of which are unrelated to human use.
- refers to WTP to maintain a particular good in existence even though there is no actual, planned or possible use.
- Existence Value: values expressed by individuals such that those values are unrelated to use of environment or future use.
- Intrinsic Value: It may mean the value that stays 'in' something and that is unrelated to human preferences or observations altogether. Hence, this value is not included in TEV. But it may be taken as a kind of altruism.
- Bequest Value: value of nature left for future generations.

Contd…

- In practice, techniques of environmental valuation can be seen as measuring either in totality or in parts.
- Some other related ways to get environmental valuation is in terms of ‘Ecosystem services 'which reflect human wellbeing.

Measuring Total Economic Value and Non-Market Valuation



OECD. (2006, p.88)

Selected Reading

- OECD.(2018). *Cost-benefit analysis and the environment: Further Developments and Policy use*. OED Publishing: Paris.
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Environmental Economics

Consumer Demand for Environmental Goods-I

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- What is so special about an environmental good?
- Measuring Demand: Revealed Preference and Stated Preference
- Ordinary versus Compensated Demand
- Consumer Demand for Environmental Goods: Restricted Demand
- Let’s Sum Up

Introduction

- One of the cornerstones in economics is the understanding of consumer preferences for goods.
- The typical way of representation of those preferences is through demand functions.
- The problem with demand curves for environmental goods is that there are typically no markets, so we have no observations on how much of an environmental good is consumed at different prices.
- Economists have developed the concepts of public bads and externalities to describe the characteristics of environmental goods that lead to market failure.

What is so special about an environmental good?

- Environmental problems really involve a trade-off between using resources (money) for conventional goods and services and using same resources for environmental protection.
- What a demand curve of environmental goods represents: How much is the consumer willing to pay for particular levels of environmental goods.
- Distinctions between environmental and ordinary goods is the existence of a market characteristics. E.g. it is difficult to generate a demand curve for clean air in an urban area.
- The absence of a market is the major factor that complicates finding the demand curve for environmental goods.

Measuring Demand

- A conventional demand curve plots quantity demanded as a function of price. Without a market there is no price.
- Because of the absence of markets for environmental goods, measuring demand is not straightforward.
- *Two Approaches to measure* demand: **Revealed preference and Stated preference.**
- Within Revealed Preference: **hedonics and household production**
- Within the Stated preference: the dominant approach is **contingent valuation.**

Revealed Preference

- **In revealed preference**, we observe a real choice in some market and infer information on the trade-offs between money and the environmental good.
- Eg, we may notice that two communities are identical except that one has high housing prices and clean air and other has lower housing prices and dirty air.
- We may infer that the **difference in housing prices reflects the value of people place on clean air.**

Contd…

- **In the hedonic approach**, the goal is to see how the price of a conventional good (e.g. house) varies as the amount of closely related environmental good changes (e.g. the air quality in the vicinity of the house). This relationship is then used to infer value.
- **The household production approach** starts with the assumption that consumers will combine private goods with environmental goods to 'produce' another good, which is real source of utility.
- For instance, soundproofing may be combined with noise that impinges on the outside of a house to obtain particular indoor noise levels.

Stated Preferences

- Stated preferences, basically involves asking people how much an environmental good is worth.
- Opinion polls and surveys are used to derive this information.
- In stated preferences, the dominant approach is **contingent valuation.**
- Contingent valuation relies on direct revelation of demand from consumers. It means "**value contingent on there being a market"—if there were a market, how much would you pay for the environmental good?**

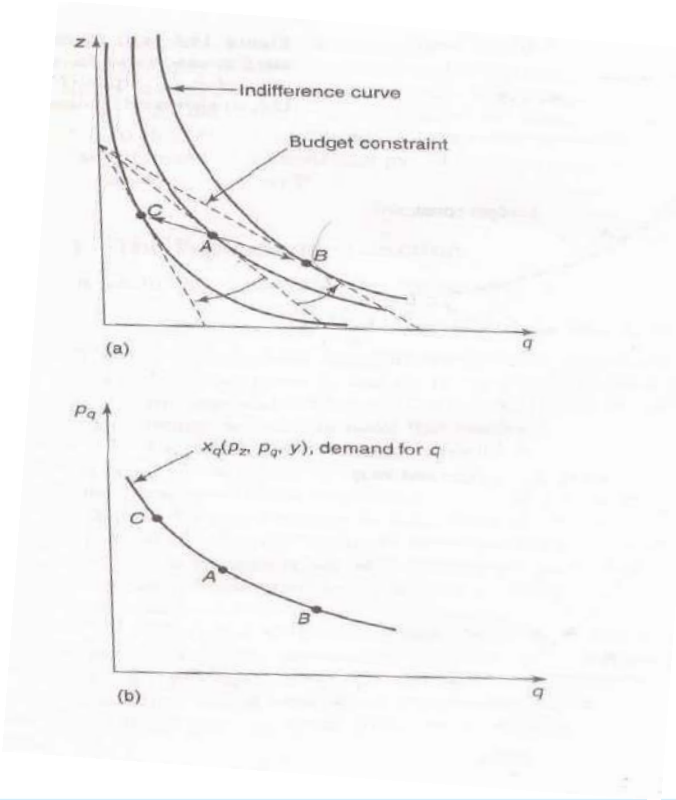
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- Demand is assessed by directly asking a sample of potential consumers of the environmental good.
- Contingent valuation is a type of **constructed market**. In a constructed market, a researcher will take a situation in which no market exists and generate a market. Constructed markets can be hypothetical or real.
- For instance, laboratory experiments.

Analysis of Consumer Demand for Market Goods

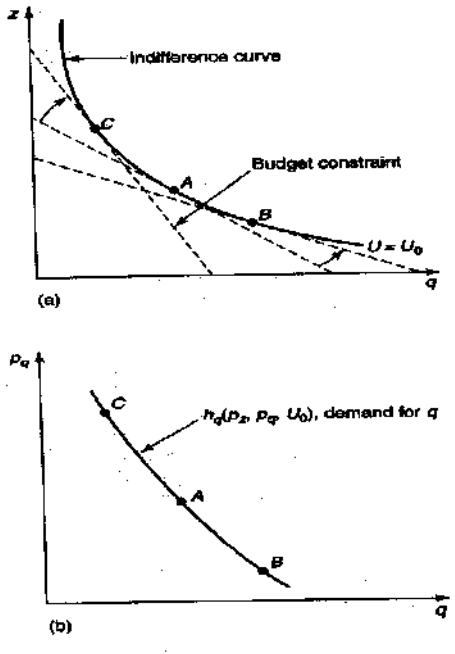
- The starting point for analysis is a utility function for a consumer. This utility function gives the level of utility for various bundles of say z and q products.
- Two types of Demand Functions:
  - Ordinary Demand Function
  - Compensated Demand Function

Ordinary Demand



- An ordinary demand function gives the amount of a good a consumer chooses as a function of prices and income: Xq (Pz, Pq, y).
- Figure illustrates the graphic process of generating a demand curve for good q from indifference curves between goods z and q.
- The utility maximizing choice of q also changes, tracing out the demand curve. This is an ordinary demand curve because **income, rather than utility is kept constant as the price of q changes.**

Compensated Demand



- An alternate way to generate a demand curve is to **keep utility constant as the price of q changes** which can be done by adjusting income so that the consumer remains on the same utility curve.

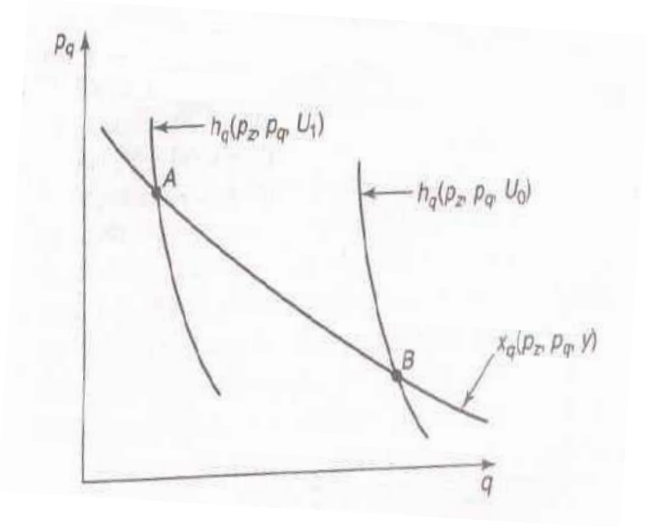
- The demand curve that traces out quantity demanded as a function of price, keeping utility constant, is called **compensated demand function: Hq (Pz,Pq, U)** where U is a particular level of utility.

Contd…

- Two points on a compensated demand curve represent two relative prices but same level of utility.
- In contrast, two points on an ordinary demand curve represents two different prices and two different utility levels (since income is constant).
- We often wish to evaluate the effects of a government policy that changes relative prices.
- To evaluate such policy, we want to examine the results of the price changes only, not income effects (since income effects can be addressed through programmes to redistribute income). In other words we want to use compensated demand function.

Relationship between two demand curves

- The figure shows a plot of an ordinary demand curve (xq) and two compensated demand curves (hq), for different utility levels.
- One point shared by the xq and hq is point A.
- At A on the ordinary demand curve, the consumer enjoys the some level of utility, call it U1. At the level of utility, we can trace out a compensated demand curve. It must pass through A.
- A similar logic would apply to point B.



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Environmental Economics

Consumer Demand for Environmental Goods-II

Prof. S. P. Singh

Department of Humanities and Social Sciences



CONTENTS

- The expenditure function
- Consumer demand for environmental goods
- Sum up

The Expenditure Function

- A useful way of rewriting the compensated demand function is in terms of the expenditure function:

  $E(P_z, P_q, U) = P_z H_z(P_z, P_q, U) + P_q H_q(P_z, P_q, U)$ (1)

  This indicates how much income is necessary to achieve utility U, facing price Pz and Pq.

The expenditure function is useful for two reasons:

1. Its units are measurable in money, so we can easily compare different values of the expenditure function.
2. The parameters of the expenditure function include prices, not quantity consumed.

Contd…

- The expenditure function is very similar to the utility function.
- A simple relationship between the expenditure function and a compensated demand function.
- Suppose you have Rs. 1000 to buy an optimal basket of goods, including a packet of bread.
- If price of bread goes up by Rs.5. How much money do you need to be able to afford the same basket (keeping utility constant)?
- Precisely, the No. of packets of bread in the basket multiplied Rs.5.

$$\Delta E(P_z, P_q, U) = H_q(P_z, P_q, U)\,\Delta P_q \quad \text{Or} \quad (2)$$

$$H_q(P_z, P_q, U) = \Delta E(P_z, P_q, U)/\Delta P_q \quad (3)$$

- That is, the ratio of the change in the expenditure function to a change in the price of good is the same as the compensated demand for good.

Consumer Demand for Environmental Goods

Restricted Demand

- The quantity of environmental goods or public goods in general is not chosen by the consumer.
- Let q be the environmental good; z be the conventional good. The ordinary demand function for z can be written as Xz (Pz, q, y) and the compensated demand function as Hz (Pz, q, U).
- Example: We might observe the demand for boat rentals (z) and see that it depends on the price of z and water quality (q).
- The expenditure function in this case is straightforward:

$$E(Pz, q, U) = Pz\ Hz(Pz, q, U) \qquad (4)$$

Contd…

- The demand functions and the expenditure functions are very similar except that the quantity of q appears rather the price of q.
- We call this type of demand and expenditure functions restricted demand and expenditure functions.
- There is an interesting relationship between q and E.
- Let us start with a grocery basket with a variety of groceries that you have chosen, including five packets of bread priced at Rs.25 each. The cost of basket is Rs. 2000: Rs.125 for bread and Rs.1875 for other items.
- If you do not know the price of bread, can you know how much the bread is worth to you only by observing your demand for the other items?

Contd…

- The answer is yes.
- We can determine the value of one more packet of bread.
- Suppose we give you one more packet of bread and observe that you do not need to spend quite as much on other grocery items to keep utility constant.
- We observe from your compensated demand function for the other grocery items how much income must be taken away to keep utility constant.
- This is your marginal valuation of a packet of bread.

Contd…

- Now consider the restricted expenditure function again. What happens if we increase q by a little say Δq? Utility will go up unless we reduce income accordingly.
- The ratio of the change in the expenditure function to the change in q is the value of q at the margin, known as marginal willingness to pay for q,(that is Pq).
- If q were offered on the market at this price, and income were supplemented by Pq q, the amount chosen voluntarily would be exactly q.
- Thus we have a connection between compensated demand and the restricted demand function:

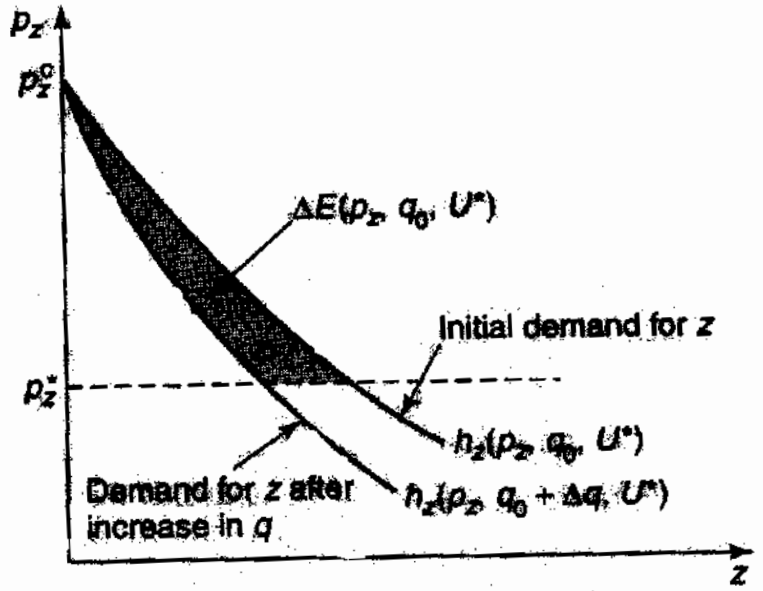
  $\Delta E (Pz,q,U)/ \Delta q = -Pq \qquad (5)$
- Think of Eq.(5) as having four unknowns, Pz, Pq, q and U. If values of three are known, we can get the value of the fourth, using Eq (5).

Condt…

- The eq.5 can be rearranged to yield q as a function of remaining variables:

  q = Hq (Pz, Pq,U)
- Which is the compensated demand function for q.
- We start with a compensated demand curve for a market good, z: hz (Pz,q,U). Suppose price of z is P*z and the amount of q is qo.
- Now suppose we increase q a bit to qo + Δq. Since q is environmental quantity, presumably, it is desirable. Thus to keep utility constant, we need to takeaway some z when q is increased. The resulting compensated demand curve for qo + Δq is shown in the figure next slide, slightly lower than the initial one.

Condt…



Compensated Demand z: Hz (Pz,q,U). price of z = P*z and the amount of q =qo.
Hz (Pz*, q0,U*) for z product
Hz (Pz*, q0+ Δq,U*) for z product, when q increases
Shaded area = income equivalent of the change in q or marginal willingness to pay for change in q.

Let's Sum Up

- Ordinary Demand and compensated demand
- Restricted Demand Function
- Expenditure Function
- Demand for environmental goods can be determined by measuring demand for ordinary goods for different levels of the environmental goods, even though there is no market for that good!

REFERENCES

- Pearce, D., Atkinson, G., & Mourato, S. (2006). *Cost-benefit analysis and the environment: recent developments*. Organisation for Economic Co-operation and development.
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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Welfare Effects of Price Change

Prof. S. P. Singh

Department of Humanities and Social Sciences



Contents

- Introduction
- Welfare Effects of Price Change: Compensating Variation and Equivalent Variation
  - Compensating Variation
  - Equivalent Variation
- Welfare Effects of Environmental Change
  - Compensating Surplus
  - Equivalent Surplus
- Let’s Sum Up

Introduction

- Utility or welfare of a consumer is affected by the change in the prices of various commodities she/he consumes.
- Utility is an ordinal measure so the consumer is better or worse off depending on decrease or increase in price of a product.
- Indifference curve (IC) show that the given level of utility can be obtained using different combinations of two products.
- The optimum combination is achieved at the point where slope of IC is equal to the price ratio (price line) of the two products.
- Higher IC indicates higher level of utility.
- Welfare effects of price change are explained by CV and EV
- Welfare effects of environment change are explained by CS and ES.

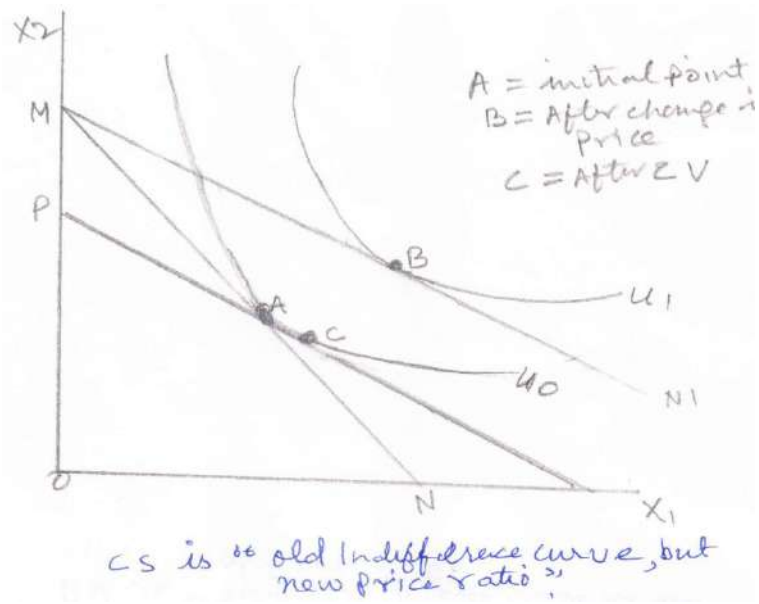
Welfare Effects of Price Change

- Compensating Variation
- Equivalent Variation

Compensating Variation

- Compensating variation (CV) measures the net revenue of a planner who must compensate the consumer for a price change after it occurs, bringing her back to the utility level at the initial level.
- It would be negative if the planner would have to pay the consumer a positive level of compensation because the price change makes her worse off.
- It measures the difference in attaining the initial utility level at the initial and subsequent prices. Compensation takes place after the price so that compensating variation uses the after change prices.
- We can easily see that the individual is worse off than originally – but how much worse off? There is one way we can get an answer to this question – by simply asking “how much money do we need to give to the individual to compensate him or her for this price rise?”

Graphic Presentation of CV

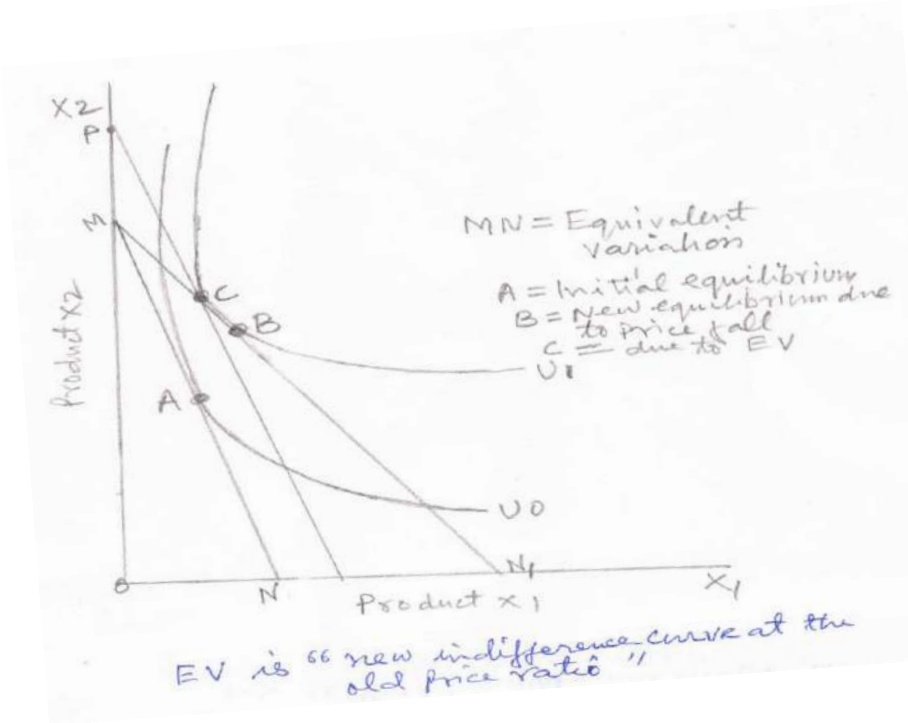


The consumer is initially at point A on the original budget line and has a utility denoted by U0. Then the price of x1 falls and the budget line rotates out (NEW Budget Line) and the consumer's new equilibrium is now B. The consumer's new utility is U1. CV is "**old indifference curve, but new price ratio**". Therefore we shift the new budget line parallel until it is tangent to the original IC at point C.

Equivalent Variation

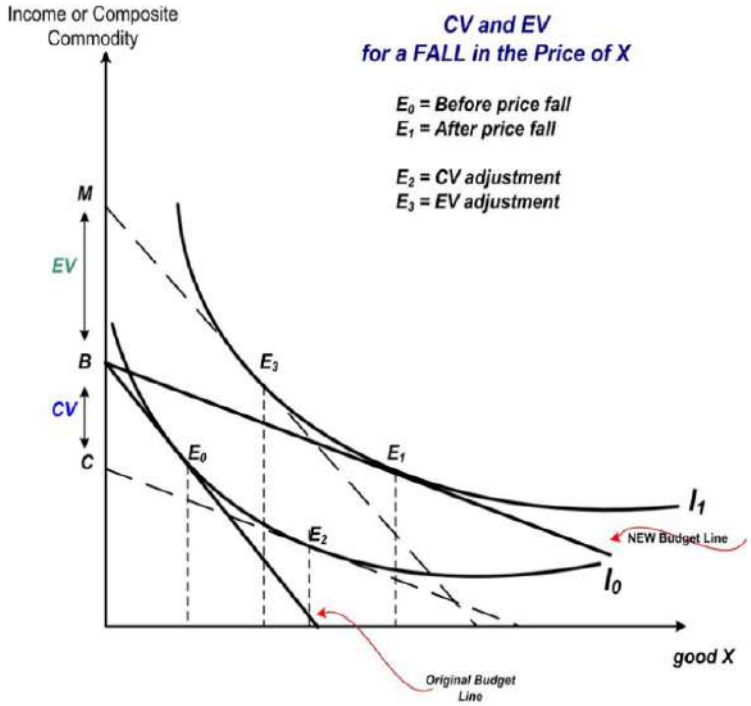
- EV, or equivalent variation is the adjustment in income that changes the consumer's utility equal to the level that would occur if the event had happened.
- In the case of a positive economic change, such as a fall in price, EV would be the increase in income that would give the consumer the same additional utility that would happen if a price did fall.
- In the case of a negative economic change, EV would be the amount of income that would be taken away to lower the consumer's utility to the level that would happen if the change had occurred.

Graphic Presentation of EV



The consumer is initially at point A on the original budget line and has a utility denoted by U0. Then the price of x1 falls and the budget line rotates out (NEW Budget Line) and the consumer's new equilibrium is now B. EV is "**new indifference curve at the old price ratio**". Therefore we shift the original budget line parallel until it is tangent to the new IC at at point C. ES = MP in the figure.

CS Versus ES



CV is "old indifference curve, but new price ratio".

EV is "new indifference curve at the old price ratio".

Welfare Effects of Environmental Change

Compensating Surplus and Equivalent Surplus

- The amount of money that would keep the individual at the original level of utility with the change in q is called the compensating surplus: CS (q0,ql) = E(Pz, q0,U0) - E(Pz, ql, U0)
- While the amount of money that would move the individual to the new level of utility without the change in q is called the equivalent surplus. ES((q0,ql) = E(Pz, q0,U1) -E(Pz, q0,U0)
- Conceptually, these two terms are very analogous to compensating and equivalent variation in the case of a market good.
- The reason for using the term "surplus" instead of "variation" is that the consumer is not free to vary the quantity of q. In all other respects, the concepts are identical.

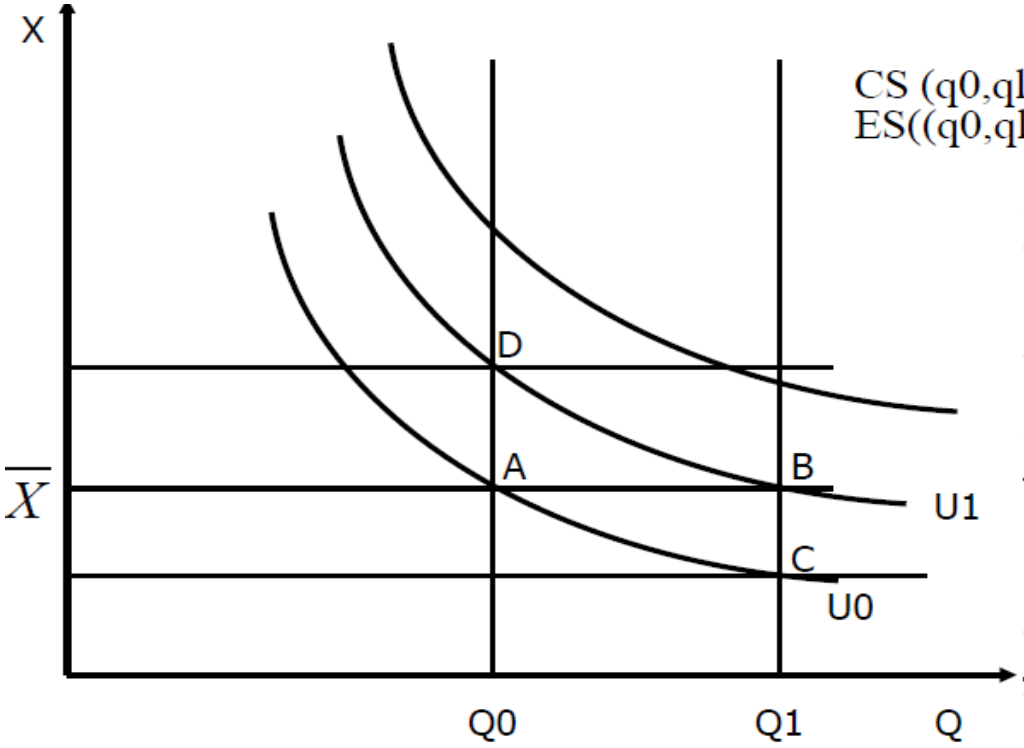
Contd…

- CS(q0,ql) is the CS of moving from q0 to q1 and ES(q0, q1) is the analogous ES:
- CS (q0,ql) = E(Pz, q0,U0) -E(Pz, ql, U0) (A)
- ES((q0,ql) = E(Pz, q0,U1) -E(Pz, q0,U0) (B)
- If the amount of q that the consumer consumes has been reduced from q0 to ql (if q1< q0). So we want to compensate consumers for the change in q by giving them an amount of money that brings them back to the same utility level.
- This is the CS defined in Eq. A as the difference in the income needed to achieve the old level of utility at the old quantity [E(Pz, q0,U0)] and the income needed to keep the same level of utility at the new quantity [E(Pz, q1,U0)].

Contd…

- Consumer surplus can be either a willingness to pay (WTP) or a willingness to accept compensation (WTAC) measure.
- An individual's willingness to pay for a change in environmental quality is based on a theory of rational choice, and is therefore a consistent estimate of preferences.
- The individual choice of all market goods and services is constrained by fixed monetary income, and the prices of these goods and services (maximizing utility s.t. income constraint).
- Q0 is fixed level of environmental good, the problem is to define the economic value of an increase in the level of the environmental good from Q0 to Q1 (remember, differences in utility are not measurable => consumer surplus).

WTP/WTAC for improved environmental services



$$CS\ (q0,ql) = E(Pz,\ q0,U0) - E(Pz,\ ql,\ U0) \quad (A)$$
$$ES((q0,ql) = E(Pz,\ q0,U1) - E(Pz,\ q0,U0) \quad (B$$

starting in A:

to increase quality (Q0 => Q1), and to keep consumption good X fixed, the individual's utility increases from U0 to U1.

two questions:

- what is the maximum she is willing to pay (WTP) to secure the change from Q0 to Q1?
answer: the individual would give up the market good (X) to reach the original level of utility (U0) (B => C) compensating surplus (CS).

- what is the minimum compensation the individual is willing to accept (WTAC) to forgo the increase in the environmental good?
answer: the individual would require an increase in the level of the composite consumption good (X) to reach U1, when Q0 would have => Q1 (A => D) (ES)

Contd…

- The ES is the change in income that would have the same ("equivalent") effect as the quantity change, without the quantity change actually occurring.
- We compare the expenditures that yield utility UI at the original quantity with the expenditure necessary to yield utility U0 at the original quantity:
- ES(q0, q1) = E(Pz, q0,U1) -E(Pz, q0,U0) (C)
- which is actually the same as Eq. (b) because E(Pz, ql,U1) = E(Pz, q0,U0)

Contd…

- Suppose X lives in a house close to the ocean with only a vacant field between him and the ocean. Y owns the vacant field and wants to build a house that will block X's view of the ocean.
- There are two ways to view the money equivalent of the damage to X from the house built by Y.
- One measure would be the maximum amount X would pay Y not to build the house.
- Clearly, X would be willing to pay an amount such that his income after payment and with a clear view of the ocean results in the same utility as his income without the payment and without the view. **This is the ES.**
- X actually will be indifferent between paying this amount to prevent the loss of the view and not paying but losing the view.

Contd...

- Another way of looking at the value would be if the town X lives in has laws that prohibit construction that blocks another's view without permission of the viewing party.
- If Y wants to build his house, he has to pay X for that right.
- How much would he have to pay X?
- He would have to pay the amount of money so that X's utility with the payment but without the view is the same as with no payment but no blocked view.
- This is the compensating surplus—Y has to compensate X for the loss of utility from taking away his view.

Let's Sum Up

- Indifference Curve
- Compensating variation and Equivalent Variation
- Welfare Effects of Change in Environmental Quality

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NPTEL ONLINE CERTIFICATION COURSE

NON-MARKET ENVIRONMNETAL VALUATION: REVEALED PREFERENCE METHODS

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



Content

- Environmental Valuation
- Why Non-Market Valuation?

- Approaches for Non-Market Valuation
- Revealed Preference Methods (RPMs)
- Theoretical Background of RPMs

- Types of RFM
- Advantages and Limitations of RPMs

Environmental Valuation

- In CBA, we quantify and value the environmental resources by weighing their usefulness and benefits.

- The usefulness can be categorized as direct use and indirect use .

- Economic valuation involves the monetary measure of the change in an individual's well-being due to a change in environmental quality. This measured value is known as Total Economic Value (TEV).
- TEV= UV+NUV, where

  UV= DUV+IUV+OV

  NUV=EV+BV

Why Non-Market Valuation?

- Public policies/projects are appraised by their impacts (tangible and intangible) on the wellbeing of individuals that are quantified in terms of costs and benefits expressed in monetary units.

- However, many impacts of policies/projects are intangible in nature.

- Hence, the value that individuals place on these intangible impacts cannot be observed in the ambit of market information like price, consumption, etc. and therefore, cannot be estimated in actual markets.

Approaches for Non-Market Valuation

- In those cases, where markets are not available, two main approaches can be used to estimate economic value of non-market goods and services, viz.
    - Revealed preference methods and
    - Stated preference methods

Revealed Preference Methods

- Revealed Preferences Method (RPM) provide methodologies for estimating environmental values in the context of consumers and producers making consumption and production choices in the market.

- Hence, these methods look at 'surrogate markets'- that analyze or infer preferences for nonmarket goods as implied by past behaviour in an associated market.

Contd…

- Surrogate market approach looks for a market in which goods or factors of production are brought and sold
- It observes that environmental benefits or costs that are frequently attributes of those goods or factors.
- It is used when one cannot directly estimate the market prices for certain environmental goods.
- In the absence of a market, information can be drawn from related marketed goods to determine the use-value of the non-marketed goods.
- Consumptive decision of marketed goods may reveal consumer's preferences for environmental non-marketed goods. Hence a surrogate market is created.

Contd…

- These are based on observations of individual choices (actual choice).
- Here, non-market goods are not traded in markets but they are related in consumption to goods that are traded in markets.
- Example???





How do you value pure water or water quality?

How do you value impact of water quality on your health??

Contd…

- Example: when an individual buys water purification systems, like Aqua guard Arrow,- this purchasing behaviour reveals the minimum amount s/he would be willing to pay for improved water quality.
- RPM is called the household production approach and is also known as market based method where impacts of non-market goods are measured by observing actual behaviour and purchases made in actual markets.
- Therefore, RPM can be defined as a method which determines the value that an individual holds for an environmental good by simply observing her/his purchase of those goods in the market that directly/indirectly relate to environmental quality.

Theoretical Background of RPM

- This method is drawn from P.A. Samuelson's Revealed preference theory (1938) which states that the preferences of individual consumers can be revealed by their purchasing behaviour, assuming income of the consumer and prices of goods remaining constant.
- Revealed preference theory is an alternative approach to the theory of consumer behaviour.
- Which one would be the best indicator of consumers' preferences/demand for goods and services?
  - Utility of goods/services (based on Jeremy Bentham's Utilitarian approach)
  - Consumers' purchasing behaviour

Overview of Revealed Preference Methods

Method	Revealed behaviour	Conceptual framework	Types of application
Hedonic pricing	Property purchased; choice of job	Demand for differentiated products	Property value and wage determinants
Travel cost	Participation in recreation activity at chosen site	Household production; complementary goods	Recreational demand
Averting behaviour/defensive expenditure	Time costs; purchases to avoid harm	Household production; substitute goods	Health: mortality and morbidity
Costs of illness	Expenditures to treat illness	Treatment costs	Health: morbidity

Source: OECD (2006)

Advantages of RPM

- These models are based on actual behaviour and hence are more credible for policy making.

- It is applicable to valuate use value of environmental goods/services.

Limitations of RPM

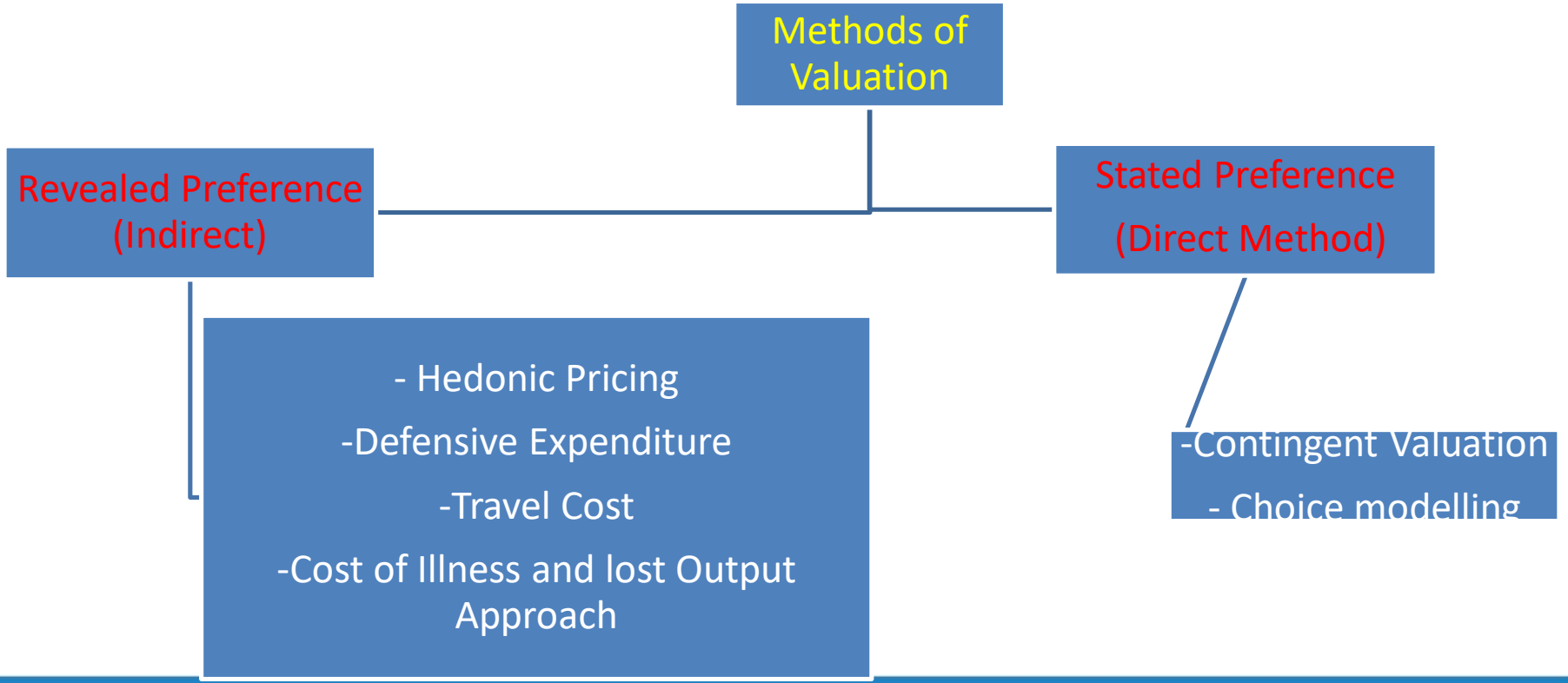
- These methods cannot estimate non-use values as they are based on use related behaviour.

- They cannot estimate values for levels of quality that have not been experienced and revealed by the market.

Stated Preference Methods

- Stated Preference Methods depend on values that are expressed by consumers and producers in the context of hypothetical market scenarios (preferred choice).

Methods of Non-Market Valuation



Selected Readings

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2003). (Eds). *A primer on nonmarket valuation*. Springer: New York. Chapter. 8.

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HEDONIC PRICING METHOD

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Contents

- What is hedonic price method?
- Applications of HPM
- HP function
- Stages of HPM
- HP Measure of Value
- Recent Developments
- Issues of HPM

What is Hedonic Price Method?

- The Hedonic Price Method (HPM) is a revealed preference method of valuation.
- It is based on our observation, i.e., the price of a large number of market goods is a function of a set of their characteristics.
- For example: price of a laptop depends on its configuration, size, colour, etc.
- This approach seeks to explain the value of a commodity as a bundle of valuable characteristics (like environmental).
- It is only applicable to measure use-values.
- It uses surrogate markets for placing a value on environmental quality.

Contd…

- It was derived from the characteristics of theory of value proposed by Lancaster (1966) and Rosen (1974), although Waugh (1926) used a kind of hedonic modelling in the study of price differences in fresh vegetables.
- It was first applied to environmental valuation by Ridker and Henning (1967).
- The HPM uses different statistical/econometric techniques to estimate the implicit price of each of these characteristics.

Uses of HPM

- The hedonic pricing method is used to estimate economic values for ecosystem or environmental services that directly affect market prices.
- It can be applicable to two broad markets. viz,.
  - A) Property Market
  - B) labour Market

- HPM is most commonly applied to variations in housing prices that reflect the value of local environmental attributes.

Contd…

- In case of property market, HPM is used to identify the value of non-market goods/bads affecting house prices.
- These non-market goods/bads which affect house prices are, for e.g. aircraft/railway noise, air pollution, water quality etc.

- The basic premise is that the price of a marketed good is related to its characteristics, or the services it provides.
- Property price= f(property variables, neighbourhood variables, accessibility variables, environmental variables)

Contd…

- The HP approach attempts to identify, how much of a property differential is due to a particular environmental difference between properties.

- Also attempts to find how much people are WTP for an improvement in the environmental quality that they face and what the social value of improvement is.

Contd…

- In case of labour market, HPM is used to estimate the value of avoiding risk of death/injury by looking for price differentials between the wages in jobs with different exposures to physical risk.

- Thus, HP process involves
  - collecting a varied number of data on prices and characteristics of properties in an area;
  - Applying statistical techniques to estimate hedonic price function, relating each characteristics of interest to the house price.

The HP Function

- In equilibrium, the hedonic price function of property is given by

$$P_h = f(S,N, Q)$$

- where P_h is the property price function,
- S is a vector of structural characteristics
- N is a vector of neighborhood characteristics and
- Q is a vector of environmental amenities

- The first order partial differential of the function with respect to Q is the marginal implicit price of amenity qi

Stages of HPM

- A hedonic price function (hpf) is estimated.
- Implicit prices are calculated for the environmental variable of interest.
- A demand curve for this variable may be estimated.

Contd…

- HPF is estimated: Need of data on house sale prices and all their characteristics.

- Implicit prices are calculated: In our example of property price case, partially differentiating the property price with respect to any characteristic gives its implicit price/ marginal cost.

Contd…

- Demand curve for the variable air quality may be estimated: (optional one): The estimation of an inverse demand curve for the environmental quality variable involves regressing calculated values of MCs against levels of the environmental variable and socioeconomic parameters.

- Example: Garrod and Willis, 1992; Bookshire et al, 1981

- Bookshire et al. study found that improving air quality from ‘poor’ to ‘fair’ gave benefits of $5800/home/yr

Recent Developments in HPM

- Use of geocoded/spatially reference data and GISs
- Use of GIS in recreational and hedonic analysis
- Use of spatial econometric methods in HP studies
- Combining stated preference method with HPM to increase efficiency.

Issues in Practical Application of HPM

- Assumption of equilibrium in housing market

- Households might not have perfect information: e.g., the workers may not be fully aware of accident risks in the job place. Therefore, their wage-risk choices do not accurately reflect their true valuation of risk.

- Omitted variable bias
- Multi-collinearity

Contd…

- Choice of functional form for the HP function: Economic theory does not specify which non-linear function is to be used for the HP equation.

- Expected vs. actual characteristic levels: House sales may be a function of expected future environmental conditions in addition to current observed conditions.

- Attitudes to risk: the HP studies may either overestimate or underestimate welfare changes.

Suggested Readings

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2003). (Eds). *A primer on nonmarket valuation*. Springer: New York.

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HOUSEHOLD PRODUCTION FUNCTION: AVERTING EXPENDITURE/DEFENSIVE EXPENDITURE METHOD-I

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



CONTENTS

- What is defensive/averting behaviour?
- Averting Expenditure method

- To what extent the household will invest in averting environment bad?

- Model of Defensive Expenditure: Theoretical Framework

- Limitations
- Recent Developments

What is Defensive Behaviour?

- Defensive behaviour generally refers to actions that people take to reduce/avoid environmental damages.
- It includes two things, viz.
  - actions that reduce exposure to pollution and
  - actions that mitigate adverse effects of exposure
- Assumption: A rational individual will take defensive behaviour till the value of the damage avoided is greater than the cost of the defensive action.
- Therefore, choices of defensive behaviour states regarding the value of avoiding environmental change.

Defensive Expenditure Method

- Defensive expenditure is an example of household production function.

- The HPF involves environmental commodities whose effect on a person can be modified using market goods.

- It involves a consumer combining a market good with an environmental good to produce a third synthetic good that actually enters consumer's utility function.

Contd…

- When market good is combined with an environmental bad like air pollution or noise, the household production model is usually referred to as the Defense Expenditure or Averting Expenditure Model.
- The basic idea is that the consumers might be able to avoid exposure to non-market bads through the purchase of a market goods.
- The value of the purchase indicates an implicit price for the non-market bad.
- By observing defensive expenditures, consumers' WTP to reduce the level of bad can be obtained.
- The observed defensive expenditure are the lower bound on willingness to pay.
- Example??

Example of Defensive Expenditure

- If the consumer is living in a busy traffic area, the noise from the traffic is a nuisance.
- If s/he is interested in quietness inside his house, he has to insulate his house from noise and neutralise the damage from the noise at least partially.
- The expenditures tell us about how damaging the noise is to the consumer.
- The more strongly one feels about the noise, the more the consumer will spend on keeping the noise away.
- This expenditure is known as defensive expenditure (defending against noise pollution) or averting expenditure (expenditure to avoid the damage from noise pollution).

To What Extent The Household Will Invest in Averting Environment Bad?

- The household suffering from the noise pollution will continue to invest in defensive measures as long as the value of the damage avoided is greater than the cost of the defensive action.

- Or until the MC of additional measures exceeds the MB from the reduction in noise.

- It is likely that the noise has not been totally neutralised by defensive expenditures and there is additional unaccounted for damage.

Model of Defensive Expenditure: Theoretical Framework

- U= U (X, L, S) (1)

  where U= individual utility

  X=consumption expenditure

  L= leisure time

  S= sick time

- We know that MU of consumption and leisure is positive; whereas the MU of sick time is negative.
- The very feature of defensive behaviour model is that illness does not just happen, rather it is influenced by behaviour.

Contd…

- Sick time is produced according to health production function as follows:

$$S=S(E,G,Z) \qquad (2)$$

where S= sick time

E= exposure to pollution

G= Mitigating activity

Z= a set of exogenous variable that affect length of illness like chronic health status , age etc.

Exposure to Pollution

- This is also influenced by behaviour. As per the production function

  $E = E(A, a)$ (3)

  where A= averting behaviour and

  a= the ambient level of pollution

- This health production function model includes two types of defensive behaviour . Viz.

  -averting behaviour that reduces exposure to pollution (A) and

  -mitigating behaviour that reduces adverse effects of exposure (G)

- Hence, averting behaviour involves changing activities or buying goods to reduce exposure and mitigating behaviour involves using medical care and medication.

Contd…

- Rewriting equation 2 by substituting equation 3 as health production function = S= S[G, E(A, a), Z] (4)

- This shows that G and A are two different ways of reducing illness for a given level of ambient pollution.

- Budget constraint= $I+W.T_w = X+P_g.G+ P_a.A+M(S)$ (5)

  Where I= non-labour income

  T_w = time spent working at 'w' wage rate

  P_g= unit price of G

  P_a= unit price of A

  M(S)= remedial expenses or medical expenditure as a function of sick time.

Thank You

Household Production Function:
Averting Expenditure/Defensive Expenditure Method-II

Contd…

- As we know that averting expenditure methods were developed for estimation of the economic benefits of public policies that target to reduce premature mortality or morbidity taking into account the theoretical underpinnings of utility maximisation and revealed preference method where observable behaviour is noticed.

- For estimating WTP or to use consumers' surplus as approximation to compensating variation for a change in pollution, we need to solve the utility maximisation problem after substituting equation 4 into utility function equation 1.

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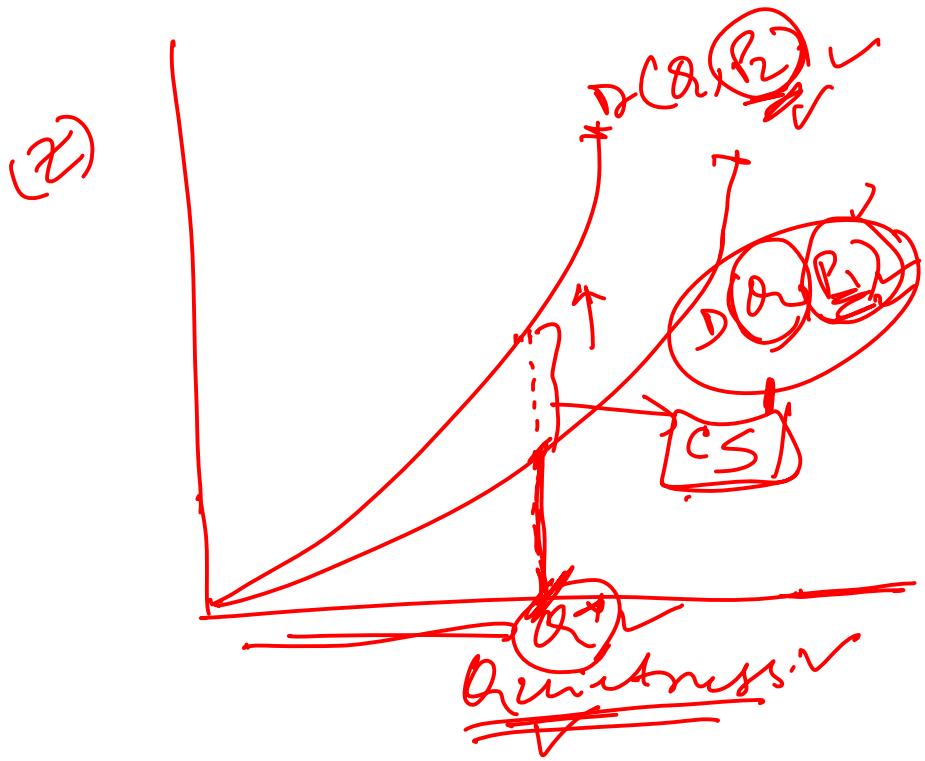
$$U = U(X, L, S) \quad (1)$$

$$\text{Health production function} = S = S[G, E(A, a), Z] \quad (4)$$

$$\text{Max } U = [X, L, S\{G, E(A, a), Z\}]$$

$$\text{S.t.} \quad I + WT = X + WL + P_g.G + P_a.A + M(S) + WS \quad (5)$$

Defensive Expenditure Function, When P Rises



- If P increases from P1 to P2, DE also increases to maintain Q
- Holding Q fixed, the DE changes, when P changes slightly, is the Marginal WTP (MWTP) to compensate for the change in P. (In this figure, to maintain Q, i.e. Q*
- when there occurs a slight change in P, from P1 to P2, the MWTP is the compensating surplus (CS) for not to increase in the level of P.

DE and WTP

- Let's suppose that level of noise pollution gets increased from P1 to P2 level.
- Then how does the observed change in DE is compared to WTP to avoid the pollution increase?
- If DE=WTP, then DE will tell something about the demand for noise pollution and
- If DE><WTP, then DE may provide an upper or lower bound on WTP

Limitations

OECD (2018) finds the following limitations ,

- Firstly, this method represents a partial or lower bound estimate of the value of the impact of the non-market bad on well-being.
- Secondly, many averting behaviour create joint products.
- Thirdly, it is not easy to assign monetary value to behavioural changes associated with defensive actions.
  - Example: keeping child indoor in order to avoid exposure to air pollution out side.
- Finally, difficulty in causally identifying the effects of pollution and of averting behaviour on the health outcome, in the presence of unobserved factors related to both the behaviour and the health.

Recent Developments

- Improvements in causal identification strategies.
- Use of econometric approaches to tackle the identification challenges.
- Combining averting expenditure with stated preference method.

Suggested Readings

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2003). (Eds). *A primer on nonmarket valuation*. Springer: New York.

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Thank You

DAMAGE COSTS/COSTS OF ILLNESS AND LOST OUTPUT APPROACH

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DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

Contents

- What is Damage cost method?
- Defensive behaviour method Vs. Damage cost method
- What is Cost of Illness approach?
- Steps for cost of illness study
- Concluding Remarks

Damage Cost Method

- The damage cost method targets to estimate the resource costs (direct and indirect costs) associated with environmental change (like pollution).
- Direct costs= expenditure to treat illness or to repair + replace or maintain damaged materials.
- Indirect costs = the opportunity costs of reduced productivity or output forgone because of environmental pollution.
- This method uses the reduction in real resource costs to measure the benefits of reduced pollution.

Defensive Behaviour Method Vs. Damage Cost Method

- The Defensive behaviour (expenditure) method analyses how human behaviour responds to changes in the environment and the impact of behaviour on the outcomes experienced.
- On the other hand, the damage cost (DC)method implicitly assumes that there is no behavioural response to environmental changes or that behavioural responses are not effective.
- The DE expenditure method is designed to estimate economic value like WTP; whereas DC method cannot estimate economic value.
- However, DCs are less than WTP and they are easier to estimate and comprehended even to layman.

Similarities between DE and DC Method

- Both of these methods are based of household production method.
- Both the methods are used to value materials and preferably health.
- Both these methods are used to value changes in outcomes of pollution. For example- lenghth of illness and to value the underlying changes in pollution that cause changes in outcome.

Application of Damage Costs

- Material damage
- Other end point damage
- Health valuation

Cost of Illness Method

- The specific damage cost method applied in the health context is known as the cost of Illness approach.

- The cost of illness = direct costs+ indirect costs

- Direct costs= the value of resources used to diagnose, treat, rehabilitate or support injured persons (special education)
- Indirect costs= the value of no output produced because of morbidity or premature mortality.
- Foregone earnings are the main components of indirect cost.

Approaches for Measuring Costs

- Two approaches for measuring costs based on different measurements of the occurrence of a disease/condition, Viz.
  - Prevalence based costs
  - Incidence based costs

Contd…

- The Prevalent population consists of all persons who have a condition at a given time;
- The incident population consists of the new cases diagnosed during a given time.

- Prevalence based costs: these are annual costs associated with cases of a condition present during a year.
- Incidence based costs: these costs focus on lifetime costs from onset of the condition till recovery /death.

Contd…

For Example:

- The prevalence approach, in case of measuring the costs of asthma, focuses on all persons alive at the beginning of the year diagnosed with asthma.

- Direct costs = direct costs of treatment like emergency room charge, physician fees, medicines, etc. during a year.

- Indirect costs = morbidity costs in terms of loss of wage and forgone household production during a year + mortality costs like present value of lost life time earnings and household production.

Contd…

- Likewise,

- The incidence approach measures lifetime costs of persons diagnosed with asthma during the year, by forecasting and discounting costs because of morbidity and mortality, over the remaining life expectancy.

- Therefore, the incidence approach is more complicated.

WTP and Damage Costs

- As we know from equation 6a.

  C(S)= M(S)+WS

- Damage cost in the health context is known as Cost of Illness.

- We can find the marginal impact of given change in illness on cost=

$$\frac{dC}{dS} = \frac{dM}{dS} + W$$

- Moreover, the WTP for a given reduction in illness exceeds the cost of illness. Because cost of illness does not take into account the utility value of health or for pain and sufferings.

Steps for Conducting Cost of Illness Study

1. Define the valuation problem.

   Specify the alternative condition.
   Determine appropriate specificity of condition valued.
   Define linkage to environmental conditions.
   Prevalence or incidence approach?

2. Estimate direct costs.

   Identify relevant categories of health-care costs.
   Obtain data on utilization and cost for each category.
   Estimate costs using prevalence or incidence methodology.
   Estimate non-medical direct costs.

3. Estimate indirect costs.

   Estimate morbidity costs.
   Estimate morality costs using value of statistical life or cost-of-illness methodology.
   Account for changes in value of home production.

4. Adjust costs to a common base year.

   Adjust for changes in prices and wages.
   Discount to present value.
   Adjust for changes in prevalence or incidence over time.

Champ, Boyle, and Brown.(2003) p.433

Limitations

- Methodological Challenges:
- Most data on direct costs report charges rather than costs.
- Direct costs usually also include fixed overhead like insurance administration rather than, only variable costs.
- Methods of indirect cost measurement from forgone household production have limitations because these cannot satisfactorily measure indirect costs for children, retired and old age persons.
- In the presence of comorbidities, the techniques for allocating/apportioning costs are arbitrary.

Cond…

- Fundamental challenge:
- This method is not a money measure of welfare change in true sense as it is giving only a picture of lower bound on WTP.

Suggested Readings

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2003). (Eds). *A primer on nonmarket valuation*. Springer: New York. Chapter 11.

- Kolstad, C.D. (2011). *Intermediate environmental economics.* OUP: India

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NPTEL ONLINE CERTIFICATION COURSE

TRAVEL COST METHOD

DIPTIMAYEE NAYAK

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Contents

- Background of Travel Cost Method (TCM)
- Origin of TCM
- Assumptions
- Dimensions
- Types
    - Single-site model
  - Factors affecting number of visits in single site model
  - Steps
    - Multi-site model and Estimation
  - Individual Travel cost model
  - Zonal Travel cost model
- Limitations
- Recent Developments

Background

- TCM is derived from the **theory of consumers' preferences** and is measured by estimating the **willingness to pay** for the recreational demanded destination.
- Then Consumer surplus is estimated, which serves as a measure of welfare.
- It is used to estimate economic use values associated with ecosystems or sites that are used for recreation.
- The basic premise of the travel cost method is that the **time** and **travel cost** that people incur to visit a site represent the "**price**" of access to the site.

Contd…

- Thus, peoples' willingness to pay to visit the site can be estimated based on the number of trips that they make at different travel costs.
- The Travel Cost serves as **a proxy for price**.
- This again also serves as a basis for welfare measurement.
- All of the demand for recreation are essentially models of *allocation of time*.
- Hence, It must meet individual's ***time constraint*** and ***income constraint***.
- Recreational demand modeling is an application of ***Becker's model of allocation of time.***

Origin

- Harold Hotelling (1949) - An Economic Study of the Monetary Valuation of Recreation in the National Parks - Washington, DC: U.S. Department of the Interior, National Park Service and Recreational Planning Division.

- In a letter to the US Park Service, Hotelling suggests to use TCM to assess the benefits obtained by visitors of a national park.

Intuitions

- You need to travel to reach a site in order to enjoy it.
- When you decide to visit a site, you need to pay a cost to reach (and enter) the site.
- The cost to reach/enter into the site and the number of visits to the site allow individuals to estimate a **lower bound** of the value of the site.
- Estimate only **use-values**

Assumptions

- The cost of the travel and of the time spent to reach and stay at the site are a proxy for the value of the recreational experience .

- Use value is assessed taking into account the number of *homogeneous* visits to the sites among respondents.

  E.g. Homogeneous visits last same amount of time;

  we cannot mix one- day visits with multi-day visits.

Contd…

- Visits to only one site-avoid visitors who in one day visit several sites.

- Weak complementarity between the environmental asset and consumption expenditure:
  - When consumption expenditure is zero, the MU of the Public good is zero.
  - Assume that the environmental quality of the site affects the number of visits to the site.
  - The weak complementarity assumption allows us to identify how the variation in a public good affects the behaviour of a person.

Dimensions of TCM

- How demand depends on quality of the environmental good. E.g. water quality and demand in a Marine recreational spot.

- The number and duration of trips during a period of time (per year)

- Treatment of substitute sites, i.e. when a visitor to a national park faces the choice of several parks.

Types of TCM

- On basis of sites of visit:
  - Single site model
  - Multiple/ site choice models

- On basic of zone of origin:
  - Zonal Travel Cost Model
  - Individual Travel Cost Model

Single Site Model

- The single site model describes the demand for recreation of a person during a year.
- The quantity demanded is the number of visits
- The price is the cost per visit
- $V = f(tc_v)$ (1)

  V = number of visits during a season

  tc_v = cost of a visit

(Intuition: who lives close to the site has a low cost per visit. S/he should visit the site more often than someone who lives further away).

Factors Affecting Number of Visits

- Other variables, such as age of the respondent, income, experience, availability of substitute sites may affect the number of visits.
- $V = f(tc_v, tc_s, y, z)$ (2)

tc_s = price of trip to substitute site s (expect a positive correlation)

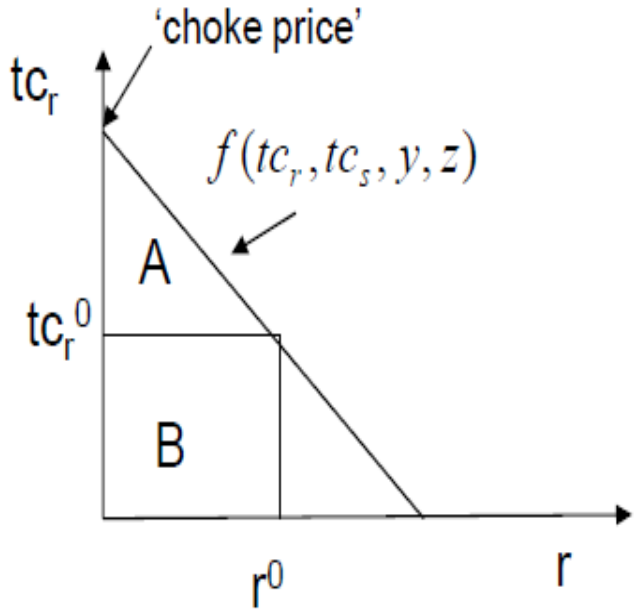
y = income

z = vector of socio-demographic characteristics of the respondent (age, gender, marital status, experience, etc.)

Consumers' Surplus

B = total trip cost
A+B = Willingness to pay
A = consumer surplus

- *The 'choke price' is the* price at which the number of trips falls to zero.



Contd…

The consumer surplus is

$$\Delta w = \int_{tc_r^0}^{tc_r^{choke}} f(tc_r, tc_s, y, \mathbf{z}) dtc_r$$

Steps in Single Site Model

1. Define the site to be valued
2. Define the recreation uses and seasons
3. Develop a sampling strategy
4. Specify the model
5. Decide on the treatment of multiple purpose trips
6. Design and implement the survey
7. Measure trip cost
8. Estimate the model
9. Calculate access value

The Multi-site Model

- The multi-site describes the demand for recreation as a **problem of choice among alternatives**.

- The model used is the random utility model (RUM).
- Choices are explained by the characteristics of the sites.
- We assume that respondents choose the sites that give them the highest level of utility.
- This model allows to analyse the behaviour of economic agents when they are facing a problem of choosing which site to visit even when the alternative sites are many

Estimation

- Identify what to valuate
- Define the population to sample
- Define the choice set
- Decide the sampling plan (on-site sampling is very troublesome)
- Specify the model
- Obtain information about the sites
- Analysis of multi-objective visits
- Data collection: need to get information on **all** sites
- Estimate of the cost per visit (to all sites)
- Model estimates
- Access value estimate

Zonal Travel Cost

- A survey is made on the visitors to the recreational area
- The visitation rates are determined for different zones around the area
- Identify the origin of visitors
- Estimate number of visitors per year from each region (zones)
- Generate demand function for visit to the recreational area
- Collect data on visitation rate, travel cost and characteristics of the zone
- Statistically determine how the visitation rate is affected by travel costs and zonal characteristics
- Choosing several admission fees for the site and resulting total attendance at the site generates points on the demand curve

Contd…

- Divide the region into 'z' number of zones.
- Assume travel time from any point in the region to the destination
- Index these zones as Z= 1, 2,….., z
- Visitation $V_z = g(\prod_z + f, Y_z, W_z)$ (5)
- The visitation rate $V_z = S_z/P_z$
- Where, P_z is population in each zone
- Y_z is average income in each zone
- W_z is other demographic characteristics of the zone
- $\prod_z$ is travel cost
- f is entry fee
- S= total number of visitors in a year
- S_z is total number of visitors from each zone
- Demand for the visit=

$$q(f) = \sum_{i=1}^{z} g(\prod_z + f, Yz, Wz) \quad (6)$$

Limitations

- Individual travel cost model does not easily take into account the presence of substitute recreational sites.
- TCM has narrow applicability in estimating recreational use values and requires large data set.
- Problem of multipurpose trips
- Problem in valuating travel time

Recent Developments

- To measure change in quality of recreation/site
- Innovations in econometric analysis
- Models for handling on-site sample
- Integrating TCM with stated preference methods

Suggested Readings

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2003). (Eds). *A primer on nonmarket valuation*. Springer: New York.

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CONTINGENT VALUATION METHOD

A STATED PREFERNCE APPROACH

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Outline

- Basic premise and background for stated preference method
- What is Contingent valuation method (CVM)?
- Mode of survey
- Components of CV Survey
- A sample of CV questionnaire
- Applications
- Example
- Issues with CV
- Advantages
- Recent Developments

Why Stated Preference Method? Basic Premise And Background

- In case of non-use values like, Intrinsic value, revealed preference method will not be applicable.

- If one values something for mere existence, unrelated to its possible use, the consumer's behavior with regard to market goods will be unaffected by whether that thing is available.

- And if consumers' behavior is unaffected, revealed preference method is difficult to apply.

Contd…

- Even in some cases there is no market .
- So constructed market valuation tools have been developed.
- Richard Carson used the term ‘constructed market’ to refer to valuation techniques that involve the construction of a market to be used to generate the value or demand for an environmental good.

Types of Constructed Market

- Hypothetical/stated preference/contingent valuation method
- Experimental method
- In CVM, consumers are directly asked what they would pay for an environmental good, if there were a market (conditional or contingent).
- In an experimental market, the researcher constructs all the characteristics of a market, involving transactions (trading money for good) and then observes behaviour.

What is CVM?

- CVM is a Stated preference method.
- Stated preference methods of valuation involve finding an individual's WTP for a good by posing a set of questions regarding preferences directly to the individual.
- It is survey based
- In CV question, the individual is asked to imagine some situation that is typically outside the individual's experience and speculate on how s/he would react in such a situation.
- This method has its application since 1960s by Davis (1963), Randall et al. (1974).

Mode of Survey

- The CV method uses a direct approach. It basically asks people what they are willing to pay for a benefit, and what they are willing to receive by way of compensation to tolerate cost.

- The process of asking the questions may involve the following:
  - Direct questionnaire/survey
  - Experimental technique

Components of CV Survey

- To accomplish or to accept the compensation or to pay, we must design, administer and interpret a survey.

- A Survey typically consists of 3 parts, viz.

  - Background Information of the study on the environmental good
  - Section on elicitation of the value
  - Section asking background information on the respondents.

Components of a successful CV Survey: Carson

- Define Market Scenario
- Choose elicitation method
- Design Market administration
- Design Sampling
- Design Experiment
- Estimate WTP function

Define Market Scenario

- What is Market Scenario?
- The market scenario is the information to be conveyed to a respondent and to place the respondent in the right frame of mind to give meaningful responses to questions.
- Steps:
- How to define the good to be valued? For e.g., if a beach has been harmed by an oil spill, what environmental good should be valued?
- Description of the market to be realistic to the respondent
- Payment mechanism - tax, user fee, entry fee
- Right context for the survey (for e.g. ground water contamination)

Choosing Elicitation Method

- It is about to decide how best to obtain the valuation response.

- Mechanisms:
- Direct Question: People may not spend much effort in deciding their WTP and probability of getting extreme responses is large.

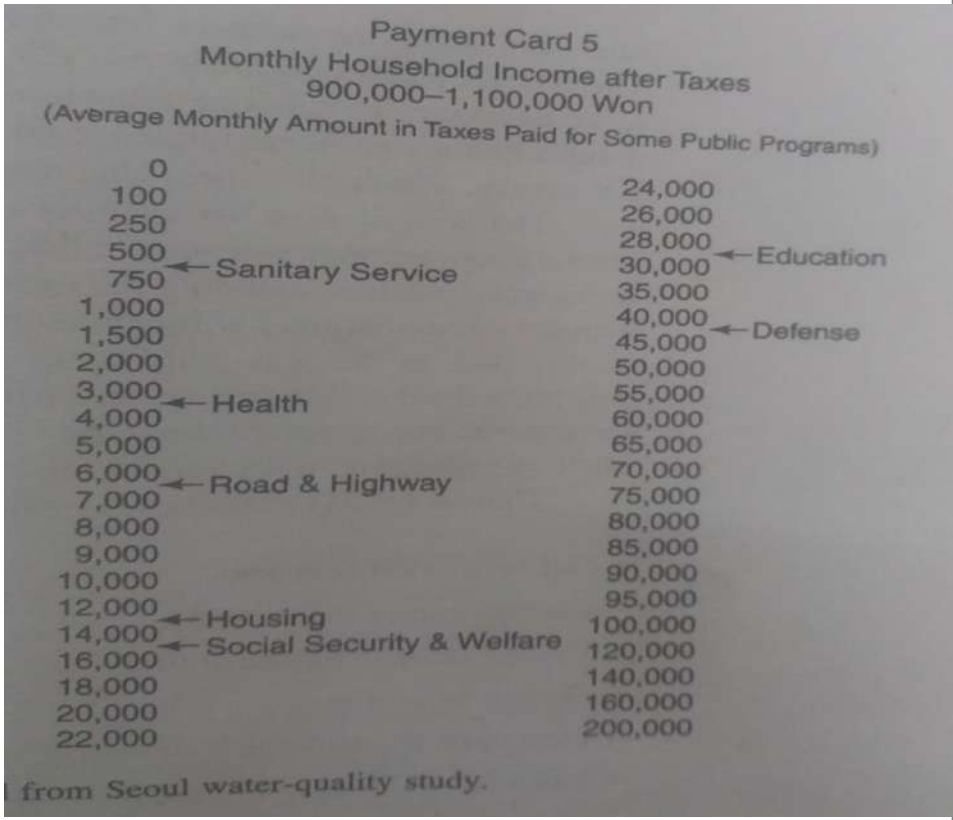

Chiru- an endangered animal

Contd…

- Bidding Game :
- It starts with a WTP number and seeks a binary response (Yes-No).
- If the response is ‘yes’, then the amount is gradually increased until a ‘no’ is received.
- On the other hand, if the reply is ‘no’, the amount is gradually decreased until a ‘yes’ is received.
- The basic problem of a bidding game is ‘starting point bias’.

Contd…

- Payment Card:
- Card containing different ranges of amounts.
- It cannot be used for telephonic survey.



Payment Card 5

Monthly Household Income after Taxes
900,000–1,100,000 Won

(Average Monthly Amount in Taxes Paid for Some Public Programs)

0		24,000
100		26,000
250		28,000 ← Education
500 ← Sanitary Service		30,000
750		35,000
1,000		40,000 ← Defense
1,500		45,000
2,000		50,000
3,000 ← Health		55,000
4,000		60,000
5,000		65,000
6,000 ← Road & Highway		70,000
7,000		75,000
8,000		80,000
9,000		85,000
10,000		90,000
12,000 ← Housing		95,000
14,000 ← Social Security & Welfare		100,000
16,000		120,000
18,000		140,000
20,000		160,000
22,000		200,000

from Seoul water-quality study.

Contd…

- Referendum Choice/discrete choice:

- In this mechanism, WTP figure is offered to the respondent who is asked if s/he would be willing to pay that amount - Yes, No.

- Again, different respondents are offered different WTP figures, chosen to span the possible range of WTP .

Design of Market Administration

- Following are the items that need to be covered.

- Respondents must complete the survey and their responses collated.

- Mechanism for survey administration

  - Mail
  - Email
  - Telephone
  - In-person

Sample Design

- Its about sampling frame- choosing the group from which sample is drawn.
- How to identify the group?
- Who are going to be interviewed? Households or individuals?
- Male/female, head of the family, other members or children

Experimental Design

- Appropriate information to be collected in an efficient manner without unintended bias.

- Statistical analysis

Estimation of WTP Function

- To analyse the survey results and estimate WTP function.

A Sample of CV Questionnaire

Table 18.1 *Excerpts from Seoul Questionnaire*[a]

Section A — Introduction
Interviewer: Introduce yourself and explain the purpose of the survey. Indicate that the interview should take less than a half hour.

Section B — Background Attitudes and Information [6 questions]
1. What is your opinion about current tap water quality in Seoul?
   (a) very good; (b) good; (c) average; (d) bad; (e) very bad.
2. In the last five years, have you or your household taken any of the following actions to improve your drinking water?
   (a) installed water filter; (b) purchased bottled water; (c) boiled tap water regularly; (d) gone to a spring to obtain spring water.
3. What is the monthly combined income of your household after taxes, last year? [If refuses to answer, show card with ranges of income and ask to identify range of income]

Section C — Value of Water Quality Improvement
Interviewer: Two minute verbal description of major pollution accident in 1991 that contaminated the drinking water of Seoul. Because the consequences were not known immediately, many people drank contaminated water. [Visual props are used to describe the accident and its consequences.]

1. If the government takes no action, how many times do you think you might experience an accident similar to the Nak-Dong phenol accident in the next five years?

Interviewer: Give one minute verbal description of continuous monitoring system that would prevent introduction of pollutants into the city water supply. Indicate that with the system, the chance of drinking polluted water will be similar to the chance "that you will be a president of Korea." Chart is presented showing how taxes are currently apportioned among various uses (such as education, roads, sanitary services, and defense). Payment card for household's income is presented.

2. What is the most your household would pay in increased monthly taxes to reach the goal attainable with the new monitoring system?
[If response is zero, unusually high, or no response (refusal to answer), ask a follow-up question. If response is zero, ask reason for zero response. If refuses to answer, repeat question and try to get a response. If payment response is unusually high (higher than education expenditures), give respondent opportunity to revise his or her response.]

Section D — Background Information [7 questions]
1. What is your age?
2. What is the last grade of regular school that you and your spouse completed?
3. How many people live in your household? How many are under age 13?
4. What is your average monthly water and sewer bill?

[a]*Questionnaire follows an exact script that is paraphrased here.*
Source: Seung-Jun Kwak, personal communication, November 1997.

Kolstad, 2011, p.197

Applications

- A contingent market is taken to include not only for an environmental good like better air quality, but also the institutional context which it would provide and the way in which it would finance (mode of finance).

Example

- The value of improving Air quality in Los Angeles by Brookshire et al (1982):
- Brookshire et al (1982) were interested in how much people would be willing to pay to increase the air quality in their community.
- They generated photographs depicting 3 different levels of air quality on the basis of visibility -poor, fair, and good.
- They sampled 290 people from communities with varying air quality.
- The people in a community, with poor air quality were shown the photos asked about their WTP to move to fair air quality.
- And people with fair air quality were asked their WTP to move to good air quality.

Contd…

- CV supporting Government Regulatory actions:

- In Australia, the Govt. debated whether to preserve or allow mining in an area known as the Kakadu Conservation Zone.

Example

- Exxon Valdez oil spill (1989)

- NOAA (National Oceanic and Atmospheric Administration) Panel agreed that CVM is valid if the following recommendations are taken into account.

  - Dichotomous choice format
  - Minimum response rate of 70%
  - In person interviewing
  - WTP rather than WTA
  - Sensitivity to scope is accounted for
  - Respondents are reminded of their budget constraints

Limitations of CVM

- It is controversial.
- The values elicited by CV survey are not based on real resource decisions and they are hypothetical.
- There is no budget constraint in a hypothetical survey and without budget constraints choices are meaningless. But its counter argument is that there may be more scatteredness in the response to a hypothetical question but the average person will be unaffected by the hypothetical nature of the exercise.

Contd..

- Issue of 'warm glow effect' for preserving an environmental good. For example, respondents may express a positive WTP because they feel good about the act of giving for a social good (referred to as the "warm glow" effect), although they believe that the good itself is unimportant.

- Embedding problem (to determine the value of a specific natural resource- say a particular park. People may value same to the substitute park outside the survey)

Advantages

- CV is flexible in that it can be used to estimate the economic value.
- However, it is best able to estimate values for goods and services that are easily identified and understood by respondents.
- It is the most widely accepted method for estimating TEV, including all types of non-use, or "passive use," values.
-
- It can estimate use values, as well as existence values , option values , and bequest values .

Recent Developments

- Borrowing insights from behavioural economics
  - Identified empirical phenomena that are not adequately explained by neoclassical economics
  - From initial focus on the identification of behavioural inconsistencies to formalising alternative theoretical conceptualisations, to fully integrating these alternatives into economic analysis and hence helping in reshape and improvement in economic principles (Rabin, 2010)
- Role of psychological factors and experiments
- Developments in technology and social media of interactions

Suggested Readings

- Bateman, I. and Wills, G. (2001) (Eds). *Valuing environmental preferences : Theory and practice of the contingent valuation method in the US, EU, and developing countries*, OUP: New York.

- Kolstad, C.D. (2011). *Intermediate environmental economics*, OUP: New Delhi.

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CHOICE EXPERIMENT METHOD

A STATED PREFERENCE APPROACH

DIPTIMAYEE NAYAK

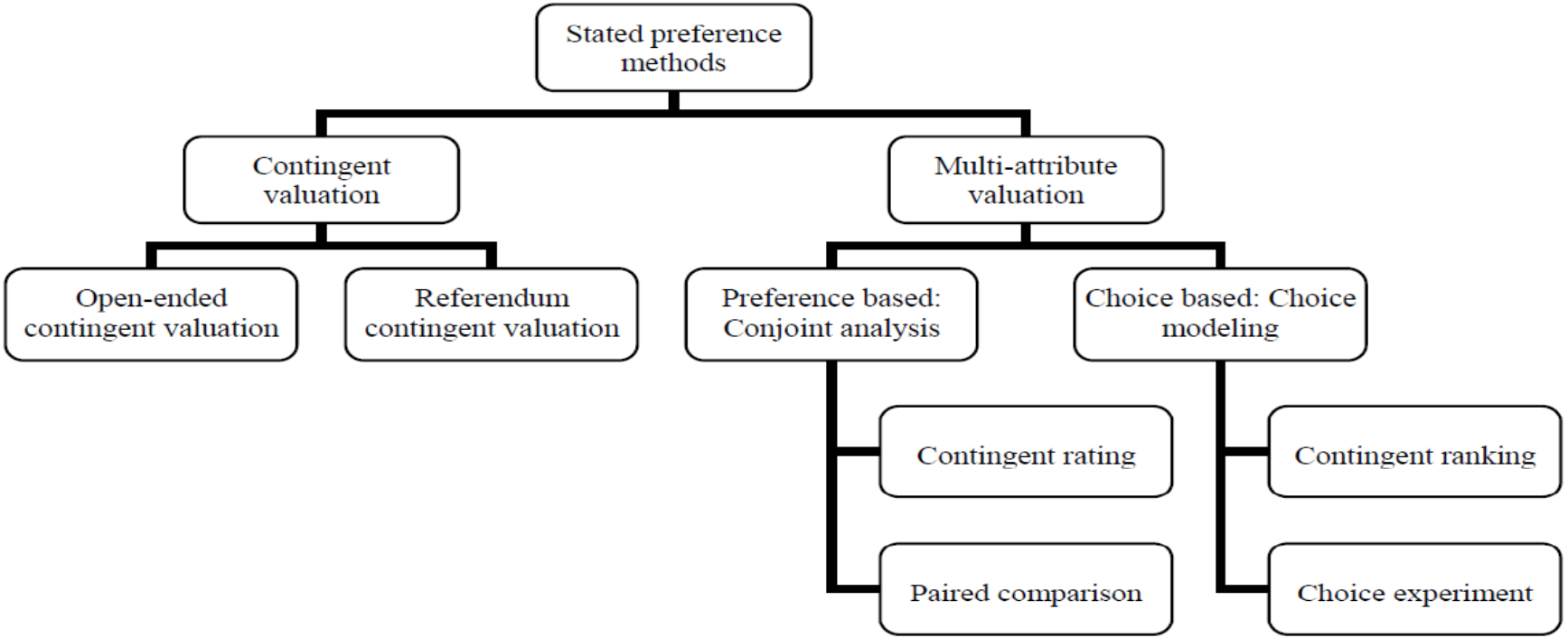
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Outline

- What is choice-experiment method?
- History of choice experiment
- Steps in conducting a choice experiment
- Differences between CVM and Choice Experiment
- Limitations

Family of Stated Preference Methods



Source: Merino-Castello (2003) as in Vega and Alpizar,2011, p. 4

Contd..

- The application of multi-attribute valuation has come up as a response to the limitations of contingent valuation method.
- In general, CVM and multi-attribute method (MAM) like choice experiment differ on the ground that the latter allows the practitioner to estimate values for multiple attributes of a product and their trade off simultaneously (ibid, 2011).
- The theoretical underpinnings are based in Lancastrian microeconomics (1966) (in which individuals derive utility from characteristics or attributes of a good) and in McFadden's (1974) random utility theory (in which utility has a deterministic and probabilistic component).

What is Choice Experiment Method?

- A choice experiment method explains how to construct an attribute-based experiment and to estimate economic values of a set of attributes of an environmental good.
- The objectives of choice experiment is to estimate the economic values for characteristics/attributes of an environmental good which is the subject of policy analysis.
- This can also create hypothetical but realistic scenarios for respondents and alternatives for the environmental good/service in discussion.
- It is the simplest of the choice based approaches in terms of cognitive requirements.

Contd…

- In this method, respondents are given with a set of choices with alternative choices related to the environmental quality/scenario in discussion.
- Then the respondents are asked to choose their preferred alternative from each of the choice sets.
- Then the good/service in discussion is decomposed according to its attributes/characteristics and the combination of various levels of the sets of attributes.
- Attributes can be quantitative or qualitative in nature and combined attributes of different nature can be possible.
- A baseline scenario (the status quo) is introduced as an alternative in the choice sets.

Models Used

- Multinomial logit model
- Conditional logit
- Random parameters logit

History of Choice Experiment

- The origins of CE are found in social science disciplines like psychology, economics, marketing , product designs.
- In economics, the conceptual foundation of CE is derived from the hedonic method (demand for attributes), hedonic regressions for automobiles (Court, 1939), hedonic price indices (Griliches, 1971) and Lancaster's theory of consumer demand (1966)- 'Consumer demand is driven by commodity attributes'.
- Random Utility (Thurstone, 1927) and Discrete choice theory (McFadden, 1974)
- Econometric model the multinomial logit (McFadden)

Steps

Step 1	Characterize the decision problem
Step 2	Identify and describe the attributes
Step 3	Develop an experimental design
Step 4	Develop the questionnaire
Step 5	Collect data
Step 6	Estimate model
Step 7	Interpret results for policy analysis or decision support

Champ, P.A., Boyle, K.J. and Brown, T.C. (2017)

Contd…

- Characterise the decision problem:
- Need to clearly identify the dimensions of the problem
  - The geographic and temporal scope of potential changes in policy attributes (whose values are to be included in the valuation- like any endangered species management ?; who will be impacted by changes in policy attributes and how will they be impacted?); location of substitute sites (if any); temporal considerations
  - The types of values that are associated with those changes (whether the choice to be examined reflects use value, passive use value, total economic value)

Contd…

- Attribute identification and description:

  – Identify and describe the relevant attributes by the help of focus groups and key informants
    - Attribute levels must be specified (like presence/absence; richness/moderately available etc.)
    - Specify price/cost levels

Contd…

- Develop an experimental design:
- Determine the number of alternatives to present in each choice set (1/2/3/4, etc.)
- And number of choice sets
- Choice questions must include a status quo/base level in each choice set and alternative indicating a change from the status quo.

Contd..

- Questionnaire Development:
- Questionnaire formats for survey-
  - Email- mail out/mail back surveys
  - Telephone recruitment
  - Social media- face book/twitter
  - Intercept survey using paper/pencil or computer assisted
  - Internet based surveys

Contd…

- Data Collection:
- Care to be taken to prevent respondents from reading ahead or going back and changing responses.
- In a paper pencil survey it this difficult to accomplish; in computer based survey (like google spread sheet) or in person it can be followed.

Contd…

- Model Estimation:
- To estimate preference parameters using RUM
- Econometric specifications to analyse choice data

Contd…

- Policy analysis and decision support:
- Most of CE applications estimate welfare measures, prediction of behaviour or both.
- To evaluate the welfare effects of multiple policy options involving a set of combinations of attributes and levels.

Limitations

- Careful design of Scenarios
- Selection of attributes and levels
- Choice of econometric specifications

Contingent Valuation Vs. Choice Experiment

- CVM produces a single value for an overall change in good in discussion (environmental good/service/quality/quantity); whereas CE estimates a value for each individual attributes of an environmental programme/policy.
- The CE is a generalisation of CVM- rather than asking respondents to choose between a base case and specific alternative, in CE we ask respondents to choose between cases characterised by different attributes and these scenarios are taken from possible scenarios.

Suggested Readings

- Adamowicz, W., Boxall, P., Williams, M. and Louviere, J. (1998). Stated preference approaches for measuring use values: Choice experiments and contingent valuation, *American Journal of agricultural Economics*, 80 (1):64-75.

- Champ, P.A., Boyle, K.J. and Brown, T.C. (2017). (Eds). *A primer on nonmarket valuation*. Springer: New York, Chapter 5.

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Market Efficiency and Optimality-I

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- Defining Market, Efficiency and Optimality
- Efficiency and Competitive Markets
- Efficiency in Exchange: Goods
- Market and Exchange
- Efficiency in Exchange: Bads
- Let’s Sum Up

Introduction

- One of the basic normative issues in economics for specific goods is: how much should be produced and who should produce and consume—the production and allocation decisions.
- This is also an important issue in environmental economics.
- How much environmental goods/bads should be produced?
- Optimization of output of goods/bads
- This lecture explains the concept of efficiency and market and also discusses the efficiency in exchange (consumption) in case of both conventional and environmental goods.

Concept of Efficiency

- Efficiency refers to the degree to which the observed use of resources to produce outputs of a given quality matches the optimal use of resources to produce outputs of a given quality.
- A firm is said to be fully efficient if and only if quantity of its input/ output cannot be reduced/increased without increasing/decreasing some of its other input/output (Pareto optimality).
- **Technical Efficiency (TE)**: It refers to the firm's ability to produce the maximum possible output from a given combination of inputs and technology.
- **Allocative Efficiency (AE)**: It refers to the ability to choose optimum input levels when input prices are given.
- **Economic Efficiency (EE)**: EE = (TE) X (AE)

Technical, Allocative and Economic Efficiencies

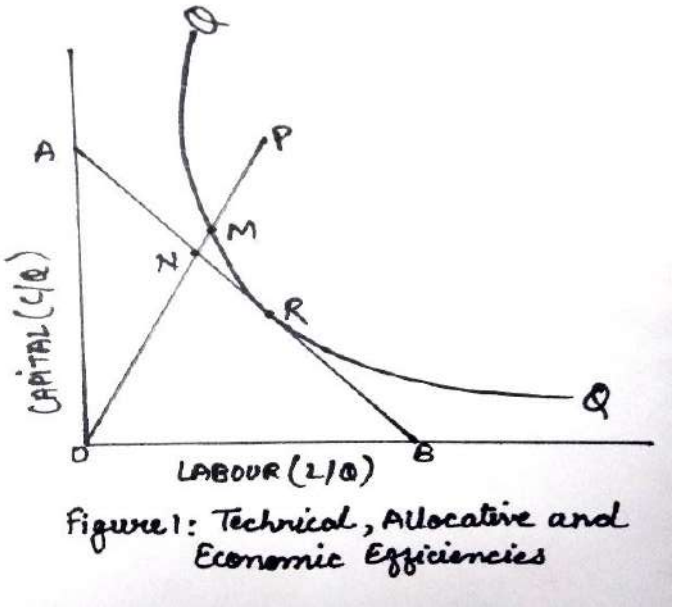


Figure 1: Technical, Allocative and Economic Efficiencies

QQ = unit isoquant

AB = Factor price line (Price Ratio)

TE of Firm "P" = OM/OP

AE of firm "P" = ON/OM

EE of firm " P" = (OM/OP) X (ON/OM) =ON/OP

"M" is allocatively inefficient but technically efficient; while "R" has 100% TE, AE and EE

Concept of Market

- A market is an institution in which buyers and sellers of goods and services carry out mutually agreed-upon exchanges.
- Presumably buyers would like to pay a low price, whereas sellers would prefer high prices. What brings all these conflicting objectives into balance is the adjustment of prices in the market.
- A market system normally produces better economic results overall than any other system through its “invisible hand”.
- If consumers consume to the point where MU=price of a good and producers produce to the point where MC = price, then resources will be optimally allocated.

Contd…

- Allocative and Creative functions of Market
  - Optimum allocation of resources (Pareto optimality)
  - It provides environment for change that expands PPF (creative function). Change means all the dynamic forces leading to technical progress, innovations and investment.
- However, the conditions required for market to perform these functions in an optimal manner are unlikely to meet.
- Market failure and distortions: adversely affect the function of market to efficiently allocate scarce resources and co-ordinate individual decision making.

Allocative and Creative Role of Market

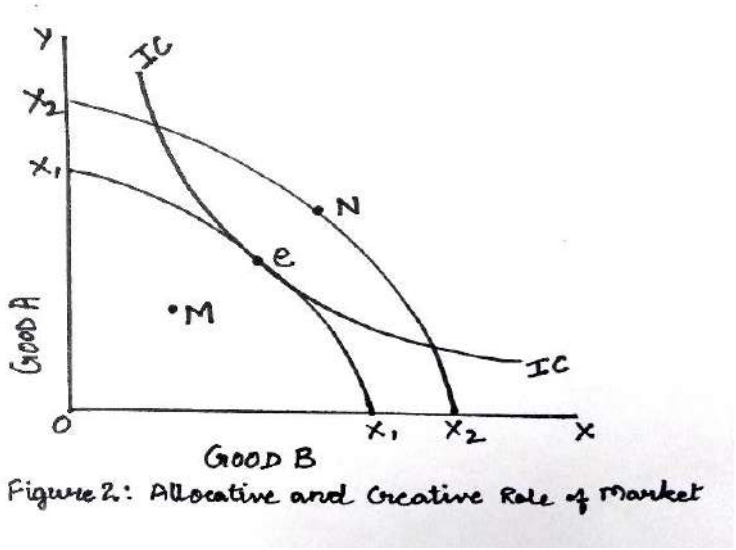


Figure 2.: Allocative and Creative Role of Market

- X_1 X_1 and X_2 X_2 are PPFs.
- Ic Ic = Society's Indifference Curve
- M= Pareto inefficient
- e = Pareto optimal or optimal allocation of resources between Good A and B. Any point left or right of "e" or inside the PPF indicate lower utility/welfare (allocative role)
- Creative Role: shifting of PPF outwards to X_2 X_2

Market Mechanism

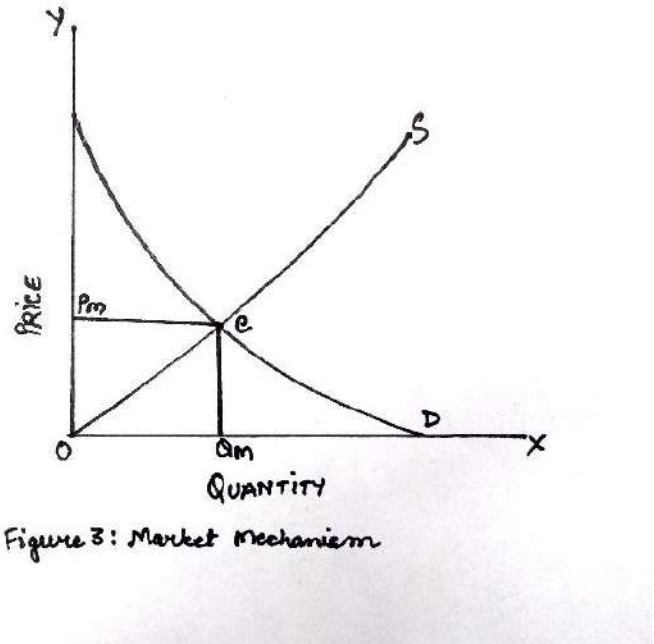


Figure 3: Market Mechanism

- Demand Curve: Buyers' willingness to pay prices for different quantities (MWTP).
- Supply curve: Supplier' willingness to accept prices for different quantities.
- Price must be allowed to adjust freely to make balance between supply and demand.
- Point e is allocatively efficient. Consumer and producers surpluses are maximum at e.

Efficiency and Competitive Markets

- Efficiency means Pareto optimality. If an economy is operating on the Pareto frontier, it is efficient.
- Typically, there are two types of inefficiencies—inefficiency in exchange and inefficiency in production.
- Inefficiency in exchange means that the resources could be moved around among individuals without making anyone worse off. (Ex. solid waste disposal)

Contd…

- The other type of inefficiency occurs in production.
- For example, suppose an environmental regulation calls for a use of a ‘scrubber’ on the smoke generated at a factory. The factory could achieve the same level of emissions reduction by changing its production process, but regulation does not allow using process changes to achieve emission targets, only the scrubber.
- In this example, resources are wasted in production with no apparent gain. Everyone would be better off by using a method of pollution control that costs less.
- Competitive markets are efficient.

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Environmental Economics

Market Efficiency and Optimality-II

Prof. S. P. Singh
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Indian Institute of Technology Roorkee



Contents

- Introduction
- Efficiency in Exchange: Goods
- Market and Exchange
- Efficiency in Exchange: Bads
- Let’s Sum Up

Efficiency in Exchange: Goods

- To explain the efficiency in exchange, let us consider two persons X and Y who have some quantity of two goods—cheese and wine and wish to enter into exchange (trade).
- First draw indifference curves for cheese and wine, reflecting X's and Y's preferences.
- Suppose X has a kg of cheese (no wine) and Y has a bottle of wine (no cheese). Both would prefer a little of each good.
- Both persons' welfare will increase if they enter in trade (under Barter System).
- The Edgeworth box is a useful graphic way of representing exchange of two commodities between two individuals.

X's Preference for Cheese and Wine

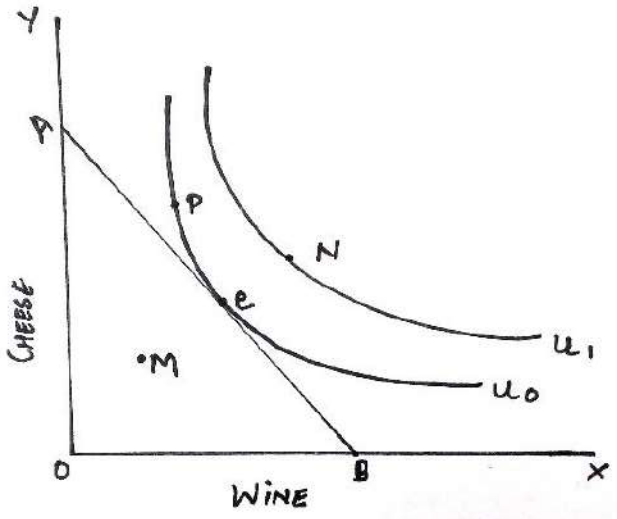


Figure 1: X's Preference for Cheese and Wine

Efficiency in Exchange through Edgeworth Box

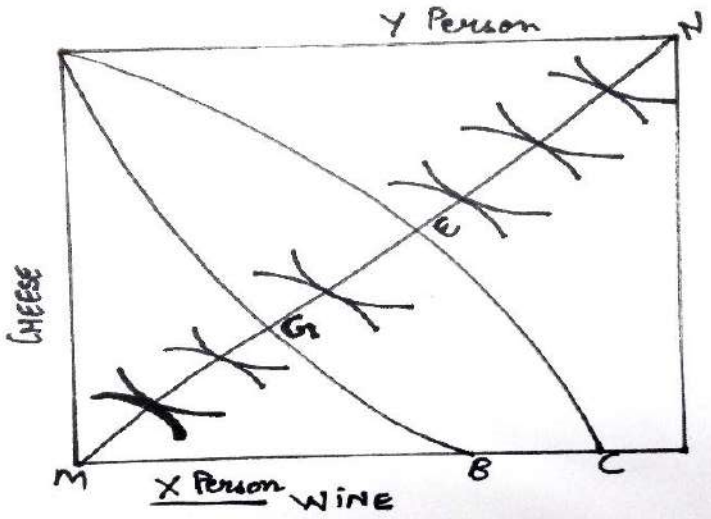


Figure 2: Efficiency in Exchange

- Any division of the cheese and wine that lies between AB line and AC line will result in increased utility for both individuals.
- Movement of X from G toward E and Y from E toward G is Pareto improvement.
- Any point in the region ABC at which the IC of X and Y are tangent to one another lies on the Pareto frontier—also called the contract curve.
- Efficiency requires that the MRS of wine for cheese be the same for both the persons.

Market and Exchange

- Let Co and Wo be X's initial endowment of cheese and wine respectively.
- Let C* and W* be the quantities of two goods X eventually end up with.
- Let prices of the two commodities be Pc and Pw, then X's budget balancing requires the following to hold for any (C, W) consumed.

$$PcCo + PwWo = PcC^* + PwW^*$$

- This equation can be written with C* on the left-hand side and W on the right-hand side.

$$C^* = [Co + (Pw/Pc)Wo] - (Pw/Pc)\,W^*$$

- The equation describe a linear line with slope -(Pw/Pc), defining the budget constraint—combination of C* and W* that are affordable.

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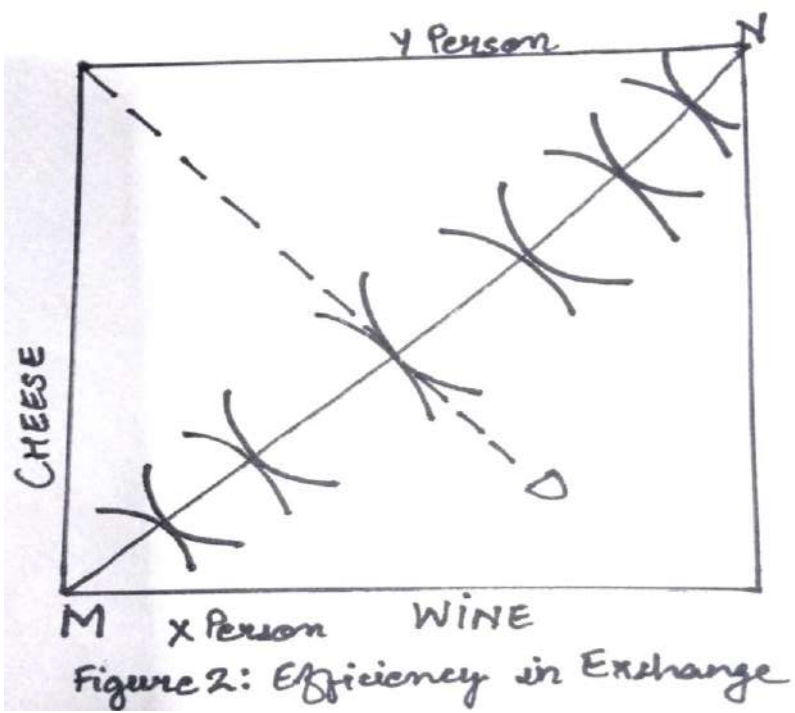


Figure 2: Efficiency in Exchange

- The budget line passes through (Co, Wo), the initial endowment, labeled point A.
- There are other allocations along the contract curve MN that are also Pareto optimal but will not result from the operation of a market.
- If the initial endowment (Point A) is different, a different allocation may result.
- In fact by changing the initial distribution of cheese and wine, any point on contract curve (Pareto frontier) can be obtained using the price system.

Efficiency in Exchange: Bads

- Consider exchange involving one good and one bad—wine and garbage.
- Think of garbage as bags of trash rather than water or air pollution.
- Garbage consumption really consists of storing the bags of trash in an out-of-the-way place until they decay into harmless compounds.
- The ICs of X's preferences regarding wine and garbage are a little different than normal because we have one good and one bad.
- The IC slopes upward, away from the origin, because as wine consumption increases, garbage consumption must also increase to keep utility constant.
- MRSgw is positive.

X' Preference

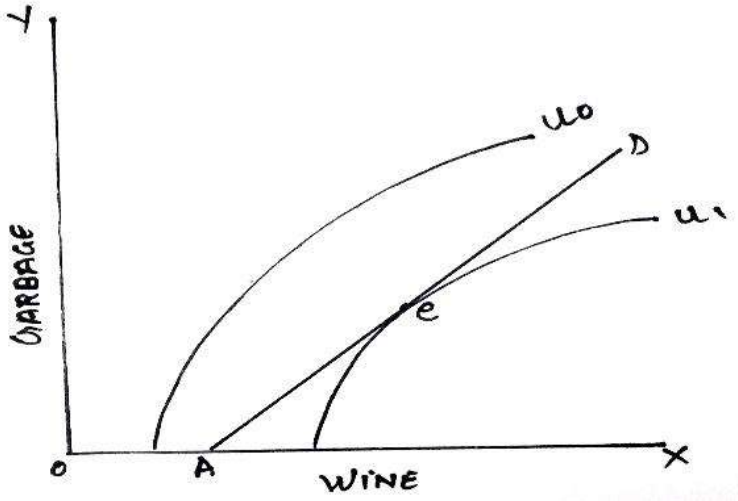


Figure 3: X's Preference for Wine and Garbage

Contd…

- How much trade occurs in this example?
- Assume X has all of the wine and all of the garbage. Y has none.
- Because we are looking at exchange, the quantities of wine and garbage are fixed.
- The Edgeworth box can be used in this case to examine what trade might take place.
- The length of the horizontal axis is the total amount of wine available and the height of the vertical axis is the total amount of garbage available. The point A in the figure is the initial endowment—everything to X and nothing to Y.

Contd…

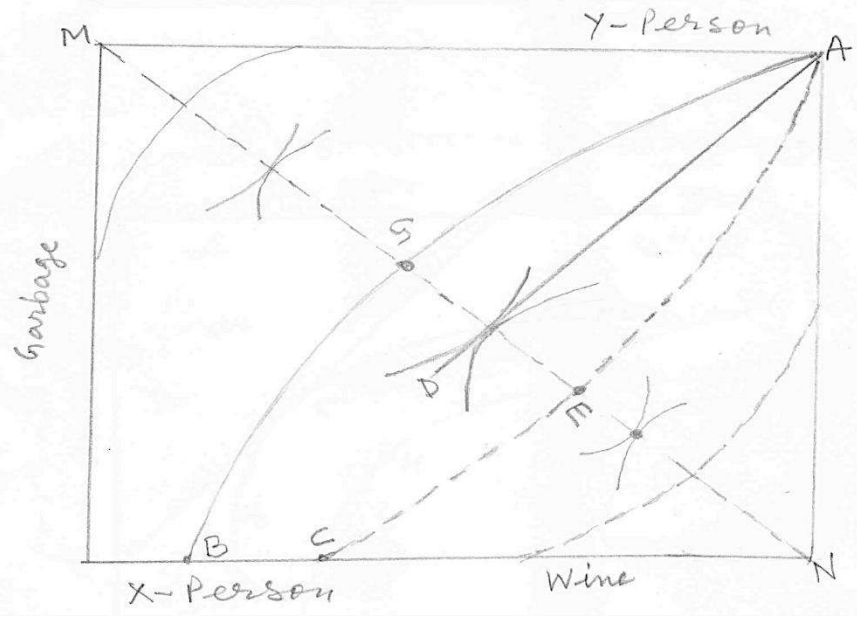
- Let Go and Wo be the initial endowment of garbage and wine possessed by X.
- Let G* and W* be the amount of garbage and wine X ends up with. Budget balancing requires:

$$P_gG_o + P_wW_o = P_gG^* + P_wW^* \quad \text{or}$$

$$W^* = [(P_g/P_w)\, G_o + W_o] - (P_g/P_w)\, G^*$$

- The slope is positive since price of garbage Pg is negative.
- A quick check will see that the initial endowment (Go Wo) satisfies the equation which means that the line passes through the initial endowments.

Efficiency in Exchange of Wine and Garbage by Edgeworth Box



- AB is IC for X and AC for Y
- MN is Contract Curve
- Any division of the garbage and wine that lies between AB and AC will be a Pareto improvement as It will lie on higher ICs for both the persons.
- The price of Garbage (G) is negative and Wine (W) is positive. Therefore price line AD has positive slope.
- The value of any bundle of W and G that X or Y ends up with must be exactly equal to the value of initial endowment of each. pers

Let's Sum Up

- Efficiency means Pareto optimality
- If an economy operates on the Pareto frontier, it is termed as efficient.
- Two types of inefficiencies: : Inefficiency in exchange and inefficiency in production.
- Inefficiency in exchange means that the resources could be moved around among individuals without making anyone worse off.
- Inefficiency in Production: Economy produces below the PPF (will be discussed in next lecture).

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Market Efficiency and Optimality-III

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- Concept of production and production efficiency
- Efficiency in Production with two goods
- Efficiency in Production in Bads

Introduction

- If consumers consume to the point where MU=price of a good and producers produce to the point where MC = price, then resources will be optimally allocated (Efficiency in Consumption and Efficiency in production).
- Marginal Private Benefits and Marginal Private costs (MPB=MPC)
- Marginal Social Benefits and Marginal Social Costs (MSB=MSC)
- Optimization of output (with and without externality)
- This lecture explains the efficiency in production with a good and a bad (pollution a by-product of good product) and derivation of demand and supply and market equilibrium in case of environmental bad product.

Concept of Production and Production Efficiency

- Production is a process of using resources to create a good or service suitable for use or exchange in the market.
- It is transforming inputs into outputs (goods and services)
- It is a flow concept and combines both material inputs and immaterial inputs (knowhow, skills, etc.)
- Three aspects of production process:
  - Quantity of a good or service produced
  - The form of the good or service created
  - The temporal or spatial distribution of the good or service produced

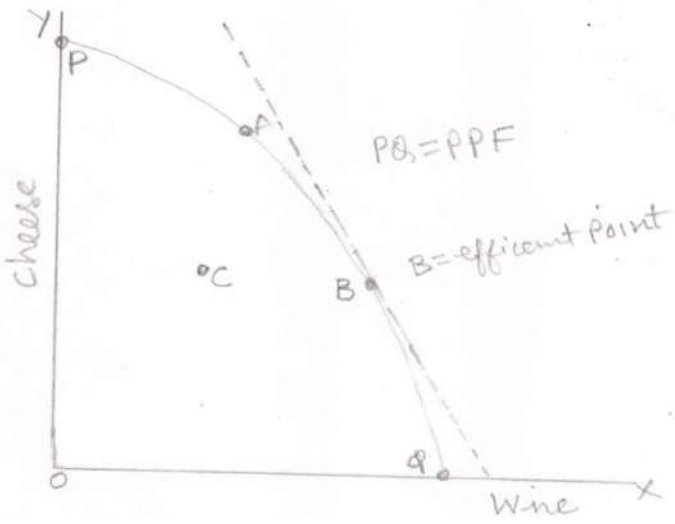
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- Efficiency in production means producing goods and services with the optimal combination of inputs to produce maximum output for the minimum cost.
- It means that the economy must produce on its production possibility frontier (PPF).
- Technical Efficiency and Allocative Efficiency in Production.
- TE implies optimal combination of inputs (labour and capital) to produce a a given level of output (i.e. cannot produce more of a good, without more inputs).
- AE implies optimal allocation of inputs (MC= Price).

Efficiency in Production with two Goods

- PPF represents the point at which an economy is most efficiently producing and allocating its goods and services.
- It shows that there are limits to production.
- To achieve efficiency, an economy must decide what combination of goods and services can be produced.
- The slope of PPF is called marginal rate of transformation (MRT).
- It is the opportunity cost of *X* in terms of *Y* at the margin.
- It is concave to the origin to represent increasing opportunity cost with increased output of a good. Thus, MRT increases in absolute size as one moves from the top left of the PPF to the bottom right of the PP.

Contd…



• Consider production of wine and Cheese

Assume a fixed amount of resources (Labour) to produce W and C.

• PG= PFF (shows trade-offs between cheese and wine)

The inputs into production (labour) are fixed resulting in cost C. but the producer does have a choice about how much wine and cheese to produce.

• Profit (π) = Total Revenue(pcC + pwW)–total Cost (C)

$$\pi = pcC + pwW - C$$

$$W = (\pi - C)/pw - pc/pwC$$

A straight line with intercept (π - C)/pw and slope -pc/pwC, which is negative

B is efficient point

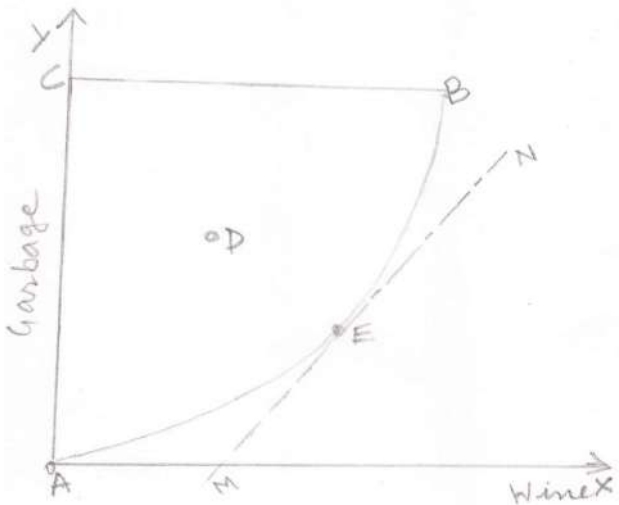
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- The slope of the production possibility frontier at any particular point is called marginal rate of transformation (MRT)
- At E the MRTcw = pc/pw.

Efficiency in Production in Bads

- Consider efficiency in production of wine with garbage as a by-product.
- Assume a fixed amount of resources (Labour) to produce wine and garbage. If we put all of our resources (Labour) into wine production, we will produce the maximum amount of wine as well as garbage.
- Allocation of some labour (L) into garbage prevention.
- With less L available for wine, we will produce less wine.
- The question is to how to split L between these two items (optimization).
- The inputs into production (labour) are fixed resulting in cost C. but the producer does have a choice about how much wine and garbage to produce.
- Profit= Total Revenue – total Cost ($\pi = pwW + pgG - C$)

Contd…



The points along line AB are the efficient combinations of wine and garbage—the Pareto frontier.

Point D in Pareto inefficient as labour is not fully utilized.

$$\pi = pwW + pgG - C$$

$$G = (\pi + C)/pg - pw/pgW$$

A straight line with intercept $(\pi+C)/pg$ and slope $-pw/pg$, which is positive because pg is negative.

Profit is highest when the intercept is as low as possible (since pg is negative).

E is profit maximization point.

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Market Efficiency and Optimality-IV

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Contents

- Introduction
- Equi-marginal Principle
- Supply and Demand for Bads
- Market Equilibrium
- Let’s Sum Up

Equi-marginal Principle

- In case of multiple producers of wine and garbage, for efficiency to hold, the MRTwg must be same for all the producers. MRTwg of A = MRTwg of B = MRTwg of N (Equi-marginal principle)
- If MRTwg of A > B and if we reduce wine production from A by one unit and increase the same from B, then wine production stays constant and overall garbage production declines. This is obviously a Pareto improvement.
- This is an important point that in fact fails to hold for most pollution regulations. Unfortunately, this principle rarely holds for actual environmental regulations.

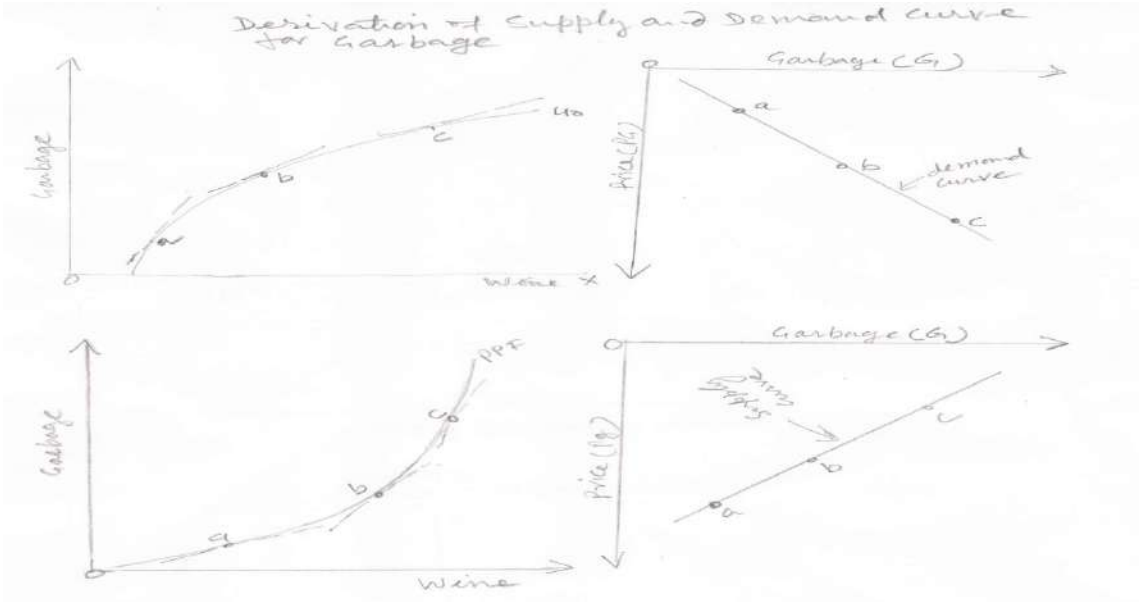
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- MRSgw (Consumption) = MRTgw (production)
- MRT of two firms be same for two commodities
- In a competitive market, market equilibrium is Pareto optimal
- Pareto optimality can be achieved in the market, provided the resources in the economy are appropriately distributed before the market is allowed to operate.
- Generation of pollution violates the assumption of a competitive market
  - Complete property rights
  - Atomistic participants (producers and consumers too small to influence market prices)
  - Complete information
  - No transaction cost

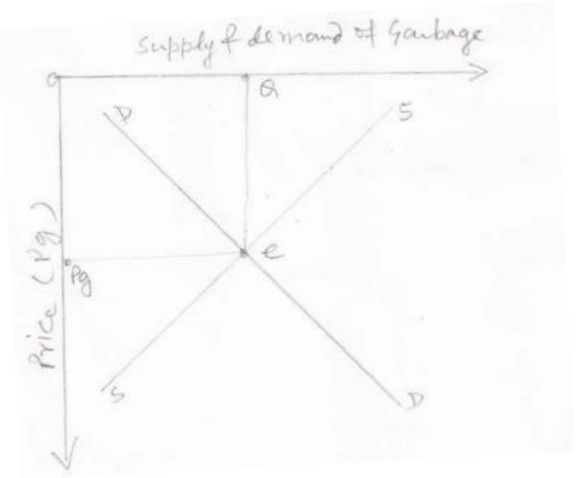
Supply and Demand for Bads (Garbage)

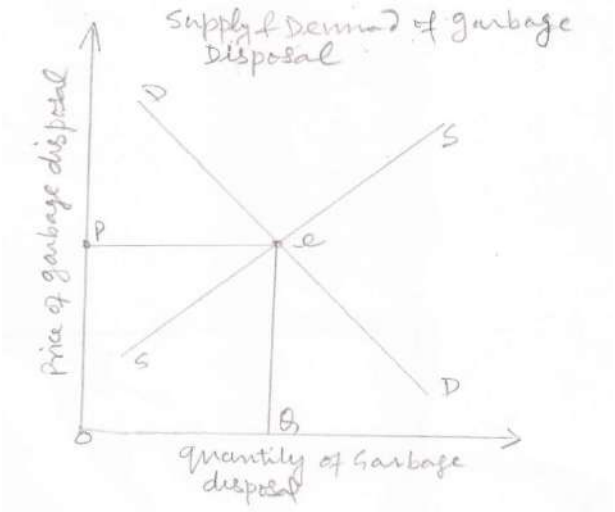
- Viewing garbage as a commodity, it has a negative price. To be willing to consume (i.e store) garbage, you must pay a negative price (i.e., receive a positive price or compensation).
- If the price of G (Compensation) is low, people will not be willing to take much. Similarly, generators of garbage will provide little if they must pay a large price.
- Thinking in terms of negative prices is not common in economics.
- Hence, instead of thinking in terms of garbage (a bad), think in terms of garbage disposal (a good).
- A consumer of garbage is a supplier of garbage disposal. A producer of garbage is a consumer of garbage disposal.

Derivation of Supply and Demand Curves for Garbage



Market Equilibrium





Let's Sum Up

- Production means transforming inputs into outputs (Good and Services)
- Three aspects: Quantity of a good or service produced; The form of the good or service created; and The temporal or spatial distribution of the good or service produced.
- PPF and production efficiency in case of both goods and bads
- Equi-marginal principles
- Derivation of demand and supply curves for bads

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Environmental Economics

Consumer and Producer Surplus-I

Prof. S. P. Singh
Humanities and Social sciences
Indian Institute of Technology Roorkee



CONTENTS

- Introduction
- Consumer surplus
- Producer surplus
- Calculating CS and PS
- Application in Environment
- Let’s Sum Up

INTRODUCTION

- Consumer and producer surplus are the tools to understand the consumer behavior and the market functioning both for conventional and environmental goods.
- CS and PS measure the overall welfare gain associated with an exchange.
- These concepts shows the difference between the value of utility expressed in terms of willingness to pay and accept and value of exchange that is expressed in terms of equilibrium market price.
- This measure of economic gain has many uses in welfare and environment economics.(Example, compensation criterion to study the optimal level of production and consumption).

CONSUMER SURPLUS

- Based on law of diminishing marginal utility.
- Since the price of the commodity is fixed, the marginal utility gained from the consumption exceeds the amount paid for initial intakes. The additional utility gained is the consumer surplus.
- According to Marshal, consumer surplus is 'excess of the price which the consumer would be willing to pay rather than go without the thing, over that which he actually does pay.'
- Thus, CS is a measure of economic welfare that people gain by consuming goods and services.

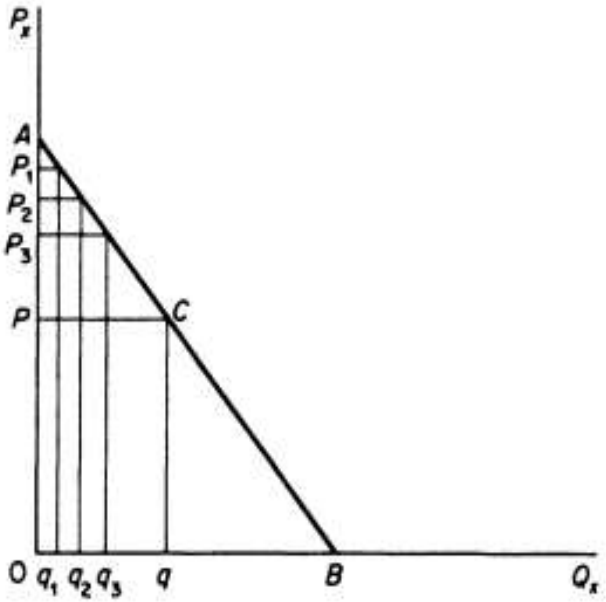
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- Assumptions:
  - The current market price do not change by quantity purchased of this commodity.
  - Marginal utility of money is constant.

- The concept can be explained through following figure:

Contd…

- At price P, the consumer buys q units of x and pays an amount (q) · (P) for it. However, he would be willing to pay P 1 for q 1, P 2 for q2, P 3 for q3 and so on.
- The fact that the price in the market is lower than the price he would be willing to pay for the initial units of x implies that his actual expenditure is less than he would be willing to spend to acquire the quantity q.
- This difference is the consumer's surplus, and is calculated by the area of the triangle PAC.

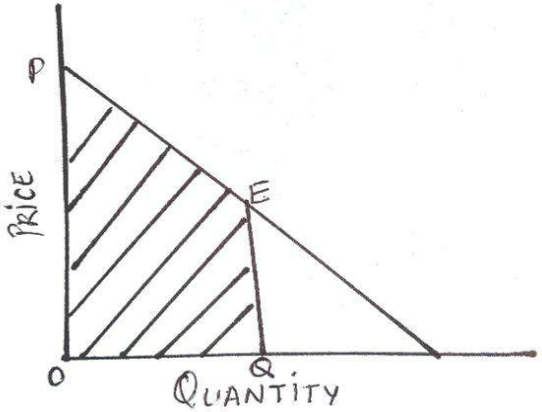


CONSUMER SURPLUS AND ENVIRONMENT GOODS AND SERVICES

- CS is useful in environmental economics for evaluating the environmental impacts of a particular project, or activity.
- Since economic value of many environmental goods and services is very low or free, the CS component in total utility could be very high.
- Consequently, it is important to know that if the environmental goods are lost or damaged, there will be immeasurable loss of human welfare.
- Many environment goods are non-marketed commodity, therefore the value of the commodity is assessed by the willingness to pay (WTP).

Contd…

- WTP is defined as the amount that a consumer would pay for a hypothetical good, service or change in some state of the environment.



- WTP for a particular quantity of a commodity is the total area lying to the left of a demand curve. In this figure, WTP for quantity Q is area shaded as OPEQ.

Contd…

- For environmental goods, the demand curve indicates the marginal willingness to pay for one more unit of the good, as a function of the total quantity of the good being consumed.
- The consumer surplus for such goods arises from both use value and non-use value of the good unlike the conventional goods where CS is obtained by use value or actual consumption only.
- For example, the determination of consumer surplus in case of non-market commodity such as water quality.
- Use value is the on-site utility that the consumer gets from better fish catch from good quality water and the non-use value part is from bequests to the future generation, environment stewardship, etc.

PRODUCER SURPLUS

- **Producer Surplus (PS)**
- Producer surplus is the difference between how much a person would be willing to accept for given quantity of a good versus how much they can receive by selling the good at the market price.
- Opportunity of cost increases with the additional unit of output supplied in the market.

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Environmental Economics

Optimal Provision of Public Goods and Bads-I

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- Types of goods
- Characteristics of public goods
- Pure public goods and bads

Introduction

- Basic difference between conventional goods and environmental goods.
- The market allocates private goods efficiently, while it fails to allocate environmental goods efficiently.
- The concepts of public bads and externalities are developed to describe the characteristics of environmental goods that lead to market failure.
- This lecture explains public and private goods, optimal and market provision of public goods/bads and externalities.

Types of Goods

- Private goods (rival and excludable)
- Public goods (non-rival and non-excludable)
- Quasi-public goods (have characteristics of both private and public goods, partial excludability, partial rivalry, such as roads, tunnels, bridges, parks)
- Merit goods (education, vaccination, healthcare, generate positive externalities)
- Demerit goods (goods that harm the consumers and generate negative externalities; smoking, drugs, alcohol drinking, etc., additional cost on society in terms of health and police expenditure)
- Free Goods (goods without scarcity and opportunity cost)

Characteristics of Public Goods

- **Non-excludability:** once supplied, potential users or consumers cannot be preventing deriving a benefit.
- **Non-rivalry**: once supplied, individuals do not need to compete with each other, given that it become 'freely available' once it is produced.
- **Non-diminishability**: when one consumer uses the good the stock available to others does not diminish.
- **Non-rejectability**: once a public good is supplied it cannot be rejected by beneficiaries.
- **Non-marginal cost:** once supplied, the marginal cost of supplying another individual is zero.

Pure Public Goods and Bads

- Bads are either rival or non-rival or excludable or non-excludable.
- Noise can be made excludable by prohibiting people from being within a certain distance of noise. That is why noise could be both—it depends on nature of noise.
- Garbage is excludable and rival. Greenhouse gases, leading to climate change are non-rival and non-excludable.
- Private goods and bads are typically provided by the market and provided efficiently.
- Pure public bads are not be provided by the market, primarily because non-exclusion of those who are not wiling to pay.

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- King, E. Singh, K., & Shishodia, A. (2007). *Environmental economics: theory and applications*. SAGE Publications India.
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Environmental Economics

Optimal Provision of Public Goods and Bads-II

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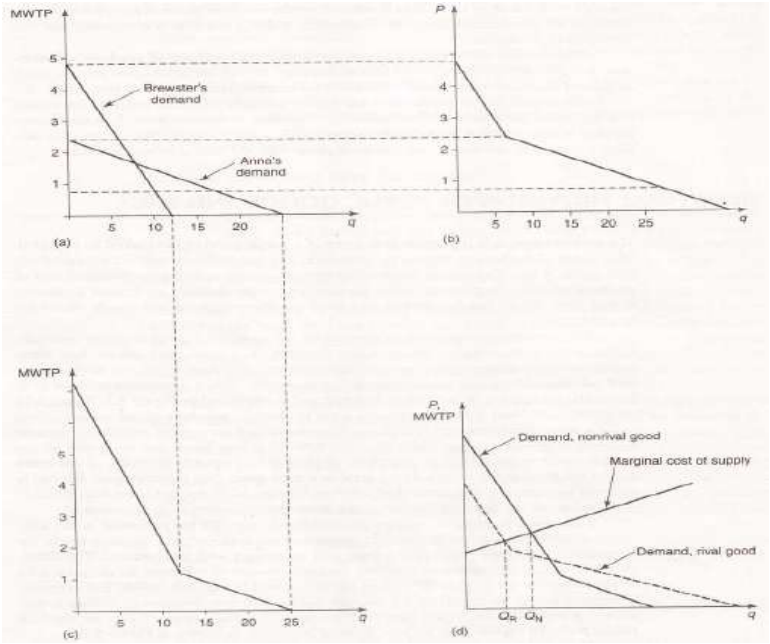
CONTENTS

- Optimal provision of public goods and bads
- Supply and demand for rival versus non-rival goods
- Market provision and efficient provision of public goods

Optimal Provision of Public Goods and Bads

- How much of a public good or bad should be provided?
- The standard response in the case of pvt. goods is that they should be provided up to the point at which the MC of production = price and where Supply = demand at that price.
- In case of rival good, market demand is the horizontal summation of individual demands.
- If a good is non-rival then we proceed differently. We are interested in the total MWTP for specific amounts of the non-rival good.
- The good is produced in the same way whether or not it is rival in consumption.
- Thus, the same supply or marginal cost curve applies to its production. (all goods rival in production)

Supply and Demand for Rival Versus Non-rival Goods



- (A) individual demands; (B) aggregate/market demand; and (C) efficient provision.
- Qr quantity of rival goods and Qn quantity of non-rival goods.
- Market Demand for Rival goods: horizontal summation
- Non-rival: Vertical summation
- Although markets rarely exist for public goods/bads, demand curves still exist.

Market Provision and Efficient Provision of Public Goods

- What happens if markets are used?
- If we let the market operate without intervention, public goods generally will be underprovided.
- To see this most simply, assume we have a society with N identical individuals. Each individual can consume two goods, a rival, excludable private good, P, and a non-rival, non-excludable public good G.
- To clarify things, the public good is non-excludable in consumption but it is, of course, excludable in production.

Contd…

- Assume that each person has income y available to them. The utility of any of the identical people is given by:

  Utility = u (P, G)

  G is out of the control of an individual as it is a public good. However, individuals are free to purchase some of the public goods.

  Let g be the amount of the good that one individual will purchase and let G^- be what is provided by everyone else: $G = G^- + g$.

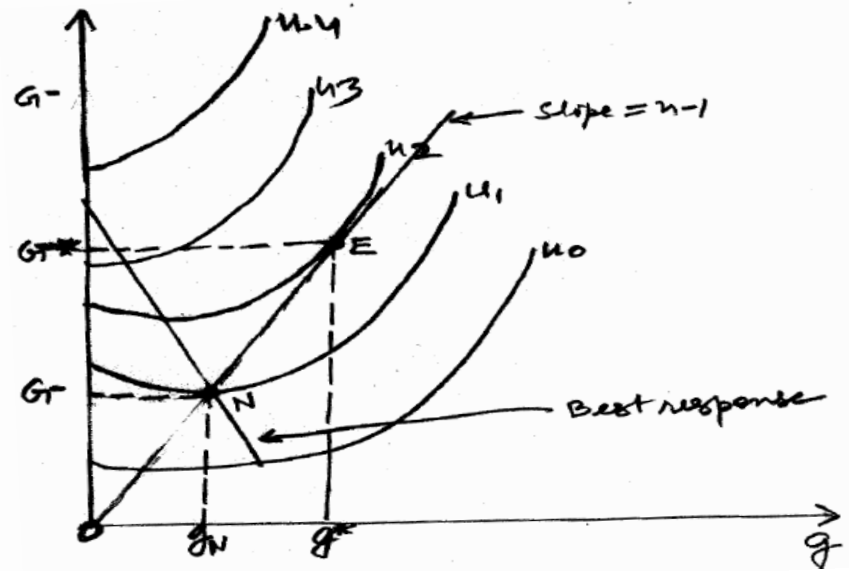
- If individual provides g of the public good, only y-g of money is available for Pvt good. Thus

  Utility = u (y-g, $G^- + g$)

Contd…

- Now the consumer must decide how much g to choose to consume. The choice depends, of course, on how much of the public good is being provided by everyone else.
- ICs for the consumer with the two goods being g and G^- with fixed (y) are drawn.
- At high levels of public goods provision by others, G^-, it is entirely plausible that no privately provided public goods are necessary. In this case G^- needs to increase to keep utility constant, as g rises from zero and ICs will be upward sloping in g.
- Note that if G^- is fixed, the best choice of g is the point at which one of the ICs just touches the horizontal line at the fixed G^- .

Market and Efficient Provisions



Point N: Market provision ($G^{-}N$; $g_N N$)

Point E: Efficient Provision($G^{-*}E$; g^*E

As G^{-} increases, the best choice g falls.

The 'best response' line indicates inverse relation between g and G^{-}. For a high enough G^{-} no g will be created.

Since it is identical society, everyone will make the same choice about g. Thus $G^{-} = (n-1)\,g$. This is a straight line of slope n-1.

Contd…

- What is the efficient level of g and G^- ?
- Non-cooperative solution at point N (Market Provision).
- Cooperative solution at point E (Efficient Provision).
- If they could be sure that when they raise their g, everyone else matches them by raising their gs. Then consumers see the full benefits of raising g. If consumer know that g and G^- will lie along the upward slope of IC, the best choice of g is the point at which an IC is just tangent to the straight line, at E point, resulting in public good provision g* and G^{-*} .
- Market provision < efficient provision, thus market under-provides the public goods.

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Environmental Economics

Optimal Provision of Public Goods and Bads-III

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CONTENTS

- Externalities
- Relationship between public goods and externalities
- Modeling a negative environmental externality
- Sum up

Externalities

- An externality occurs when one person' or firm's actions affect another entity without permission (consumption and production externalities)
- Negative Externalities
- Positive Externalities
- Why should we be worried about externalities?
- The basic problem is that the generator of the externality is deciding how much of the externality to produce but is not taken into account the effects of the externality on others.
- Example: Steel Mill and Laundry

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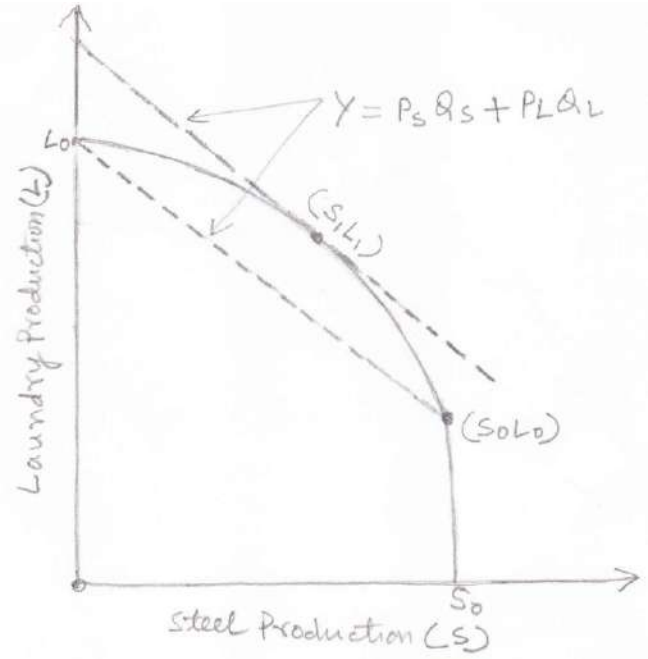


Fig. illustrates the case of the steel mill and the laundry with a modest externality in the form of smoke from the steel mill. Q = output; p= Price S=Steel, L= Laundry; Y = total value of output

S_0L_0 = PPF

$Y = P_S Q_S + P_L Q_L$

Max Possible $Q_L = Y/P_L$; when $Q_S = 0$; (L_0)

Max Possible $Q_S = Y/P_S$; when $Q_L = 0$; (S_0)

Initial output at $S_0 L_0$

After Merger: $S_1 L_1$ (Pareto Improvement)

Contd…

- The PPF of steel and laundry (maximum possible output $S_o L_o$)
- Assuming price of steel and laundry (P_S and P_L), the total value of output will be as: $Y = P_S Q_S + P_L Q_L$, where Q is the quantity and P is the price.
- Initial maximum output of laundry = Y_0 / P_L ; steel = Y_0 / P_S.
- Total value of output ($S_o L_o$) is $Y_0 = P_S Q_{S0} + P_L Q_{L0}$
- To solve for Y_0 . If we plot $Y = P_S Q_S + P_L Q_L$ and pass it through ($S_o L_o$), then that line will intersect the vertical axis at $L = Y_0 / P_L$.
- $S_o L_o$ indicate Pareto relevance of externality (no coordination)
- Merging the two eliminates externality.
- New level of output value (Y_1) is at $S_1 L_1$. (without externality)
- $Y_1 > Y_0$.

Relationship between Public Goods and Externalities

- Although public goods and externalities are not the same concept, they are closely related
  - If the externality affects a broad segment of society and if its effects are non-rival and non-excludable, the externality is itself a public good (such as climate change and global warning)
  - If the externality affects a narrower group of individuals or firms, those effects are more properly modeled as an externality (such as sewage)

Modeling a Negative Environmental Externality (Sugar Mills Example)

- Consider Sugar Market
  - Assume the market is competitive (Perfect Competition)
  - Supply is the marginal *private* cost (MPC)
  - Demand is the marginal *private* benefit (MPB)
  - Sugar production generates pollution, modeled as a marginal *external* cost (MEC)
- **Problem**: Producers have no incentive to consider the externality
- **Result**: Competitive solution is inefficient

Finding a Competitive Solution

- S: $P = 100.0 + 0.50Q$
- D: $P = 500-0.75Q$,
- where

  Q = Quantity of sugar produced; p = price per tons in $

- Since S is MPC and D is MPB, rewrite as:

  $MPC = 100.0 + 0.50Q$

  $MPB = 500.0 - 0.75Q$

- Competitive solution?

Contd…

- Set MPB = MPC

  500 - 0.75Q = 100 + 0.50Q

- Solve:

  Q = 320 tons P = $260 per ton

- Analysis:
  - This ignores external costs from pollution
  - MPC undervalues opportunity costs of production; Q is too high; P is too low

Contd…

- Let Marginal External Cost (MEC) = 0.10Q
- Marginal Social Cost (MSC) = MPC + MEC
  - MSC = 100.0 + 0.50Q + 0.10Q
    = 100.0 + 0.60Q
- Marginal Social Benefit (MSB) = MPB + MEB
  - Assuming no external benefits, MEB= 0, so MSB = MPB
- Set MSC = MSB
  - 100.0 + 0.60Q = 500 - 0.75Q
  - Solving: Q = 296.3 tons P = $277.8/ ton
- Results of externality: Q too high and P too low (Market distortion)

Using Mπ and MEC Criteria

- Competitive firm maximizes profit where

  Marginal Profit (Mπ) = 0; MPB – MPC = 0, or

  Mπ = 0

- Efficient firm produces where: MSB = MSC or MPB + MEB = MPC + MEC or MPB - MPC = MEC, if MEB = 0,

  Mπ = MEC

- Mπ = MPB - MPC

  = (500 - 0.75Q) - (100 + 0.50Q) so

- Mπ = 400 - 0.125Q
- MEC = 0.10Q

Contd…

- **Competitive Solution**

  Set Mp = 0; 400 - 0.125Q = 0, so Q = 320

  Find P by substituting into MPB or MPC

  Using MPB, P = 500 – 0.75(320) = $260

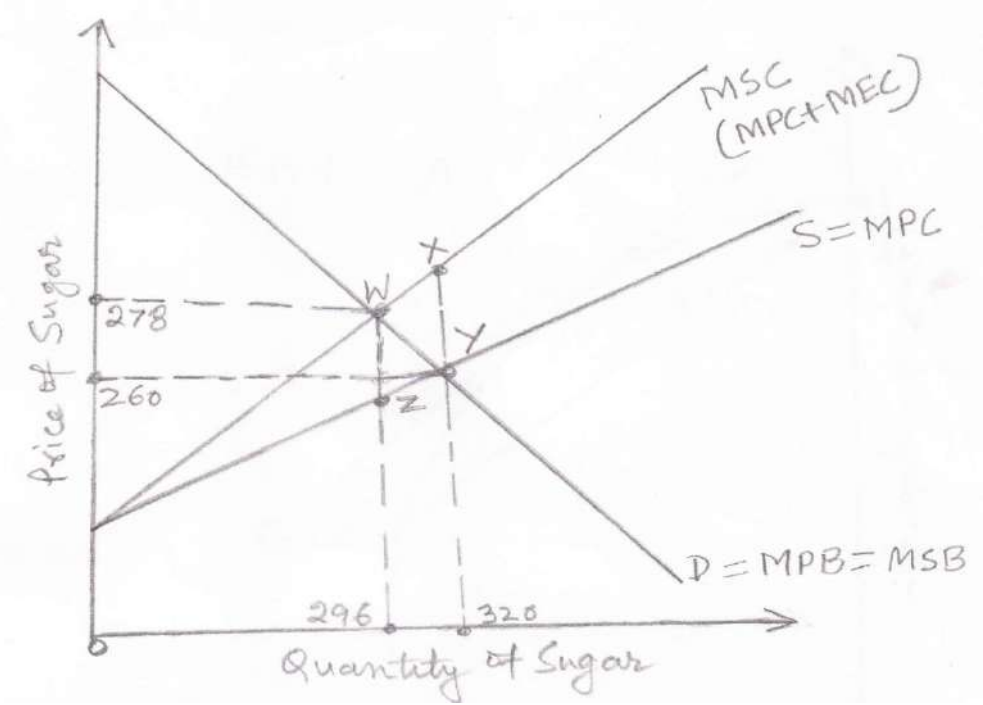
- **Efficient Solution**

  Set Mp = MEC, or 400 - 0.125Q = 0.10Q, so Q= 296.3

  Find P by substituting into MPB or MPC

  Using MPB, P = 500 – 0.75(296.3) = $277.8

Measuring Society's Net Gain from Restoring Efficiency



WXY = producer Loss= WYZ due to reduction in Q

Societal Gain= WXYZ due to reduction in MEC for each unit of Q reduction

Net Gain = Area WXYZ –Area WYZ = Area WXY

Let's Sum Up

- Private/public goods (rival/non-rival and excludable/non-excludable).
- Aggregate demand for rival and non-rival goods.
- Supply curve for rival and non- rival goods: same as both are rival in production.
- Market provision < the optimal provision of public goods/bads.
- Externalities and public goods.
- Societal gains if externalities are internalized.

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EXTERNALITY AND MARKET FAILURE

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



Contents

- Market Failure
- Externalities and Market failure
- Types of Externality
- How to internalize externality?

Market Failure

- Market failure occurs in the presence of misallocation or inefficient allocation of resources causing waste or loss of value.

- What is efficient allocation: the allocation of resources under perfectly competitive market is said to be efficient where P=MC.

- Four important sources of market failure:
  - Imperfect market structure/non-competitive behaviour
  - The existence of public goods
  - The presence of external costs or benefits
  - Imperfect information

Externalities and Market Failure

- Externalities as a source of inefficiency:
- When we are transacting or making some activities/decisions and its consequences fall on second/third party and we do not have any incentive to consider those consequences.
- Examples - actions of power plants, manufacturing firm
- Consequences: air/water pollution
- An externality exists when actions/decisions of one person or groups impose a cost or bestow a benefit on others. Therefore, it is also known as spill overs/third part effect/neighbourhood effect.
- Inefficient decisions occur when decision makers fail to consider social costs and benefits.

Contd…

- An externality occurs when one person's or firm's actions affect another entity without the permission of latter.
- It exists when the consumption or production choices of one person or firm enters the utility or production function of another entity without that entity's permission or compensation.
- Exceptions: It does not include the consumption or production of goods and services that are transacted, intentional doing of any good (altruism) or bad to others (murder).

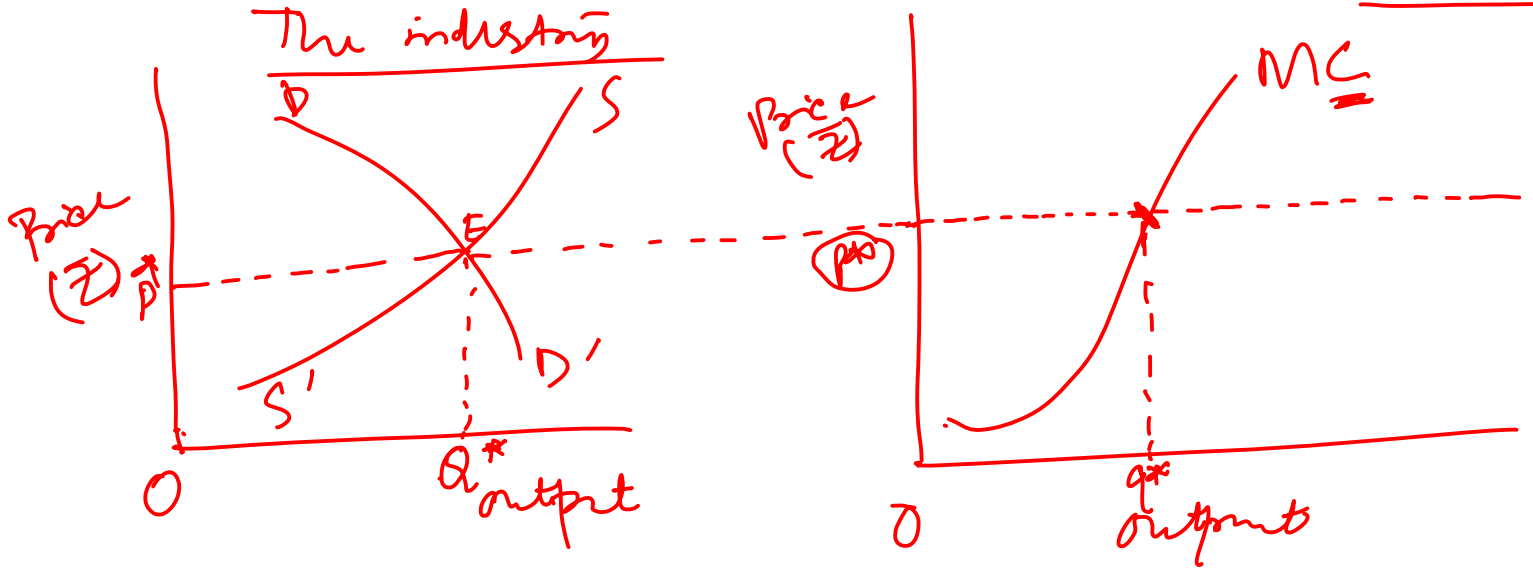
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- Case of A profit maximising firm- Absence of externalities:

- In the absence of externalities, a firm takes price and MC into account to decide output (P=MC) and weighs the full benefits of additional production to society against the full costs of that production to society.
- Price of a product is an accepted measure of additional units of product
- Those consumers who value the additional product worth of price will pay for the price to consume and obviously consumers who value less worth than price will not buy the product.
- If MC= summation of all costs to society, additional production will be efficient till P=MC and each unit of production reaps more benefits over costs.
- Profit maximising perfectly competitive form will produce up to the point at which P=MC.

- Let's explain this graphically…

Figure 1: A Profit Maximising Firm in the Absence of Externality



A Profit Maximising Firm in the Presence of Externality

- When the production of a firm generates externality, this external costs if not taken into account by the firm, it is likely that the firm will overproduce.
- Let's explain graphically what happens when we add externality in the decision making.
- Here MSC is the sum of the MCs of producing the product and correctly measured external costs generated in production process.

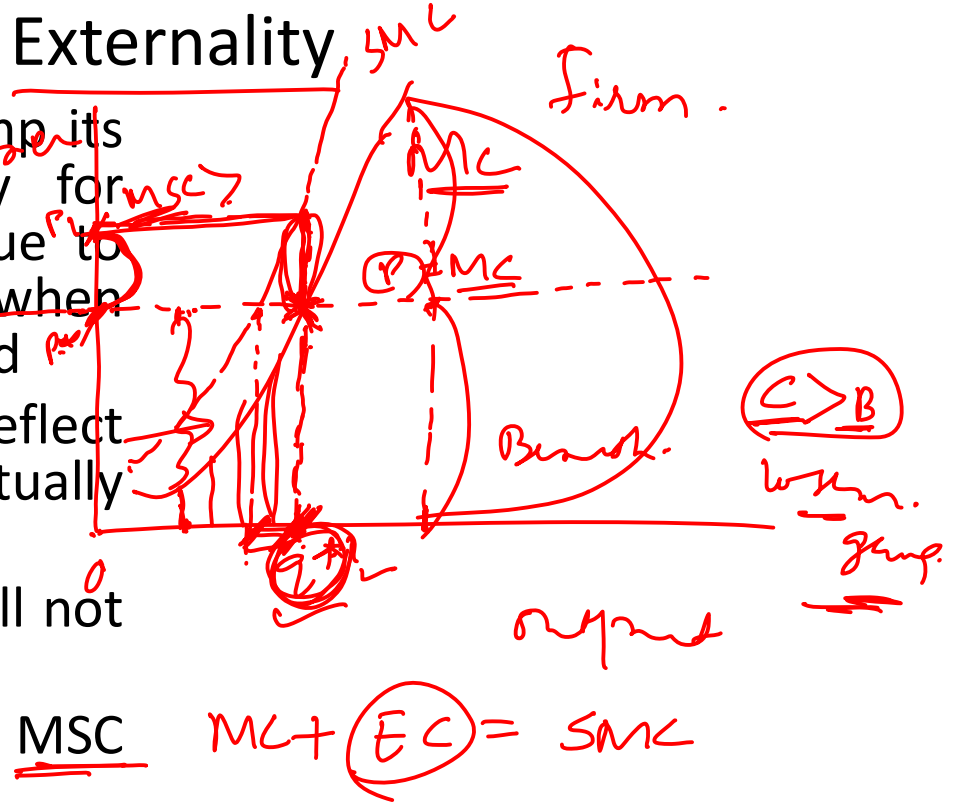
External cost

$TC = \Sigma MC + EC \rightarrow$ firm

$MC + EC \rightarrow MSC$

Figure 2 A Profit Maximising Firm in the Presence of Externality

- If the firm is allowed to dump its wastes and does not pay for damage, then it will continue to produce at q* as before (when externality was not there) and
- price p* will continue to reflect the costs that the firm actually pays to produce its product.
- However, market price p* will not reflect the cost of externality.
- Therefore, at equilibrium q*, MSC >P



Contd…

- In case of positive externality also, the individuals/firms/groups have too little incentive to engage in the activity which generates external benefits.

- Example – vaccination to combat communicable diseases

- As a result of positive externality in this case too few parents would vaccinate their kids if there is no such compulsion from outside , let's say from school.

Types

- Negative externality
- Positive externality
- Production externality
- Consumption externality

Contd…

- Production Externality:

- This externality may occur when one firm's profits are involuntarily affected by another's action.

- For example – by use of production function: smoke is not an input for production of laundry's output. However, laundry's output is affected by smoke emitted which is from a steel factory. Smoke is an externality here.

Contd…

- Consumption Externality:
- A consumption externality exists when a good or bad which is not chosen by the consumer for its utility function but however utility functions or satisfactions are affected by the unchosen items.

- This is an example of consumption externality.

- For example: the externality is generated by a paper mill that discharges pollution into a river that is used for swimming.

How to Internalise the Externality?

- Internalisation is a process/mechanism that provides decision makers with incentives to weigh the external costs and benefits of their decisions.
- There are five approaches commonly used to solve the problem of externalities, Viz.
  - Private bargaining and negotiation (Coase theorem)
  - Legal rules and procedures
  - Govt. imposed taxes and subsidies
  - Sale auctioning of rights to impose externalities
  - Direct Govt. regulation

Suggested Readings

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PIGOUVIAN FEES

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



Contents

- Introduction - What is Pigouvian tax?
- Optimal Pigouvian tax
- Issues of Pigouvian tax

Introduction

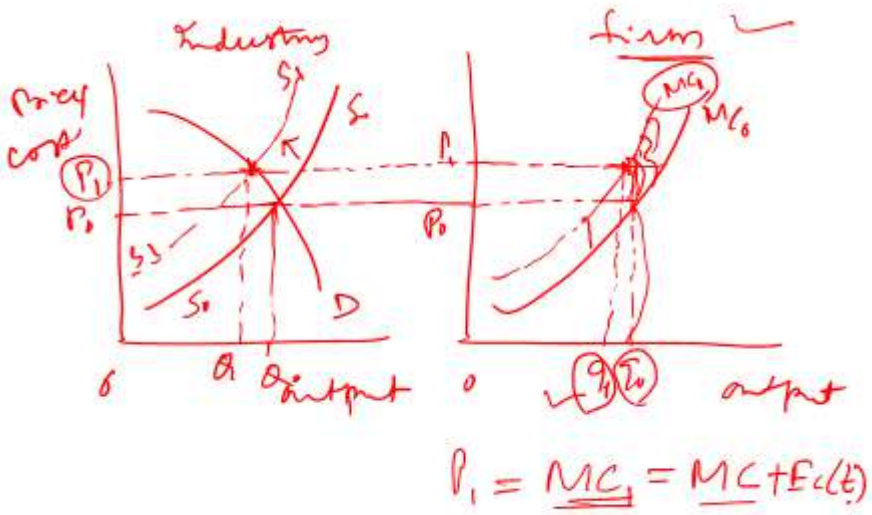
- How to intervene in case of externality in order to reach the social optimum?
- When private bargaining and negotiations fail, marginal taxes and subsidies are direct mechanisms of forcing firms to internalise externalities.
- A tax to be imposed on the polluter based on the estimated damage done/external costs: Such a tax is known as Pigovian tax.
- Arthur C Pigou (1877-1969), in his ‘ Economics of Welfare’ (1920), had proposed a tax as an instrument of equating PC= SC
- PC+Externality= SC
- PC+tax= SC

Pigouvian Tax/Fee

- When a firm produces and imposes a social cost, a per-unit tax can be imposed equal to damages of each successive unit of output produced by the firm.
- Tax = MCs or marginal damage costs
- The damage costs are paid by the firm in form of a per-unit tax, i.e., tax = MDC
- When tax is imposed MC = MSC

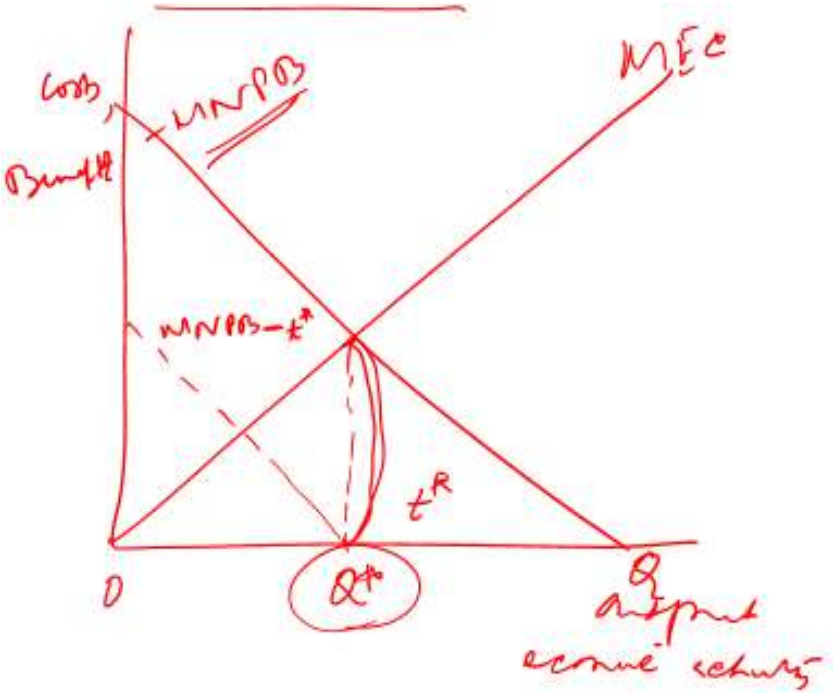
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- When tax is added into MC_0 of firm, the supply curve of the industry shifts to the left, and price is increased from p_0 to p_1 level.
- Now, the efficient level of output (q_1) of the firm is achieved, where $P = MC_1 = MSC$



Theoretically 'Optimal'/ 'Ideal' Pigovian Tax

- Let see how optimal tax is imposed when tax is charge one per unit of the output which is generating pollution:
- Now tax = t* is imposed and as a result MNPB shifts leftward to MNPB-t*.
- Here, t* is the optimal tax as it achieves the social optimum at Q* level of output.



How t* is Determined?

- t*= MEC at the ‘social optimum level of output’

- Hence, t* is the optimal Pigouvian tax = MEC (marginal pollution damage) at the optimal level of pollution.

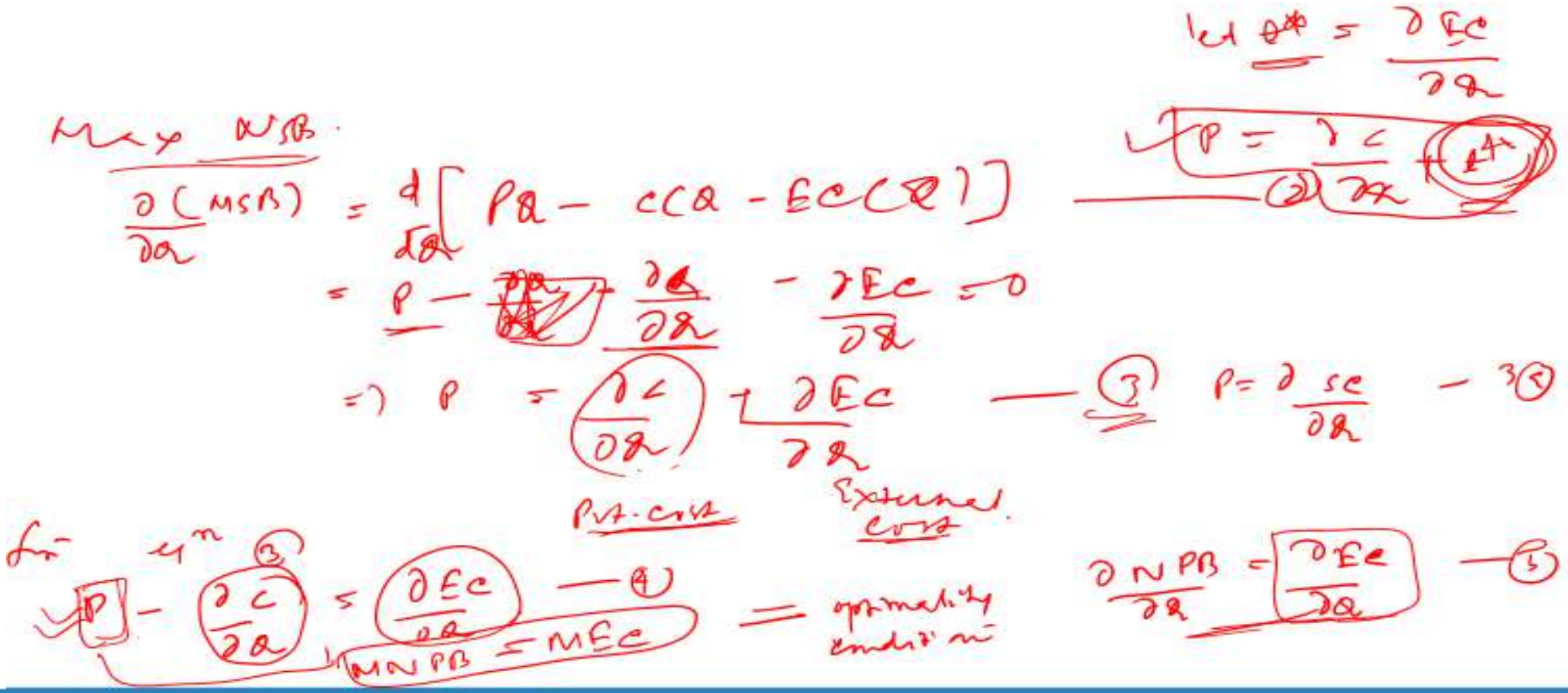
- A damage function tells us ‘how pollution damage varies with the level of pollution emitted and what the monetary value of that damage is’.

Mathematical Representation of Optimal Pigouvian Tax

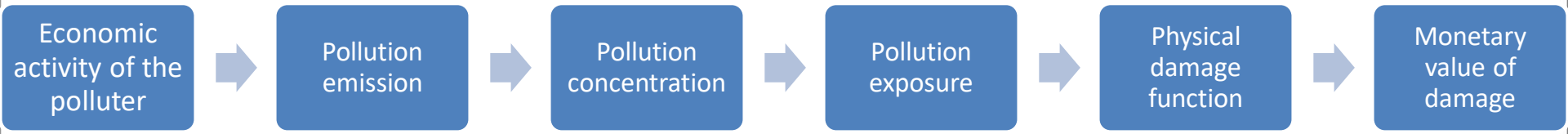
- We know that,
- NSB = gross benefits of polluting activity (P.Q) - Pvt. Cost (C) – external costs (EC)

- NSB = PQ - C(Q) - EC (Q) (1)
- Where, P= price; Q= output

Contd…


Let $\theta^* = \frac{\partial EC}{\partial Q}$
$P = \frac{\partial C}{\partial Q} + \theta^*$ — (2)
Max NSB.
$\frac{\partial (NSB)}{\partial Q} = \frac{d}{dQ}[PQ - C(Q) - EC(Q)]$
$= P - \frac{\partial C}{\partial Q} - \frac{\partial EC}{\partial Q} = 0$
$\Rightarrow P = \frac{\partial C}{\partial Q} + \frac{\partial EC}{\partial Q}$ — (3) $P = \frac{\partial SC}{\partial Q}$ — 3*
Pvt. cost
External cost
for 4th (3)
$P - \frac{\partial C}{\partial Q} = \frac{\partial EC}{\partial Q}$ — (4)
MNPB = MEC
= optimality condition
$\frac{\partial NPB}{\partial Q} = \frac{\partial EC}{\partial Q}$ — (5)

The Sequence for Finding Monetary Value of Damage

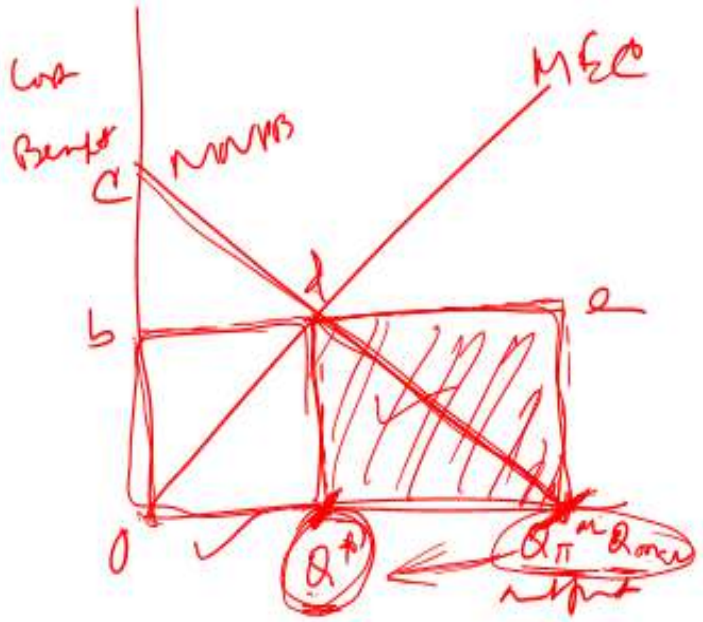


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- For understanding emission-damage sequence, we need to understand the components of MEC and MNPB.
- Because of commercial confidentiality of information, a polluter firm may not disclose information related to MEC and MNPB.
- Therefore, a government/govt.-agency as a regulator may not be in a strong position to extract the related information.
- This asymmetry of information between the polluter and the regulator is often regarded as an objection to any form of government intervention.

Issues of Pigouvian Tax

- Pollution charges and Property rights:
- If the polluter continued to pollute at Qmax, it would be liable for a total pollution tax bill = ObdQ*+ Q*deQmax.
- However, Q*deQmax amount is not feasible to be paid as tax bill is more than the net private benefit.
- So, pollution/output will be reduced to Q*, to avoid tax.
- Even after reaching Q*, the polluter firm still pays ObdQ* of tax bill despite the fact that it is emitting the optimal level of pollution.
- It seems the polluter is penalised twice: by cutting back his output and paying tax bill ObdQ*

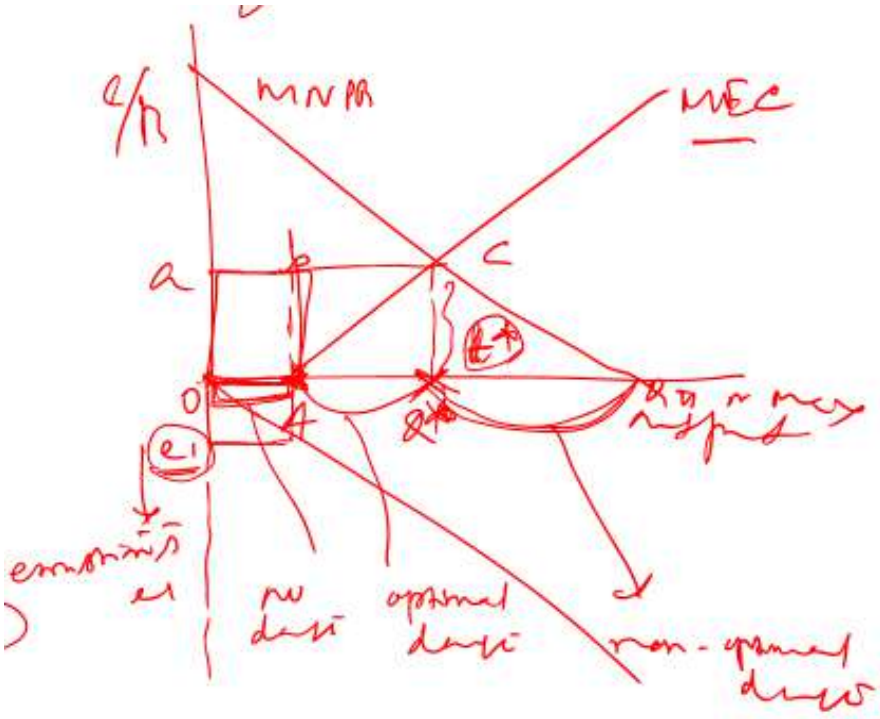


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- The proper Pigouvian fee is related to the monetary value of damage done at the optimum.
- However, if they are to be introduced at all, they are generally levied on emissions or ambient concentrations which are measured in physical terms.
- We cannot assume emissions and concentrations are related in a one-to-one fashion.
- We need to consider Assimilative Capacity of the nature

Pollution Charge and Assimilative Capacity

- The environment can assimilate the waste up to emission level e_1.
- After this point, further emissions cause physical changes which may lead to immediate economic damage.
- Then only MEC begins to rise at point A.
- Now if tax is imposed, bill is amounting OacQ*, that means even though there is no impact of emissions, still the firms needs to pay OabA tax.
- It leads to unfairness in tax imposition.
- Therefore, this tax may be called as use of a scarce resource not use of environment.



Suggested Readings

- Field, B.C. & Field, M.K. (2017). *Environmental Economics: An Introduction*, McGraw Hills, USA.

- Pearce, D. W. and Turner, R. K. (1990). *Economics of Environment and Natural Resources*. Harvester Wheatsheaf: New York. Chapter 6.

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NPTEL ONLINE CERTIFICATION COURSE

PROPERTY RIGHTS & COASE THEOREM-I

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



CONTENTS

- What is property rights?
- Types
- Characteristics of property rights
- Optimal Externality
- Coase Theorem and property right
- Why the Coase Theorem may not work?

Property Rights

- Property rights are a set of entitlements that define the owners' privileges and obligations for the use of the resources or the assets.

- PR exercises the exclusive authority to determine how a resource is used, whether that resource is owned by government or by individuals.

- It can be perceived as an attribute of an economic good.

Types of Property Right

- Public property right: If the resource is owned by the government, the agent who determines its use has to operate under a set of rules determined.
- Private property right: is the exclusive right to the services of the resource.
- Common/collective property right: if the property right is held by group of individuals or groups in common, is known as common property right. By this right, properties are jointly owned and access, use and exclusion are controlled by joint decisions.

Characteristics of Property Right

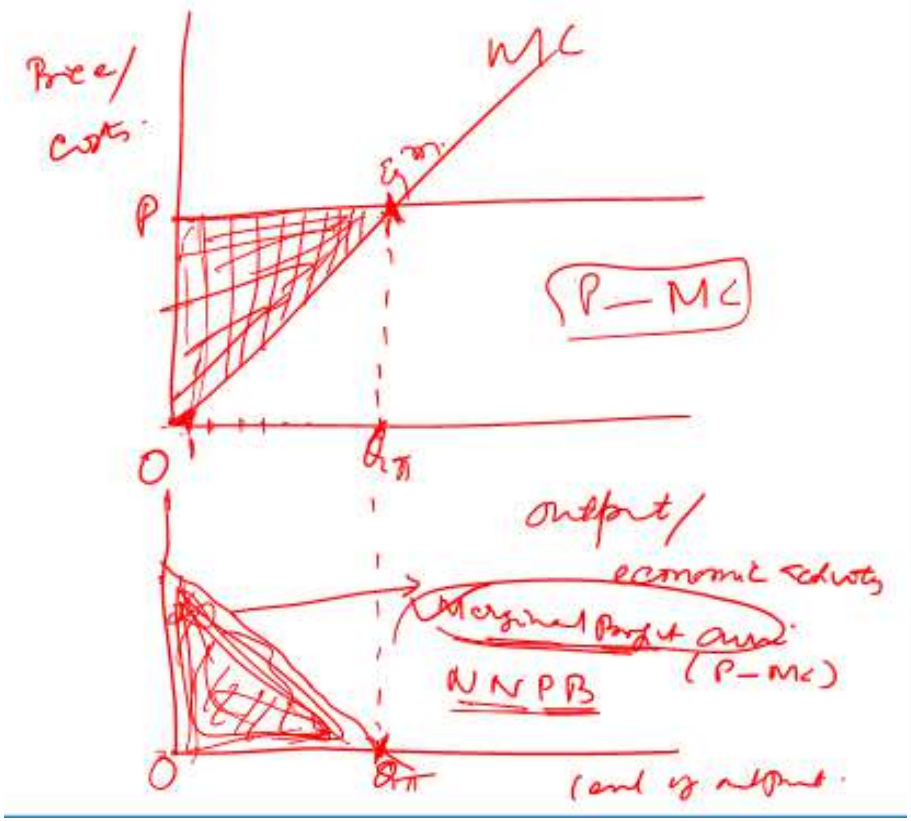
- **Exclusivity**: which states that all benefits and costs of having a right over resources must accrue to the owner and to the owner only.
- **Transferability:** states that the holder of the resource who enjoys property right, also enjoys the privilege of transferring the resource through voluntary exchange.
- **Enforceability:** The property right should be secured for involuntary seizure and encroachment.
- **Comprehensibility**: The property right should be owned either privately or collectively.

Optimal Externality

- Marginal Net Private Benefit (MNPB):
- **The firm is the polluter:**
- It will incur costs in undertaking activity that happens to give rise to the pollution and receive benefits in the form of revenue.
- NPB (Net Private Benefit) = diff. between Total Revenue and Total Costs
- MNPB (Marginal Net Private Benefit)= the extra net benefit from changing the level of activity by one unit.
- Marginal External Cost (MEC): the value of the extra damage done by pollution arising from the increased activity measured by Q.

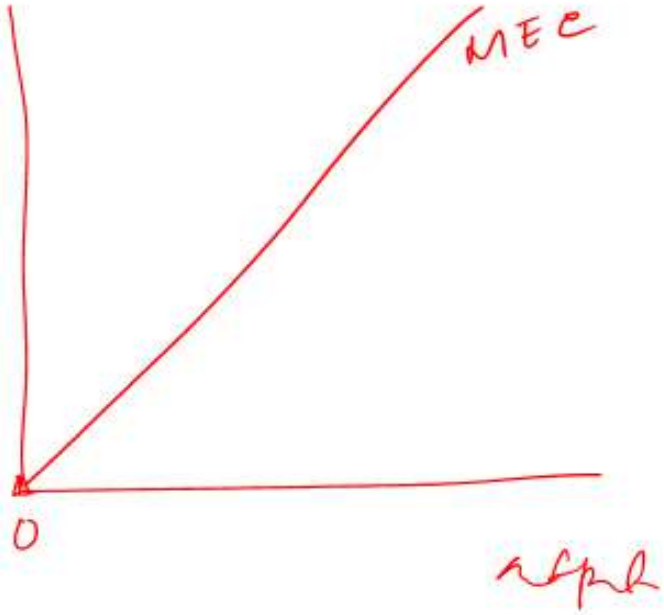
Marginal Net Private Benefit

- Total profit is the area under the marginal profit curve.
- Total profit is maximised when marginal profit is zero.
- Marginal profit is equivalent to MNPB.



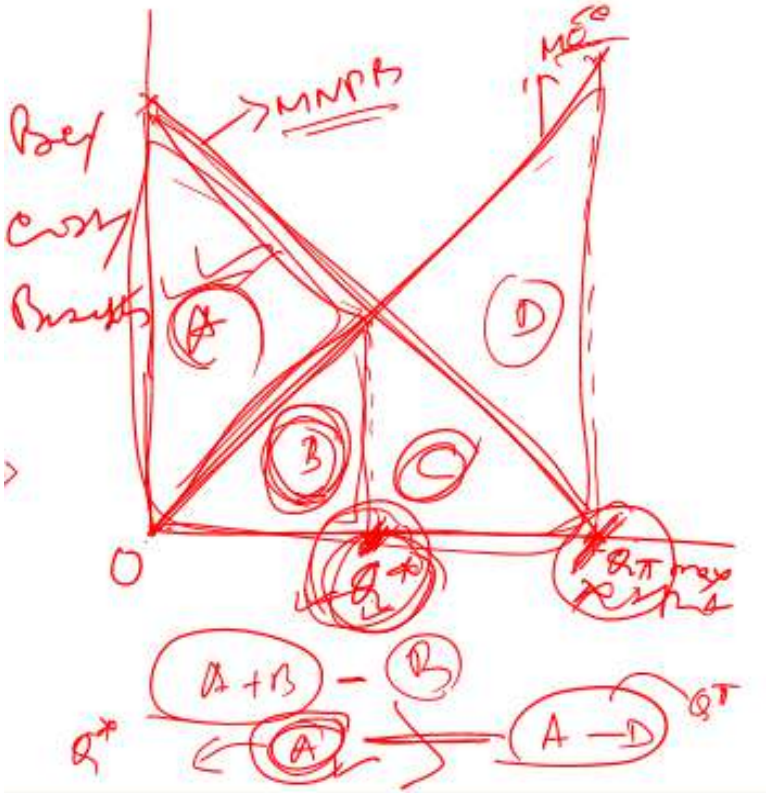
Marginal External Costs

- Marginal external costs are the extra damage caused to the environment by increasing output by the polluter firm by one unit.
- MEC = f(Q); f > 0
- The larger the economic activity, the more will be the MEC.
- MECs are proportion to the level of output.
- So the MEC is a linear function passing through the origin.



Optimal Externality

- Since both the curves are marginal curves, the area under them are total magnitudes.
- The area under MNPB is the polluter's total net private benefit.
- The area under MEC is the total external costs.
- The objective of society is to Maximise (B- C)
- If the producer produces at Q* of economic activity, then the largest area of net benefit obtainable = A
- So Q* is the optimal level of activity
- The optimal level of pollution is the one, that corresponds to the optimal level of activity, i.e., B.



Contd…

- For producing Q_π level of output,

TEC= B+C+D

MNPB= A+B+C

At Q_π, the net social benefit= MNPB –TEC

= A –D

At Q* level of output,

TEC=B

MNPB=A+B

the net social benefit= A

A>A-D

So, Q* is the optimal output level and optimal pollution corresponding to it is B

Contd…

- At Q*, MNPB = MEC
- We know that MNPB = P - MC
- So, P – MC = MEC
    - P = MC + MEC = MSC (Marginal Social Cost)
    - P = MSC is the condition for Pareto optimality

Suggested Readings

- Coase, R. H. (1960). The Problem of Social Cost. *The Journal of Law and Economics,* 3:1–44.

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PROPERTY RIGHTS & COASE THEOREM-II

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Coase Theorem And Achievement of Optimal Externality

- Ronald H. Coase (1960) 'Problem of Social cost';
- He bagged 1991 Nobel Prize in Economic Sciences, for the significance of **transaction costs** and property rights for the institutional structure and functioning of the economy.
- He proposed that if property rights are well-defined, the problems of **externalities can be overcome.**
- The potential of market bargains

  If the victim has the property right

  If the polluter has the property right

Assumptions

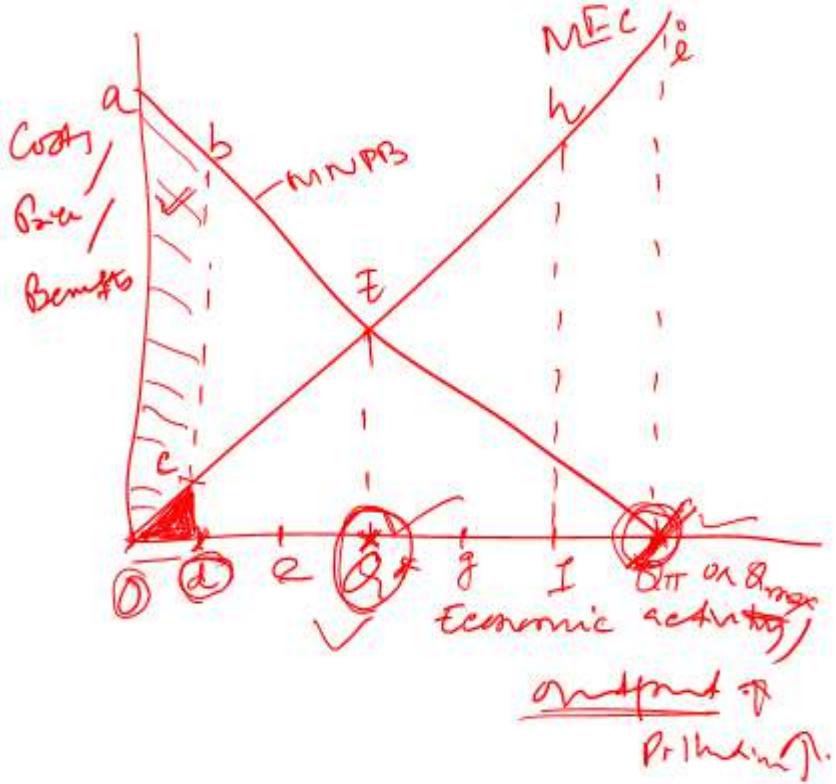
- Some producers or consumers are subject to externalities generated by other producers or consumers.
- Perfect information
- Consumers and producers are price takers
- Costless court system for enforcing agreements
- No transaction costs
- Producers maximise profits
- Consumers maximise utility

The Bargain: If The Victim Has The Property Right

- Now the victim has the right not to be polluted and the polluter does not have the right to pollute.
- The victim will not prefer any pollution to take place
- Whether potential bargain is feasible over the level of externality?

The Bargain

- Left unregulated, the polluter will try to operate at Qmax, where profits are maximised.
- But the social optimum is Q*
- So both are incompatible
- If there is a bargain over the level of externality, then??



The Bargain: **Will They Move to Any Point Like ‘d’?**

- If they move from **0** to **d,** the polluter would gain oabd amount in total profit.
- But the victim would lose ocd amount.

- Since oabd > ocd, there is a potential for bargain.

- The polluter could compensate and offer any amount more than ocd and less than oabd.

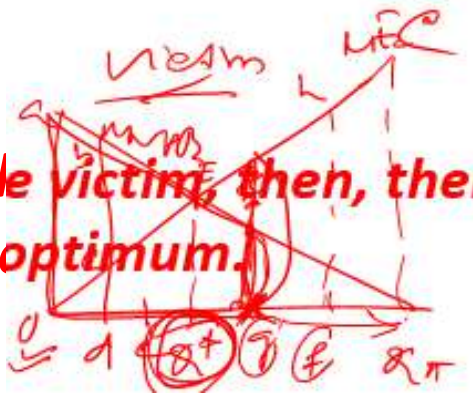
- Bargain amount > ocd, but < oabd.

- The polluter will still have a net profit and the victim will better off.

Contd…

- If such bargain works, the move to **d, would be seen as a Pareto improvement**
- **Like wise,** move to e and further move to Q* is also social improvement
- **Any move right to Q*** is not feasible.

- ***If we start at 0 and the property right is with the victim, then, there is a natural tendency to move to Q*, the social optimum.***



The Bargain: If The Polluter Has The Property Right



- The starting point is Q_π
- Now is there any possibility to bargain over reducing pollution?
- Now let's consider a move from $\boldsymbol{Q_\pi}$ back to '**f**'
- The victim can compensate the polluter to give up a certain amount of activity.
- The victim would have to tolerate a loss of **fhi**$\boldsymbol{Q_\pi}$, if the move to 'f' does not take place.
- To make this move happen the bargain amount need to be greater than fgQ_π, but less than fhiQ_π

Contd…

- If the move to ‘f’ is a social improvement, so is the move from ‘f’ to ‘g’ and to Q*, the social optimal output level.

- *So long as a bargain gets established between this two parties, the market will lead to social optimum output and pollution level.*

- *Regardless of who holds the property rights, there is an automatic tendency to approach the ‘social optimum’.*

Why The Coase Theorem May Not Work?

- Transaction Costs: costs of monitoring, enforcement, and negotiation
- Observability (uncertainty): must have full knowledge of benefit function.
- Lots of paper works associated with trade rights
- Political power of the polluters sometimes makes difficulty for ‘polluters pay principle’
- Victims may dislike to pay for reducing pollution.

Contd…

- When policymakers have only limited information regarding environmental costs or benefits, they may lack sufficient knowledge to design optimal, or first-best outcomes (i.e., a policy which ensures that MSB = MSC).
- Instead, they may be forced to develop a second-best policy.
- For example, as a second-best policy alternative, a regulator may decide **to minimize the cost of achieving some target level of pollution**. Such a target may be set by experts who might make educated guesses as to the optimal level while further research is being done (Baumol and Oats).

Suggested Readings

- Coase, R. H. (1960). The Problem of Social Cost. *The Journal of Law and Economics,* 3:1–44.

- Pearce, D. W. and Turner, R. K. (1990). Economics of Environment and Natural Resources. Harvester Wheatsheaf: New York. Chapter 5.

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Environmental Regulation and Basic Regulatory Instruments-I

Prof. S. P. Singh
Department of Humanities and Social Sciences
Indian Institute of Technology Roorkee



Contents

- Introduction
- Rationale for Regulation
- Public Interest Theory
- Interest Group Theory
- A Political Model of Regulation

Introduction

- Human beings are performing many activities for the purposes of their survival, economic and material growth, safety, and security.
- These anthropogenic activities generate waste/pollution.
- Limited capacity of environment to absorb wastes.
- When the wastes are produced beyond its capacity to absorb, it adversely impacts on both human beings and natural systems.
- This gives rise to the need for environmental regulation.

Rationale for Regulation

- Environmental regulation is a special and recent example of economic regulation.
- Economic regulation involves the govt. intervention in a variety of ways in the private actions of firms and individuals.
- Two basic theories of regulation: the public interest theory and the interest group theory
- The former views the purpose of regulation as the promotion of the public interest while the later views the purpose of regulation as promoting the narrow interests of particular groups in society such as an industry.

Public Interest Theory

- To a certain extent, the public interest theory of regulation is a normative theory.
- Three general reasons why regulation might exist:
  - Imperfect competition (Natural Monopoly; Market Power, Collusion & Mergers)
  - Imperfect information (information asymmetric)
  - Externalities
- These three factors, commonly associated with market economy, necessitate government regulations and controls.

Imperfect Competition

- Imperfect competition is the traditional normative justification for govt. regulation. In the case of natural monopoly (such as an electricity distribution firm), economic efficiency calls for a single firm to operate.
- The role of govt. is to guarantee a monopoly to a particular firm and control prices to protect consumers from monopoly pricing.
- Prevent undue concentration of power in markets (Competition Law).
- Prevent collusion and restrict mergers.

Imperfect Information

- The second major rationale for govt. regulation is imperfect information
- Acquiring information is costly and consumers may not always have complete information on items such as product quality.
- The costs of acquiring information may far exceed the benefits.
- This is a justification for the govt. to step in to compensate for incomplete information.
- Govt. can set liability rules to encourage the provision of safety-related quality. Such rules induce firms to provide an efficient level of safety-related quality.
- Quality Standards (ISI, Hall Mark, BIS, Star Ratings, etc.).

Externalities

- A third rationale for govt. regulation is in the area of the provision of public goods and bads.
- When there are elements of non- rivalry or non-excludability, private markets are inefficient.
- Govt. may step into provide goods or bads, effectively eliminating the private market, but mostly public goods and rarely public bads.
- In public bads, the approach is for govt. to define a set of institutions and regulations to govern the provision of these public bads
- Emission Standards

The Interest Group Theory

- According to this theory, regulation is required to protect the interest of specific groups or regions. E.G., in order to attract more pvt. investment in clean energy generation, govt can provide input subsidies and tax exemptions to promote investment in alternative sources of energy.
- 5% mix of ethanol in petrol in some states of India is another example.
- This theory maintains that rent seeking is the primary rationale for regulation.
- Groups that benefit from such regulation lobby govt. for regulations that provide them with rents that would not exist in a competitive market.

A Political Economy Model of Regulation

- The basic problem of environmental regulation involves the govt. trying to induce a polluter to take socially desirable actions, which may not be in the best interest of the polluter.

- The govt. faces a complex problem of determining exactly what level of pollution is best for society.

- In reality, the government faces pressures from consumers and polluters (lobbyists).

REFERENCES

- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. *Sustainable Human Development Review*, *105*.
- Kolstad, C. (2011). Intermediate Environmental Economics: International Edition. *OUP Catalogue*.

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Environmental Economics

Environmental Regulation and Basic Regulatory Instruments-II

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CONTENTS

- A political economy model of regulation
- Interaction among government, polluting firms and consumer citizens
- Basic regulatory instruments
- Sum up

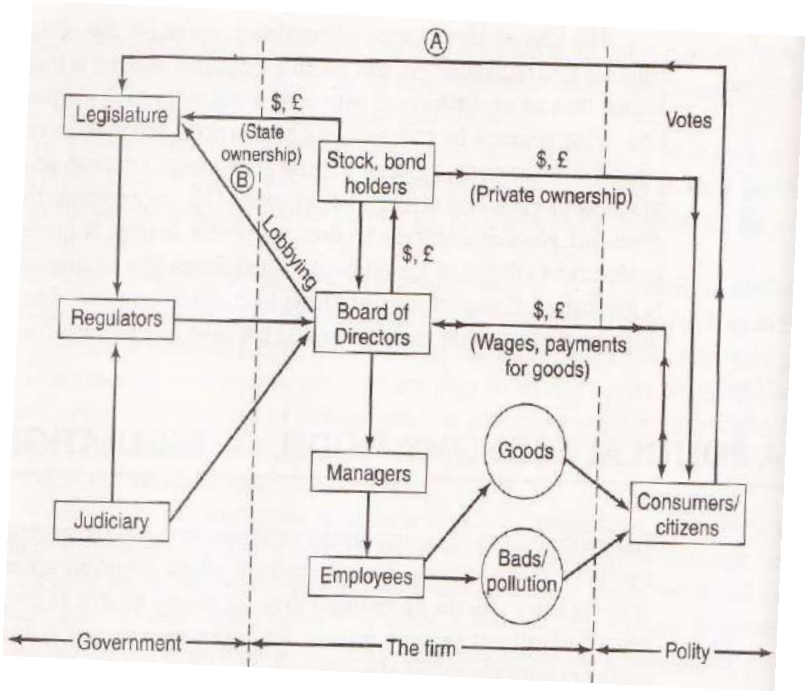
A Political Economy Model of Regulation

- The basic problem of environmental regulation involves the govt trying to induce a polluter to take socially desirable actions, which may not be in the best interest of the polluter.

- The govt. faces a complex problem of determining exactly what level of pollution is best for society.

- In reality, the government faces pressures from consumers and polluters (lobbyists).

Interactions among Govt., Polluting firms and Consumer Citizens



- The govt. consists of three branches.
- The legislature passes laws defining what the regulators are to do in controlling pollution.
- The regulators are charged with the detailed implementation of the legislature's laws.
- The regulator's actions are tempered by the judiciary.
- The legislature may have difficulty in achieving its goals since it does not directly influence the polluter .
- The regulator may have different goals. For instance, regulators may be interested in job security or the size of the bureaucracy they control as well as in reducing pollution levels.

Contd…

- There are two important lessons from the figure:
  - Imperfect links between the legislature and the pollution generating process. Since the legislature cannot physically control pollution directly, it must rely on indirect means to obtain its ends, and often these indirect means may be less than perfect.
  - Regulation may be excessively costly, may result in considerable cheating, may result in excessive time delays, and may result in too much pollution.
- The legislature is influenced only by citizen welfare (line A) but also by polluters (B).
- Normative theory and positive theory
- Endogenous politics (inclusion of B) and exogenous politics (omission of B).

Contd…

- Typically, the objective of the regulator is viewed as maximizing the sum of producer and consumer surplus.
- The positive interest group theory is consistent with the endogenous political model of regulation. With endogenous politics, we are conscious of the ability of interest groups to influence the regulators, legislators, and perhaps the judiciary.
- Environmental regulation is influenced by interest groups such as "browns." and "greens,".
- Need to understand how the browns and greens interact with the legislature and regulatory agencies.

Basic Regulatory Instruments

- Basic regulatory instruments can be classified into two categories:
  - Command and Control (CC) (Direct instruments)
  - Economic Incentives (Indirect Instruments)
- CC Approach is a direction by the government in the way in which production and consumption activities are carried out.
- The government enacts rules and regulations, institute enforcement authorities at different levels to implement them and places a system of fine and penalties for dealing with cases of violation of the rules and regulations.
- Economic Incentives (EIs): (i) subsidies, (ii) pollution taxes/fees, (iii) marketable permits, (iv) liabilities (or damages) and (v) refundable deposits.

Let's Sum Up

- Two theories for justifying the government regulation: Public Interest Theory and Interest Group Theory (Normative theory and positive theory)
- Political economy of regulation (Govt. is influenced by both "browns" and "greens" groups).
- Two basic regulatory instruments: C&C and EIs.

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- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. *Sustainable Human Development Review*, *105*.
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Environmental Economics

Command and Control Approach

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Indian Institute of Technology Roorkee



Contents

- Introduction
- What is Command & Control?
- Forms of C&C
- Key Features of C&C
- Advantages and Disadvantages of C&C
- Let’s Sum Up

Introduction

- C&C is a dominant form of environmental regulation in the world today
- The basic concept of C&C is for the regulator to specify the steps individual polluters must take to solve a pollution problem.
- The essence of C&C is that the regulator collects the information necessary to decide the physical actions to control pollution.
- The regulator then commands the polluter to take specific physical steps to control the pollution.
- Regulator directs the polluter what to do and what not to do.
- Govt. sets up institutional framework to implement C&C system.

What is Command & Control?

- Under the CAC approach, govt. enacts environmental laws and ordinances to:
  - (i) regulate the production and consumption of certain activities;
  - (ii) prompt people and business entities to bring about a behaviour considered socially desirable; and
  - (iii) ensure the compliance through enforcement mechanism (court, police, fines, etc.).
- The CAC approach sets specific limits for pollution emissions and/or specific pollution-control technologies that must be used.
- For instance, setting emission standard means legally allowing polluters a certain level of emission beyond which it becomes punishable by law.

Contd….

- Under the C&C, the govt. directly intervenes in the activities of polluters to make them perform in a standard manner.
- It comprises both laws that specify allowable quantities of pollution and laws that detail which pollution-control technologies must be used.
- It directs the business entities to install specific pollution treatment equipment or technology.
- The regulator/authority prescribes how much pollution a source is allowed to emit.
- It also specifies steps and standards and monitors/implements so that individual polluters must take action to generate pollution below the prescribed limit.

Forms of C&C

- One way of conveying the nature of C&C is by example :the Clean Air Act in the US requires the regulator to determine the minimum pollution control "performance" of new sources of pollution.
- The regulator is required to specify for each new category of source (e.g., new power plants or new tyre factories) what pollution controls and emission rates are deemed acceptable**.**
- In tyre manufacture, the regulator may hire engineers to examine the process of tyre manufacture, generating a "Control Technology Guideline".
- Power plants (producing electricity) may be required to use certain technologies to reduce emissions of sulfur dioxide (Technology standards).
- Specific pollution control equipment requirements can be specified.

Contd…

- Alternatively, the regulation may specify an emission limit for particular types of plants and particular pollutants (emission standards).
- Every new car may emit no more than x grams of carbon monoxide per mile driven. Furthermore, all new cars are required to have a specifically defined system for capturing vapor that might escape from the petrol tank during refueling.
- In the case of power plants, fuel quality may be limited (e.g., sulfur content cannot exceed 1%) or emissions per unit of fuel use may be limited.
- C&C may be combined with significant fines and penalties associated with non-compliance.

Key Features of C&C Approach

- Restricted choice for the polluter.
- No mechanism for equalizing marginal control costs among multiple polluters.
- Preferred over other instruments when the ‘marginal abatement costs’ (MACs) of pollution treatment are uniform across all firms in the regulated industry.
- However, due to asymmetric information, it could be difficult for a regulator to get the knowledge of such costs. Further, a firm may manipulate information strategically.
- One thing that characterizes all C&C regulation is a centralization of some of the pollution control decisions that could be made by the polluter.

Advantages of C&C

- More flexibility in regulating complex environmental processes.
  - In an urban area with factories at different locations contributing differently to overall levels of urban pollution, it can be difficult to fashion a workable set of emission taxes or other incentives to ensure a certain level of pollution.
- Greater certainty in how much pollution control results from regulations
  - Better than economic incentives in an atmosphere of uncertainty.
- Easy monitoring of compliance.
  - For instance, if a CAC regulation states that a particular piece of pollution control equipment must be used, then the task of regulator is to see whether it is installed or not.

Disadvantages of C&C

- High cost of information collection and possibility of getting distorted information from polluters.
- No incentive for R&D
- Difficulty in applying the equi-marginal principle.
- CAC measures are considered inflexible since it applies the same standard for all polluters often with the same pollution-control technology.
- Since the CAC regulations are framed by the legislators, they are subject to compromises in the political process due to influence of lobbyists.
- The polluter pays only pollution control, not residual damage from the pollution (paper manufacture from virgin wood/recycled paper).
- Difficult to implement effectively in the changing level of technology.

Let's Sum Up

- Two approaches of Regulation: C&C and EIs
- Direct interventions and indirect interventions through incentives or disincentives
- Key features of C&C : restricted choice for the polluter in deciding how to reduce pollution and lack of equating MAC to MAB; Asymmetric information and costly affair
- Advantages: More flexible and certain in its outcome; easy monitoring compliance
- Disadvantages: high implementation cost; lack of incentive for R&D; application of equi-marginal principles not possible

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- Field, B. C., & Field, M. K. (1997). Environmental economics: an introduction. *Sustainable Human Development Review*, *105*.
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Environmental Economics

Command and Control Approach: Types of Standards-I

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Contents

- Introduction
- Types of Environmental Standards
  - Ambient Standards
  - Emission standards
  - Technology Standards

Introduction

- Command and control policy approach of environmental regulation includes the setting up of environmental standards for different activities, prescribing rules and regulations for the production process and ensuring their enforcement for controlling the generation of pollution by fixing an acceptable limit of the pollution level, etc.
- As an instrument to regulate the polluters' activities, the CAC approach uses three types of Environmental standards. These are:
  - Ambient Standards
  - Emission Standards
  - Technology Standards.

Types of Environmental Standards

- In general, a standard is simply a mandated level of environmental performance that is enforced in law.
- It is defined as a policy guideline to regulate the effect of anthropogenic activities on the environment. E.G. level of BOD and COD in river water or ‘total dissolved solids’ (TDS) level in water (500 mg/l to 2000 mg per litre).
- Both formulation and enforcement of the standards are important.
- Some standards may only be in rulebook due to lack of political will to effectively execute them.
- Three Standards: ambient, emission and technology

Ambient Standards

- An *ambient standard* denotes maximum allowable level of a pollutant in the receiving medium (air, water, or soil).
- It is a ceiling on the ‘never-exceed level’. It could be the ambient quality of air over a particular city or the ambient quality of water in a particular river.
- Ambient standards refer to the quality of the surrounding environment and are expressed in terms of average concentration levels of pollutants.
- Setting ambient standards require an explicit agreement on the desired environmental quality objectives and the cost that the society is willing to bear to meet the same.
  - For example CPCB notifies ‘ambient air quality standards’ to regulate the quantity of 12 major pollutants such as sulphur dioxide, nitrogen dioxide, particulate matter, ozone, lead, carbon monoxide, etc. in the air).

Contd…

- It sets different limits of these pollutants in two categories:
  - ‘Industrial, residential, rural and other areas’
  - ‘Ecologically sensitive area’.
- Ambient standards can be set at different levels for different locations to the protect ecosystems.
- They cannot, however, be enforced directly. What can be enforced are the various emissions that lead to ambient quality levels.
- For instance, to ensure that dissolved oxygen never falls below a certain level in the river, we must know how the emissions of the various sources on the river contribute to changes in this measure and then introduce some means of controlling these sources.

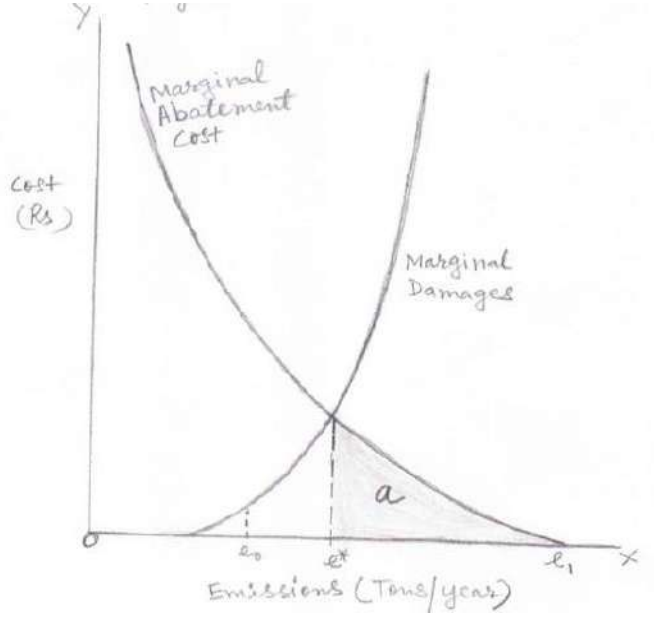
Emission Standards

- An emission standard is a max. rate of emission legally allowed by a source.
- Also known as ‘performance standards’, they may be established in terms of what can be achieved with available technology by estimating the discharges that are compatible with the ambient standards defined for the pollutant.
- For this, considerable information on both the pollution sources and the ambient environment are required.
- In India, the CPCB prescribes the ceiling on quantities of emissions for pollution sources. For instance, Bharat Stage Emission Standards are the standards set for motor vehicles in India.
  - Supreme Court ordered to ban the sale and registration of motor vehicles not complying with the Bharat Stage IV emission standards with effect from April 1, 2020.

Contd…

- *An 'emission standard'* differs from an 'ambient standard' in the sense that its use does not determine the ambient level of a pollutant in the environment. It rather attempts to reduce the overall amount of a pollutant released on a firm-by-firm basis.
- Setting emission standards does not necessarily mean meeting ambient standards.
- For instance, emission standard could be imposed on firms but if there is no control on the number of polluting firms established, aggregate environment quality (ambient standards) cannot be curtailed.

Emission Standard: MAC Versus MDC



- Regulator is assumed to have full information on MAC and MDC relating to the rate at which some production residual is emitted into the environment.
- No regulation: Max. level of emission = e_1
- Emission e_1 > the efficient rate e^*.
- To achieve e^*, the regulator sets an emission standard at e^* level (socially desirable level)
- At e_0, MAC > MDC, would prefer to pay fine
- If the firm reduces emissions in accordance with this standard, it will incur an amount equivalent to the shaded area '*a*' per year as its abatement cost.
- The MAC is the compliance cost of meeting the standard.

Key Concerns related to Emission Standards

- Difficulty in setting the socially desirable standards as incomplete information on MAC and MDC (Trail and error method and competing voices of stakeholders).
- Flexibility in changes in the standards (frequent change may affect the investment decisions).
- Public reactions
- Large bureaucracy to implement and monitor and high transection cost.
- Attaining economic efficiency (typically applied *uniformly* across emission sources).

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NPTEL ONLINE CERTIFICATION COURSE

Environmental Economics

Command and Control Approach: Types of Standards-II

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Department of Humanities and Social Sciences
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CONTENTS

- Technology standards
- Equi-marginal principle and C&C approach
- Economics of enforcement of C&C approach
- Sum up

Technology Standards

- Under the C&C, the regulator can direct the polluting firms to use specific technologies/techniques/practice/design/ process for achieving the environment targets and then periodically monitor the compliance.
- Known as ‘technology standards’, these are based on the knowledge of what can be achieved with current equipment and practices.
- These standards would compel the polluting firm to use a particular pollution control technology (such as installing scrubbers on smokestacks).
- Easy monitoring as the inspector could simply observe whether or not the equipment has been installed.

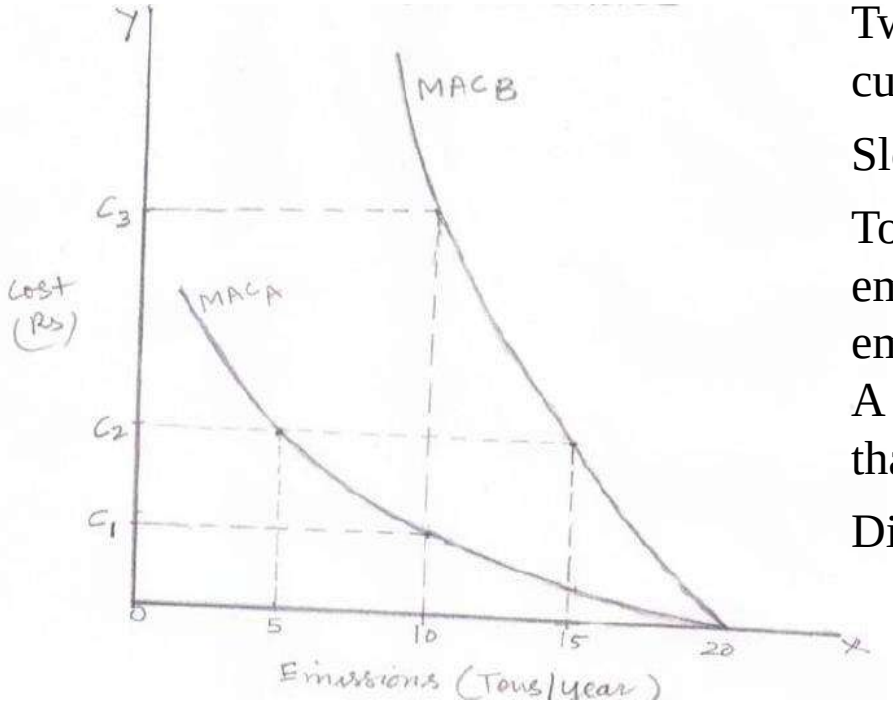
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- In technology standard, no scope for any other cost-effective measure to achieve the same environmental target. It does not provide opportunity to the polluter to explore some cheaper methods/ processes/ technologies to attain the same level of emission.
- In case of emission standards, it stipulates the maximum emissions allowed per unit of economic activity to control pollution allowing polluters to choose among available alternatives to obey the prescribed standard. (flexibility in choosing the best mechanism among the available alternatives).
- No such discretion for the industries under technology standards.
- Emission standards can be combined with technology standards

Equi-Marginal Principle and C&C Approach

- The MAC of pollution varies across firms within the same industry.
- The equi-marginal principle states that in order to get the optimum level of emissions, different sources of emissions must be controlled in such a way that they have the same MAC.
- This implies that different sources of a pollutant need to be controlled by different degrees, depending on the shape of the marginal abatement cost curve at each source.
- Generally, therefore, standards are designed to be applied uniformly across all polluting firms within the industry.
- Greater the differences in MAC among sources, the lower will be the performance of the equal-standards approach.

Contd….



Two firms A and B with different MAC curves

Slope of MAC for A < B.

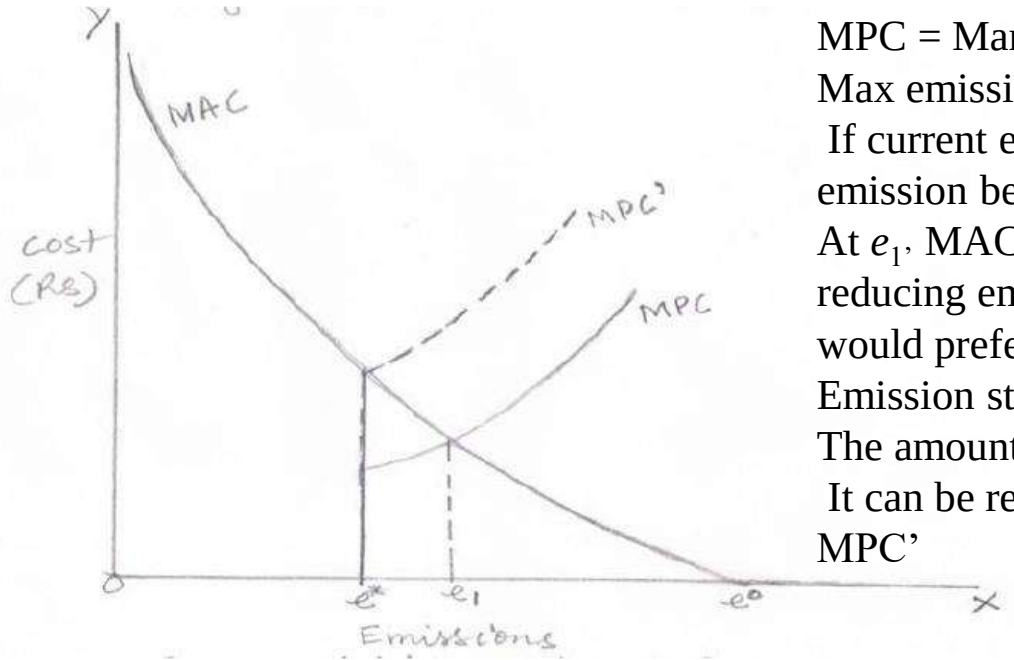
To get an overall reduction of 20 tons of emissions per month, the regulator can set emission reduction target of 5 tons/month for A and 15 tons/month for B as the MAC of A = that of B.

Difficult to do for the regulator.

Economics of Enforcement of the C&C Approach

- There are two primary dimensions of enforcement: monitoring and sanctioning.
- The ‘marginal abatement cost’ (MAC) is the cost of one additional unit of emission abated, while ‘marginal penalty cost’ (MPC) represents the expected penalties that the firms face for violating an emission standard.
- MPC arises when the firms are detected to be exceeding their emission standard and penalties are levied as a result.
- MPC will be zero if emission is less than the set standard as the firm complies the standard.
- The MPC would increase as the size of the violation increases.

Economics of Enforcement



MAC= Marginal Abatement cost
MPC = Marginal Penalty
Max emission = e^o
If current emission is e^o, the firm will reduce the emission because MAC< MPC.
At e_1, MAC = MPC; if But the firms will stop reducing emissions if it is less than e_1 the firm would prefer to pay penalty.
Emission standards = e^*
The amount of non-compliance = e^*e_1.
It can be reduced to zero by raising the MPC to MPC'

Contd…

- The only way to reduce noncompliance is to raise the penalty function by raising monitoring activities so as to be better able to detect non-compliance, or raising fines for those who have been detected as non-compliant.
- This analysis shows several things:
  - There is a trade-off in enforcement; to get higher levels of compliance authorities will normally have to devote more resources to enforcement.
  - There may be some trade-off also between monitoring and fines. The MPC curve can be raised or lowered by changes in either one of these enforcement elements.

Let's Sum Up

- The C&C basically involves setting three types of standards viz. ambient, emission and technology standards.
- An *'ambient standard'* sets the maximum allowable levels of a pollutant in the receiving medium (i.e. air, water, or soil).
- An emission standard is a maximum rate of emission that is legally allowed by a plant or a source.
- In contrast to emission standards, technology standards impose on polluters certain decisions on technologies to be used. This is therefore a form of "technology forcing' on polluting industries to adopt technological change in order to meet the environment standards.
- The major drawback of CAC approach is that it does not provide opportunity to the polluters for innovation.
- Market based Approach (EIs) is considered more effective than C&C System.

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ENVIRONMNETAL REGULATIONS AND BASIC REGULATORY INSTRUMENTS-
MARKET-BASED INSTRUMENTS/APPROACHES-I

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



CONTENTS

- What is market-based instrument?
- Types of market-based instruments (MBIs)
  - Pollution tax
  - Subsidies
  - Marketable permits
  - Liabilities
  - Refundable deposits
- Efficiency of MBIs in controlling pollution

Basic Regulatory Instruments

- Two broad basic regulatory instruments to control pollution
  - Prescriptive regulatory instruments/Command and control (CAC), and
  - Market-based instruments (economic incentives): are indirect instruments which provide rewards polluters to do what is perceived to be in public interest.
  - MBIs provide flexibilities and incentives to the polluters in using alternative technologies and production practices to reduce the given level of emission and thus they change the behavior of polluters towards more efficient use of resources.

Market-based Instruments and their Economic Incentives

- Market-based instruments are said to offer incentives to private firms in choosing,
  - cost-effective pollution control technologies, mechanisms and strategies by expanding investment on R&D, which is largely not found in CAC mechanism of controlling pollution.
- In the case of pollution fees- the firms are charged for each unit of emissions but are allowed to emit as they desire.
- Since generating additional units of emissions implies more cost to a firm in terms of pollution fees; it would try to reduce its emissions by using all forms of information regarding cost controls of alternative technologies.

Pollution Fees/Taxes

- Pollution fees are the payment paid per unit of pollution emitted by the polluter to a regulatory body.
- Theoretically, the concept of pricing pollution dates back to Arthur C. Pigou in 1920.
- However, the practical application of taxing pollution in terms of pollution fees has been developed by the environmental economist Alan Kneese in 1962.
- The genesis of the pollution fees lies in the fact that firms pollute because they are not accountable for the social damage they create.

Two Mechanisms of Pollution Fees

- The authorities may control the pollution by charging price for emissions by two ways:
  - per unit emission charge, and
  - subsidizing each unit of emissions that the firm reduces.

Per Unit Emission Charge

- Let's take an example of a power generating firm which produces electricity and generates pollution (x) as well.

- If the emission fee is 'p' and the polluter emits 'x' units of pollution, the amount of payment to be paid by the polluter to the regulator is 'px'.

- Now, the question is 'how much the polluter would emit?'

Contd…

- There are two components of pollution cost:
  - cost of pollution abatement, and
  - cost of pollution fees imposed by the regulator per unit of pollution emitted.

- Thus, total cost of pollution incurred by a firm = pollution abatement cost+ costs of the pollution fee:

$$Tc(x) = C(x) + px \qquad (1)$$

Contd…

- To minimise total costs, we need to take marginal conditions by taking into account the first order derivative of Tc(x) =

$$\frac{d[Tc(x)]}{dx} = 0$$

$$= \frac{d[C(x)]}{dx} + p = 0$$

$$= MAC + p = 0$$

$$= p = -MAC$$

$$= p = -MC$$

Some Concepts

- Marginal abatement costs:
- When pollution (x) increases, cost incurred by the firm decreases.
- This is because the firm does not have to reduce pollution by incurring expenditure.
- Therefore, when pollution is increased, with no intervention, cost to firm is decreased.
- Because of this rationality, when pollution is increased, MC or MAC is negative.

Contd…

- Marginal Savings:
- When pollution increases, and firm does not incur costs to reduce pollution level, then firm will save its expenditure by not spending on reducing the pollution level.
- This is known as savings from generating pollution in the absence of pollution abatement, and
- Savings from emitting one more unit of pollution is known as marginal saving (MS).

Features of MS and MC

- MS function:
- When emission fee is zero (price for emission is zero) that means there is no intervention in terms of pollution regulation, the level of emissions would be the highest and vice-versa.
- This very feature of MS function tells us that the MS function is having a negative slope, when we measure the level of emission in the horizontal axis and price per emission in the vertical axis.

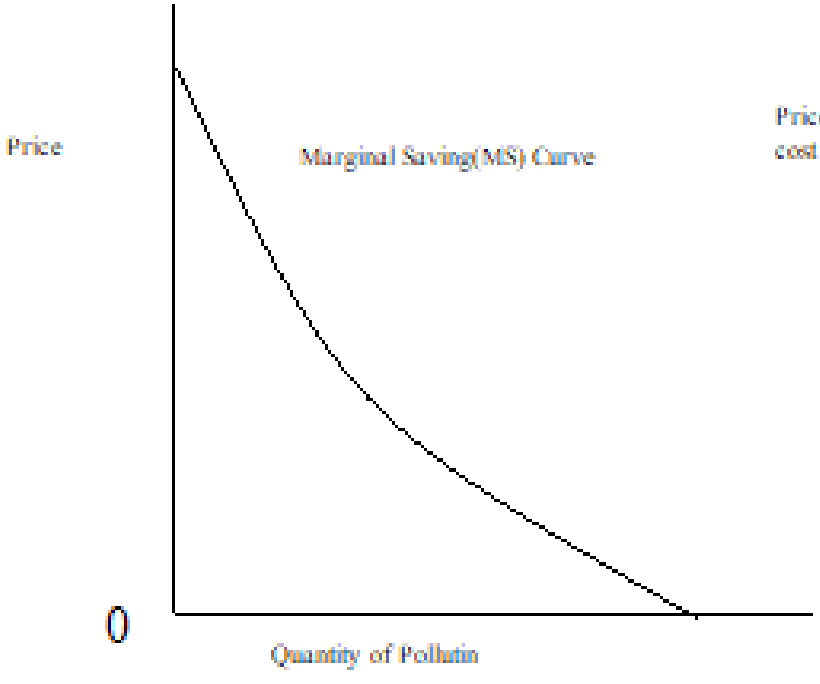
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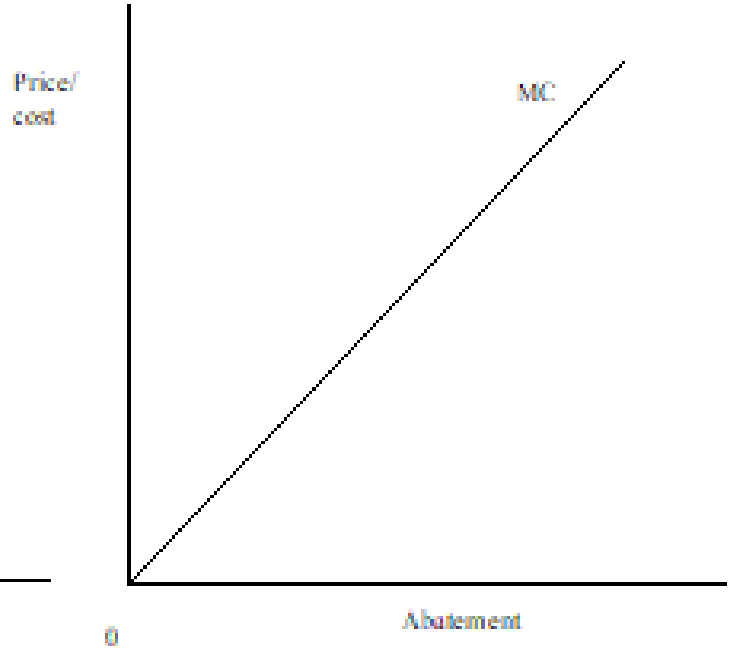
- MC function:
- MC function is increasing when the amount of abatement is increasing or pollution level is decreasing.
- It will be having a positive slope from zero origin, when the amount of abatement is measured on the horizontal axis and price or unit cost of abatement on the vertical axis.
- Therefore, marginal saving is equal to the negative of marginal costs

Contd…

- Marginal Savings:
- When pollution increases, and firm does not incur costs to reduce pollution level.
- This means the firm will save its expenditure by not spending on reducing the pollution level.
- This is known as savings from generating pollution in the absence of pollution abatement and
- Savings from emitting one more unit of pollution is known as marginal saving (MS).
- $P = - MC(x) = MS(x)$

MS and MC function





Where Does the Firm Operate?

- With an emission fee, firms will abate pollution up to the point where MC of abatement or Marginal Abatement Cost (MAC) is equal to MS from emitting and also equal to the emission fee. And, this is the point where the firm's total costs are the lowest.

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MARKET-BASED INSTRUMENTS/APPROACHES-II

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Contents

- Pigouvian fee a special kind of emission fee
- Pigouvian fee with a single polluter
- Pigouvian fee with more than one polluters

Pigouvian Fee, A Special Pollution Fee

- A Pigovian fee is a special kind of emissions fee which is set at the marginal damage of pollution.

- It is an emission fee exactly equal to the aggregate marginal damage when evaluated at the efficient level of pollution.

Pigouvian Fee: A Single Polluter Case

- We know that MS (X) = - MC (X).

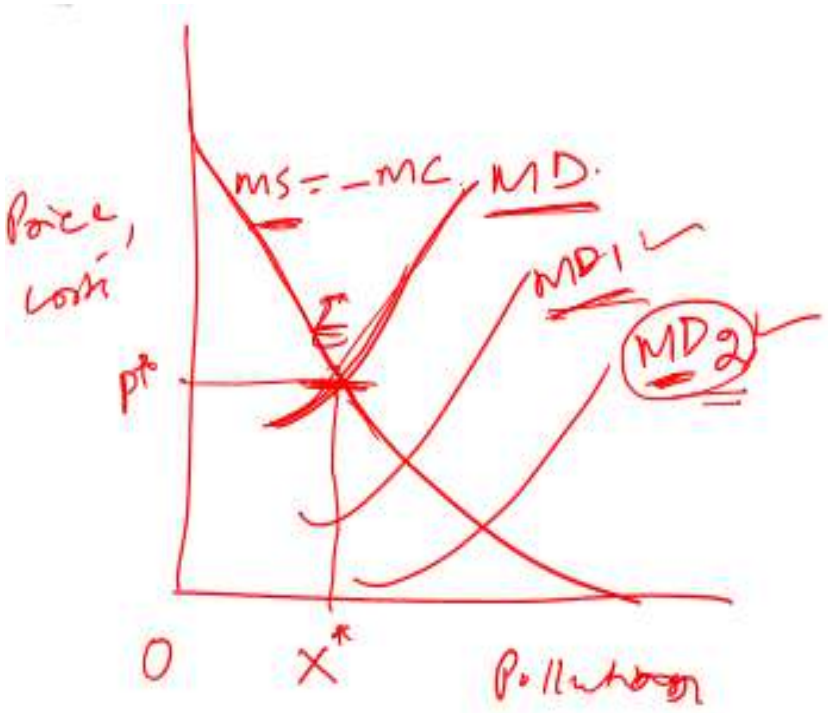
- Given the total damage costs,

TC (X) = Abatement cost + damage cost

= C(x) + D(x)

Pigouvian Fee with A Single Polluter

- MD1 is the marginal damage to the first victim and MD2 is the marginal damage cost to the second victim.
- As pollution damage is a social bad, the aggregate damage to society (MD) is obtained from vertical summation of individual damage costs.
- The optimal amount of pollution is reached by the intersection point of MD and MS curve at point E.



Contd…

- At efficient/optimal level of pollution, the firm will minimise total costs and damages.
- That means at x* optimal level of pollution, MC(x*) + MD(x*) = 0
- The optimal amount of pollution is at x* , where MD(x) = = -MC(x) = MS(x)
- The pigouvian fee is p*.
- If the polluter is charged p* per unit of pollution, MC(x*) = -p* or MS(x*) =p*
- Therefore, Pigouvian fee is defined as MS from pollution at the optimal level of pollution.

Contd…

- If we are not at the optimum level of pollution, Pigouvian fee will be neither the current MC of pollution control nor the MD from the pollution.

- Therefore, the Pigouvian fee is not any emission fee, rather it is the marginal savings from pollution at the optimal pollution level.

- We can prove this statement by taking the marginal benefit and marginal cost concepts.

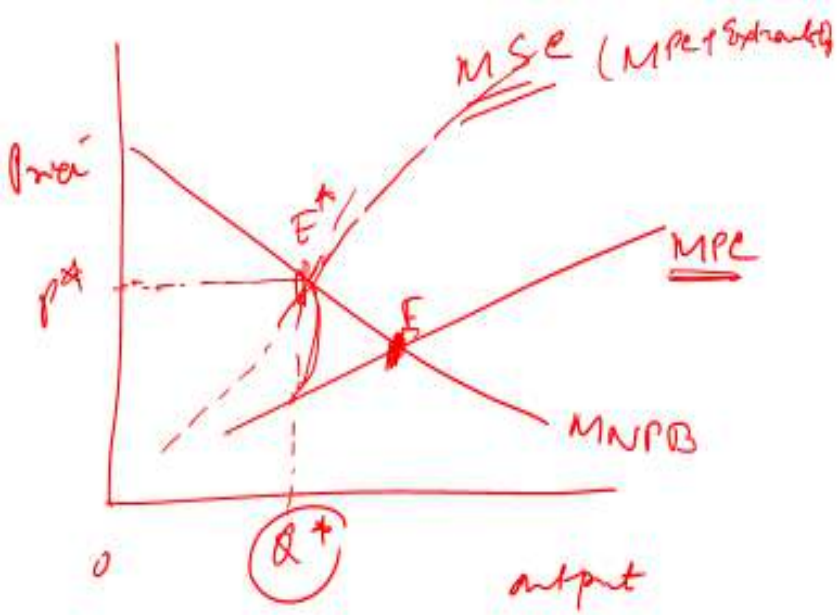
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- Marginal benefits can be taken as the demand curve and marginal costs as the supply curve,
- The intersection point is said to give us the efficient level of production and corresponding market price as well.

- In case of a negative externality like pollution, marginal social cost (MSC) is higher than the marginal private benefit (MPB) and a tax amounting to externality is imposed in order to make MSC equivalent to marginal private cost (MPC) or to compensate the MSC and be equivalent to (MPC).

Pigouvian Tax

- Without any kind of intervention, the firm would like to produce at equilibrium point E, where marginal benefit (MPB) is equating with the private marginal cost curve (PMC).
- However, this is not the socially efficient quantity of production as the firm generates negative externality in terms of pollution which is not reflected in the cost of production of the firm.
- Socially efficient quantity is produced at Q* corresponding to the equilibrium point E*.

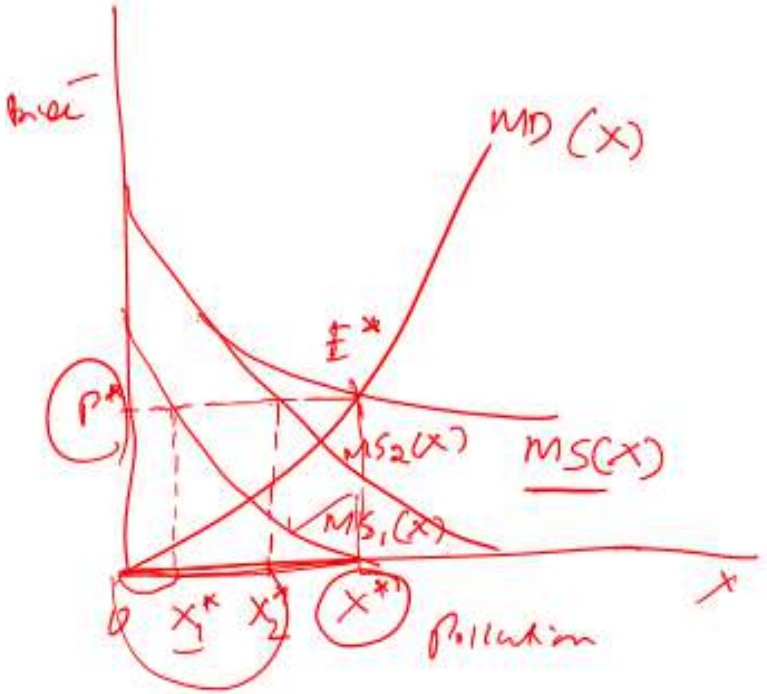


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- Now let's see what would be the level of Pigouvian tax for controlling emission.
- The amount of tax which is equal to marginal external cost (MEC) is decided by the difference between the MSC and PMC at the point of socially efficient production only.
- This means the tax is not equal to the MEC at other quantities except at the level of socially efficient quantity.
- Although a tax is introduced, but still it is not compensating the externalities at other quantities of production except Q*
- Therefore, if the firm produces beyond Q* level, there will still be some externalities which are not taxed per unit of pollution.
- Therefore, Pigouvian fee is not an emission fee, rather a special case of pollution tax.

Pigouvian Fee with Multiple Polluters

- In case of multiple polluters, equi-marginal cost conditions are sought, where MS (x) tells us how much x in total will be emitted and each MS_i (x) tells us how much each firm will contribute to that total.
- Since MS is private good for polluting firms, total MS is found by horizontal summation of individual MS curves.
- The efficient amount of pollution x* is determined by intersection point of MS and MD. This also determines the MS to polluters P* at x* and P* is the correct Pigovian fee.



- At this fee level, firm 1 will generate x1* and firm 2 will generate x2* of emission.

Contd…

- Each firm operates so that MS from polluting is set equal to the Pigouvian fee= $MS_1(X_1^*) = MS_2(X_2^*) = P^*$ and $MS(X^*) = P^*$.

- The benefit of Pigouvian fee is that all firms will control pollution at the same level of MCs following equi-marginal principles of pollution control.

Suggested Readings

- Field, B.C. & Field, M.K. (2017). *Environmental Economics: An Introduction*, McGraw Hills, USA.

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MARKET-BASED INSTRUMENTS/APPROACHES-III

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CONTENTS

- What is a subsidy?
- Subsidies as an incentive in controlling pollution
- Mechanisms
- Example
- Advantages
- Limitations

What Is A Subsidy?

- Subsidy is an incentive which may be in form of grants, cash or tax rebate provided by a state authority/public body/regulating body to help an industry or business keep the price of a commodity/service low and improve the supply of that commodity/service.

- It can also be used as an instrument to reduce pollution.

- For example, if it is given on green and clean sources of energy, it would provide an incentive to the producers to change their mode of production by shifting to green technology and hence pollution gets reduced.

Mechanisms of Subsidy

- The mechanism of pollution fee is based on placing a price on environmental assets where pollution is emitted.
- But in case of subsidy, the regulating authority would pay a polluter, an amount for every tons of emissions it reduced from a benchmark level.
- Thus, subsidy acts as a reward for reducing emissions.
- Moreover, it can be interpreted as an opportunity cost as when the polluter continues to generate pollution, it forgoes the subsidy payment which can be claimed.

Contd…

- Emission reduction incentive as positive externality: It is very likely that the market may produce less than the socially efficient level of production.
- But provision of subsidy would restore socially efficient level of production Subsidies may be in the form of
  - direct cash payment, tax rebate or in the form of subsidised inputs.

- Whatever may be the mechanisms of subsidies, each of them provides incentive to firms to cut down their pollution level or to expand green production.

Example-I

- Abatement subsidies:
- Here a regulator pays a polluters per tons of emissions it reduced from a benchmark level.
- Let's assume that the firm's base level of emission is set at its current emission rate , i.e., 10 tons/month.
- It is paid rupees 120/ton for every ton of emission it abates from this base level.
- How the subsidy is going to reduce pollution??

Abatement Subsidy – An Example

Emissions (tons/month)	Marginal Abatement Cost	Total Abatement Cost	Total Subsidy at $120/Ton	Total Subsidy Minus Total Abatement Costs
10	0	0	0	0
9	15	15	120	105
8	30	45	240	195
7	50	95	360	265
6	70	165	480	315
5	95	260	600	340
4	120	375	720	345
3	150	525	840	315
2	185	710	960	250
1	230	940	1,080	140
0	290	1,230	1,200	–30

(Field and Field, 2017, p.241)

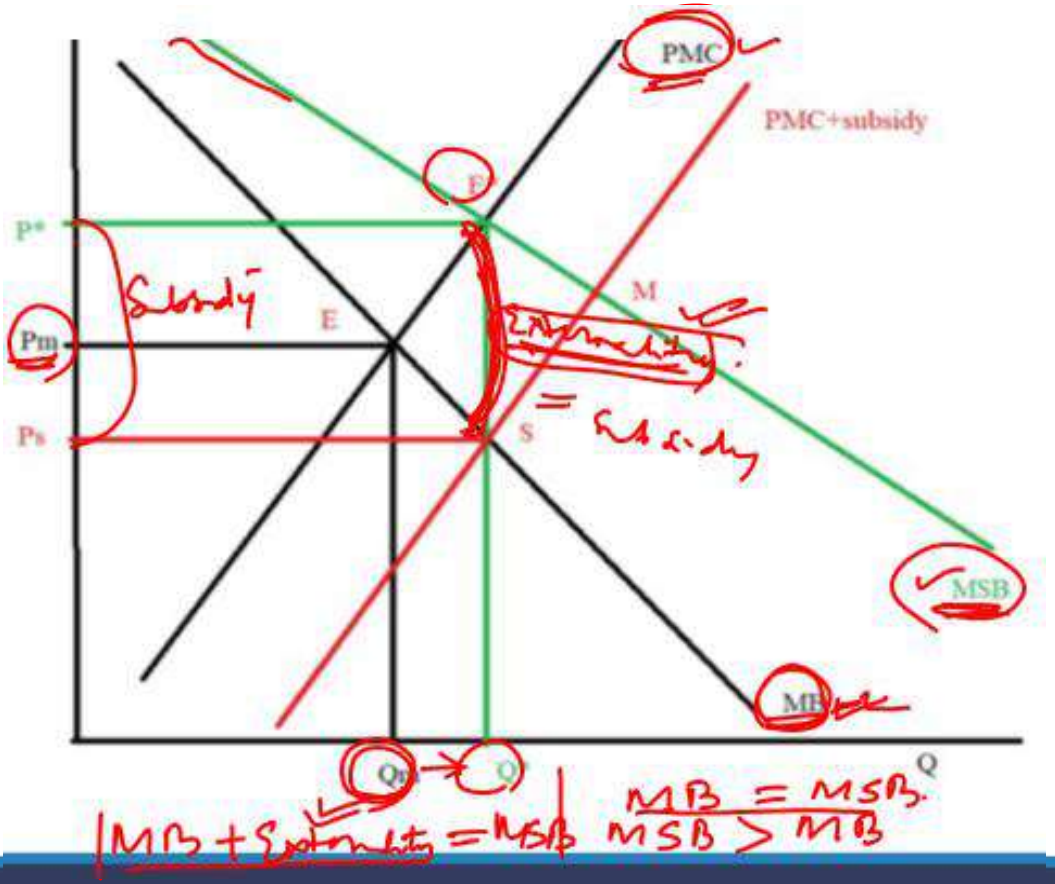
Example-II

- Lets' take the case of renewable energy producer like wind power as competitor to traditional coal firms to produce energy.
- Subsidy may make wind power more competitive to develop a market.
- The case of subsidies in emission reduction can be explained graphically.

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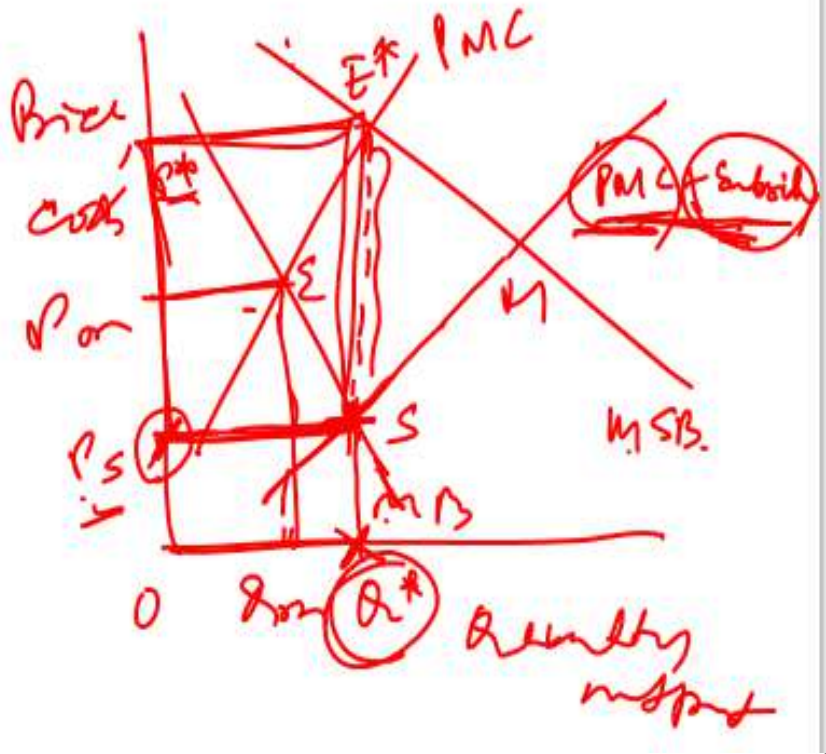
- Given the marginal cost and private marginal benefit, the firm will produce Qm level of output at a price Pm.
- This level of production is much below the socially efficient level of production which is established at Q*, by the intersection point of MSB and PMC.

- Now both the quantity produced as well as price charged per unit of quantity is high in comparison to competitive market.



Contd…

- Positive externality related to this production is E*S,
- It is regarded as subsidy that needs to be provided to the renewable energy producing firms.
- The total amount of subsidy would be $P_sSE^*P^*$.
- Because of the subsidy the supply curve of the firm shifts rightward, thereby maintaining the socially efficient level of output at a low price P_s per unit.



Advantages

- Subsidies enable greater social efficiency.
- Consumers pay the socially efficient price.
- It may help to change preferences of consumers to use energy generated from wind power.
- More output is produced and consumers would be encouraged to stick to renewable form of energy production.
- Subsidies can also be used as tax credits which help in reducing the tax liability of a firm.

Limitations

- Subsidies may be misused which may end up wasting public money with insignificant reduction of emissions.
- The cost of subsidies are paid through taxation.
- Although, emission per firm may be reduced under subsidy option, but the total emissions may increase as the industry may set up more number of firms to obtain net revenue rather than incurring cost. This is a major limitation of the per unit type of subsidy.
- Subsidies may encourage firms to be inefficient as they would depend more on government subsidies to expand production and charge less price to be competent, instead of improving their efficiency by R & D activities.

Contd…

- Shortage of supply: although we explained that subsidies increase the supply of commodity/service; it may also cause shortage in supply. Because lower prices may lead to sudden increase in demand which many firms struggle to match.

- It is difficult to quantify the success of subsidy policy.

Suggested Readings

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MARKET-BASED INSTRUMENTS/APPROACHES-IV

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Deposit-Refund System (DRS)

- Robert Solow (1971) and Edwin S. Mills (1972) separately advanced the idea of a materials- use fee to be levied.
- This fee was seen to be equivalent to refundable deposit to encourage recycling.
- This is the combination of a tax (deposit) and subsidy (refund).
- This deposit-refund systems work well where a product is widely dispersed when it is purchased and used, but it is difficult to monitor, detect and observe environmental damage for its disposal.
- The very objective of the subsidy is to incentivise people to refrain from disposing of those items in environmentally damaging manner.

Contd…

- Example: a subsidy is paid to consumers when they return and item to a designated collection point.

- In case of some sectors, it may be difficult as well as costly for the regulator to monitor the activities of polluting companies-, i.e., dumping of waste in to the deep sea by the shipping companies.

- In this case, potential offenders should be required to leave an appropriate deposit with the environmental authorities.

Contd…

- DRS can also be used to encourage the consumers to return back the used containers, bottles, packaging, products or their residues.

- It requires paying a deposit on the purchase of potentially polluting products, which is refunded when the products or their residues are returned for recycling or disposal.

- It aims to give a financial incentive for consumers to return the product or the wastes back to retailers or producers for reuse, recycling or disposal.

Some Examples

- A number of states in the US have enacted DRS for beverage containers to reduce litter and encourage recycling.
- Same is also practiced in Europe.
- Germany also has DRS on waste lubricating oil. Each year large quantities of waste oil are disposed improperly threatening the very components of the environment.
- New lubricating oil face tax and is channelized into a fund.
- This fund is then used to subsidize a waste oil recovery and reprocessing system.

Contd…

- DRS have been instituted for cars in Sweden and Norway.
- New car buyers pay deposit at the time of purchase and can refund if these cars after use are sent to an authorised junk dealer.
- It can also be applicable for consumer products having hazardous substance like mobile batteries , laptops, electronic gadgets, etc.

Contd…

- How to generate this refund?
- It can be voluntary or mandated by legislation.
- The funds which are used for paying subsidy are raised by levying taxes on those items when they are purchased.
- Here the very purpose of this tax (deposit) is to generate money to pay the subsidy.
- It does not target to reduce the consumption of those items.
- For example: deposit-refund systems for beverage containers in the US to reduce litter and encourage recycling.

Contd..

- If the deposit return is higher than the cost of deposit this can increase the incentive to return the item and reduce consumer resistance to the scheme where there is a long period between paying the deposit and receiving the refund.

Suggested Readings

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ENVIRONMNETAL REGULATIONS AND BASIC REGULATORY INSTRUMENTS- MARKET TRADING SYSTEMS-I

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- Emission Trading-
- What is Tradable Emission Permit?
- How is initial permit decided?
  - Auction pollution permit
  - Free distribution
    - Grand parenting
    - Benchmarking
- Types of trading systems
  - offset trading, emission rate trading, cap and trade

What Is Emission Trading?

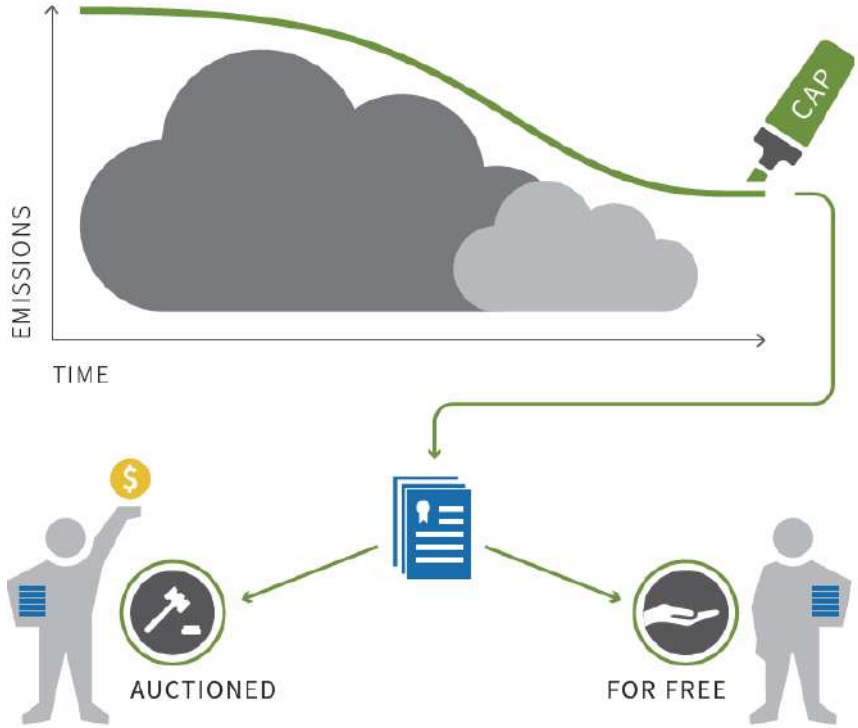
- Emission trading is a mechanism where the government sets an overall limit of emissions and defines permits/allowances to emit up to the overall limit.

Tradable Permit

- A **pollution permit** offers firms a legal right to pollute a certain amount.
- **A tradable *permit*** allows polluter to buy and sell the right to pollute and pollution control is expressed by holding the number of emission permits.
- If a firm has to emit pollution, it has to purchase permits to pollute. The price of the permits can be represented as a tax on emission.
- Tradable permits are a hybrid of quota and Pigouvian tax system because holding the number of permits is similar to quota system and buying permits acts as a tax system on pollution.
- If a firm has to pollute more than the initial permit, it has to purchase more permit and has to pay for polluting more.

How Is The Initial Permit Decided?

- Two basic mechanisms for distribution of permits, viz.,.
  - auctioned pollution permits
  - freely distributed transferrable permits.



ICAP (2017), ETS brief 6, May

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- Under auctioned pollution permits, the government auctions off a fixed number of ex-ante rights to emit a unit of pollution.

- This method is considered as a straight forward and efficient means of getting permits to companies/firms who value them the highest.

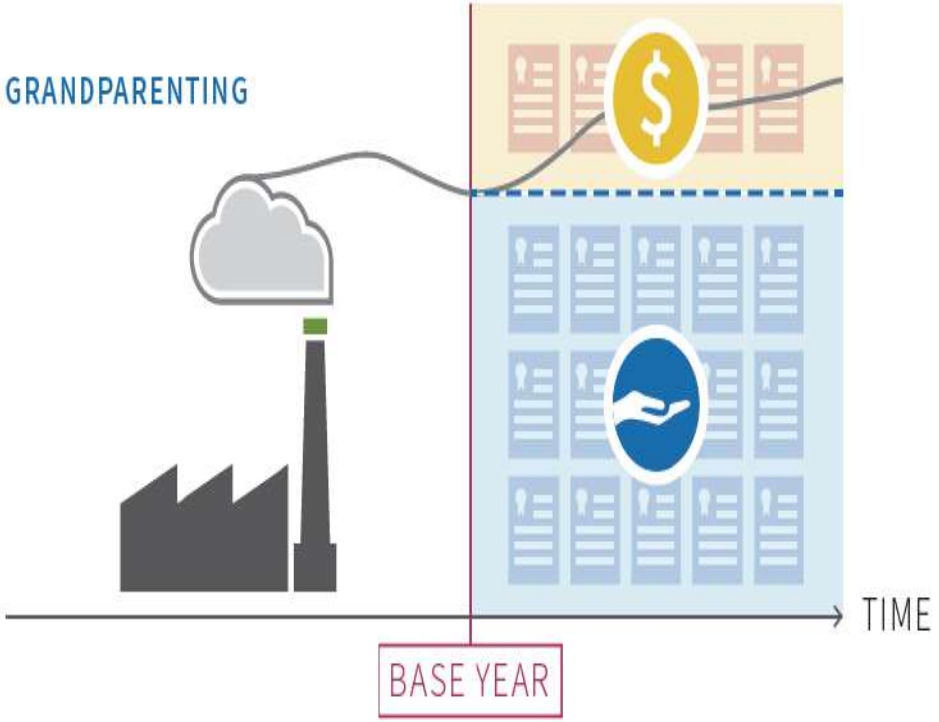
- Moreover, this method can generate revenue, rewards early action and promotes active carbon market by revealing a carbon price and encouraging trading.

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- In case of freely distributed transferrable permits, the government sets the limit of pollution and distributes permits by fixing the aggregate pollution goals.
- Here the firms do not have right to unlimited pollution but to emit a limited pollution without any charge.
- Once the initial endowment of distribution is over, trading becomes possible.
- Different methods of free allocation: Grand parenting; Benchmarking

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- Grand Parenting: Here companies receive free allowances based on their historical emissions from a specified period.

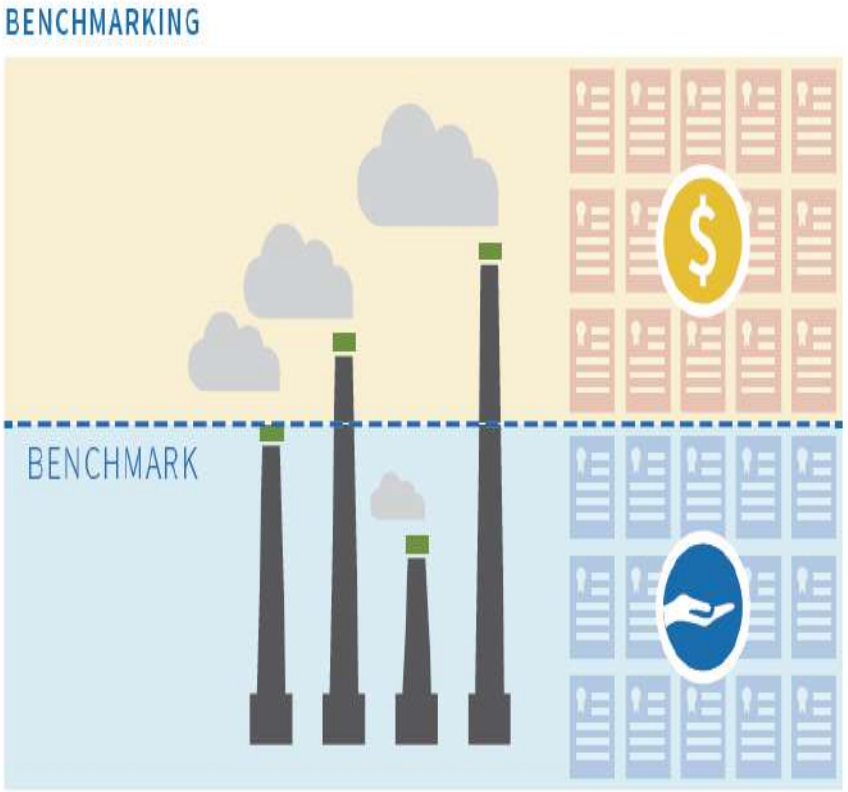


ICAP (2017), ETS brief 6, May

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- Benchmarking: In this method, companies receive free allowances depending on a set of performances standards, based on the emissions intensity of a product or across a sector.

- It may address fairness issues.



ICAP (2017), ETS brief 6, May

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- Merits of free permit distribution: When the entities get pollution permit for free, they are compensated for their existing carbon intensive infrastructure and processes.
- This free allocation of permits may be used to protect entities from the potential loss of competitiveness.
- Even though these entities receive permits for free, they are incentivised to invest in low-carbon technology.
  - If they reduce their emission they can sale their extra permits and if they increase their emissions they will face extra cost for buying permits.

Types of Trading Systems

- Offset trading, emission rate trading, cap and trade
- Offset trading: Its objective is how to achieve pollution reduction and continue economic growth by expanding output and production.
- Its mechanism targets the new firms to pay existing firms to reduce their emissions below standard so as to offset the added emissions of the new firms.
- Thus, trade between two firms transacting emissions following regulatory control or voluntary agreement is known as offset trading/credit trading.
- For example, clean development mechanisms (CDMs) under Kyoto Protocol.

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- Emission rate trading: It is expressed in terms of the rate that a pollutant constitutes in total output.

- For instance, greenhouse gas emission rate is defined ‘as tons of CO2 per 1000 megawatt hours of power production.’

- Once the base rate is set, trading is possible between sources either voluntarily or through regulations.

- The emitters, who can emit below the base rate, can sell their credits to those who want to emit above their apportioned base rate.

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- Cap and Trade: The regulator sets a cap (a limit) on overall emissions and allows trading among polluters to determine who emits what amount of pollution.
- This system is more centralized (states are responsible for their own national emission cap)
- Trading involves a price or value on a permit to pollute.
- This trading becomes an incentive to reduce pollution as firms realize polluting to be an expensive activity.
- For polluting firms, less pollution means fewer permits needed to be purchased. There is also an opportunity cost of emitting, by not emitting, the firm can sell more permits.

Suggested Readings

- ICAP .(2017). Allocation: How emission permits are distributed. brief 6, May,pp.1-2.

- Field, B.C. & Field, M.K. (2017). *Environmental Economics: An Introduction*, McGraw Hills, USA. Chapter 13.

Thank You

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ENVIRONMNETAL REGULATIONS AND BASIC REGULATORY INSTRUMENTS- MARKET TRADING SYSTEMS-II

DIPTIMAYEE NAYAK

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES



CONTENTS

- Cap and Trade-Example
- Marginal savings from polluting functions
- How to determine permit price
- Emission tax Vs. Cap and Trade
- Liability
- Mechanisms of Liability and Economic Incentives

Cap and Trade

- Suppose the government has introduced a cap and trade scheme to reduce the amount of any greenhouse gases emitted from power plants.
- The current total emissions= 1,50,000 tons/year and the policy makers target to reduce the level to 100,000 tons/year.
- As there are many firms in an industry, let's take the case of one single power plant which is currently emitting 5,000 tons /yr of the emission.
- Under the cap and trade provision, this particular firm has got the permit of 2500 tons/ year to emit.

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- What would be the strategy?

- In this case, the plant manager now has three alternative choices:
  - to reduce the emission to the initial permits level, i.e., to 2500 tons/yr.;
  - to buy additional permit of 2500 tons/yr. and emit at initial level of pollution (i.e., at 5000 tons/yr.); and
  - to sell the surplus permits available by reducing emission below the allotted permits of 2500 tons/yr.

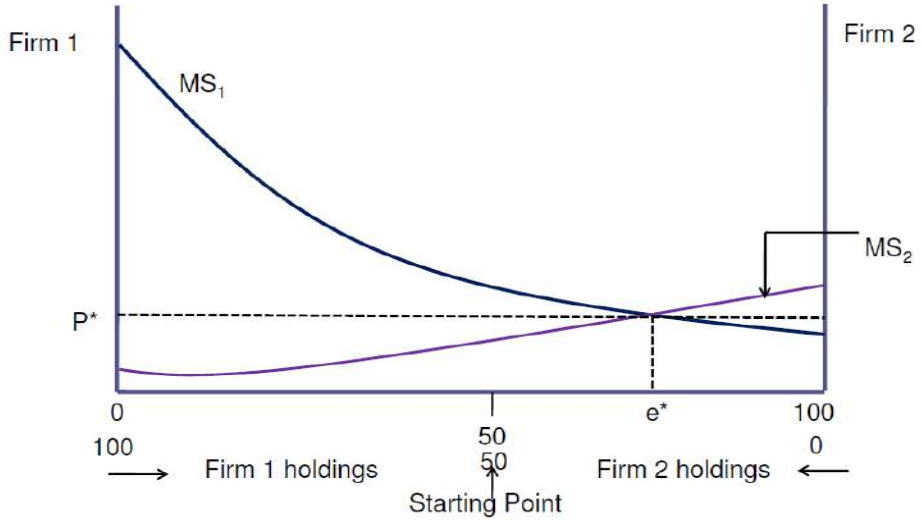
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- Let’s take another example:

- Assume a situation where two polluters exist and
- We are interested in allowing 100 units of pollution in total.
- Let’s start by giving each firm 50 permits.

Marginal Savings from Polluting Functions for - Firm 1 and Firm 2

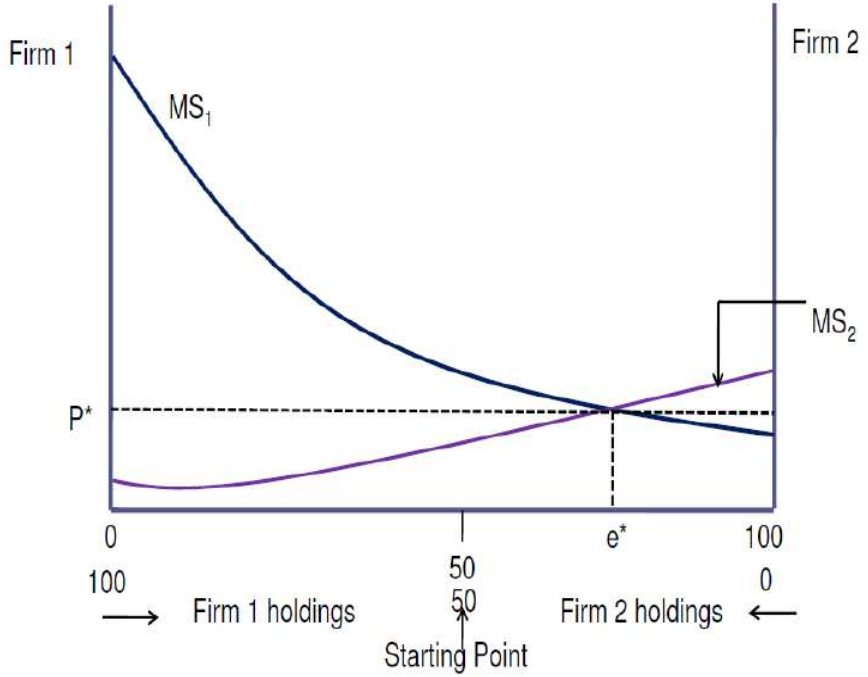
- The equilibrium price of permit is p*.
- MS_1 is the Marginal savings from emitting for firm 1; MS_2 Marginal savings from emitting for firm 2;
- e* is equilibrium holding of permits; p* is equilibrium price of permits. An emission fee of P* would achieve exactly the same outcome as the marketable permit.



Kolstad (2016)

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- Firm 1 is emitting more than the allotted permits at e*, while Firm 2 is emitting less than the allotted permits.
- Thus, Firm 1 can purchase extra permits (e*-50) from Firm 2.
- Trading can be continued till firm 1 holds e* and firm 2 holds 100-e* permits.



Kolstad (2016)

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- As we know that if marginal abatement cost (MAC) > permit price (P), the firm will buy extra permits and vice versa.

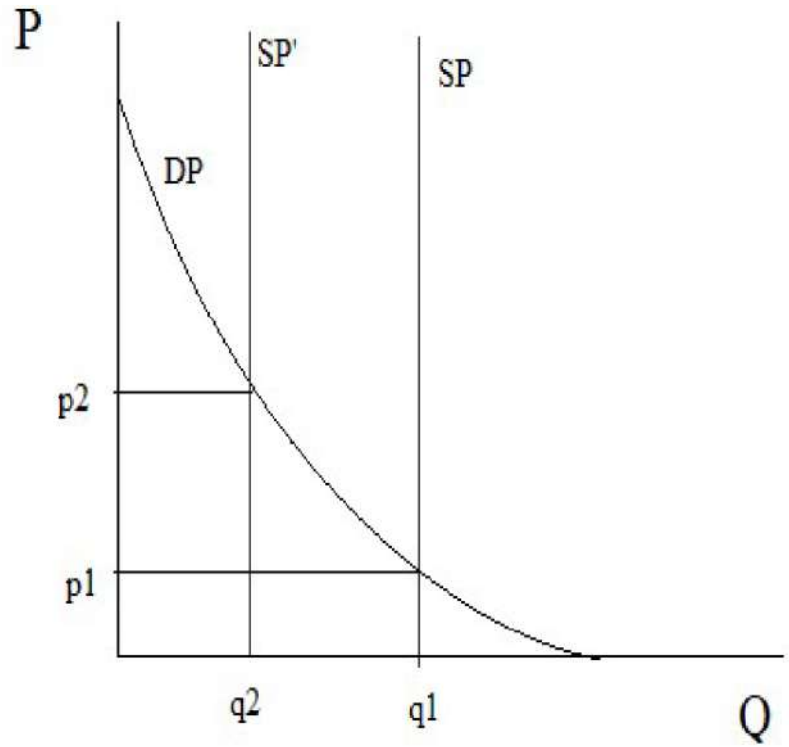
- If there is a large number of firms in the industry, by buying and selling, the firm will reach a point where Aggregate MACs = price of the permit.

- The aggregate MAC constitutes the demand for permits and the quantity of permits fixed by the regulating authority becomes the supply side in the transaction.

Price and Quantity Determination in A Permit Market

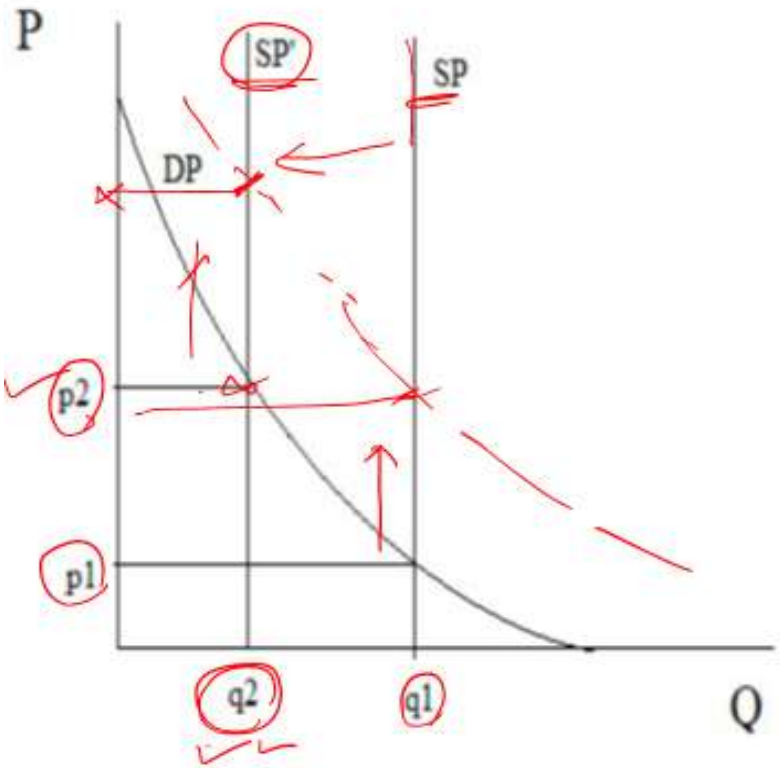
- Demand for permits: The aggregate MAC
-
- Supply for permits: the quantity of permits fixed by the regulating authority.

- In this case q_1 is the supply of permits (SP) allotted by the regulating body and demand curve (DP) is the aggregate of MACs of plants and therefore their intersecting point would determine the permit price p_1.



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- If there would be more restrictive quantity further like q_2, the permit price would rise to p_2.
- Similarly, if demand curve for permit shifts upward, the permit price will go up.
- The cost effectiveness in trading in this case requires that there would be a single market and single price of permit and all the transacting plants would equate their MACs with that single price.



Emission Tax Vs. Cap and Trade

- Emission fee: the marginal cost of emission control will be precisely known; less sure about the quantity of pollution.

- Marketable permit: exact amount of pollution can be known, but less sure about the marginal cost of pollution control.

- Why: under cap and trade, the quantity of restriction is fixed by the regulator and firms adjust their emission which determine the permit price;
- Whereas, emission price per unit of emission is fixed by the regulator and firms adjust their emission level.

Liability

- The basic idea is that if you harm someone, you must compensate that person for damage.
- For example, let's take the case of hazardous waste storage facility ('dump').
- The dump can take steps to minimize the risk of hazardous wastes leaking into the environment through 'Precaution'.
- But precaution is expensive and other things given constant, the dump would prefer to take little precaution.

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- Damage to society also depends upon the level of precaution.
- Both costs - to dump and damage to society are the functions of the level of precaution.
- The socially desirable level of precaution is the point at which the marginal costs of taking more precaution are just offset by the reduction in marginal damage from taking more basic regulatory mechanisms and economic incentives by the reduction in marginal damage from taking more precaution.

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- Liability and Economic Incentives: Negligence of liability works in the sense that the fear of being responsible from accident damages acts as sufficient incentives for firms to take the socially desirable amount of precaution.

- Because of the very principle of liability, polluters would get economic incentives to compensate for the damages they have created.

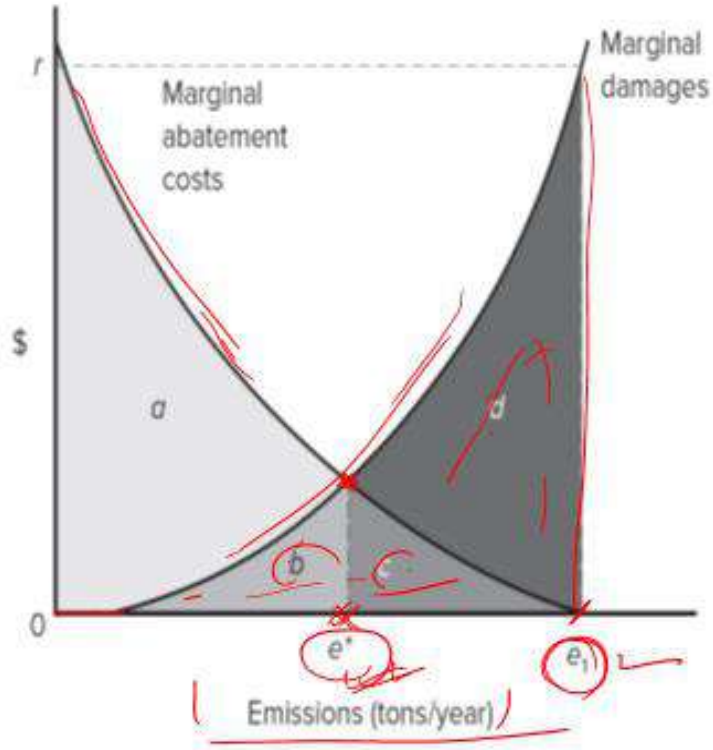
Mechanisms of Liability And Economic Incentives

- Let's take an example how liability principle works by taking the demand and supply concepts of liability.

- Here MAC is the demand curve for liability and MD is the supply curve for liability.

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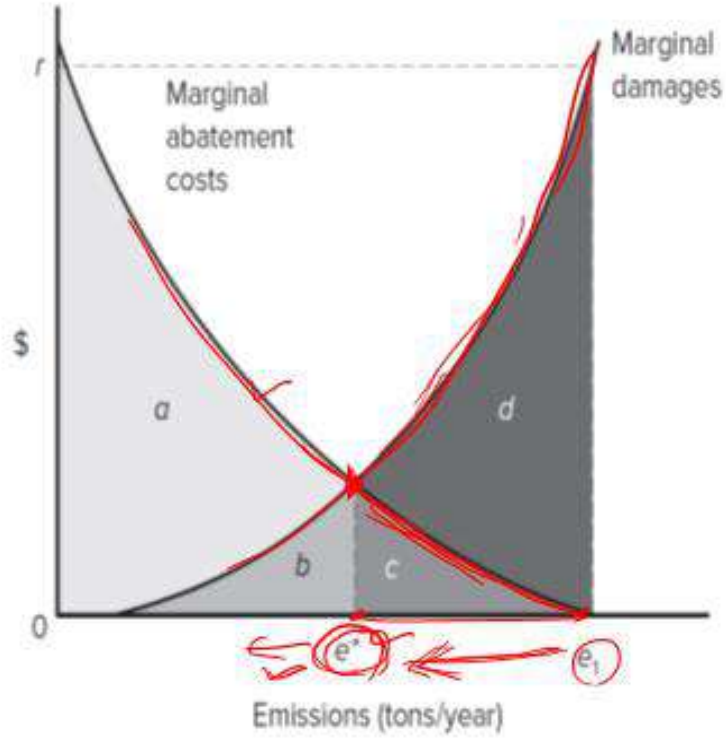
- The efficient level of emission is determined by the intersection point of demand and supply curve for liability at e*.
- However, let's take that the actual emission is at e_1, which is more than the efficient level e*.
- In the presence of liability principle, the cost of the damage has to be internalized.
- At e_1, total compensation cost the polluters need to pay is the area of damage i.e., b+c+d.



(Field and Field, 2017)

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- How to decrease compensation costs?
- The polluters could decrease the compensation costs by reducing emissions and increasing abatement costs.
- It can reduce the emission as long as MAC is less than the marginal damage costs and
- Till MAC < MD, liability would act as an economic incentive by reducing pollution to automatically achieve the efficient level of pollution e*.



(Field and Field, 2017)

Suggested Readings

- Field, B.C. & Field, M.K. (2017). *Environmental Economics: An Introduction*, McGraw Hills, USA.

- Kolstad, C. D. (2016). *Intermediate Environmental Economics*, Oxford University Press, New Delhi.

Thank You